Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 12:04:59

Measurement Details:

- **Sample:** Muestra6
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

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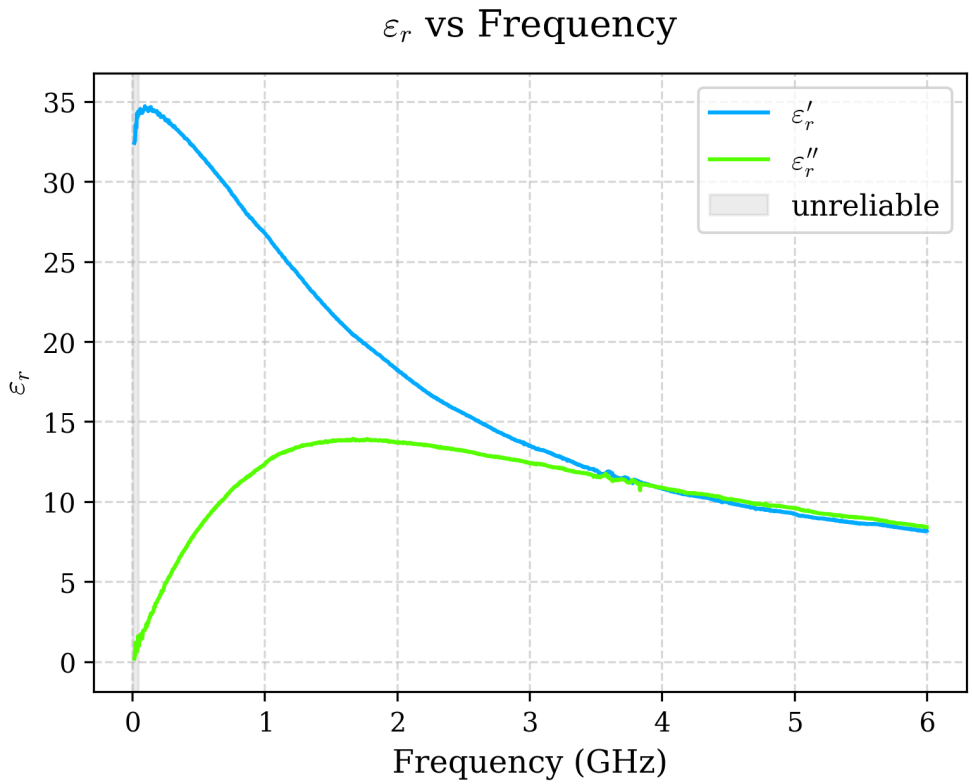
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	32.4095	0.1924	0.0059
0.020	32.5372	0.3995	0.0123
0.025	33.5269	1.1960	0.0357
0.030	33.3611	1.0250	0.0307
0.035	34.2536	0.6162	0.0180
0.040	34.0272	0.9425	0.0277
0.045	34.3771	1.6126	0.0469
0.050	34.1946	0.9834	0.0288
0.055	34.3685	1.4843	0.0432
0.060	34.5366	1.3547	0.0392
0.065	34.3625	1.7246	0.0502
0.070	34.3780	1.6025	0.0466
0.075	34.2803	1.4499	0.0423
0.080	34.3727	1.9146	0.0557
0.085	34.4671	1.9661	0.0570

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	34.4916	1.9554	0.0567
0.095	34.7054	1.9834	0.0572
0.100	34.5971	2.1064	0.0609
0.105	34.5426	2.3415	0.0678
0.110	34.5327	2.1926	0.0635
0.115	34.5695	2.4575	0.0711
0.120	34.3948	2.4233	0.0705
0.125	34.5732	2.6071	0.0754
0.130	34.4794	2.7802	0.0806
0.135	34.6263	2.8034	0.0810
0.140	34.6541	3.0221	0.0872
0.145	34.4734	2.9597	0.0859
0.150	34.4138	3.0662	0.0891
0.155	34.4733	3.0787	0.0893
0.160	34.3950	3.2157	0.0935
0.165	34.4488	3.4348	0.0997
0.170	34.3673	3.3982	0.0989
0.175	34.3772	3.6444	0.1060
0.180	34.3922	3.6828	0.1071
0.185	34.4079	3.6982	0.1075
0.190	34.2807	3.8314	0.1118
0.195	34.1709	3.9115	0.1145
0.200	34.1562	3.9903	0.1168
0.205	34.2593	4.0228	0.1174
0.210	34.1899	4.1797	0.1222
0.215	34.0253	4.1997	0.1234
0.220	34.0816	4.4229	0.1298
0.225	34.0360	4.3524	0.1279
0.230	34.0532	4.4172	0.1297
0.235	34.0765	4.5066	0.1322
0.240	33.9979	4.6658	0.1372
0.245	33.9161	4.8464	0.1429
0.250	33.8168	4.9060	0.1451
0.255	33.8714	4.9300	0.1456
0.260	33.8620	4.9922	0.1474
0.265	33.7910	5.1091	0.1512
0.270	33.7123	5.1764	0.1535
0.275	33.8247	5.2949	0.1565
0.280	33.6198	5.2990	0.1576
0.285	33.6608	5.3880	0.1601
0.290	33.6205	5.4395	0.1618
0.295	33.5579	5.5681	0.1659
0.300	33.5043	5.6333	0.1681
0.305	33.4832	5.7593	0.1720
0.310	33.4128	5.8039	0.1737
0.315	33.4778	5.8173	0.1738

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	33.4456	5.9004	0.1764
0.325	33.3425	6.0413	0.1812
0.330	33.2865	6.0582	0.1820
0.335	33.2565	6.1248	0.1842
0.340	33.2299	6.2257	0.1874
0.345	33.2133	6.2544	0.1883
0.350	33.1412	6.3771	0.1924
0.355	33.0786	6.5068	0.1967
0.360	33.0607	6.5260	0.1974
0.365	33.0243	6.5979	0.1998
0.370	32.9548	6.6437	0.2016
0.375	32.9181	6.7783	0.2059
0.380	32.9078	6.8033	0.2067
0.385	32.8488	6.8732	0.2092
0.390	32.8131	6.9540	0.2119
0.395	32.7160	6.9867	0.2136
0.400	32.6886	7.0365	0.2153
0.405	32.6901	7.1455	0.2186
0.410	32.6748	7.1516	0.2189
0.415	32.6194	7.2628	0.2227
0.420	32.5765	7.3447	0.2255
0.425	32.5333	7.4046	0.2276
0.430	32.4742	7.4982	0.2309
0.435	32.3680	7.5054	0.2319
0.440	32.3797	7.5956	0.2346
0.445	32.2920	7.7045	0.2386
0.450	32.2919	7.7480	0.2399
0.455	32.2322	7.7437	0.2402
0.460	32.2233	7.8780	0.2445
0.465	32.1706	7.8807	0.2450
0.470	32.1049	7.9832	0.2487
0.475	32.0420	8.0360	0.2508
0.480	32.0141	8.0987	0.2530
0.485	31.9398	8.1353	0.2547
0.490	31.9023	8.2144	0.2575
0.495	31.8516	8.2706	0.2597
0.500	31.8201	8.3067	0.2611
0.505	31.7781	8.3606	0.2631
0.510	31.7154	8.4123	0.2652
0.515	31.6322	8.4928	0.2685
0.520	31.6320	8.5019	0.2688
0.525	31.5648	8.5809	0.2719
0.530	31.5064	8.6503	0.2746
0.535	31.4789	8.7079	0.2766
0.540	31.4363	8.7737	0.2791
0.545	31.4143	8.8365	0.2813

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	31.3419	8.8456	0.2822
0.555	31.3074	8.9068	0.2845
0.560	31.1988	8.9560	0.2871
0.565	31.1800	9.0126	0.2891
0.570	31.1187	9.0835	0.2919
0.575	31.1155	9.1239	0.2932
0.580	31.0510	9.1771	0.2955
0.585	30.9727	9.1981	0.2970
0.590	30.9468	9.2381	0.2985
0.595	30.8805	9.3408	0.3025
0.600	30.8684	9.3837	0.3040
0.605	30.7938	9.4224	0.3060
0.610	30.7317	9.4551	0.3077
0.615	30.6859	9.5296	0.3106
0.620	30.6240	9.5253	0.3110
0.625	30.5572	9.6173	0.3147
0.630	30.5470	9.6359	0.3154
0.635	30.4727	9.7277	0.3192
0.640	30.4458	9.7491	0.3202
0.645	30.3719	9.7796	0.3220
0.650	30.3230	9.8452	0.3247
0.655	30.2793	9.9188	0.3276
0.660	30.2333	9.9074	0.3277
0.665	30.1935	9.9654	0.3301
0.670	30.1248	10.0362	0.3332
0.675	30.0949	10.0371	0.3335
0.680	29.9968	10.1080	0.3370
0.685	29.9766	10.1233	0.3377
0.690	29.9590	10.1532	0.3389
0.695	29.8706	10.1903	0.3411
0.700	29.7967	10.2466	0.3439
0.705	29.7864	10.3378	0.3471
0.710	29.7237	10.3299	0.3475
0.715	29.6809	10.4129	0.3508
0.720	29.6333	10.4844	0.3538
0.725	29.5767	10.5078	0.3553
0.730	29.5480	10.5273	0.3563
0.735	29.4262	10.5892	0.3599
0.740	29.4147	10.6524	0.3621
0.745	29.3520	10.7014	0.3646
0.750	29.2391	10.7274	0.3669
0.755	29.1860	10.7571	0.3686
0.760	29.1559	10.8224	0.3712
0.765	29.1055	10.8665	0.3733
0.770	29.0489	10.9154	0.3758
0.775	29.0322	10.9159	0.3760

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	28.9378	10.9665	0.3790
0.785	28.8680	11.0053	0.3812
0.790	28.8434	10.9939	0.3812
0.795	28.7656	11.0494	0.3841
0.800	28.6925	11.0683	0.3858
0.805	28.6241	11.1497	0.3895
0.810	28.5748	11.1710	0.3909
0.815	28.5519	11.2404	0.3937
0.820	28.4856	11.2632	0.3954
0.825	28.4297	11.2515	0.3958
0.830	28.4051	11.3110	0.3982
0.835	28.3116	11.3357	0.4004
0.840	28.2431	11.3440	0.4017
0.845	28.2395	11.3722	0.4027
0.850	28.1440	11.4074	0.4053
0.855	28.0836	11.4766	0.4087
0.860	28.0137	11.4576	0.4090
0.865	27.9482	11.5492	0.4132
0.870	27.9341	11.5865	0.4148
0.875	27.9347	11.5684	0.4141
0.880	27.8359	11.5577	0.4152
0.885	27.7750	11.6081	0.4179
0.890	27.7673	11.6877	0.4209
0.895	27.6951	11.6751	0.4216
0.900	27.7114	11.6985	0.4222
0.905	27.6605	11.7752	0.4257
0.910	27.5991	11.8033	0.4277
0.915	27.5120	11.8265	0.4299
0.920	27.4792	11.8268	0.4304
0.925	27.4542	11.9290	0.4345
0.930	27.3873	11.9399	0.4360
0.935	27.3060	11.9516	0.4377
0.940	27.2946	12.0145	0.4402
0.945	27.2320	12.0165	0.4413
0.950	27.2076	12.0270	0.4420
0.955	27.1572	12.0788	0.4448
0.960	27.0967	12.0884	0.4461
0.965	27.0580	12.1554	0.4492
0.970	27.0349	12.1543	0.4496
0.975	26.9746	12.1666	0.4510
0.980	26.9569	12.2281	0.4536
0.985	26.9288	12.2449	0.4547
0.990	26.8554	12.2793	0.4572
0.995	26.8266	12.3160	0.4591
1.000	26.7837	12.3162	0.4598
1.005	26.7236	12.4032	0.4641

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	26.6763	12.4606	0.4671
1.015	26.6003	12.4541	0.4682
1.020	26.5593	12.5008	0.4707
1.025	26.5511	12.5539	0.4728
1.030	26.4736	12.5743	0.4750
1.035	26.4353	12.6157	0.4772
1.040	26.3694	12.6553	0.4799
1.045	26.3040	12.6716	0.4817
1.050	26.2461	12.7131	0.4844
1.055	26.1579	12.6978	0.4854
1.060	26.1368	12.7802	0.4890
1.065	26.0761	12.7330	0.4883
1.070	26.0354	12.7963	0.4915
1.075	25.9423	12.8258	0.4944
1.080	25.9063	12.8201	0.4949
1.085	25.8708	12.8341	0.4961
1.090	25.8275	12.8964	0.4993
1.095	25.7526	12.9213	0.5017
1.100	25.7220	12.9587	0.5038
1.105	25.6509	12.9484	0.5048
1.110	25.5798	12.9742	0.5072
1.115	25.5361	13.0045	0.5093
1.120	25.5133	13.0078	0.5098
1.125	25.4738	13.0213	0.5112
1.130	25.3958	13.0594	0.5142
1.135	25.3171	13.0504	0.5155
1.140	25.2929	13.0367	0.5154
1.145	25.2403	13.1068	0.5193
1.150	25.2352	13.0941	0.5189
1.155	25.1446	13.1399	0.5226
1.160	25.0752	13.0982	0.5224
1.165	25.0315	13.1661	0.5260
1.170	24.9850	13.1462	0.5262
1.175	24.8938	13.1642	0.5288
1.180	24.8758	13.2214	0.5315
1.185	24.8587	13.2036	0.5311
1.190	24.7852	13.2036	0.5327
1.195	24.7439	13.2463	0.5353
1.200	24.7017	13.2731	0.5373
1.205	24.6108	13.2160	0.5370
1.210	24.5955	13.2767	0.5398
1.215	24.5311	13.2965	0.5420
1.220	24.5326	13.3089	0.5425
1.225	24.4879	13.3259	0.5442
1.230	24.4407	13.3746	0.5472
1.235	24.3603	13.3579	0.5483

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	24.3417	13.3914	0.5501
1.245	24.2788	13.4018	0.5520
1.250	24.1941	13.3895	0.5534
1.255	24.1439	13.3991	0.5550
1.260	24.1155	13.4091	0.5560
1.265	24.0363	13.4447	0.5593
1.270	24.0156	13.4456	0.5599
1.275	23.9739	13.4799	0.5623
1.280	23.9171	13.4841	0.5638
1.285	23.8853	13.5408	0.5669
1.290	23.8178	13.4789	0.5659
1.295	23.7391	13.5092	0.5691
1.300	23.6931	13.5222	0.5707
1.305	23.6498	13.5334	0.5722
1.310	23.5959	13.5455	0.5741
1.315	23.5590	13.5548	0.5754
1.320	23.5011	13.5364	0.5760
1.325	23.4456	13.5694	0.5788
1.330	23.3868	13.5633	0.5800
1.335	23.3355	13.5606	0.5811
1.340	23.3114	13.5748	0.5823
1.345	23.2270	13.5570	0.5837
1.350	23.1789	13.5860	0.5861
1.355	23.1354	13.5456	0.5855
1.360	23.1233	13.6129	0.5887
1.365	23.0684	13.6283	0.5908
1.370	23.0283	13.6342	0.5921
1.375	23.0073	13.6467	0.5931
1.380	22.9239	13.6417	0.5951
1.385	22.8628	13.6605	0.5975
1.390	22.8157	13.6399	0.5978
1.395	22.7579	13.6818	0.6012
1.400	22.7528	13.6850	0.6015
1.405	22.6700	13.6822	0.6035
1.410	22.6591	13.7123	0.6052
1.415	22.6024	13.6706	0.6048
1.420	22.5700	13.7218	0.6080
1.425	22.5250	13.6694	0.6069
1.430	22.4347	13.6951	0.6104
1.435	22.4043	13.6867	0.6109
1.440	22.3459	13.7001	0.6131
1.445	22.3082	13.7059	0.6144
1.450	22.2901	13.7386	0.6164
1.455	22.2302	13.7451	0.6183
1.460	22.1674	13.7854	0.6219
1.465	22.1299	13.7720	0.6223

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	22.0881	13.7493	0.6225
1.475	22.0564	13.7025	0.6213
1.480	22.0187	13.7605	0.6249
1.485	21.9518	13.7517	0.6264
1.490	21.9288	13.7655	0.6277
1.495	21.8677	13.7643	0.6294
1.500	21.8070	13.7889	0.6323
1.505	21.7612	13.7907	0.6337
1.510	21.7278	13.7623	0.6334
1.515	21.7079	13.8405	0.6376
1.520	21.6281	13.8113	0.6386
1.525	21.5855	13.8259	0.6405
1.530	21.5601	13.7948	0.6398
1.535	21.5180	13.7975	0.6412
1.540	21.4663	13.7975	0.6427
1.545	21.4002	13.8390	0.6467
1.550	21.4051	13.8296	0.6461
1.555	21.3595	13.8178	0.6469
1.560	21.3041	13.8362	0.6495
1.565	21.2475	13.8439	0.6516
1.570	21.2068	13.8220	0.6518
1.575	21.1631	13.8436	0.6541
1.580	21.1621	13.8302	0.6535
1.585	21.1050	13.8143	0.6545
1.590	21.0521	13.8346	0.6572
1.595	21.0168	13.8379	0.6584
1.600	20.9745	13.8397	0.6598
1.605	20.9417	13.8433	0.6610
1.610	20.9029	13.8127	0.6608
1.615	20.8478	13.8312	0.6634
1.620	20.8131	13.8652	0.6662
1.625	20.7725	13.8662	0.6675
1.630	20.7289	13.8199	0.6667
1.635	20.6924	13.8640	0.6700
1.640	20.6727	13.8682	0.6708
1.645	20.6109	13.8602	0.6725
1.650	20.5682	13.8197	0.6719
1.655	20.5407	13.8325	0.6734
1.660	20.4629	13.8615	0.6774
1.665	20.4602	13.9291	0.6808
1.670	20.4154	13.8838	0.6801
1.675	20.3644	13.8589	0.6805
1.680	20.3723	13.8929	0.6820
1.685	20.3494	13.8114	0.6787
1.690	20.2969	13.8759	0.6836
1.695	20.2764	13.8493	0.6830

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	20.2332	13.8482	0.6844
1.705	20.1818	13.8538	0.6864
1.710	20.1570	13.8368	0.6865
1.715	20.1546	13.8098	0.6852
1.720	20.0685	13.8360	0.6894
1.725	20.0604	13.8244	0.6891
1.730	20.0526	13.8588	0.6911
1.735	19.9909	13.8525	0.6929
1.740	19.9517	13.8504	0.6942
1.745	19.8844	13.8587	0.6970
1.750	19.9123	13.8205	0.6941
1.755	19.8708	13.8678	0.6979
1.760	19.8358	13.8533	0.6984
1.765	19.8186	13.8834	0.7005
1.770	19.7585	13.8620	0.7016
1.775	19.7296	13.9151	0.7053
1.780	19.7121	13.8672	0.7035
1.785	19.6397	13.8765	0.7066
1.790	19.6393	13.8489	0.7052
1.795	19.6065	13.8697	0.7074
1.800	19.5766	13.8320	0.7066
1.805	19.5179	13.8297	0.7086
1.810	19.5081	13.8371	0.7093
1.815	19.4808	13.8304	0.7099
1.820	19.4577	13.8385	0.7112
1.825	19.4233	13.8467	0.7129
1.830	19.3658	13.8306	0.7142
1.835	19.3309	13.8488	0.7164
1.840	19.2956	13.8438	0.7175
1.845	19.2664	13.8364	0.7182
1.850	19.2297	13.8304	0.7192
1.855	19.2100	13.8238	0.7196
1.860	19.1877	13.8054	0.7195
1.865	19.1790	13.8342	0.7213
1.870	19.1490	13.8405	0.7228
1.875	19.0959	13.8451	0.7250
1.880	19.0675	13.8267	0.7251
1.885	19.0098	13.8017	0.7260
1.890	18.9867	13.7941	0.7265
1.895	18.9431	13.7942	0.7282
1.900	18.8947	13.8269	0.7318
1.905	18.8866	13.8033	0.7308
1.910	18.8560	13.7969	0.7317
1.915	18.8123	13.8168	0.7345
1.920	18.7967	13.7977	0.7340
1.925	18.7605	13.8253	0.7369

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	18.7396	13.7651	0.7345
1.935	18.6975	13.7871	0.7374
1.940	18.6578	13.7217	0.7354
1.945	18.5853	13.7653	0.7407
1.950	18.5612	13.7494	0.7408
1.955	18.5340	13.7676	0.7428
1.960	18.5275	13.7472	0.7420
1.965	18.4672	13.7392	0.7440
1.970	18.4529	13.7340	0.7443
1.975	18.4079	13.7607	0.7475
1.980	18.3680	13.7088	0.7463
1.985	18.3608	13.7013	0.7462
1.990	18.3071	13.7037	0.7485
1.995	18.2731	13.7017	0.7498
2.000	18.2132	13.7115	0.7528
2.005	18.2314	13.7025	0.7516
2.010	18.1632	13.7303	0.7559
2.015	18.1341	13.6706	0.7539
2.020	18.0961	13.6724	0.7555
2.025	18.0673	13.6977	0.7581
2.030	18.0757	13.6755	0.7566
2.035	18.0190	13.6861	0.7595
2.040	17.9972	13.6863	0.7605
2.045	17.9303	13.6441	0.7610
2.050	17.9175	13.7196	0.7657
2.055	17.9116	13.6970	0.7647
2.060	17.8425	13.6885	0.7672
2.065	17.8513	13.6884	0.7668
2.070	17.7795	13.6501	0.7677
2.075	17.7362	13.6264	0.7683
2.080	17.7142	13.6495	0.7705
2.085	17.6900	13.6523	0.7718
2.090	17.6742	13.6567	0.7727
2.095	17.6481	13.6437	0.7731
2.100	17.5801	13.6716	0.7777
2.105	17.5707	13.6485	0.7768
2.110	17.5537	13.6527	0.7778
2.115	17.4969	13.6337	0.7792
2.120	17.4650	13.5986	0.7786
2.125	17.4150	13.6053	0.7812
2.130	17.4027	13.6053	0.7818
2.135	17.3800	13.6115	0.7832
2.140	17.3551	13.6119	0.7843
2.145	17.3132	13.5899	0.7849
2.150	17.3146	13.5962	0.7852
2.155	17.2631	13.5834	0.7868

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	17.2214	13.5813	0.7886
2.165	17.2094	13.5907	0.7897
2.170	17.1532	13.5735	0.7913
2.175	17.1379	13.5864	0.7928
2.180	17.0999	13.5718	0.7937
2.185	17.0827	13.5756	0.7947
2.190	17.0483	13.5692	0.7959
2.195	16.9889	13.5388	0.7969
2.200	16.9691	13.5519	0.7986
2.205	16.9656	13.5239	0.7971
2.210	16.9269	13.5645	0.8014
2.215	16.8987	13.5264	0.8004
2.220	16.8656	13.5270	0.8020
2.225	16.8174	13.5428	0.8053
2.230	16.7924	13.5311	0.8058
2.235	16.7810	13.5368	0.8067
2.240	16.7734	13.5056	0.8052
2.245	16.7023	13.5212	0.8095
2.250	16.6953	13.5000	0.8086
2.255	16.6794	13.4931	0.8090
2.260	16.6129	13.4996	0.8126
2.265	16.5982	13.4993	0.8133
2.270	16.5948	13.4544	0.8108
2.275	16.5392	13.4521	0.8133
2.280	16.5345	13.4453	0.8132
2.285	16.5079	13.4721	0.8161
2.290	16.4767	13.4526	0.8165
2.295	16.4929	13.4339	0.8145
2.300	16.4536	13.3955	0.8141
2.305	16.3829	13.4269	0.8196
2.310	16.3959	13.3754	0.8158
2.315	16.3772	13.4083	0.8187
2.320	16.3357	13.4069	0.8207
2.325	16.3093	13.3554	0.8189
2.330	16.2734	13.4329	0.8254
2.335	16.2542	13.3945	0.8241
2.340	16.2699	13.3654	0.8215
2.345	16.1947	13.3657	0.8253
2.350	16.1737	13.3689	0.8266
2.355	16.1652	13.3241	0.8242
2.360	16.1327	13.3658	0.8285
2.365	16.1122	13.3662	0.8296
2.370	16.0988	13.3352	0.8283
2.375	16.0959	13.3062	0.8267
2.380	16.0503	13.3478	0.8316
2.385	16.0141	13.3276	0.8322

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	16.0139	13.3388	0.8330
2.395	15.9516	13.3360	0.8360
2.400	15.9273	13.3168	0.8361
2.405	15.9317	13.2747	0.8332
2.410	15.9066	13.2962	0.8359
2.415	15.8695	13.2830	0.8370
2.420	15.8481	13.3094	0.8398
2.425	15.8354	13.2652	0.8377
2.430	15.8214	13.2598	0.8381
2.435	15.8191	13.2676	0.8387
2.440	15.7788	13.2504	0.8398
2.445	15.7553	13.2472	0.8408
2.450	15.7413	13.2512	0.8418
2.455	15.7068	13.2249	0.8420
2.460	15.6957	13.2317	0.8430
2.465	15.6798	13.2530	0.8452
2.470	15.6535	13.2111	0.8440
2.475	15.6350	13.2123	0.8450
2.480	15.6062	13.2050	0.8461
2.485	15.5885	13.2203	0.8481
2.490	15.5621	13.1966	0.8480
2.495	15.5288	13.1944	0.8497
2.500	15.5396	13.1920	0.8489
2.505	15.5079	13.1675	0.8491
2.510	15.4855	13.1749	0.8508
2.515	15.4582	13.1602	0.8513
2.520	15.4334	13.1633	0.8529
2.525	15.3878	13.1300	0.8533
2.530	15.3789	13.1377	0.8543
2.535	15.3807	13.1448	0.8546
2.540	15.3512	13.1159	0.8544
2.545	15.3378	13.1036	0.8543
2.550	15.2980	13.1026	0.8565
2.555	15.3018	13.0988	0.8560
2.560	15.2637	13.0979	0.8581
2.565	15.2415	13.0877	0.8587
2.570	15.2282	13.0769	0.8587
2.575	15.2049	13.0755	0.8600
2.580	15.1722	13.0475	0.8600
2.585	15.1353	13.0667	0.8633
2.590	15.1305	13.0266	0.8609
2.595	15.1000	13.0400	0.8636
2.600	15.0744	13.0109	0.8631
2.605	15.0423	12.9888	0.8635
2.610	15.0614	13.0066	0.8636
2.615	15.0381	12.9715	0.8626

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	14.9812	12.9731	0.8660
2.625	14.9576	12.9527	0.8660
2.630	14.9446	12.9679	0.8677
2.635	14.9454	12.9484	0.8664
2.640	14.8886	12.9395	0.8691
2.645	14.8914	12.9378	0.8688
2.650	14.8427	12.9290	0.8711
2.655	14.8517	12.9068	0.8690
2.660	14.8055	12.9128	0.8722
2.665	14.8226	12.9087	0.8709
2.670	14.7977	12.8721	0.8699
2.675	14.7486	12.8724	0.8728
2.680	14.7405	12.8642	0.8727
2.685	14.6990	12.8578	0.8747
2.690	14.6843	12.8447	0.8747
2.695	14.6669	12.8549	0.8765
2.700	14.6452	12.8535	0.8777
2.705	14.6313	12.8410	0.8776
2.710	14.5920	12.8559	0.8810
2.715	14.5872	12.8240	0.8791
2.720	14.5535	12.8262	0.8813
2.725	14.5101	12.8012	0.8822
2.730	14.4832	12.7962	0.8835
2.735	14.4661	12.8219	0.8863
2.740	14.4839	12.7775	0.8822
2.745	14.4353	12.8211	0.8882
2.750	14.4076	12.8020	0.8886
2.755	14.3770	12.7887	0.8895
2.760	14.3835	12.7980	0.8898
2.765	14.3501	12.7614	0.8893
2.770	14.3389	12.8025	0.8928
2.775	14.3106	12.7477	0.8908
2.780	14.3003	12.7512	0.8917
2.785	14.2759	12.7590	0.8937
2.790	14.2774	12.7382	0.8922
2.795	14.2343	12.7523	0.8959
2.800	14.2151	12.7309	0.8956
2.805	14.1999	12.7374	0.8970
2.810	14.1536	12.7025	0.8975
2.815	14.1474	12.7129	0.8986
2.820	14.1439	12.6991	0.8979
2.825	14.1215	12.6807	0.8980
2.830	14.1089	12.6969	0.8999
2.835	14.1005	12.6940	0.9002
2.840	14.0560	12.6886	0.9027
2.845	14.0358	12.6843	0.9037

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	14.0477	12.6385	0.8997
2.855	14.0147	12.6642	0.9036
2.860	13.9808	12.6284	0.9033
2.865	13.9407	12.6296	0.9060
2.870	13.9309	12.6210	0.9060
2.875	13.9251	12.6073	0.9054
2.880	13.8990	12.6308	0.9088
2.885	13.8973	12.6244	0.9084
2.890	13.8419	12.6338	0.9127
2.895	13.8421	12.5863	0.9093
2.900	13.8387	12.5872	0.9096
2.905	13.7996	12.6057	0.9135
2.910	13.7785	12.6024	0.9146
2.915	13.7757	12.5752	0.9129
2.920	13.7711	12.5822	0.9137
2.925	13.7818	12.5520	0.9108
2.930	13.7443	12.5544	0.9134
2.935	13.7146	12.5401	0.9144
2.940	13.7205	12.5458	0.9144
2.945	13.6854	12.5371	0.9161
2.950	13.6516	12.5136	0.9166
2.955	13.7094	12.5219	0.9134
2.960	13.5735	12.5166	0.9221
2.965	13.6507	12.4880	0.9148
2.970	13.5985	12.4516	0.9157
2.975	13.5884	12.4577	0.9168
2.980	13.5879	12.4710	0.9178
2.985	13.5493	12.4376	0.9180
2.990	13.5073	12.4531	0.9220
2.995	13.5210	12.4167	0.9183
3.000	13.4671	12.4228	0.9225
3.005	13.4890	12.4230	0.9210
3.010	13.4728	12.3729	0.9184
3.015	13.4486	12.3727	0.9200
3.020	13.3940	12.4120	0.9267
3.025	13.3889	12.3755	0.9243
3.030	13.4018	12.3559	0.9220
3.035	13.3919	12.3865	0.9249
3.040	13.3489	12.3660	0.9264
3.045	13.3324	12.3469	0.9261
3.050	13.2877	12.3157	0.9269
3.055	13.2953	12.3815	0.9313
3.060	13.2780	12.3719	0.9318
3.065	13.3138	12.3532	0.9279
3.070	13.2476	12.3683	0.9336
3.075	13.2583	12.3671	0.9328

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	13.2459	12.3694	0.9338
3.085	13.2266	12.3191	0.9314
3.090	13.2225	12.3369	0.9330
3.095	13.2185	12.3410	0.9336
3.100	13.2279	12.3307	0.9322
3.105	13.2020	12.2935	0.9312
3.110	13.1927	12.3119	0.9332
3.115	13.1848	12.2913	0.9322
3.120	13.1552	12.2961	0.9347
3.125	13.1434	12.2652	0.9332
3.130	13.1206	12.2506	0.9337
3.135	13.0938	12.2318	0.9342
3.140	13.1025	12.2310	0.9335
3.145	13.0648	12.2024	0.9340
3.150	13.0632	12.2083	0.9346
3.155	13.0407	12.2079	0.9361
3.160	13.0473	12.1881	0.9341
3.165	12.9895	12.1830	0.9379
3.170	13.0104	12.1583	0.9345
3.175	12.9615	12.1310	0.9359
3.180	12.9611	12.1366	0.9364
3.185	12.9124	12.1511	0.9410
3.190	12.9250	12.1400	0.9393
3.195	12.8894	12.1396	0.9418
3.200	12.8618	12.1235	0.9426
3.205	12.8538	12.1408	0.9445
3.210	12.8393	12.1170	0.9437
3.215	12.8420	12.0923	0.9416
3.220	12.8562	12.1206	0.9428
3.225	12.7911	12.1345	0.9487
3.230	12.8208	12.0779	0.9421
3.235	12.7881	12.1091	0.9469
3.240	12.7559	12.0785	0.9469
3.245	12.7698	12.0599	0.9444
3.250	12.6792	12.0996	0.9543
3.255	12.6806	12.0419	0.9496
3.260	12.7102	12.0449	0.9477
3.265	12.6704	12.0659	0.9523
3.270	12.6600	11.9808	0.9463
3.275	12.6457	11.9849	0.9477
3.280	12.6479	12.0022	0.9490
3.285	12.5510	11.9716	0.9538
3.290	12.6096	11.9493	0.9476
3.295	12.5607	11.9302	0.9498
3.300	12.5851	11.9465	0.9493
3.305	12.5459	11.9519	0.9527

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	12.5293	11.9303	0.9522
3.315	12.5011	11.9071	0.9525
3.320	12.5010	11.9308	0.9544
3.325	12.5052	11.9098	0.9524
3.330	12.4571	11.9106	0.9561
3.335	12.4243	11.8582	0.9544
3.340	12.4268	11.8799	0.9560
3.345	12.3977	11.8445	0.9554
3.350	12.3609	11.8399	0.9578
3.355	12.3757	11.8365	0.9564
3.360	12.3218	11.8375	0.9607
3.365	12.3137	11.8286	0.9606
3.370	12.2873	11.8260	0.9625
3.375	12.3103	11.8471	0.9624
3.380	12.2940	11.8344	0.9626
3.385	12.2587	11.8173	0.9640
3.390	12.2491	11.8012	0.9634
3.395	12.2410	11.8012	0.9641
3.400	12.1786	11.7989	0.9688
3.405	12.1863	11.7776	0.9665
3.410	12.1828	11.7992	0.9685
3.415	12.1391	11.7777	0.9702
3.420	12.1590	11.7706	0.9681
3.425	12.1589	11.7665	0.9677
3.430	12.1534	11.7647	0.9680
3.435	12.1244	11.7838	0.9719
3.440	12.1374	11.7297	0.9664
3.445	12.0964	11.7253	0.9693
3.450	12.0411	11.7004	0.9717
3.455	12.0643	11.7221	0.9716
3.460	12.0423	11.7028	0.9718
3.465	12.0343	11.6882	0.9712
3.470	12.0425	11.6798	0.9699
3.475	12.0382	11.6479	0.9676
3.480	12.0317	11.6149	0.9654
3.485	12.0420	11.5464	0.9588
3.490	11.9558	11.6274	0.9725
3.495	11.9362	11.6140	0.9730
3.500	11.9452	11.5678	0.9684
3.505	11.9641	11.5989	0.9695
3.510	11.9397	11.5309	0.9658
3.515	11.8871	11.5658	0.9730
3.520	11.8716	11.5245	0.9708
3.525	11.7603	11.5464	0.9818
3.530	11.8102	11.5742	0.9800
3.535	11.7621	11.5145	0.9789

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	11.7107	11.5504	0.9863
3.545	11.7379	11.5911	0.9875
3.550	11.6960	11.5976	0.9916
3.555	11.7136	11.5878	0.9893
3.560	11.7220	11.6300	0.9922
3.565	11.7545	11.6219	0.9887
3.570	11.6846	11.6368	0.9959
3.575	11.6645	11.7423	1.0067
3.580	11.7479	11.7117	0.9969
3.585	11.8626	11.6680	0.9836
3.590	11.8704	11.5523	0.9732
3.595	11.8902	11.5099	0.9680
3.600	11.8346	11.5302	0.9743
3.605	11.8750	11.4500	0.9642
3.610	11.8486	11.4348	0.9651
3.615	11.7685	11.4005	0.9687
3.620	11.7282	11.3570	0.9683
3.625	11.7875	11.2980	0.9585
3.630	11.6135	11.2787	0.9712
3.635	11.6066	11.3193	0.9752
3.640	11.6154	11.3528	0.9774
3.645	11.5823	11.3568	0.9805
3.650	11.5403	11.3529	0.9838
3.655	11.5239	11.3375	0.9838
3.660	11.4988	11.3799	0.9897
3.665	11.5041	11.3506	0.9867
3.670	11.4640	11.3662	0.9915
3.675	11.4757	11.3436	0.9885
3.680	11.4633	11.3630	0.9913
3.685	11.4520	11.3838	0.9940
3.690	11.4530	11.3783	0.9935
3.695	11.4460	11.4010	0.9961
3.700	11.4635	11.3989	0.9944
3.705	11.4855	11.3895	0.9916
3.710	11.4626	11.3777	0.9926
3.715	11.4932	11.3758	0.9898
3.720	11.5289	11.3390	0.9835
3.725	11.4992	11.3356	0.9858
3.730	11.5012	11.2919	0.9818
3.735	11.4806	11.2326	0.9784
3.740	11.4771	11.2151	0.9772
3.745	11.4476	11.2002	0.9784
3.750	11.3776	11.1757	0.9823
3.755	11.3362	11.1861	0.9868
3.760	11.2863	11.2038	0.9927
3.765	11.2427	11.2339	0.9992

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	11.2692	11.3145	1.0040
3.775	11.3310	11.2867	0.9961
3.780	11.1477	11.3547	1.0186
3.785	11.3222	11.3205	0.9999
3.790	11.3717	11.2359	0.9881
3.795	11.3492	11.2292	0.9894
3.800	11.3129	11.2816	0.9972
3.805	11.3445	11.2039	0.9876
3.810	11.2934	11.1809	0.9900
3.815	11.2775	11.1205	0.9861
3.820	11.2850	11.1624	0.9891
3.825	11.2645	11.1071	0.9860
3.830	11.2531	11.1305	0.9891
3.835	11.0468	10.7109	0.9696
3.840	11.2096	11.1139	0.9915
3.845	11.1829	11.1125	0.9937
3.850	11.1833	11.0943	0.9920
3.855	11.1560	11.0976	0.9948
3.860	11.1388	11.0674	0.9936
3.865	11.0938	11.0657	0.9975
3.870	11.1394	11.0761	0.9943
3.875	11.1013	11.0606	0.9963
3.880	11.1189	11.0598	0.9947
3.885	11.0792	11.0563	0.9979
3.890	11.0804	11.0694	0.9990
3.895	11.0891	11.0218	0.9939
3.900	11.0597	11.0117	0.9957
3.905	11.0098	11.0250	1.0014
3.910	11.0364	11.0085	0.9975
3.915	11.0220	11.0006	0.9981
3.920	11.0007	11.0156	1.0014
3.925	10.9938	10.9836	0.9991
3.930	10.9868	10.9773	0.9991
3.935	10.9904	10.9690	0.9981
3.940	10.9579	10.9657	1.0007
3.945	10.9500	10.9475	0.9998
3.950	10.9307	10.9325	1.0002
3.955	10.9098	10.9379	1.0026
3.960	10.9072	10.9155	1.0008
3.965	10.9064	10.9160	1.0009
3.970	10.8811	10.9170	1.0033
3.975	10.8695	10.9031	1.0031
3.980	10.8634	10.8940	1.0028
3.985	10.8582	10.8998	1.0038
3.990	10.8554	10.8925	1.0034
3.995	10.8350	10.8710	1.0033

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	10.8168	10.8645	1.0044
4.005	10.8109	10.8555	1.0041
4.010	10.7824	10.8433	1.0056
4.015	10.7785	10.8562	1.0072
4.020	10.7738	10.8354	1.0057
4.025	10.7935	10.8274	1.0031
4.030	10.7479	10.8137	1.0061
4.035	10.7283	10.8184	1.0084
4.040	10.7350	10.8119	1.0072
4.045	10.7341	10.7883	1.0051
4.050	10.7109	10.7712	1.0056
4.055	10.7011	10.7564	1.0052
4.060	10.6769	10.7592	1.0077
4.065	10.6517	10.7489	1.0091
4.070	10.6619	10.7626	1.0094
4.075	10.6453	10.7419	1.0091
4.080	10.6272	10.7571	1.0122
4.085	10.6253	10.7395	1.0107
4.090	10.6200	10.7262	1.0100
4.095	10.5884	10.7301	1.0134
4.100	10.5945	10.7387	1.0136
4.105	10.5949	10.6624	1.0064
4.110	10.5888	10.7089	1.0113
4.115	10.5646	10.6951	1.0123
4.120	10.5441	10.6809	1.0130
4.125	10.5630	10.6869	1.0117
4.130	10.5587	10.6963	1.0130
4.135	10.5196	10.6433	1.0118
4.140	10.5369	10.6196	1.0078
4.145	10.5801	10.6399	1.0057
4.150	10.4902	10.6344	1.0137
4.155	10.4699	10.6338	1.0157
4.160	10.4960	10.6343	1.0132
4.165	10.4597	10.5952	1.0130
4.170	10.4553	10.5805	1.0120
4.175	10.4788	10.5546	1.0072
4.180	10.4690	10.5793	1.0105
4.185	10.4574	10.5412	1.0080
4.190	10.4245	10.5362	1.0107
4.195	10.4236	10.5308	1.0103
4.200	10.4433	10.5430	1.0095
4.205	10.4337	10.5403	1.0102
4.210	10.3809	10.5266	1.0140
4.215	10.4157	10.5326	1.0112
4.220	10.3927	10.5310	1.0133
4.225	10.4380	10.5279	1.0086

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	10.3906	10.5130	1.0118
4.235	10.3704	10.5067	1.0131
4.240	10.3629	10.5490	1.0180
4.245	10.3883	10.5267	1.0133
4.250	10.3689	10.5087	1.0135
4.255	10.3685	10.5250	1.0151
4.260	10.3360	10.5055	1.0164
4.265	10.3661	10.5021	1.0131
4.270	10.3306	10.4872	1.0152
4.275	10.3170	10.4828	1.0161
4.280	10.3146	10.4933	1.0173
4.285	10.3064	10.4932	1.0181
4.290	10.2937	10.4841	1.0185
4.295	10.2924	10.4759	1.0178
4.300	10.2853	10.4881	1.0197
4.305	10.2741	10.4755	1.0196
4.310	10.2674	10.4650	1.0192
4.315	10.2563	10.4572	1.0196
4.320	10.2598	10.4511	1.0186
4.325	10.2484	10.4520	1.0199
4.330	10.2577	10.4533	1.0191
4.335	10.2504	10.4519	1.0197
4.340	10.2561	10.4360	1.0175
4.345	10.2350	10.4253	1.0186
4.350	10.2287	10.4169	1.0184
4.355	10.2102	10.4151	1.0201
4.360	10.2032	10.3929	1.0186
4.365	10.1800	10.3866	1.0203
4.370	10.1959	10.3897	1.0190
4.375	10.1638	10.3901	1.0223
4.380	10.1743	10.3889	1.0211
4.385	10.1307	10.3573	1.0224
4.390	10.1482	10.3462	1.0195
4.395	10.1189	10.3443	1.0223
4.400	10.1014	10.3439	1.0240
4.405	10.0676	10.3344	1.0265
4.410	10.0849	10.3372	1.0250
4.415	10.0776	10.3167	1.0237
4.420	10.0545	10.3388	1.0283
4.425	10.0718	10.3473	1.0273
4.430	10.0581	10.3144	1.0255
4.435	10.0318	10.2925	1.0260
4.440	10.0377	10.3074	1.0269
4.445	9.9972	10.2762	1.0279
4.450	10.0092	10.2429	1.0233
4.455	9.9538	10.2578	1.0305

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	9.9846	10.2117	1.0227
4.465	9.9930	10.1774	1.0185
4.470	9.9990	10.2079	1.0209
4.475	10.0177	10.1867	1.0169
4.480	10.0276	10.2030	1.0175
4.485	9.9770	10.1537	1.0177
4.490	9.9854	10.1750	1.0190
4.495	9.9844	10.1657	1.0182
4.500	9.9590	10.1563	1.0198
4.505	9.9412	10.1392	1.0199
4.510	9.8986	10.1335	1.0237
4.515	9.9007	10.1697	1.0272
4.520	9.8942	10.1646	1.0273
4.525	9.9167	10.1651	1.0251
4.530	9.8746	10.1387	1.0267
4.535	9.8673	10.1541	1.0291
4.540	9.8743	10.1300	1.0259
4.545	9.8447	10.1124	1.0272
4.550	9.8498	10.0876	1.0241
4.555	9.8286	10.0961	1.0272
4.560	9.8279	10.0987	1.0276
4.565	9.8149	10.1045	1.0295
4.570	9.8124	10.0791	1.0272
4.575	9.8063	10.0760	1.0275
4.580	9.7985	10.0842	1.0292
4.585	9.7845	10.0667	1.0288
4.590	9.7805	10.0527	1.0278
4.595	9.7661	10.0607	1.0302
4.600	9.7606	10.0458	1.0292
4.605	9.7514	10.0381	1.0294
4.610	9.7445	10.0384	1.0302
4.615	9.7351	10.0221	1.0295
4.620	9.7325	10.0086	1.0284
4.625	9.7214	10.0158	1.0303
4.630	9.7179	9.9946	1.0285
4.635	9.7023	9.9806	1.0287
4.640	9.6839	9.9776	1.0303
4.645	9.6904	9.9701	1.0289
4.650	9.6789	9.9747	1.0306
4.655	9.6851	9.9390	1.0262
4.660	9.6831	9.9663	1.0292
4.665	9.6725	9.9554	1.0293
4.670	9.6228	9.9292	1.0318
4.675	9.6289	9.8995	1.0281
4.680	9.6288	9.9025	1.0284
4.685	9.6275	9.8943	1.0277

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	9.6256	9.9175	1.0303
4.695	9.6268	9.8658	1.0248
4.700	9.5986	9.8888	1.0302
4.705	9.5997	9.8235	1.0233
4.710	9.5277	9.8656	1.0355
4.715	9.5606	9.8688	1.0322
4.720	9.5582	9.8421	1.0297
4.725	9.5523	9.8697	1.0332
4.730	9.5554	9.8506	1.0309
4.735	9.5641	9.8707	1.0321
4.740	9.5612	9.8531	1.0305
4.745	9.5223	9.8341	1.0327
4.750	9.5555	9.8327	1.0290
4.755	9.5402	9.8555	1.0330
4.760	9.5298	9.8218	1.0306
4.765	9.5577	9.8420	1.0297
4.770	9.5438	9.8465	1.0317
4.775	9.4879	9.7787	1.0307
4.780	9.5140	9.7981	1.0299
4.785	9.4946	9.8808	1.0407
4.790	9.5143	9.7666	1.0265
4.795	9.4787	9.7735	1.0311
4.800	9.4872	9.7772	1.0306
4.805	9.4974	9.7285	1.0243
4.810	9.4536	9.7802	1.0345
4.815	9.4873	9.7218	1.0247
4.820	9.4246	9.7502	1.0345
4.825	9.4611	9.7593	1.0315
4.830	9.4206	9.7522	1.0352
4.835	9.4207	9.7663	1.0367
4.840	9.4287	9.7582	1.0350
4.845	9.4350	9.7428	1.0326
4.850	9.3955	9.7409	1.0368
4.855	9.4201	9.7566	1.0357
4.860	9.4063	9.7360	1.0350
4.865	9.4379	9.7337	1.0313
4.870	9.4001	9.7230	1.0343
4.875	9.3999	9.7356	1.0357
4.880	9.3949	9.7482	1.0376
4.885	9.3908	9.7184	1.0349
4.890	9.4125	9.7479	1.0356
4.895	9.3766	9.7128	1.0359
4.900	9.3761	9.7236	1.0371
4.905	9.3752	9.7118	1.0359
4.910	9.3523	9.6955	1.0367
4.915	9.3478	9.6933	1.0370

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	9.3461	9.6821	1.0360
4.925	9.3440	9.6767	1.0356
4.930	9.3304	9.6670	1.0361
4.935	9.3330	9.6636	1.0354
4.940	9.3313	9.6575	1.0350
4.945	9.3166	9.6509	1.0359
4.950	9.3211	9.6449	1.0347
4.955	9.3176	9.6460	1.0352
4.960	9.3075	9.6249	1.0341
4.965	9.3051	9.6247	1.0343
4.970	9.2996	9.6298	1.0355
4.975	9.3087	9.6346	1.0350
4.980	9.2824	9.6271	1.0371
4.985	9.2808	9.6067	1.0351
4.990	9.2738	9.6160	1.0369
4.995	9.2321	9.6071	1.0406
5.000	9.2353	9.5943	1.0389
5.005	9.2213	9.5932	1.0403
5.010	9.2122	9.5808	1.0400
5.015	9.2017	9.5998	1.0433
5.020	9.1951	9.5712	1.0409
5.025	9.1594	9.5602	1.0438
5.030	9.1381	9.5396	1.0439
5.035	9.1413	9.5573	1.0455
5.040	9.1172	9.5283	1.0451
5.045	9.1213	9.5086	1.0425
5.050	9.1082	9.4700	1.0397
5.055	9.1300	9.4878	1.0392
5.060	9.1132	9.4708	1.0392
5.065	9.1259	9.4510	1.0356
5.070	9.1047	9.4704	1.0402
5.075	9.1017	9.4505	1.0383
5.080	9.0841	9.4282	1.0379
5.085	9.0789	9.4105	1.0365
5.090	9.0774	9.4268	1.0385
5.095	9.0827	9.4360	1.0389
5.100	9.0811	9.4234	1.0377
5.105	9.0588	9.4578	1.0440
5.110	9.0582	9.4064	1.0384
5.115	9.0475	9.3992	1.0389
5.120	9.0436	9.3938	1.0387
5.125	9.0335	9.3930	1.0398
5.130	9.0349	9.3735	1.0375
5.135	9.0080	9.3593	1.0390
5.140	9.0077	9.3668	1.0399
5.145	8.9988	9.3295	1.0367

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	8.9913	9.3523	1.0401
5.155	8.9886	9.3300	1.0380
5.160	8.9850	9.3241	1.0377
5.165	8.9757	9.3218	1.0386
5.170	8.9756	9.3058	1.0368
5.175	8.9607	9.2914	1.0369
5.180	8.9618	9.2870	1.0363
5.185	8.9494	9.2731	1.0362
5.190	8.9525	9.2682	1.0353
5.195	8.9519	9.2585	1.0343
5.200	8.9523	9.2626	1.0347
5.205	8.9529	9.2598	1.0343
5.210	8.9494	9.2537	1.0340
5.215	8.9514	9.2598	1.0345
5.220	8.9367	9.2433	1.0343
5.225	8.9241	9.2372	1.0351
5.230	8.9359	9.2320	1.0331
5.235	8.9119	9.2349	1.0362
5.240	8.9088	9.2228	1.0352
5.245	8.9105	9.2281	1.0356
5.250	8.8949	9.2300	1.0377
5.255	8.8744	9.2228	1.0393
5.260	8.9060	9.2112	1.0343
5.265	8.8995	9.2291	1.0370
5.270	8.8762	9.2059	1.0371
5.275	8.8640	9.1855	1.0363
5.280	8.8651	9.2039	1.0382
5.285	8.8754	9.1865	1.0351
5.290	8.8583	9.1758	1.0358
5.295	8.8844	9.1484	1.0297
5.300	8.8483	9.1650	1.0358
5.305	8.8707	9.1682	1.0335
5.310	8.8146	9.1494	1.0380
5.315	8.8376	9.1551	1.0359
5.320	8.8198	9.1273	1.0349
5.325	8.8272	9.1389	1.0353
5.330	8.8170	9.1362	1.0362
5.335	8.7912	9.1301	1.0386
5.340	8.7855	9.1438	1.0408
5.345	8.7676	9.1542	1.0441
5.350	8.8092	9.1199	1.0353
5.355	8.7651	9.1132	1.0397
5.360	8.7678	9.0671	1.0341
5.365	8.7993	9.1085	1.0351
5.370	8.7485	9.0660	1.0363
5.375	8.7563	9.0935	1.0385

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	8.7644	9.0963	1.0379
5.385	8.7426	9.0694	1.0374
5.390	8.7349	9.0954	1.0413
5.395	8.7603	9.0884	1.0375
5.400	8.7285	9.0744	1.0396
5.405	8.7287	9.0653	1.0386
5.410	8.7193	9.0635	1.0395
5.415	8.7203	9.0279	1.0353
5.420	8.6999	9.0586	1.0412
5.425	8.7017	9.0671	1.0420
5.430	8.7187	9.0478	1.0377
5.435	8.7115	9.0380	1.0375
5.440	8.6785	9.0509	1.0429
5.445	8.7087	9.0432	1.0384
5.450	8.6715	9.0260	1.0409
5.455	8.6849	9.0394	1.0408
5.460	8.6743	9.0178	1.0396
5.465	8.6813	9.0467	1.0421
5.470	8.6417	9.0145	1.0431
5.475	8.6626	9.0083	1.0399
5.480	8.6467	9.0213	1.0433
5.485	8.6571	9.0062	1.0403
5.490	8.6215	9.0165	1.0458
5.495	8.6496	8.9980	1.0403
5.500	8.6289	9.0067	1.0438
5.505	8.6319	9.0093	1.0437
5.510	8.6319	8.9961	1.0422
5.515	8.6361	8.9923	1.0413
5.520	8.6242	8.9864	1.0420
5.525	8.6317	8.9871	1.0412
5.530	8.6247	8.9802	1.0412
5.535	8.6288	8.9733	1.0399
5.540	8.6108	8.9744	1.0422
5.545	8.6003	8.9551	1.0413
5.550	8.6013	8.9535	1.0409
5.555	8.6083	8.9460	1.0392
5.560	8.6024	8.9464	1.0400
5.565	8.6063	8.9469	1.0396
5.570	8.6029	8.9444	1.0397
5.575	8.6049	8.9398	1.0389
5.580	8.6015	8.9487	1.0404
5.585	8.5975	8.9395	1.0398
5.590	8.6060	8.9323	1.0379
5.595	8.5986	8.9260	1.0381
5.600	8.6021	8.9252	1.0376
5.605	8.6140	8.9128	1.0347

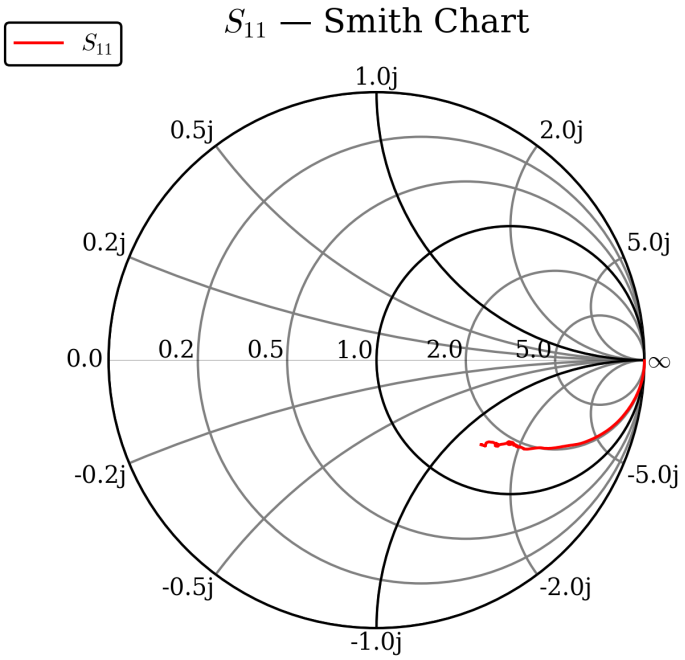
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	8.5971	8.8981	1.0350
5.615	8.5965	8.8807	1.0331
5.620	8.5924	8.8851	1.0341
5.625	8.5925	8.8988	1.0356
5.630	8.5802	8.8667	1.0334
5.635	8.5805	8.8542	1.0319
5.640	8.5808	8.8298	1.0290
5.645	8.5570	8.8229	1.0311
5.650	8.5623	8.8383	1.0322
5.655	8.5478	8.8273	1.0327
5.660	8.5424	8.8317	1.0339
5.665	8.5783	8.8299	1.0293
5.670	8.5101	8.8211	1.0365
5.675	8.5371	8.7972	1.0305
5.680	8.5195	8.7917	1.0319
5.685	8.5401	8.7808	1.0282
5.690	8.5127	8.7770	1.0310
5.695	8.5075	8.8009	1.0345
5.700	8.5033	8.7521	1.0293
5.705	8.4693	8.7429	1.0323
5.710	8.5062	8.7437	1.0279
5.715	8.4861	8.7149	1.0270
5.720	8.4762	8.7274	1.0296
5.725	8.4493	8.7176	1.0317
5.730	8.4506	8.7076	1.0304
5.735	8.4523	8.7010	1.0294
5.740	8.4439	8.7073	1.0312
5.745	8.4466	8.7107	1.0313
5.750	8.4266	8.6929	1.0316
5.755	8.4349	8.6901	1.0303
5.760	8.4087	8.6656	1.0305
5.765	8.4326	8.6627	1.0273
5.770	8.4199	8.6431	1.0265
5.775	8.4125	8.6533	1.0286
5.780	8.3912	8.6499	1.0308
5.785	8.3840	8.6414	1.0307
5.790	8.3719	8.6504	1.0333
5.795	8.3773	8.6545	1.0331
5.800	8.3634	8.6419	1.0333
5.805	8.3611	8.6265	1.0317
5.810	8.3553	8.6360	1.0336
5.815	8.3501	8.6309	1.0336
5.820	8.3474	8.6123	1.0317
5.825	8.3350	8.6073	1.0327
5.830	8.3216	8.6025	1.0337
5.835	8.3269	8.5997	1.0328

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	8.3222	8.6006	1.0335
5.845	8.3118	8.6034	1.0351
5.850	8.3163	8.5928	1.0332
5.855	8.3123	8.5829	1.0325
5.860	8.2915	8.5813	1.0349
5.865	8.2993	8.5779	1.0336
5.870	8.2878	8.5570	1.0325
5.875	8.2821	8.5555	1.0330
5.880	8.2778	8.5480	1.0326
5.885	8.2884	8.5333	1.0295
5.890	8.2479	8.5521	1.0369
5.895	8.2550	8.5381	1.0343
5.900	8.2656	8.5231	1.0311
5.905	8.2565	8.5422	1.0346
5.910	8.2483	8.4977	1.0302
5.915	8.2378	8.5063	1.0326
5.920	8.2359	8.4939	1.0313
5.925	8.2358	8.4806	1.0297
5.930	8.2003	8.4881	1.0351
5.935	8.1992	8.4711	1.0332
5.940	8.1977	8.4835	1.0349
5.945	8.2261	8.4747	1.0302
5.950	8.1929	8.4774	1.0347
5.955	8.1937	8.4823	1.0352
5.960	8.1809	8.4691	1.0352
5.965	8.1774	8.4709	1.0359
5.970	8.1739	8.4753	1.0369
5.975	8.1947	8.4266	1.0283
5.980	8.2291	8.4589	1.0279
5.985	8.1409	8.4478	1.0377
5.990	8.1493	8.4259	1.0339
5.995	8.1571	8.4119	1.0312
6.000	8.1748	8.4129	1.0291

1.2 S_{11} — Smith Chart



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	0.9999	-0.0023	-0.00
0.010	1.0000	-0.0044	0.00
0.015	1.0000	-0.0066	-0.00
0.020	0.9999	-0.0089	-0.00
0.025	0.9997	-0.0111	-0.00
0.030	0.9997	-0.0134	-0.00
0.035	0.9997	-0.0158	-0.00
0.040	0.9995	-0.0179	-0.00
0.045	0.9993	-0.0201	-0.00
0.050	0.9992	-0.0223	-0.00
0.055	0.9989	-0.0246	-0.01
0.060	0.9988	-0.0268	-0.01
0.065	0.9986	-0.0291	-0.01
0.070	0.9984	-0.0313	-0.01
0.075	0.9983	-0.0333	-0.01
0.080	0.9979	-0.0356	-0.01
0.085	0.9977	-0.0380	-0.01
0.090	0.9973	-0.0402	-0.02
0.095	0.9970	-0.0423	-0.02
0.100	0.9967	-0.0445	-0.02
0.105	0.9963	-0.0468	-0.02
0.110	0.9961	-0.0490	-0.02
0.115	0.9957	-0.0512	-0.03

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9955	-0.0533	-0.03
0.125	0.9949	-0.0556	-0.03
0.130	0.9944	-0.0576	-0.03
0.135	0.9941	-0.0599	-0.04
0.140	0.9937	-0.0622	-0.04
0.145	0.9932	-0.0643	-0.04
0.150	0.9927	-0.0664	-0.04
0.155	0.9925	-0.0685	-0.05
0.160	0.9918	-0.0707	-0.05
0.165	0.9912	-0.0727	-0.05
0.170	0.9908	-0.0749	-0.06
0.175	0.9902	-0.0771	-0.06
0.180	0.9895	-0.0792	-0.06
0.185	0.9891	-0.0813	-0.07
0.190	0.9884	-0.0835	-0.07
0.195	0.9878	-0.0855	-0.07
0.200	0.9872	-0.0876	-0.08
0.205	0.9866	-0.0898	-0.08
0.210	0.9860	-0.0918	-0.09
0.215	0.9853	-0.0939	-0.09
0.220	0.9845	-0.0960	-0.09
0.225	0.9840	-0.0981	-0.10
0.230	0.9833	-0.1003	-0.10
0.235	0.9826	-0.1024	-0.11
0.240	0.9818	-0.1043	-0.11
0.245	0.9810	-0.1063	-0.12
0.250	0.9803	-0.1083	-0.12
0.255	0.9795	-0.1104	-0.13
0.260	0.9787	-0.1125	-0.13
0.265	0.9778	-0.1145	-0.14
0.270	0.9771	-0.1165	-0.14
0.275	0.9763	-0.1186	-0.14
0.280	0.9754	-0.1205	-0.15
0.285	0.9745	-0.1225	-0.16
0.290	0.9739	-0.1244	-0.16
0.295	0.9728	-0.1263	-0.17
0.300	0.9720	-0.1282	-0.17
0.305	0.9710	-0.1303	-0.18
0.310	0.9702	-0.1322	-0.18
0.315	0.9693	-0.1342	-0.19
0.320	0.9682	-0.1361	-0.20
0.325	0.9674	-0.1380	-0.20
0.330	0.9664	-0.1399	-0.21
0.335	0.9655	-0.1417	-0.21
0.340	0.9644	-0.1435	-0.22
0.345	0.9635	-0.1454	-0.23

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9625	-0.1472	-0.23
0.355	0.9616	-0.1491	-0.24
0.360	0.9605	-0.1509	-0.24
0.365	0.9595	-0.1526	-0.25
0.370	0.9584	-0.1545	-0.26
0.375	0.9573	-0.1563	-0.26
0.380	0.9564	-0.1581	-0.27
0.385	0.9553	-0.1598	-0.28
0.390	0.9543	-0.1615	-0.28
0.395	0.9532	-0.1632	-0.29
0.400	0.9522	-0.1650	-0.30
0.405	0.9512	-0.1668	-0.30
0.410	0.9501	-0.1684	-0.31
0.415	0.9490	-0.1702	-0.32
0.420	0.9478	-0.1718	-0.32
0.425	0.9468	-0.1734	-0.33
0.430	0.9457	-0.1748	-0.34
0.435	0.9447	-0.1764	-0.35
0.440	0.9435	-0.1781	-0.35
0.445	0.9425	-0.1796	-0.36
0.450	0.9413	-0.1814	-0.37
0.455	0.9402	-0.1830	-0.37
0.460	0.9391	-0.1845	-0.38
0.465	0.9382	-0.1861	-0.39
0.470	0.9369	-0.1875	-0.40
0.475	0.9357	-0.1891	-0.40
0.480	0.9345	-0.1904	-0.41
0.485	0.9335	-0.1920	-0.42
0.490	0.9323	-0.1935	-0.43
0.495	0.9312	-0.1951	-0.43
0.500	0.9300	-0.1965	-0.44
0.505	0.9289	-0.1979	-0.45
0.510	0.9277	-0.1993	-0.46
0.515	0.9267	-0.2007	-0.46
0.520	0.9255	-0.2021	-0.47
0.525	0.9244	-0.2034	-0.48
0.530	0.9234	-0.2047	-0.48
0.535	0.9222	-0.2062	-0.49
0.540	0.9211	-0.2075	-0.50
0.545	0.9199	-0.2090	-0.51
0.550	0.9187	-0.2103	-0.51
0.555	0.9176	-0.2118	-0.52
0.560	0.9164	-0.2130	-0.53
0.565	0.9152	-0.2143	-0.54
0.570	0.9142	-0.2155	-0.54
0.575	0.9130	-0.2168	-0.55

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.9119	-0.2180	-0.56
0.585	0.9108	-0.2193	-0.57
0.590	0.9096	-0.2207	-0.58
0.595	0.9083	-0.2218	-0.58
0.600	0.9072	-0.2231	-0.59
0.605	0.9060	-0.2243	-0.60
0.610	0.9050	-0.2255	-0.61
0.615	0.9037	-0.2266	-0.61
0.620	0.9027	-0.2278	-0.62
0.625	0.9015	-0.2289	-0.63
0.630	0.9003	-0.2301	-0.64
0.635	0.8991	-0.2313	-0.65
0.640	0.8979	-0.2325	-0.65
0.645	0.8968	-0.2337	-0.66
0.650	0.8956	-0.2348	-0.67
0.655	0.8943	-0.2358	-0.68
0.660	0.8934	-0.2369	-0.68
0.665	0.8922	-0.2379	-0.69
0.670	0.8909	-0.2390	-0.70
0.675	0.8898	-0.2401	-0.71
0.680	0.8886	-0.2410	-0.72
0.685	0.8874	-0.2422	-0.73
0.690	0.8862	-0.2433	-0.73
0.695	0.8851	-0.2442	-0.74
0.700	0.8839	-0.2451	-0.75
0.705	0.8826	-0.2462	-0.76
0.710	0.8816	-0.2472	-0.77
0.715	0.8803	-0.2483	-0.78
0.720	0.8791	-0.2492	-0.78
0.725	0.8778	-0.2502	-0.79
0.730	0.8766	-0.2513	-0.80
0.735	0.8754	-0.2522	-0.81
0.740	0.8741	-0.2531	-0.82
0.745	0.8730	-0.2540	-0.83
0.750	0.8719	-0.2549	-0.83
0.755	0.8707	-0.2557	-0.84
0.760	0.8694	-0.2566	-0.85
0.765	0.8682	-0.2576	-0.86
0.770	0.8668	-0.2586	-0.87
0.775	0.8657	-0.2595	-0.88
0.780	0.8645	-0.2605	-0.89
0.785	0.8631	-0.2614	-0.90
0.790	0.8620	-0.2624	-0.91
0.795	0.8607	-0.2631	-0.91
0.800	0.8595	-0.2639	-0.92
0.805	0.8581	-0.2648	-0.93

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.810	0.8568	-0.2656	-0.94
0.815	0.8556	-0.2664	-0.95
0.820	0.8544	-0.2674	-0.96
0.825	0.8530	-0.2683	-0.97
0.830	0.8518	-0.2691	-0.98
0.835	0.8505	-0.2699	-0.99
0.840	0.8492	-0.2707	-1.00
0.845	0.8480	-0.2716	-1.01
0.850	0.8467	-0.2723	-1.02
0.855	0.8455	-0.2730	-1.03
0.860	0.8441	-0.2739	-1.04
0.865	0.8428	-0.2747	-1.05
0.870	0.8415	-0.2756	-1.06
0.875	0.8401	-0.2764	-1.07
0.880	0.8389	-0.2771	-1.08
0.885	0.8376	-0.2778	-1.09
0.890	0.8361	-0.2786	-1.10
0.895	0.8349	-0.2794	-1.11
0.900	0.8335	-0.2800	-1.12
0.905	0.8323	-0.2809	-1.13
0.910	0.8310	-0.2817	-1.14
0.915	0.8297	-0.2824	-1.15
0.920	0.8284	-0.2831	-1.16
0.925	0.8270	-0.2838	-1.17
0.930	0.8257	-0.2843	-1.18
0.935	0.8245	-0.2851	-1.19
0.940	0.8230	-0.2859	-1.20
0.945	0.8218	-0.2865	-1.21
0.950	0.8204	-0.2872	-1.22
0.955	0.8191	-0.2880	-1.23
0.960	0.8178	-0.2886	-1.24
0.965	0.8165	-0.2892	-1.25
0.970	0.8153	-0.2899	-1.26
0.975	0.8139	-0.2905	-1.27
0.980	0.8125	-0.2911	-1.28
0.985	0.8112	-0.2918	-1.29
0.990	0.8098	-0.2925	-1.30
0.995	0.8086	-0.2932	-1.31
1.000	0.8072	-0.2937	-1.32
1.005	0.8058	-0.2944	-1.33
1.010	0.8047	-0.2949	-1.34
1.015	0.8034	-0.2954	-1.35
1.020	0.8022	-0.2960	-1.36
1.025	0.8007	-0.2965	-1.37
1.030	0.7995	-0.2972	-1.38
1.035	0.7982	-0.2977	-1.39

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.040	0.7969	-0.2983	-1.40
1.045	0.7956	-0.2989	-1.41
1.050	0.7943	-0.2994	-1.42
1.055	0.7930	-0.2999	-1.43
1.060	0.7918	-0.3004	-1.44
1.065	0.7906	-0.3008	-1.45
1.070	0.7895	-0.3014	-1.46
1.075	0.7884	-0.3019	-1.47
1.080	0.7871	-0.3023	-1.48
1.085	0.7859	-0.3028	-1.49
1.090	0.7845	-0.3033	-1.50
1.095	0.7833	-0.3038	-1.51
1.100	0.7821	-0.3043	-1.52
1.105	0.7810	-0.3047	-1.53
1.110	0.7798	-0.3051	-1.54
1.115	0.7787	-0.3054	-1.55
1.120	0.7776	-0.3059	-1.56
1.125	0.7763	-0.3064	-1.57
1.130	0.7752	-0.3068	-1.58
1.135	0.7740	-0.3072	-1.59
1.140	0.7728	-0.3076	-1.60
1.145	0.7717	-0.3080	-1.61
1.150	0.7706	-0.3083	-1.62
1.155	0.7696	-0.3086	-1.63
1.160	0.7687	-0.3089	-1.64
1.165	0.7676	-0.3092	-1.64
1.170	0.7666	-0.3096	-1.65
1.175	0.7655	-0.3099	-1.66
1.180	0.7642	-0.3104	-1.67
1.185	0.7632	-0.3107	-1.68
1.190	0.7622	-0.3109	-1.69
1.195	0.7612	-0.3111	-1.70
1.200	0.7603	-0.3114	-1.71
1.205	0.7594	-0.3116	-1.71
1.210	0.7584	-0.3118	-1.72
1.215	0.7572	-0.3120	-1.73
1.220	0.7562	-0.3123	-1.74
1.225	0.7553	-0.3127	-1.75
1.230	0.7542	-0.3129	-1.76
1.235	0.7532	-0.3132	-1.77
1.240	0.7524	-0.3133	-1.78
1.245	0.7515	-0.3135	-1.78
1.250	0.7506	-0.3137	-1.79
1.255	0.7497	-0.3139	-1.80
1.260	0.7488	-0.3140	-1.81
1.265	0.7479	-0.3141	-1.82

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.7470	-0.3144	-1.83
1.275	0.7459	-0.3145	-1.84
1.280	0.7451	-0.3148	-1.84
1.285	0.7442	-0.3150	-1.85
1.290	0.7433	-0.3152	-1.86
1.295	0.7423	-0.3153	-1.87
1.300	0.7415	-0.3154	-1.88
1.305	0.7406	-0.3156	-1.88
1.310	0.7398	-0.3158	-1.89
1.315	0.7389	-0.3160	-1.90
1.320	0.7379	-0.3162	-1.91
1.325	0.7370	-0.3162	-1.92
1.330	0.7362	-0.3164	-1.92
1.335	0.7356	-0.3164	-1.93
1.340	0.7348	-0.3165	-1.94
1.345	0.7340	-0.3167	-1.94
1.350	0.7332	-0.3167	-1.95
1.355	0.7324	-0.3168	-1.96
1.360	0.7314	-0.3170	-1.97
1.365	0.7304	-0.3171	-1.98
1.370	0.7296	-0.3172	-1.99
1.375	0.7289	-0.3174	-1.99
1.380	0.7281	-0.3174	-2.00
1.385	0.7273	-0.3174	-2.01
1.390	0.7266	-0.3176	-2.01
1.395	0.7257	-0.3175	-2.02
1.400	0.7249	-0.3177	-2.03
1.405	0.7238	-0.3180	-2.04
1.410	0.7228	-0.3181	-2.05
1.415	0.7219	-0.3182	-2.06
1.420	0.7209	-0.3183	-2.07
1.425	0.7203	-0.3184	-2.07
1.430	0.7194	-0.3185	-2.08
1.435	0.7185	-0.3186	-2.09
1.440	0.7178	-0.3187	-2.10
1.445	0.7168	-0.3190	-2.11
1.450	0.7158	-0.3191	-2.12
1.455	0.7149	-0.3194	-2.12
1.460	0.7139	-0.3194	-2.13
1.465	0.7129	-0.3196	-2.14
1.470	0.7121	-0.3198	-2.15
1.475	0.7112	-0.3199	-2.16
1.480	0.7103	-0.3200	-2.17
1.485	0.7094	-0.3201	-2.18
1.490	0.7084	-0.3204	-2.19
1.495	0.7074	-0.3205	-2.20

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.500	0.7064	-0.3207	-2.21
1.505	0.7055	-0.3208	-2.21
1.510	0.7043	-0.3210	-2.23
1.515	0.7033	-0.3212	-2.23
1.520	0.7024	-0.3214	-2.24
1.525	0.7014	-0.3215	-2.25
1.530	0.7005	-0.3216	-2.26
1.535	0.6995	-0.3218	-2.27
1.540	0.6984	-0.3220	-2.28
1.545	0.6973	-0.3221	-2.29
1.550	0.6963	-0.3223	-2.30
1.555	0.6951	-0.3225	-2.31
1.560	0.6940	-0.3227	-2.32
1.565	0.6929	-0.3230	-2.33
1.570	0.6919	-0.3230	-2.34
1.575	0.6909	-0.3233	-2.35
1.580	0.6898	-0.3234	-2.36
1.585	0.6886	-0.3236	-2.37
1.590	0.6875	-0.3239	-2.38
1.595	0.6865	-0.3240	-2.39
1.600	0.6855	-0.3243	-2.40
1.605	0.6845	-0.3244	-2.41
1.610	0.6835	-0.3245	-2.42
1.615	0.6825	-0.3247	-2.43
1.620	0.6814	-0.3249	-2.44
1.625	0.6804	-0.3252	-2.45
1.630	0.6793	-0.3252	-2.46
1.635	0.6781	-0.3253	-2.47
1.640	0.6769	-0.3256	-2.49
1.645	0.6758	-0.3257	-2.50
1.650	0.6747	-0.3260	-2.51
1.655	0.6737	-0.3261	-2.52
1.660	0.6726	-0.3263	-2.53
1.665	0.6714	-0.3264	-2.54
1.670	0.6704	-0.3266	-2.55
1.675	0.6694	-0.3268	-2.56
1.680	0.6682	-0.3270	-2.57
1.685	0.6672	-0.3273	-2.58
1.690	0.6660	-0.3273	-2.59
1.695	0.6652	-0.3273	-2.60
1.700	0.6640	-0.3275	-2.61
1.705	0.6631	-0.3275	-2.62
1.710	0.6621	-0.3277	-2.63
1.715	0.6609	-0.3278	-2.64
1.720	0.6598	-0.3279	-2.65
1.725	0.6586	-0.3282	-2.66

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.730	0.6577	-0.3282	-2.67
1.735	0.6566	-0.3283	-2.69
1.740	0.6556	-0.3285	-2.69
1.745	0.6545	-0.3287	-2.70
1.750	0.6537	-0.3287	-2.71
1.755	0.6528	-0.3288	-2.72
1.760	0.6519	-0.3288	-2.73
1.765	0.6509	-0.3290	-2.74
1.770	0.6501	-0.3291	-2.75
1.775	0.6492	-0.3291	-2.76
1.780	0.6483	-0.3292	-2.77
1.785	0.6475	-0.3290	-2.78
1.790	0.6466	-0.3292	-2.79
1.795	0.6457	-0.3290	-2.80
1.800	0.6449	-0.3291	-2.81
1.805	0.6442	-0.3293	-2.81
1.810	0.6432	-0.3293	-2.82
1.815	0.6425	-0.3295	-2.83
1.820	0.6416	-0.3296	-2.84
1.825	0.6408	-0.3296	-2.85
1.830	0.6401	-0.3296	-2.85
1.835	0.6394	-0.3295	-2.86
1.840	0.6387	-0.3296	-2.87
1.845	0.6381	-0.3295	-2.88
1.850	0.6374	-0.3293	-2.88
1.855	0.6367	-0.3292	-2.89
1.860	0.6358	-0.3292	-2.90
1.865	0.6350	-0.3292	-2.91
1.870	0.6342	-0.3293	-2.92
1.875	0.6333	-0.3293	-2.93
1.880	0.6329	-0.3294	-2.93
1.885	0.6324	-0.3294	-2.94
1.890	0.6321	-0.3293	-2.94
1.895	0.6315	-0.3292	-2.95
1.900	0.6309	-0.3291	-2.96
1.905	0.6302	-0.3291	-2.96
1.910	0.6295	-0.3292	-2.97
1.915	0.6288	-0.3293	-2.98
1.920	0.6282	-0.3293	-2.98
1.925	0.6277	-0.3292	-2.99
1.930	0.6274	-0.3292	-2.99
1.935	0.6270	-0.3291	-3.00
1.940	0.6266	-0.3290	-3.00
1.945	0.6262	-0.3290	-3.01
1.950	0.6258	-0.3290	-3.01
1.955	0.6251	-0.3290	-3.02

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.960	0.6247	-0.3289	-3.02
1.965	0.6242	-0.3289	-3.03
1.970	0.6237	-0.3288	-3.04
1.975	0.6234	-0.3288	-3.04
1.980	0.6231	-0.3286	-3.04
1.985	0.6228	-0.3284	-3.05
1.990	0.6225	-0.3283	-3.05
1.995	0.6220	-0.3283	-3.06
2.000	0.6214	-0.3282	-3.06
2.005	0.6210	-0.3282	-3.07
2.010	0.6205	-0.3281	-3.07
2.015	0.6202	-0.3280	-3.08
2.020	0.6198	-0.3279	-3.08
2.025	0.6194	-0.3277	-3.09
2.030	0.6191	-0.3278	-3.09
2.035	0.6188	-0.3278	-3.10
2.040	0.6182	-0.3276	-3.10
2.045	0.6177	-0.3276	-3.11
2.050	0.6173	-0.3276	-3.11
2.055	0.6169	-0.3276	-3.12
2.060	0.6164	-0.3274	-3.12
2.065	0.6161	-0.3275	-3.13
2.070	0.6157	-0.3272	-3.13
2.075	0.6154	-0.3271	-3.14
2.080	0.6148	-0.3271	-3.14
2.085	0.6143	-0.3271	-3.15
2.090	0.6136	-0.3271	-3.16
2.095	0.6130	-0.3271	-3.16
2.100	0.6125	-0.3271	-3.17
2.105	0.6121	-0.3271	-3.17
2.110	0.6116	-0.3270	-3.18
2.115	0.6110	-0.3269	-3.19
2.120	0.6105	-0.3270	-3.19
2.125	0.6101	-0.3269	-3.20
2.130	0.6093	-0.3269	-3.20
2.135	0.6087	-0.3270	-3.21
2.140	0.6080	-0.3270	-3.22
2.145	0.6074	-0.3270	-3.22
2.150	0.6068	-0.3271	-3.23
2.155	0.6062	-0.3271	-3.24
2.160	0.6054	-0.3272	-3.25
2.165	0.6049	-0.3272	-3.25
2.170	0.6043	-0.3272	-3.26
2.175	0.6036	-0.3273	-3.27
2.180	0.6030	-0.3272	-3.27
2.185	0.6022	-0.3274	-3.28

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.190	0.6013	-0.3274	-3.29
2.195	0.6006	-0.3277	-3.30
2.200	0.5997	-0.3277	-3.31
2.205	0.5991	-0.3278	-3.31
2.210	0.5983	-0.3278	-3.32
2.215	0.5976	-0.3278	-3.33
2.220	0.5970	-0.3279	-3.34
2.225	0.5961	-0.3279	-3.35
2.230	0.5953	-0.3280	-3.35
2.235	0.5943	-0.3282	-3.36
2.240	0.5936	-0.3284	-3.37
2.245	0.5928	-0.3282	-3.38
2.250	0.5920	-0.3285	-3.39
2.255	0.5914	-0.3285	-3.39
2.260	0.5906	-0.3285	-3.40
2.265	0.5898	-0.3287	-3.41
2.270	0.5890	-0.3288	-3.42
2.275	0.5881	-0.3289	-3.43
2.280	0.5872	-0.3291	-3.44
2.285	0.5864	-0.3293	-3.45
2.290	0.5855	-0.3292	-3.46
2.295	0.5848	-0.3295	-3.46
2.300	0.5841	-0.3294	-3.47
2.305	0.5832	-0.3295	-3.48
2.310	0.5824	-0.3295	-3.49
2.315	0.5816	-0.3297	-3.50
2.320	0.5807	-0.3299	-3.51
2.325	0.5799	-0.3301	-3.51
2.330	0.5790	-0.3301	-3.52
2.335	0.5781	-0.3303	-3.53
2.340	0.5774	-0.3305	-3.54
2.345	0.5766	-0.3307	-3.55
2.350	0.5759	-0.3308	-3.56
2.355	0.5750	-0.3308	-3.56
2.360	0.5743	-0.3308	-3.57
2.365	0.5733	-0.3309	-3.58
2.370	0.5726	-0.3310	-3.59
2.375	0.5719	-0.3312	-3.60
2.380	0.5710	-0.3311	-3.61
2.385	0.5703	-0.3313	-3.62
2.390	0.5696	-0.3314	-3.62
2.395	0.5689	-0.3316	-3.63
2.400	0.5681	-0.3317	-3.64
2.405	0.5675	-0.3319	-3.64
2.410	0.5668	-0.3318	-3.65
2.415	0.5661	-0.3319	-3.66

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.420	0.5652	-0.3320	-3.67
2.425	0.5646	-0.3321	-3.67
2.430	0.5639	-0.3322	-3.68
2.435	0.5632	-0.3322	-3.69
2.440	0.5627	-0.3322	-3.70
2.445	0.5622	-0.3323	-3.70
2.450	0.5617	-0.3322	-3.71
2.455	0.5611	-0.3321	-3.72
2.460	0.5604	-0.3322	-3.72
2.465	0.5598	-0.3322	-3.73
2.470	0.5591	-0.3322	-3.74
2.475	0.5586	-0.3322	-3.74
2.480	0.5581	-0.3322	-3.75
2.485	0.5576	-0.3321	-3.75
2.490	0.5571	-0.3320	-3.76
2.495	0.5566	-0.3320	-3.77
2.500	0.5561	-0.3320	-3.77
2.505	0.5557	-0.3321	-3.78
2.510	0.5551	-0.3320	-3.78
2.515	0.5546	-0.3320	-3.79
2.520	0.5542	-0.3320	-3.79
2.525	0.5538	-0.3320	-3.80
2.530	0.5532	-0.3319	-3.81
2.535	0.5528	-0.3319	-3.81
2.540	0.5523	-0.3319	-3.82
2.545	0.5518	-0.3320	-3.82
2.550	0.5514	-0.3319	-3.83
2.555	0.5510	-0.3318	-3.83
2.560	0.5505	-0.3317	-3.84
2.565	0.5502	-0.3314	-3.85
2.570	0.5499	-0.3311	-3.85
2.575	0.5499	-0.3307	-3.85
2.580	0.5499	-0.3305	-3.86
2.585	0.5500	-0.3304	-3.85
2.590	0.5500	-0.3301	-3.86
2.595	0.5499	-0.3302	-3.86
2.600	0.5500	-0.3304	-3.85
2.605	0.5499	-0.3305	-3.86
2.610	0.5499	-0.3306	-3.86
2.615	0.5498	-0.3305	-3.86
2.620	0.5499	-0.3304	-3.86
2.625	0.5499	-0.3305	-3.86
2.630	0.5497	-0.3304	-3.86
2.635	0.5496	-0.3303	-3.86
2.640	0.5493	-0.3303	-3.86
2.645	0.5491	-0.3303	-3.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	0.5491	-0.3303	-3.87
2.655	0.5490	-0.3303	-3.87
2.660	0.5487	-0.3301	-3.87
2.665	0.5487	-0.3301	-3.87
2.670	0.5485	-0.3301	-3.87
2.675	0.5484	-0.3299	-3.88
2.680	0.5483	-0.3300	-3.88
2.685	0.5480	-0.3299	-3.88
2.690	0.5478	-0.3301	-3.88
2.695	0.5475	-0.3300	-3.89
2.700	0.5475	-0.3300	-3.89
2.705	0.5474	-0.3299	-3.89
2.710	0.5474	-0.3299	-3.89
2.715	0.5475	-0.3297	-3.89
2.720	0.5473	-0.3296	-3.89
2.725	0.5472	-0.3293	-3.89
2.730	0.5470	-0.3293	-3.90
2.735	0.5468	-0.3290	-3.90
2.740	0.5465	-0.3288	-3.91
2.745	0.5463	-0.3286	-3.91
2.750	0.5462	-0.3283	-3.91
2.755	0.5460	-0.3281	-3.92
2.760	0.5457	-0.3279	-3.92
2.765	0.5454	-0.3277	-3.93
2.770	0.5452	-0.3275	-3.93
2.775	0.5448	-0.3274	-3.94
2.780	0.5445	-0.3271	-3.94
2.785	0.5440	-0.3270	-3.95
2.790	0.5438	-0.3269	-3.95
2.795	0.5435	-0.3266	-3.96
2.800	0.5432	-0.3264	-3.96
2.805	0.5429	-0.3263	-3.97
2.810	0.5425	-0.3261	-3.97
2.815	0.5422	-0.3261	-3.98
2.820	0.5418	-0.3261	-3.98
2.825	0.5413	-0.3260	-3.99
2.830	0.5409	-0.3260	-3.99
2.835	0.5404	-0.3259	-4.00
2.840	0.5401	-0.3259	-4.00
2.845	0.5397	-0.3258	-4.01
2.850	0.5394	-0.3258	-4.01
2.855	0.5392	-0.3257	-4.01
2.860	0.5389	-0.3256	-4.02
2.865	0.5387	-0.3255	-4.02
2.870	0.5384	-0.3254	-4.03
2.875	0.5380	-0.3253	-4.03

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.880	0.5375	-0.3251	-4.04
2.885	0.5374	-0.3248	-4.04
2.890	0.5369	-0.3247	-4.05
2.895	0.5367	-0.3242	-4.05
2.900	0.5361	-0.3241	-4.06
2.905	0.5356	-0.3237	-4.07
2.910	0.5353	-0.3234	-4.08
2.915	0.5346	-0.3235	-4.08
2.920	0.5338	-0.3234	-4.09
2.925	0.5331	-0.3235	-4.10
2.930	0.5324	-0.3231	-4.11
2.935	0.5321	-0.3232	-4.12
2.940	0.5315	-0.3233	-4.12
2.945	0.5310	-0.3235	-4.13
2.950	0.5306	-0.3234	-4.13
2.955	0.5300	-0.3237	-4.14
2.960	0.5296	-0.3237	-4.14
2.965	0.5291	-0.3240	-4.15
2.970	0.5290	-0.3240	-4.15
2.975	0.5286	-0.3242	-4.15
2.980	0.5284	-0.3243	-4.15
2.985	0.5285	-0.3244	-4.15
2.990	0.5285	-0.3243	-4.15
2.995	0.5283	-0.3244	-4.15
3.000	0.5281	-0.3243	-4.16
3.005	0.5266	-0.3240	-4.18
3.010	0.5264	-0.3243	-4.18
3.015	0.5264	-0.3243	-4.18
3.020	0.5264	-0.3244	-4.18
3.025	0.5265	-0.3245	-4.17
3.030	0.5267	-0.3245	-4.17
3.035	0.5269	-0.3245	-4.17
3.040	0.5270	-0.3242	-4.17
3.045	0.5270	-0.3240	-4.17
3.050	0.5271	-0.3234	-4.17
3.055	0.5272	-0.3232	-4.17
3.060	0.5270	-0.3228	-4.18
3.065	0.5268	-0.3222	-4.19
3.070	0.5267	-0.3218	-4.19
3.075	0.5264	-0.3212	-4.20
3.080	0.5262	-0.3209	-4.20
3.085	0.5258	-0.3203	-4.21
3.090	0.5253	-0.3199	-4.22
3.095	0.5247	-0.3196	-4.23
3.100	0.5242	-0.3196	-4.24
3.105	0.5237	-0.3193	-4.25

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.110	0.5233	-0.3193	-4.25
3.115	0.5227	-0.3192	-4.26
3.120	0.5223	-0.3192	-4.26
3.125	0.5219	-0.3195	-4.27
3.130	0.5217	-0.3196	-4.27
3.135	0.5214	-0.3199	-4.27
3.140	0.5212	-0.3201	-4.27
3.145	0.5209	-0.3204	-4.27
3.150	0.5207	-0.3207	-4.27
3.155	0.5206	-0.3210	-4.27
3.160	0.5208	-0.3213	-4.27
3.165	0.5209	-0.3215	-4.26
3.170	0.5213	-0.3214	-4.26
3.175	0.5217	-0.3217	-4.25
3.180	0.5219	-0.3219	-4.25
3.185	0.5220	-0.3216	-4.25
3.190	0.5222	-0.3216	-4.25
3.195	0.5224	-0.3214	-4.25
3.200	0.5224	-0.3211	-4.25
3.205	0.5228	-0.3207	-4.25
3.210	0.5230	-0.3201	-4.25
3.215	0.5233	-0.3197	-4.25
3.220	0.5230	-0.3191	-4.26
3.225	0.5224	-0.3187	-4.27
3.230	0.5220	-0.3187	-4.27
3.235	0.5214	-0.3187	-4.28
3.240	0.5213	-0.3189	-4.28
3.245	0.5212	-0.3187	-4.28
3.250	0.5209	-0.3187	-4.28
3.255	0.5208	-0.3188	-4.28
3.260	0.5209	-0.3194	-4.28
3.265	0.5208	-0.3192	-4.28
3.270	0.5207	-0.3192	-4.28
3.275	0.5208	-0.3198	-4.28
3.280	0.5206	-0.3197	-4.28
3.285	0.5209	-0.3198	-4.28
3.290	0.5206	-0.3201	-4.28
3.295	0.5209	-0.3202	-4.27
3.300	0.5207	-0.3201	-4.28
3.305	0.5207	-0.3202	-4.27
3.310	0.5206	-0.3204	-4.27
3.315	0.5208	-0.3204	-4.27
3.320	0.5209	-0.3205	-4.27
3.325	0.5209	-0.3208	-4.27
3.330	0.5210	-0.3206	-4.27
3.335	0.5210	-0.3206	-4.27

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.340	0.5211	-0.3206	-4.27
3.345	0.5212	-0.3204	-4.27
3.350	0.5213	-0.3203	-4.27
3.355	0.5214	-0.3201	-4.27
3.360	0.5215	-0.3200	-4.27
3.365	0.5215	-0.3198	-4.27
3.370	0.5213	-0.3197	-4.27
3.375	0.5212	-0.3194	-4.28
3.380	0.5210	-0.3192	-4.28
3.385	0.5211	-0.3189	-4.28
3.390	0.5208	-0.3187	-4.29
3.395	0.5208	-0.3183	-4.29
3.400	0.5205	-0.3179	-4.29
3.405	0.5203	-0.3176	-4.30
3.410	0.5199	-0.3173	-4.31
3.415	0.5195	-0.3169	-4.31
3.420	0.5190	-0.3168	-4.32
3.425	0.5188	-0.3164	-4.33
3.430	0.5183	-0.3163	-4.33
3.435	0.5178	-0.3159	-4.34
3.440	0.5174	-0.3160	-4.35
3.445	0.5170	-0.3157	-4.35
3.450	0.5164	-0.3153	-4.36
3.455	0.5159	-0.3154	-4.37
3.460	0.5152	-0.3154	-4.38
3.465	0.5145	-0.3153	-4.39
3.470	0.5138	-0.3158	-4.39
3.475	0.5135	-0.3156	-4.40
3.480	0.5128	-0.3160	-4.40
3.485	0.5129	-0.3162	-4.40
3.490	0.5123	-0.3165	-4.41
3.495	0.5121	-0.3165	-4.41
3.500	0.5119	-0.3170	-4.41
3.505	0.5119	-0.3174	-4.40
3.510	0.5117	-0.3173	-4.41
3.515	0.5118	-0.3174	-4.40
3.520	0.5118	-0.3175	-4.40
3.525	0.5120	-0.3175	-4.40
3.530	0.5120	-0.3178	-4.40
3.535	0.5131	-0.3166	-4.39
3.540	0.5132	-0.3160	-4.40
3.545	0.5136	-0.3153	-4.40
3.550	0.5132	-0.3140	-4.41
3.555	0.5131	-0.3128	-4.42
3.560	0.5123	-0.3116	-4.44
3.565	0.5116	-0.3105	-4.46

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.570	0.5102	-0.3092	-4.49
3.575	0.5083	-0.3072	-4.53
3.580	0.5071	-0.3069	-4.54
3.585	0.5051	-0.3067	-4.57
3.590	0.5033	-0.3071	-4.59
3.595	0.5015	-0.3079	-4.61
3.600	0.5001	-0.3090	-4.61
3.605	0.4990	-0.3105	-4.62
3.610	0.4984	-0.3118	-4.61
3.615	0.4988	-0.3129	-4.60
3.620	0.4993	-0.3138	-4.59
3.625	0.5000	-0.3144	-4.57
3.630	0.5008	-0.3146	-4.56
3.635	0.5020	-0.3144	-4.55
3.640	0.5022	-0.3139	-4.55
3.645	0.5024	-0.3135	-4.55
3.650	0.5027	-0.3129	-4.55
3.655	0.5026	-0.3120	-4.56
3.660	0.5026	-0.3114	-4.57
3.665	0.5024	-0.3109	-4.57
3.670	0.5022	-0.3101	-4.58
3.675	0.5021	-0.3097	-4.58
3.680	0.5020	-0.3088	-4.59
3.685	0.5017	-0.3083	-4.60
3.690	0.5011	-0.3077	-4.61
3.695	0.5006	-0.3070	-4.62
3.700	0.4999	-0.3064	-4.64
3.705	0.4992	-0.3059	-4.65
3.710	0.4981	-0.3055	-4.67
3.715	0.4972	-0.3050	-4.68
3.720	0.4961	-0.3049	-4.70
3.725	0.4948	-0.3051	-4.71
3.730	0.4932	-0.3052	-4.73
3.735	0.4919	-0.3059	-4.74
3.740	0.4908	-0.3071	-4.75
3.745	0.4900	-0.3084	-4.75
3.750	0.4897	-0.3099	-4.74
3.755	0.4898	-0.3115	-4.72
3.760	0.4907	-0.3131	-4.70
3.765	0.4921	-0.3144	-4.67
3.770	0.4938	-0.3146	-4.65
3.775	0.4960	-0.3153	-4.62
3.780	0.4982	-0.3130	-4.61
3.785	0.4984	-0.3111	-4.62
3.790	0.4985	-0.3091	-4.63
3.795	0.4976	-0.3075	-4.66

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.800	0.4963	-0.3063	-4.68
3.805	0.4955	-0.3057	-4.70
3.810	0.4945	-0.3054	-4.71
3.815	0.4932	-0.3053	-4.73
3.820	0.4923	-0.3054	-4.74
3.825	0.4912	-0.3058	-4.75
3.830	0.4904	-0.3062	-4.76
3.835	0.4899	-0.3084	-4.75
3.840	0.4897	-0.3086	-4.75
3.845	0.4898	-0.3086	-4.75
3.850	0.4896	-0.3088	-4.75
3.855	0.4894	-0.3090	-4.75
3.860	0.4894	-0.3089	-4.75
3.865	0.4891	-0.3089	-4.75
3.870	0.4888	-0.3092	-4.76
3.875	0.4886	-0.3092	-4.76
3.880	0.4883	-0.3093	-4.76
3.885	0.4880	-0.3094	-4.76
3.890	0.4878	-0.3096	-4.77
3.895	0.4876	-0.3095	-4.77
3.900	0.4874	-0.3097	-4.77
3.905	0.4871	-0.3096	-4.77
3.910	0.4867	-0.3098	-4.78
3.915	0.4862	-0.3100	-4.78
3.920	0.4858	-0.3103	-4.79
3.925	0.4854	-0.3104	-4.79
3.930	0.4851	-0.3105	-4.79
3.935	0.4848	-0.3106	-4.79
3.940	0.4843	-0.3107	-4.80
3.945	0.4839	-0.3107	-4.81
3.950	0.4835	-0.3110	-4.81
3.955	0.4831	-0.3110	-4.81
3.960	0.4827	-0.3112	-4.82
3.965	0.4822	-0.3113	-4.82
3.970	0.4817	-0.3114	-4.83
3.975	0.4812	-0.3115	-4.83
3.980	0.4806	-0.3116	-4.84
3.985	0.4799	-0.3118	-4.85
3.990	0.4794	-0.3119	-4.85
3.995	0.4788	-0.3121	-4.86
4.000	0.4783	-0.3123	-4.86
4.005	0.4776	-0.3126	-4.87
4.010	0.4772	-0.3125	-4.88
4.015	0.4765	-0.3125	-4.88
4.020	0.4758	-0.3127	-4.89
4.025	0.4751	-0.3128	-4.90

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.030	0.4745	-0.3130	-4.91
4.035	0.4739	-0.3131	-4.91
4.040	0.4732	-0.3131	-4.92
4.045	0.4726	-0.3132	-4.93
4.050	0.4721	-0.3133	-4.93
4.055	0.4713	-0.3136	-4.94
4.060	0.4706	-0.3136	-4.95
4.065	0.4700	-0.3135	-4.96
4.070	0.4691	-0.3138	-4.97
4.075	0.4685	-0.3138	-4.98
4.080	0.4679	-0.3139	-4.98
4.085	0.4670	-0.3139	-4.99
4.090	0.4664	-0.3139	-5.00
4.095	0.4657	-0.3142	-5.01
4.100	0.4650	-0.3142	-5.02
4.105	0.4640	-0.3147	-5.03
4.110	0.4634	-0.3148	-5.03
4.115	0.4626	-0.3148	-5.04
4.120	0.4619	-0.3151	-5.05
4.125	0.4613	-0.3151	-5.06
4.130	0.4603	-0.3157	-5.07
4.135	0.4596	-0.3158	-5.07
4.140	0.4586	-0.3158	-5.09
4.145	0.4572	-0.3161	-5.10
4.150	0.4562	-0.3161	-5.11
4.155	0.4554	-0.3159	-5.13
4.160	0.4545	-0.3158	-5.14
4.165	0.4539	-0.3156	-5.15
4.170	0.4533	-0.3156	-5.16
4.175	0.4527	-0.3156	-5.16
4.180	0.4522	-0.3153	-5.17
4.185	0.4516	-0.3152	-5.18
4.190	0.4512	-0.3150	-5.19
4.195	0.4507	-0.3151	-5.19
4.200	0.4501	-0.3151	-5.20
4.205	0.4498	-0.3153	-5.20
4.210	0.4494	-0.3152	-5.21
4.215	0.4491	-0.3155	-5.21
4.220	0.4488	-0.3153	-5.22
4.225	0.4484	-0.3158	-5.22
4.230	0.4482	-0.3156	-5.22
4.235	0.4478	-0.3156	-5.23
4.240	0.4472	-0.3158	-5.23
4.245	0.4468	-0.3160	-5.24
4.250	0.4466	-0.3160	-5.24
4.255	0.4465	-0.3161	-5.24

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.260	0.4461	-0.3161	-5.24
4.265	0.4460	-0.3163	-5.24
4.270	0.4460	-0.3164	-5.24
4.275	0.4458	-0.3165	-5.24
4.280	0.4455	-0.3166	-5.25
4.285	0.4452	-0.3168	-5.25
4.290	0.4450	-0.3167	-5.25
4.295	0.4448	-0.3168	-5.26
4.300	0.4446	-0.3168	-5.26
4.305	0.4446	-0.3170	-5.26
4.310	0.4445	-0.3168	-5.26
4.315	0.4445	-0.3168	-5.26
4.320	0.4443	-0.3168	-5.26
4.325	0.4440	-0.3169	-5.26
4.330	0.4436	-0.3171	-5.27
4.335	0.4433	-0.3173	-5.27
4.340	0.4431	-0.3175	-5.27
4.345	0.4431	-0.3176	-5.27
4.350	0.4432	-0.3178	-5.27
4.355	0.4433	-0.3179	-5.26
4.360	0.4433	-0.3182	-5.26
4.365	0.4433	-0.3185	-5.26
4.370	0.4432	-0.3186	-5.26
4.375	0.4432	-0.3191	-5.25
4.380	0.4431	-0.3194	-5.25
4.385	0.4434	-0.3197	-5.25
4.390	0.4436	-0.3197	-5.24
4.395	0.4440	-0.3199	-5.24
4.400	0.4445	-0.3200	-5.23
4.405	0.4448	-0.3199	-5.23
4.410	0.4450	-0.3201	-5.22
4.415	0.4451	-0.3200	-5.22
4.420	0.4454	-0.3201	-5.22
4.425	0.4454	-0.3200	-5.22
4.430	0.4454	-0.3201	-5.22
4.435	0.4454	-0.3199	-5.22
4.440	0.4455	-0.3199	-5.22
4.445	0.4457	-0.3197	-5.22
4.450	0.4461	-0.3196	-5.21
4.455	0.4463	-0.3193	-5.21
4.460	0.4463	-0.3193	-5.21
4.465	0.4465	-0.3193	-5.21
4.470	0.4465	-0.3191	-5.21
4.475	0.4468	-0.3194	-5.21
4.480	0.4472	-0.3194	-5.20
4.485	0.4477	-0.3198	-5.19

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.490	0.4482	-0.3200	-5.18
4.495	0.4488	-0.3204	-5.17
4.500	0.4496	-0.3207	-5.16
4.505	0.4501	-0.3211	-5.15
4.510	0.4506	-0.3213	-5.14
4.515	0.4510	-0.3215	-5.13
4.520	0.4514	-0.3216	-5.13
4.525	0.4516	-0.3216	-5.12
4.530	0.4521	-0.3216	-5.12
4.535	0.4524	-0.3213	-5.12
4.540	0.4527	-0.3212	-5.11
4.545	0.4531	-0.3211	-5.11
4.550	0.4534	-0.3211	-5.11
4.555	0.4535	-0.3211	-5.10
4.560	0.4536	-0.3209	-5.10
4.565	0.4537	-0.3209	-5.10
4.570	0.4539	-0.3209	-5.10
4.575	0.4540	-0.3206	-5.10
4.580	0.4541	-0.3205	-5.10
4.585	0.4543	-0.3205	-5.10
4.590	0.4544	-0.3203	-5.10
4.595	0.4542	-0.3202	-5.10
4.600	0.4542	-0.3201	-5.10
4.605	0.4539	-0.3201	-5.11
4.610	0.4539	-0.3199	-5.11
4.615	0.4538	-0.3199	-5.11
4.620	0.4537	-0.3196	-5.12
4.625	0.4536	-0.3196	-5.12
4.630	0.4537	-0.3196	-5.12
4.635	0.4535	-0.3194	-5.12
4.640	0.4533	-0.3193	-5.12
4.645	0.4532	-0.3192	-5.12
4.650	0.4529	-0.3191	-5.13
4.655	0.4528	-0.3194	-5.13
4.660	0.4528	-0.3190	-5.13
4.665	0.4527	-0.3189	-5.13
4.670	0.4530	-0.3189	-5.13
4.675	0.4530	-0.3186	-5.13
4.680	0.4530	-0.3187	-5.13
4.685	0.4531	-0.3185	-5.13
4.690	0.4530	-0.3187	-5.13
4.695	0.4534	-0.3185	-5.13
4.700	0.4534	-0.3183	-5.13
4.705	0.4527	-0.3184	-5.14
4.710	0.4531	-0.3178	-5.14
4.715	0.4529	-0.3177	-5.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.720	0.4526	-0.3174	-5.15
4.725	0.4526	-0.3173	-5.15
4.730	0.4527	-0.3167	-5.15
4.735	0.4523	-0.3166	-5.16
4.740	0.4521	-0.3162	-5.17
4.745	0.4518	-0.3162	-5.17
4.750	0.4512	-0.3159	-5.18
4.755	0.4506	-0.3157	-5.19
4.760	0.4500	-0.3157	-5.20
4.765	0.4496	-0.3155	-5.20
4.770	0.4490	-0.3157	-5.21
4.775	0.4488	-0.3155	-5.21
4.780	0.4479	-0.3156	-5.23
4.785	0.4479	-0.3159	-5.22
4.790	0.4477	-0.3157	-5.23
4.795	0.4473	-0.3155	-5.23
4.800	0.4472	-0.3154	-5.24
4.805	0.4477	-0.3151	-5.23
4.810	0.4475	-0.3144	-5.24
4.815	0.4476	-0.3142	-5.24
4.820	0.4474	-0.3137	-5.25
4.825	0.4473	-0.3135	-5.25
4.830	0.4470	-0.3130	-5.26
4.835	0.4469	-0.3125	-5.27
4.840	0.4471	-0.3121	-5.27
4.845	0.4471	-0.3117	-5.27
4.850	0.4471	-0.3115	-5.27
4.855	0.4473	-0.3111	-5.27
4.860	0.4471	-0.3108	-5.28
4.865	0.4469	-0.3109	-5.28
4.870	0.4469	-0.3107	-5.28
4.875	0.4469	-0.3106	-5.28
4.880	0.4467	-0.3105	-5.29
4.885	0.4467	-0.3105	-5.29
4.890	0.4466	-0.3107	-5.29
4.895	0.4467	-0.3107	-5.29
4.900	0.4470	-0.3107	-5.28
4.905	0.4469	-0.3108	-5.28
4.910	0.4472	-0.3109	-5.28
4.915	0.4473	-0.3111	-5.28
4.920	0.4474	-0.3112	-5.27
4.925	0.4474	-0.3115	-5.27
4.930	0.4472	-0.3117	-5.27
4.935	0.4472	-0.3121	-5.27
4.940	0.4473	-0.3123	-5.26
4.945	0.4473	-0.3126	-5.26

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	0.4473	-0.3129	-5.26
4.955	0.4474	-0.3131	-5.25
4.960	0.4475	-0.3134	-5.25
4.965	0.4475	-0.3136	-5.25
4.970	0.4473	-0.3139	-5.25
4.975	0.4470	-0.3139	-5.25
4.980	0.4469	-0.3143	-5.25
4.985	0.4470	-0.3147	-5.25
4.990	0.4469	-0.3149	-5.24
4.995	0.4471	-0.3150	-5.24
5.000	0.4471	-0.3152	-5.24
5.005	0.4470	-0.3156	-5.24
5.010	0.4467	-0.3158	-5.24
5.015	0.4464	-0.3159	-5.24
5.020	0.4464	-0.3161	-5.24
5.025	0.4463	-0.3162	-5.24
5.030	0.4463	-0.3162	-5.24
5.035	0.4463	-0.3158	-5.24
5.040	0.4466	-0.3155	-5.24
5.045	0.4469	-0.3152	-5.24
5.050	0.4472	-0.3150	-5.24
5.055	0.4473	-0.3148	-5.24
5.060	0.4477	-0.3146	-5.24
5.065	0.4480	-0.3147	-5.23
5.070	0.4486	-0.3142	-5.23
5.075	0.4490	-0.3142	-5.22
5.080	0.4496	-0.3138	-5.22
5.085	0.4501	-0.3138	-5.21
5.090	0.4506	-0.3135	-5.21
5.095	0.4508	-0.3134	-5.21
5.100	0.4512	-0.3135	-5.20
5.105	0.4512	-0.3133	-5.20
5.110	0.4513	-0.3135	-5.20
5.115	0.4513	-0.3135	-5.20
5.120	0.4517	-0.3136	-5.20
5.125	0.4518	-0.3136	-5.19
5.130	0.4521	-0.3137	-5.19
5.135	0.4523	-0.3135	-5.19
5.140	0.4526	-0.3136	-5.18
5.145	0.4527	-0.3137	-5.18
5.150	0.4528	-0.3137	-5.18
5.155	0.4528	-0.3137	-5.18
5.160	0.4527	-0.3139	-5.18
5.165	0.4525	-0.3138	-5.18
5.170	0.4525	-0.3139	-5.18
5.175	0.4525	-0.3138	-5.18

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.180	0.4525	-0.3137	-5.18
5.185	0.4526	-0.3137	-5.18
5.190	0.4525	-0.3136	-5.18
5.195	0.4523	-0.3135	-5.19
5.200	0.4520	-0.3135	-5.19
5.205	0.4519	-0.3135	-5.19
5.210	0.4515	-0.3134	-5.20
5.215	0.4510	-0.3134	-5.21
5.220	0.4510	-0.3132	-5.21
5.225	0.4508	-0.3131	-5.21
5.230	0.4503	-0.3130	-5.22
5.235	0.4502	-0.3127	-5.22
5.240	0.4497	-0.3126	-5.23
5.245	0.4493	-0.3123	-5.24
5.250	0.4489	-0.3122	-5.24
5.255	0.4485	-0.3119	-5.25
5.260	0.4481	-0.3117	-5.26
5.265	0.4477	-0.3115	-5.27
5.270	0.4471	-0.3113	-5.28
5.275	0.4466	-0.3110	-5.28
5.280	0.4459	-0.3107	-5.30
5.285	0.4455	-0.3106	-5.30
5.290	0.4449	-0.3106	-5.31
5.295	0.4444	-0.3106	-5.32
5.300	0.4439	-0.3102	-5.33
5.305	0.4433	-0.3104	-5.33
5.310	0.4427	-0.3101	-5.34
5.315	0.4423	-0.3101	-5.35
5.320	0.4417	-0.3098	-5.36
5.325	0.4414	-0.3097	-5.36
5.330	0.4408	-0.3096	-5.37
5.335	0.4402	-0.3097	-5.38
5.340	0.4398	-0.3096	-5.39
5.345	0.4395	-0.3094	-5.39
5.350	0.4390	-0.3091	-5.40
5.355	0.4387	-0.3088	-5.41
5.360	0.4384	-0.3086	-5.41
5.365	0.4378	-0.3082	-5.43
5.370	0.4373	-0.3078	-5.44
5.375	0.4367	-0.3074	-5.45
5.380	0.4360	-0.3072	-5.46
5.385	0.4351	-0.3071	-5.47
5.390	0.4344	-0.3068	-5.48
5.395	0.4337	-0.3068	-5.49
5.400	0.4332	-0.3064	-5.50
5.405	0.4324	-0.3065	-5.51

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.410	0.4319	-0.3065	-5.52
5.415	0.4316	-0.3065	-5.53
5.420	0.4305	-0.3063	-5.54
5.425	0.4297	-0.3065	-5.55
5.430	0.4289	-0.3063	-5.56
5.435	0.4284	-0.3064	-5.57
5.440	0.4277	-0.3063	-5.58
5.445	0.4272	-0.3064	-5.59
5.450	0.4269	-0.3063	-5.59
5.455	0.4264	-0.3061	-5.60
5.460	0.4262	-0.3062	-5.60
5.465	0.4254	-0.3061	-5.61
5.470	0.4251	-0.3060	-5.62
5.475	0.4245	-0.3057	-5.63
5.480	0.4239	-0.3057	-5.64
5.485	0.4231	-0.3056	-5.65
5.490	0.4227	-0.3054	-5.66
5.495	0.4222	-0.3052	-5.66
5.500	0.4216	-0.3049	-5.67
5.505	0.4209	-0.3049	-5.69
5.510	0.4203	-0.3050	-5.69
5.515	0.4195	-0.3050	-5.70
5.520	0.4187	-0.3052	-5.71
5.525	0.4184	-0.3052	-5.72
5.530	0.4179	-0.3055	-5.72
5.535	0.4175	-0.3055	-5.72
5.540	0.4172	-0.3056	-5.73
5.545	0.4169	-0.3058	-5.73
5.550	0.4166	-0.3058	-5.73
5.555	0.4162	-0.3060	-5.74
5.560	0.4158	-0.3062	-5.74
5.565	0.4154	-0.3062	-5.75
5.570	0.4151	-0.3063	-5.75
5.575	0.4147	-0.3064	-5.75
5.580	0.4143	-0.3065	-5.76
5.585	0.4137	-0.3066	-5.76
5.590	0.4132	-0.3067	-5.77
5.595	0.4127	-0.3070	-5.78
5.600	0.4123	-0.3072	-5.78
5.605	0.4117	-0.3076	-5.78
5.610	0.4111	-0.3078	-5.79
5.615	0.4105	-0.3083	-5.79
5.620	0.4099	-0.3088	-5.79
5.625	0.4094	-0.3094	-5.80
5.630	0.4094	-0.3098	-5.79
5.635	0.4093	-0.3102	-5.79

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.640	0.4092	-0.3108	-5.78
5.645	0.4090	-0.3111	-5.78
5.650	0.4090	-0.3117	-5.78
5.655	0.4086	-0.3119	-5.78
5.660	0.4082	-0.3124	-5.78
5.665	0.4079	-0.3127	-5.78
5.670	0.4078	-0.3133	-5.78
5.675	0.4076	-0.3137	-5.78
5.680	0.4075	-0.3143	-5.77
5.685	0.4073	-0.3144	-5.77
5.690	0.4073	-0.3145	-5.77
5.695	0.4070	-0.3151	-5.77
5.700	0.4069	-0.3156	-5.76
5.705	0.4070	-0.3160	-5.76
5.710	0.4067	-0.3164	-5.76
5.715	0.4068	-0.3168	-5.75
5.720	0.4067	-0.3169	-5.75
5.725	0.4069	-0.3173	-5.75
5.730	0.4068	-0.3176	-5.74
5.735	0.4068	-0.3181	-5.74
5.740	0.4067	-0.3185	-5.74
5.745	0.4068	-0.3189	-5.73
5.750	0.4065	-0.3191	-5.73
5.755	0.4066	-0.3193	-5.73
5.760	0.4067	-0.3195	-5.73
5.765	0.4068	-0.3198	-5.72
5.770	0.4068	-0.3199	-5.72
5.775	0.4069	-0.3200	-5.72
5.780	0.4067	-0.3202	-5.72
5.785	0.4065	-0.3204	-5.72
5.790	0.4063	-0.3204	-5.72
5.795	0.4061	-0.3205	-5.72
5.800	0.4060	-0.3205	-5.73
5.805	0.4060	-0.3206	-5.72
5.810	0.4059	-0.3206	-5.73
5.815	0.4059	-0.3205	-5.73
5.820	0.4059	-0.3205	-5.73
5.825	0.4058	-0.3203	-5.73
5.830	0.4056	-0.3203	-5.73
5.835	0.4053	-0.3203	-5.74
5.840	0.4048	-0.3202	-5.74
5.845	0.4045	-0.3201	-5.75
5.850	0.4041	-0.3200	-5.76
5.855	0.4038	-0.3200	-5.76
5.860	0.4036	-0.3195	-5.77
5.865	0.4034	-0.3196	-5.77

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	0.4031	-0.3193	-5.78
5.875	0.4028	-0.3192	-5.78
5.880	0.4024	-0.3191	-5.79
5.885	0.4019	-0.3191	-5.80
5.890	0.4013	-0.3189	-5.81
5.895	0.4009	-0.3187	-5.81
5.900	0.4002	-0.3187	-5.82
5.905	0.3998	-0.3188	-5.83
5.910	0.3997	-0.3188	-5.83
5.915	0.3995	-0.3186	-5.83
5.920	0.3992	-0.3186	-5.84
5.925	0.3988	-0.3186	-5.84
5.930	0.3985	-0.3183	-5.85
5.935	0.3982	-0.3180	-5.86
5.940	0.3974	-0.3176	-5.87
5.945	0.3970	-0.3172	-5.88
5.950	0.3966	-0.3171	-5.89
5.955	0.3959	-0.3166	-5.90
5.960	0.3956	-0.3162	-5.91
5.965	0.3948	-0.3157	-5.93
5.970	0.3940	-0.3159	-5.93
5.975	0.3933	-0.3154	-5.95
5.980	0.3927	-0.3157	-5.95
5.985	0.3922	-0.3152	-5.97
5.990	0.3913	-0.3153	-5.98
5.995	0.3908	-0.3157	-5.98
6.000	0.3902	-0.3156	-5.99

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

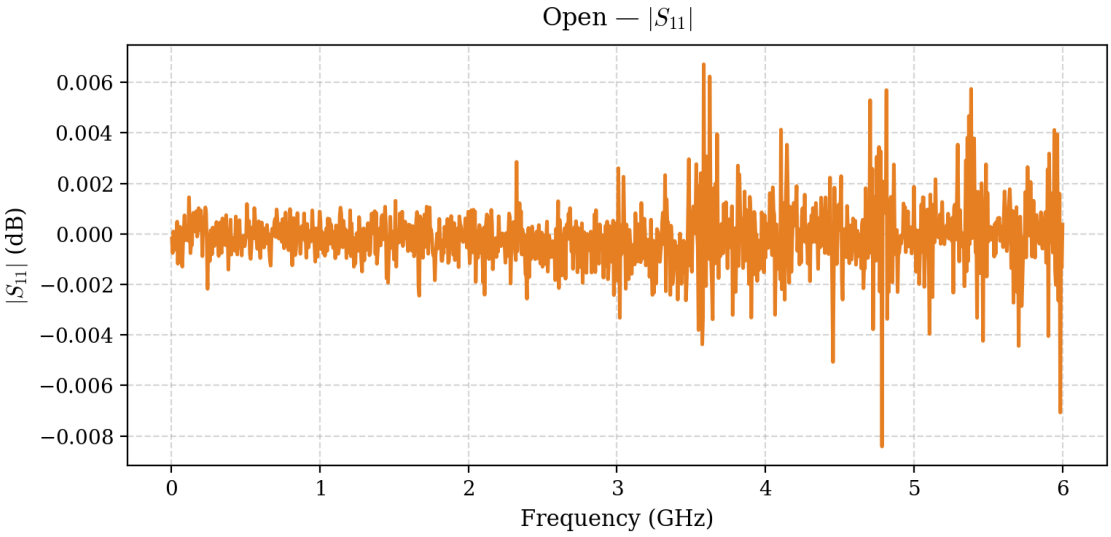
- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.

3 Calibration Standard Measurements

3.1 Open



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0000	+0.0001	-0.00
0.010	0.9999	-0.0000	-0.00
0.015	1.0000	+0.0001	0.00
0.020	1.0000	+0.0000	-0.00
0.025	0.9999	+0.0000	-0.00
0.030	1.0000	+0.0001	-0.00
0.035	0.9999	+0.0001	-0.00
0.040	1.0001	+0.0000	0.00
0.045	0.9999	+0.0000	-0.00
0.050	1.0000	+0.0000	0.00
0.055	0.9999	+0.0000	-0.00
0.060	1.0000	-0.0001	0.00
0.065	1.0000	+0.0000	0.00
0.070	0.9999	-0.0000	-0.00
0.075	0.9999	-0.0000	-0.00
0.080	1.0000	+0.0001	-0.00
0.085	1.0000	-0.0000	0.00
0.090	1.0001	+0.0001	0.00
0.095	1.0000	+0.0000	-0.00
0.100	1.0000	-0.0000	0.00
0.105	1.0000	+0.0000	0.00
0.110	1.0000	+0.0001	-0.00
0.115	1.0000	+0.0001	-0.00
0.120	1.0002	+0.0001	0.00
0.125	1.0000	-0.0000	0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.130	0.9999	-0.0001	-0.00
0.135	1.0000	-0.0001	-0.00
0.140	0.9999	-0.0000	-0.00
0.145	1.0001	+0.0000	0.00
0.150	1.0001	-0.0000	0.00
0.155	1.0000	+0.0002	0.00
0.160	1.0001	+0.0001	0.00
0.165	1.0000	+0.0000	0.00
0.170	1.0001	+0.0000	0.00
0.175	1.0000	+0.0001	-0.00
0.180	1.0000	-0.0001	0.00
0.185	1.0001	+0.0000	0.00
0.190	1.0000	+0.0001	0.00
0.195	1.0000	+0.0001	0.00
0.200	1.0001	+0.0000	0.00
0.205	1.0001	-0.0001	0.00
0.210	0.9999	+0.0001	-0.00
0.215	1.0000	+0.0000	0.00
0.220	1.0000	-0.0000	0.00
0.225	1.0000	+0.0001	0.00
0.230	1.0001	+0.0001	0.00
0.235	1.0001	-0.0000	0.00
0.240	1.0000	+0.0001	-0.00
0.245	0.9997	+0.0000	-0.00
0.250	1.0000	+0.0002	-0.00
0.255	1.0000	-0.0001	-0.00
0.260	0.9999	+0.0000	-0.00
0.265	1.0000	+0.0000	0.00
0.270	1.0000	+0.0001	0.00
0.275	0.9999	+0.0000	-0.00
0.280	1.0000	+0.0001	0.00
0.285	1.0000	-0.0001	-0.00
0.290	0.9999	+0.0000	-0.00
0.295	1.0000	-0.0000	-0.00
0.300	1.0000	-0.0000	0.00
0.305	1.0001	+0.0001	0.00
0.310	1.0000	+0.0000	-0.00
0.315	1.0000	-0.0000	-0.00
0.320	0.9999	-0.0000	-0.00
0.325	1.0000	+0.0001	0.00
0.330	1.0001	+0.0000	0.00
0.335	0.9999	-0.0001	-0.00
0.340	0.9999	+0.0000	-0.00
0.345	1.0001	-0.0001	0.00
0.350	1.0000	+0.0001	0.00
0.355	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.360	1.0000	+0.0000	-0.00
0.365	0.9999	-0.0001	-0.00
0.370	1.0001	-0.0000	0.00
0.375	1.0000	+0.0001	-0.00
0.380	1.0000	+0.0000	0.00
0.385	0.9998	+0.0001	-0.00
0.390	1.0000	-0.0000	-0.00
0.395	1.0000	+0.0000	-0.00
0.400	1.0001	+0.0001	0.00
0.405	0.9999	+0.0001	-0.00
0.410	1.0000	-0.0001	-0.00
0.415	0.9999	+0.0001	-0.00
0.420	0.9999	-0.0000	-0.00
0.425	0.9999	-0.0001	-0.00
0.430	0.9999	-0.0000	-0.00
0.435	1.0001	+0.0000	0.00
0.440	1.0000	+0.0000	-0.00
0.445	0.9999	+0.0000	-0.00
0.450	1.0000	+0.0000	-0.00
0.455	1.0000	+0.0000	0.00
0.460	1.0001	+0.0000	0.00
0.465	0.9999	+0.0001	-0.00
0.470	0.9999	+0.0002	-0.00
0.475	0.9999	+0.0001	-0.00
0.480	1.0000	-0.0000	-0.00
0.485	1.0000	-0.0000	-0.00
0.490	1.0000	+0.0001	-0.00
0.495	0.9999	+0.0001	-0.00
0.500	1.0000	+0.0000	-0.00
0.505	1.0000	-0.0000	-0.00
0.510	1.0001	+0.0001	0.00
0.515	1.0000	+0.0001	0.00
0.520	1.0001	-0.0001	0.00
0.525	1.0001	+0.0000	0.00
0.530	0.9999	+0.0001	-0.00
0.535	0.9999	+0.0001	-0.00
0.540	0.9999	+0.0000	-0.00
0.545	1.0000	+0.0000	0.00
0.550	1.0000	-0.0000	0.00
0.555	1.0000	-0.0001	-0.00
0.560	1.0001	-0.0000	0.00
0.565	1.0001	+0.0000	0.00
0.570	1.0000	+0.0000	-0.00
0.575	1.0000	-0.0001	-0.00
0.580	0.9999	-0.0000	-0.00
0.585	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.590	1.0001	-0.0000	0.00
0.595	0.9999	+0.0000	-0.00
0.600	0.9999	-0.0000	-0.00
0.605	1.0000	-0.0000	-0.00
0.610	1.0001	+0.0000	0.00
0.615	0.9999	+0.0001	-0.00
0.620	1.0000	-0.0000	0.00
0.625	0.9999	-0.0000	-0.00
0.630	1.0000	-0.0001	-0.00
0.635	1.0000	+0.0001	-0.00
0.640	1.0000	-0.0000	-0.00
0.645	1.0000	+0.0000	0.00
0.650	1.0000	+0.0000	-0.00
0.655	1.0000	+0.0001	-0.00
0.660	0.9999	-0.0000	-0.00
0.665	0.9999	+0.0000	-0.00
0.670	0.9999	+0.0000	-0.00
0.675	1.0000	-0.0000	-0.00
0.680	1.0000	-0.0000	0.00
0.685	1.0000	+0.0001	0.00
0.690	1.0000	-0.0001	-0.00
0.695	1.0001	+0.0000	0.00
0.700	1.0001	+0.0001	0.00
0.705	1.0000	+0.0000	-0.00
0.710	1.0001	-0.0001	0.00
0.715	1.0001	-0.0000	0.00
0.720	1.0000	+0.0001	0.00
0.725	1.0000	+0.0000	-0.00
0.730	1.0000	+0.0000	0.00
0.735	1.0000	+0.0001	0.00
0.740	1.0000	+0.0000	-0.00
0.745	1.0000	-0.0000	-0.00
0.750	1.0001	+0.0001	0.00
0.755	1.0000	-0.0001	0.00
0.760	0.9999	+0.0000	-0.00
0.765	1.0000	+0.0000	0.00
0.770	1.0001	+0.0000	0.00
0.775	1.0000	-0.0001	0.00
0.780	1.0000	-0.0000	0.00
0.785	0.9999	-0.0000	-0.00
0.790	0.9999	+0.0000	-0.00
0.795	1.0000	-0.0000	-0.00
0.800	1.0001	-0.0001	0.00
0.805	1.0001	+0.0000	0.00
0.810	0.9999	+0.0001	-0.00
0.815	0.9999	+0.0000	-0.00

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.820	1.0000	+0.0000	-0.00
0.825	1.0001	+0.0000	0.00
0.830	1.0000	+0.0000	0.00
0.835	0.9999	-0.0000	-0.00
0.840	0.9999	-0.0001	-0.00
0.845	1.0000	-0.0001	0.00
0.850	1.0001	+0.0000	0.00
0.855	1.0000	+0.0000	0.00
0.860	1.0000	+0.0001	0.00
0.865	1.0000	+0.0001	-0.00
0.870	0.9999	+0.0001	-0.00
0.875	1.0000	-0.0000	-0.00
0.880	1.0000	+0.0001	0.00
0.885	1.0000	+0.0001	0.00
0.890	1.0000	+0.0001	-0.00
0.895	1.0001	+0.0000	0.00
0.900	1.0000	-0.0000	-0.00
0.905	1.0000	+0.0000	-0.00
0.910	0.9999	+0.0001	-0.00
0.915	1.0000	+0.0001	0.00
0.920	1.0000	+0.0001	-0.00
0.925	0.9999	+0.0001	-0.00
0.930	1.0000	+0.0001	-0.00
0.935	1.0000	+0.0001	0.00
0.940	1.0000	+0.0001	-0.00
0.945	1.0000	-0.0000	-0.00
0.950	1.0001	+0.0000	0.00
0.955	1.0000	+0.0001	0.00
0.960	1.0000	+0.0000	-0.00
0.965	0.9999	-0.0000	-0.00
0.970	0.9999	+0.0001	-0.00
0.975	1.0000	+0.0001	0.00
0.980	1.0000	+0.0001	-0.00
0.985	0.9999	+0.0000	-0.00
0.990	1.0000	+0.0001	-0.00
0.995	1.0000	+0.0001	0.00
1.000	1.0001	-0.0000	0.00
1.005	1.0001	+0.0001	0.00
1.010	0.9999	-0.0000	-0.00
1.015	1.0000	+0.0000	0.00
1.020	1.0000	+0.0001	-0.00
1.025	0.9999	+0.0001	-0.00
1.030	1.0000	+0.0000	0.00
1.035	0.9999	-0.0001	-0.00
1.040	1.0000	+0.0001	-0.00
1.045	1.0000	+0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.050	0.9999	+0.0001	-0.00
1.055	1.0001	+0.0001	0.00
1.060	1.0000	+0.0002	-0.00
1.065	1.0001	+0.0000	0.00
1.070	1.0000	+0.0001	-0.00
1.075	1.0000	+0.0001	0.00
1.080	1.0000	+0.0000	0.00
1.085	1.0000	+0.0000	0.00
1.090	0.9999	+0.0001	-0.00
1.095	0.9999	+0.0001	-0.00
1.100	1.0000	+0.0001	0.00
1.105	0.9999	+0.0000	-0.00
1.110	1.0000	+0.0002	0.00
1.115	1.0000	+0.0002	-0.00
1.120	1.0000	+0.0000	-0.00
1.125	0.9999	-0.0000	-0.00
1.130	0.9999	+0.0002	-0.00
1.135	1.0001	+0.0000	0.00
1.140	1.0001	-0.0000	0.00
1.145	1.0001	+0.0001	0.00
1.150	1.0000	-0.0000	0.00
1.155	0.9999	+0.0001	-0.00
1.160	1.0001	-0.0000	0.00
1.165	1.0000	+0.0000	0.00
1.170	1.0000	+0.0000	0.00
1.175	1.0000	+0.0001	0.00
1.180	0.9999	-0.0000	-0.00
1.185	1.0000	+0.0001	0.00
1.190	1.0000	+0.0000	0.00
1.195	1.0000	+0.0001	-0.00
1.200	0.9999	+0.0001	-0.00
1.205	1.0001	+0.0000	0.00
1.210	0.9999	-0.0000	-0.00
1.215	1.0000	+0.0000	0.00
1.220	0.9999	+0.0000	-0.00
1.225	1.0000	+0.0000	-0.00
1.230	0.9999	+0.0000	-0.00
1.235	1.0000	+0.0001	-0.00
1.240	0.9999	-0.0001	-0.00
1.245	1.0000	+0.0000	-0.00
1.250	1.0000	+0.0001	0.00
1.255	1.0000	+0.0000	-0.00
1.260	0.9999	-0.0001	-0.00
1.265	1.0001	+0.0000	0.00
1.270	1.0000	+0.0001	0.00
1.275	1.0000	-0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.280	0.9999	+0.0000	-0.00
1.285	0.9998	+0.0000	-0.00
1.290	1.0000	+0.0000	-0.00
1.295	1.0000	+0.0001	0.00
1.300	0.9999	-0.0000	-0.00
1.305	1.0000	-0.0000	0.00
1.310	0.9999	+0.0000	-0.00
1.315	1.0000	+0.0000	-0.00
1.320	1.0000	+0.0001	0.00
1.325	1.0000	-0.0000	0.00
1.330	1.0000	+0.0001	-0.00
1.335	0.9999	+0.0001	-0.00
1.340	0.9999	+0.0001	-0.00
1.345	1.0000	+0.0002	0.00
1.350	1.0001	+0.0000	0.00
1.355	1.0001	+0.0001	0.00
1.360	1.0000	-0.0000	0.00
1.365	1.0000	-0.0000	0.00
1.370	1.0001	+0.0001	0.00
1.375	0.9999	+0.0000	-0.00
1.380	1.0001	+0.0001	0.00
1.385	0.9999	+0.0001	-0.00
1.390	0.9999	-0.0000	-0.00
1.395	1.0000	-0.0000	-0.00
1.400	1.0000	-0.0001	-0.00
1.405	1.0000	+0.0002	0.00
1.410	0.9999	+0.0001	-0.00
1.415	1.0001	+0.0001	0.00
1.420	1.0001	-0.0001	0.00
1.425	1.0000	-0.0000	-0.00
1.430	1.0001	+0.0001	0.00
1.435	1.0000	+0.0002	0.00
1.440	1.0001	+0.0002	0.00
1.445	1.0000	+0.0001	0.00
1.450	0.9998	+0.0000	-0.00
1.455	1.0000	+0.0001	-0.00
1.460	0.9998	+0.0001	-0.00
1.465	0.9999	+0.0001	-0.00
1.470	1.0000	+0.0001	-0.00
1.475	1.0001	-0.0001	0.00
1.480	0.9999	+0.0001	-0.00
1.485	1.0000	-0.0000	-0.00
1.490	1.0000	-0.0000	-0.00
1.495	1.0000	+0.0001	0.00
1.500	0.9999	+0.0001	-0.00
1.505	1.0000	+0.0001	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.510	1.0002	+0.0001	0.00
1.515	0.9999	+0.0001	-0.00
1.520	1.0000	+0.0001	-0.00
1.525	1.0000	+0.0001	-0.00
1.530	1.0000	+0.0000	0.00
1.535	1.0000	+0.0001	-0.00
1.540	1.0001	-0.0000	0.00
1.545	1.0000	+0.0002	-0.00
1.550	1.0000	+0.0000	0.00
1.555	1.0001	-0.0000	0.00
1.560	0.9999	+0.0000	-0.00
1.565	1.0000	+0.0001	0.00
1.570	1.0000	+0.0001	0.00
1.575	1.0001	+0.0001	0.00
1.580	1.0000	+0.0001	-0.00
1.585	1.0000	-0.0000	0.00
1.590	1.0000	+0.0000	-0.00
1.595	0.9999	-0.0000	-0.00
1.600	1.0000	+0.0001	0.00
1.605	1.0000	+0.0000	0.00
1.610	1.0000	+0.0001	0.00
1.615	1.0000	+0.0001	0.00
1.620	0.9999	+0.0001	-0.00
1.625	1.0000	+0.0000	0.00
1.630	1.0000	-0.0000	0.00
1.635	0.9999	+0.0001	-0.00
1.640	0.9999	+0.0001	-0.00
1.645	1.0000	+0.0001	-0.00
1.650	1.0001	-0.0000	0.00
1.655	1.0001	+0.0000	0.00
1.660	1.0000	+0.0002	0.00
1.665	0.9998	+0.0001	-0.00
1.670	0.9997	+0.0001	-0.00
1.675	1.0001	+0.0001	0.00
1.680	1.0000	+0.0000	-0.00
1.685	1.0000	-0.0001	0.00
1.690	1.0001	+0.0000	0.00
1.695	0.9999	-0.0000	-0.00
1.700	1.0000	-0.0001	0.00
1.705	1.0001	+0.0001	0.00
1.710	0.9999	-0.0000	-0.00
1.715	1.0000	-0.0001	-0.00
1.720	1.0001	-0.0000	0.00
1.725	1.0001	+0.0001	0.00
1.730	0.9999	-0.0001	-0.00
1.735	1.0001	+0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.740	1.0001	+0.0001	0.00
1.745	1.0000	+0.0002	-0.00
1.750	1.0000	-0.0001	0.00
1.755	1.0000	-0.0001	-0.00
1.760	0.9999	-0.0000	-0.00
1.765	0.9999	-0.0000	-0.00
1.770	0.9999	+0.0002	-0.00
1.775	0.9998	+0.0001	-0.00
1.780	0.9999	-0.0000	-0.00
1.785	0.9999	+0.0001	-0.00
1.790	1.0000	+0.0000	0.00
1.795	0.9999	-0.0001	-0.00
1.800	1.0000	-0.0001	-0.00
1.805	1.0001	+0.0001	0.00
1.810	1.0000	-0.0000	-0.00
1.815	1.0000	-0.0000	-0.00
1.820	0.9999	+0.0000	-0.00
1.825	0.9999	-0.0001	-0.00
1.830	0.9999	+0.0000	-0.00
1.835	0.9999	-0.0000	-0.00
1.840	1.0000	+0.0000	0.00
1.845	1.0000	-0.0001	0.00
1.850	1.0000	+0.0001	0.00
1.855	0.9999	-0.0001	-0.00
1.860	1.0001	+0.0001	0.00
1.865	0.9999	+0.0000	-0.00
1.870	0.9999	+0.0001	-0.00
1.875	1.0000	+0.0000	-0.00
1.880	0.9999	-0.0000	-0.00
1.885	1.0001	+0.0000	0.00
1.890	1.0000	-0.0001	0.00
1.895	1.0000	-0.0000	0.00
1.900	0.9999	+0.0001	-0.00
1.905	1.0000	+0.0001	0.00
1.910	0.9999	+0.0000	-0.00
1.915	1.0000	+0.0001	-0.00
1.920	1.0000	+0.0000	-0.00
1.925	1.0000	-0.0000	-0.00
1.930	1.0000	+0.0000	0.00
1.935	0.9998	+0.0000	-0.00
1.940	1.0001	-0.0001	0.00
1.945	1.0000	+0.0001	0.00
1.950	1.0000	+0.0001	0.00
1.955	1.0000	+0.0001	0.00
1.960	1.0000	+0.0000	-0.00
1.965	1.0000	+0.0000	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	$\lvert S_{11}\rvert$ **(dB)**
1.970	1.0000	-0.0001	-0.00
1.975	1.0000	+0.0000	-0.00
1.980	1.0001	+0.0000	0.00
1.985	1.0001	-0.0001	0.00
1.990	0.9999	-0.0001	-0.00
1.995	1.0000	+0.0001	-0.00
2.000	1.0001	+0.0001	0.00
2.005	1.0000	+0.0000	-0.00
2.010	0.9999	+0.0002	-0.00
2.015	0.9999	+0.0000	-0.00
2.020	1.0001	+0.0001	0.00
2.025	1.0000	+0.0000	-0.00
2.030	0.9999	-0.0001	-0.00
2.035	1.0001	-0.0001	0.00
2.040	1.0000	-0.0000	0.00
2.045	1.0001	+0.0001	0.00
2.050	0.9998	+0.0001	-0.00
2.055	0.9999	+0.0001	-0.00
2.060	1.0000	+0.0001	0.00
2.065	0.9999	+0.0000	-0.00
2.070	1.0000	-0.0001	0.00
2.075	1.0000	+0.0000	0.00
2.080	1.0000	+0.0000	0.00
2.085	0.9999	-0.0001	-0.00
2.090	1.0000	+0.0000	-0.00
2.095	0.9999	-0.0000	-0.00
2.100	0.9998	+0.0000	-0.00
2.105	1.0000	-0.0000	0.00
2.110	0.9997	+0.0000	-0.00
2.115	1.0000	+0.0001	-0.00
2.120	1.0001	+0.0001	0.00
2.125	1.0001	+0.0001	0.00
2.130	1.0001	+0.0000	0.00
2.135	1.0000	+0.0001	-0.00
2.140	0.9999	-0.0001	-0.00
2.145	1.0000	-0.0000	-0.00
2.150	0.9999	-0.0002	-0.00
2.155	1.0001	-0.0001	0.00
2.160	1.0000	+0.0001	0.00
2.165	0.9998	-0.0000	-0.00
2.170	0.9999	+0.0000	-0.00
2.175	0.9999	+0.0001	-0.00
2.180	0.9999	+0.0000	-0.00
2.185	1.0000	+0.0000	-0.00
2.190	0.9999	+0.0001	-0.00
2.195	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.200	0.9999	+0.0001	-0.00
2.205	1.0001	-0.0000	0.00
2.210	1.0000	+0.0001	-0.00
2.215	1.0000	+0.0001	-0.00
2.220	0.9999	+0.0001	-0.00
2.225	1.0000	+0.0002	-0.00
2.230	1.0000	+0.0001	0.00
2.235	1.0000	-0.0001	-0.00
2.240	1.0000	-0.0000	-0.00
2.245	0.9999	+0.0001	-0.00
2.250	0.9999	-0.0000	-0.00
2.255	1.0000	+0.0001	0.00
2.260	0.9999	+0.0002	-0.00
2.265	0.9999	+0.0001	-0.00
2.270	1.0000	-0.0000	0.00
2.275	1.0001	+0.0001	0.00
2.280	1.0001	+0.0000	0.00
2.285	0.9998	+0.0001	-0.00
2.290	0.9999	+0.0000	-0.00
2.295	1.0000	-0.0001	0.00
2.300	1.0000	+0.0000	-0.00
2.305	1.0001	+0.0001	0.00
2.310	1.0000	-0.0001	-0.00
2.315	0.9999	-0.0001	-0.00
2.320	0.9999	+0.0000	-0.00
2.325	1.0003	-0.0001	0.00
2.330	0.9999	+0.0001	-0.00
2.335	1.0000	+0.0001	0.00
2.340	1.0000	-0.0000	-0.00
2.345	1.0001	+0.0002	0.00
2.350	1.0000	+0.0001	0.00
2.355	1.0001	-0.0001	0.00
2.360	0.9999	+0.0000	-0.00
2.365	1.0000	+0.0002	0.00
2.370	0.9999	+0.0001	-0.00
2.375	1.0000	+0.0000	-0.00
2.380	0.9999	+0.0000	-0.00
2.385	0.9998	+0.0002	-0.00
2.390	0.9998	+0.0001	-0.00
2.395	0.9997	+0.0002	-0.00
2.400	1.0000	+0.0001	-0.00
2.405	1.0000	+0.0001	0.00
2.410	0.9998	-0.0001	-0.00
2.415	1.0000	+0.0001	0.00
2.420	0.9999	+0.0001	-0.00
2.425	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.430	1.0000	+0.0002	0.00
2.435	1.0000	-0.0000	-0.00
2.440	0.9999	+0.0001	-0.00
2.445	0.9999	-0.0001	-0.00
2.450	0.9999	-0.0001	-0.00
2.455	0.9999	-0.0001	-0.00
2.460	1.0000	-0.0001	0.00
2.465	0.9999	-0.0000	-0.00
2.470	1.0000	+0.0001	-0.00
2.475	1.0000	+0.0001	0.00
2.480	1.0000	+0.0001	-0.00
2.485	0.9999	-0.0000	-0.00
2.490	0.9999	+0.0000	-0.00
2.495	1.0000	+0.0000	0.00
2.500	0.9998	+0.0000	-0.00
2.505	1.0000	-0.0000	-0.00
2.510	0.9999	-0.0000	-0.00
2.515	1.0000	+0.0000	-0.00
2.520	0.9999	+0.0000	-0.00
2.525	1.0000	+0.0002	-0.00
2.530	1.0001	+0.0001	0.00
2.535	0.9998	+0.0000	-0.00
2.540	0.9999	+0.0001	-0.00
2.545	1.0000	-0.0000	-0.00
2.550	1.0000	+0.0001	-0.00
2.555	0.9999	-0.0000	-0.00
2.560	0.9999	+0.0001	-0.00
2.565	0.9999	+0.0000	-0.00
2.570	1.0000	+0.0000	-0.00
2.575	0.9999	-0.0001	-0.00
2.580	0.9999	+0.0000	-0.00
2.585	0.9998	+0.0002	-0.00
2.590	1.0000	+0.0000	0.00
2.595	1.0000	-0.0001	-0.00
2.600	1.0000	-0.0000	-0.00
2.605	1.0001	+0.0002	0.00
2.610	0.9998	-0.0001	-0.00
2.615	1.0001	-0.0002	0.00
2.620	1.0000	-0.0000	0.00
2.625	1.0000	+0.0000	-0.00
2.630	0.9998	-0.0001	-0.00
2.635	1.0000	+0.0001	-0.00
2.640	0.9999	+0.0001	-0.00
2.645	0.9999	-0.0000	-0.00
2.650	1.0000	+0.0002	0.00
2.655	0.9998	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.660	1.0000	+0.0001	0.00
2.665	1.0000	-0.0000	0.00
2.670	1.0000	+0.0002	-0.00
2.675	1.0001	-0.0001	0.00
2.680	1.0000	-0.0000	0.00
2.685	1.0001	+0.0001	0.00
2.690	1.0000	+0.0003	0.00
2.695	0.9999	+0.0001	-0.00
2.700	1.0000	+0.0000	-0.00
2.705	0.9998	+0.0000	-0.00
2.710	1.0000	+0.0001	-0.00
2.715	1.0000	-0.0002	0.00
2.720	0.9999	+0.0000	-0.00
2.725	1.0001	+0.0002	0.00
2.730	1.0000	+0.0003	0.00
2.735	0.9998	+0.0003	-0.00
2.740	1.0000	-0.0000	0.00
2.745	1.0000	+0.0000	-0.00
2.750	0.9998	+0.0002	-0.00
2.755	0.9999	+0.0002	-0.00
2.760	0.9999	+0.0002	-0.00
2.765	1.0000	+0.0001	0.00
2.770	0.9998	+0.0001	-0.00
2.775	1.0000	+0.0001	-0.00
2.780	1.0000	+0.0001	0.00
2.785	0.9999	+0.0001	-0.00
2.790	1.0000	+0.0001	-0.00
2.795	0.9998	+0.0000	-0.00
2.800	1.0000	+0.0001	0.00
2.805	1.0000	-0.0000	-0.00
2.810	1.0000	+0.0002	-0.00
2.815	1.0000	+0.0000	-0.00
2.820	0.9999	+0.0000	-0.00
2.825	1.0000	+0.0002	0.00
2.830	0.9998	+0.0002	-0.00
2.835	0.9999	+0.0001	-0.00
2.840	0.9999	+0.0003	-0.00
2.845	0.9999	+0.0000	-0.00
2.850	1.0001	+0.0001	0.00
2.855	0.9998	+0.0002	-0.00
2.860	1.0001	+0.0000	0.00
2.865	0.9999	+0.0003	-0.00
2.870	1.0000	+0.0001	-0.00
2.875	1.0000	+0.0001	-0.00
2.880	0.9999	+0.0002	-0.00
2.885	1.0000	-0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.890	1.0000	+0.0004	-0.00
2.895	0.9999	+0.0002	-0.00
2.900	1.0000	+0.0001	-0.00
2.905	1.0000	+0.0002	-0.00
2.910	0.9998	+0.0001	-0.00
2.915	1.0000	+0.0001	-0.00
2.920	0.9999	+0.0001	-0.00
2.925	1.0000	-0.0001	0.00
2.930	0.9999	+0.0002	-0.00
2.935	1.0001	+0.0001	0.00
2.940	0.9998	+0.0001	-0.00
2.945	0.9998	+0.0000	-0.00
2.950	1.0000	+0.0000	0.00
2.955	1.0000	-0.0001	-0.00
2.960	0.9999	+0.0003	-0.00
2.965	0.9999	+0.0000	-0.00
2.970	0.9999	+0.0001	-0.00
2.975	1.0001	+0.0001	0.00
2.980	0.9997	-0.0002	-0.00
2.985	0.9999	+0.0002	-0.00
2.990	0.9998	+0.0002	-0.00
2.995	1.0000	-0.0000	0.00
3.000	0.9999	+0.0002	-0.00
3.005	0.9998	+0.0003	-0.00
3.010	1.0003	-0.0002	0.00
3.015	1.0000	+0.0002	-0.00
3.020	0.9996	+0.0004	-0.00
3.025	1.0000	+0.0001	0.00
3.030	0.9999	+0.0002	-0.00
3.035	0.9999	-0.0000	-0.00
3.040	0.9999	+0.0002	-0.00
3.045	1.0003	+0.0000	0.00
3.050	1.0001	+0.0001	0.00
3.055	1.0000	+0.0000	0.00
3.060	0.9997	+0.0001	-0.00
3.065	1.0000	-0.0003	0.00
3.070	0.9999	+0.0003	-0.00
3.075	0.9998	+0.0001	-0.00
3.080	0.9998	+0.0001	-0.00
3.085	1.0000	-0.0000	0.00
3.090	1.0000	+0.0001	-0.00
3.095	0.9999	+0.0001	-0.00
3.100	0.9998	-0.0001	-0.00
3.105	0.9999	-0.0001	-0.00
3.110	0.9999	-0.0001	-0.00
3.115	1.0000	-0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.120	0.9999	+0.0001	-0.00
3.125	1.0001	+0.0001	0.00
3.130	1.0000	+0.0001	0.00
3.135	1.0000	+0.0000	-0.00
3.140	0.9999	+0.0000	-0.00
3.145	1.0000	+0.0001	0.00
3.150	0.9999	+0.0001	-0.00
3.155	0.9999	-0.0000	-0.00
3.160	0.9999	-0.0000	-0.00
3.165	1.0001	+0.0003	0.00
3.170	0.9999	-0.0000	-0.00
3.175	0.9999	+0.0002	-0.00
3.180	1.0000	-0.0001	-0.00
3.185	1.0001	-0.0001	0.00
3.190	1.0000	+0.0002	0.00
3.195	0.9999	+0.0001	-0.00
3.200	0.9999	+0.0002	-0.00
3.205	1.0000	+0.0000	-0.00
3.210	1.0000	+0.0002	-0.00
3.215	1.0000	+0.0000	-0.00
3.220	0.9998	+0.0001	-0.00
3.225	0.9999	+0.0000	-0.00
3.230	1.0001	-0.0002	0.00
3.235	0.9998	-0.0000	-0.00
3.240	0.9997	+0.0001	-0.00
3.245	1.0000	+0.0000	-0.00
3.250	0.9997	+0.0003	-0.00
3.255	0.9999	+0.0002	-0.00
3.260	0.9999	-0.0001	-0.00
3.265	0.9998	-0.0001	-0.00
3.270	1.0000	+0.0000	0.00
3.275	1.0000	+0.0001	0.00
3.280	1.0000	-0.0001	-0.00
3.285	0.9999	+0.0001	-0.00
3.290	0.9999	+0.0001	-0.00
3.295	0.9999	+0.0001	-0.00
3.300	1.0000	-0.0003	-0.00
3.305	1.0001	+0.0001	0.00
3.310	1.0001	+0.0002	0.00
3.315	1.0000	+0.0000	-0.00
3.320	1.0000	+0.0001	-0.00
3.325	1.0003	-0.0000	0.00
3.330	0.9997	-0.0001	-0.00
3.335	1.0002	+0.0001	0.00
3.340	1.0000	-0.0000	0.00
3.345	1.0001	-0.0001	0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.350	0.9999	+0.0002	-0.00
3.355	1.0000	-0.0001	0.00
3.360	0.9997	+0.0003	-0.00
3.365	1.0000	+0.0003	0.00
3.370	1.0000	+0.0001	-0.00
3.375	0.9998	+0.0001	-0.00
3.380	0.9998	+0.0001	-0.00
3.385	0.9999	+0.0001	-0.00
3.390	1.0000	+0.0001	-0.00
3.395	0.9999	-0.0000	-0.00
3.400	0.9999	+0.0003	-0.00
3.405	0.9999	+0.0002	-0.00
3.410	0.9997	+0.0002	-0.00
3.415	0.9999	+0.0004	-0.00
3.420	0.9999	+0.0003	-0.00
3.425	1.0000	+0.0001	0.00
3.430	0.9999	+0.0001	-0.00
3.435	0.9998	+0.0002	-0.00
3.440	1.0000	+0.0002	-0.00
3.445	0.9999	+0.0003	-0.00
3.450	1.0000	+0.0003	-0.00
3.455	0.9998	+0.0003	-0.00
3.460	1.0000	+0.0003	-0.00
3.465	0.9997	+0.0002	-0.00
3.470	0.9998	+0.0003	-0.00
3.475	0.9999	+0.0000	-0.00
3.480	1.0001	+0.0002	0.00
3.485	1.0003	-0.0000	0.00
3.490	1.0000	+0.0002	-0.00
3.495	0.9999	-0.0000	-0.00
3.500	1.0001	+0.0001	0.00
3.505	0.9998	+0.0000	-0.00
3.510	1.0000	+0.0000	0.00
3.515	1.0001	+0.0002	0.00
3.520	1.0000	+0.0003	0.00
3.525	1.0000	+0.0005	0.00
3.530	1.0001	+0.0000	0.00
3.535	1.0003	+0.0001	0.00
3.540	1.0000	+0.0006	0.00
3.545	0.9999	+0.0000	-0.00
3.550	0.9996	+0.0004	-0.00
3.555	1.0002	-0.0002	0.00
3.560	1.0000	+0.0001	-0.00
3.565	1.0003	-0.0004	0.00
3.570	1.0000	+0.0003	-0.00
3.575	0.9995	+0.0005	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.580	0.9996	+0.0006	-0.00
3.585	1.0008	-0.0003	0.01
3.590	1.0000	-0.0000	-0.00
3.595	1.0003	-0.0002	0.00
3.600	0.9998	+0.0003	-0.00
3.605	1.0004	+0.0001	0.00
3.610	1.0003	+0.0003	0.00
3.615	1.0000	+0.0001	0.00
3.620	0.9999	+0.0000	-0.00
3.625	1.0007	-0.0009	0.01
3.630	1.0002	+0.0005	0.00
3.635	0.9999	+0.0001	-0.00
3.640	0.9998	+0.0000	-0.00
3.645	0.9996	+0.0001	-0.00
3.650	1.0001	+0.0005	0.00
3.655	1.0002	+0.0001	0.00
3.660	0.9999	+0.0003	-0.00
3.665	1.0002	+0.0001	0.00
3.670	0.9999	+0.0004	-0.00
3.675	1.0005	+0.0004	0.00
3.680	1.0000	+0.0002	0.00
3.685	0.9998	+0.0003	-0.00
3.690	1.0000	+0.0001	0.00
3.695	0.9999	+0.0000	-0.00
3.700	0.9999	+0.0001	-0.00
3.705	1.0000	-0.0000	0.00
3.710	1.0000	+0.0001	0.00
3.715	0.9999	+0.0000	-0.00
3.720	1.0001	-0.0001	0.00
3.725	0.9999	+0.0001	-0.00
3.730	1.0000	+0.0001	0.00
3.735	1.0001	+0.0002	0.00
3.740	1.0000	+0.0002	-0.00
3.745	0.9998	-0.0000	-0.00
3.750	1.0000	+0.0001	0.00
3.755	1.0001	+0.0000	0.00
3.760	1.0000	-0.0000	0.00
3.765	0.9999	+0.0003	-0.00
3.770	1.0000	+0.0002	0.00
3.775	1.0001	+0.0001	0.00
3.780	0.9996	+0.0011	-0.00
3.785	0.9999	+0.0005	-0.00
3.790	0.9998	+0.0000	-0.00
3.795	1.0000	-0.0001	0.00
3.800	0.9997	-0.0000	-0.00
3.805	0.9999	-0.0003	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.810	1.0000	+0.0001	-0.00
3.815	1.0003	-0.0000	0.00
3.820	0.9999	+0.0001	-0.00
3.825	1.0003	+0.0002	0.00
3.830	1.0001	+0.0001	0.00
3.835	1.0000	+0.0001	0.00
3.840	1.0001	-0.0001	0.00
3.845	1.0000	+0.0001	-0.00
3.850	0.9998	-0.0001	-0.00
3.855	0.9999	+0.0003	-0.00
3.860	1.0002	+0.0002	0.00
3.865	0.9999	+0.0005	-0.00
3.870	1.0001	+0.0000	0.00
3.875	0.9999	+0.0001	-0.00
3.880	1.0000	-0.0000	0.00
3.885	0.9999	+0.0001	-0.00
3.890	0.9998	-0.0002	-0.00
3.895	1.0002	-0.0003	0.00
3.900	1.0001	+0.0002	0.00
3.905	0.9996	+0.0002	-0.00
3.910	0.9999	+0.0003	-0.00
3.915	1.0000	+0.0000	-0.00
3.920	0.9998	+0.0001	-0.00
3.925	1.0001	+0.0001	0.00
3.930	1.0001	-0.0000	0.00
3.935	1.0000	+0.0001	-0.00
3.940	1.0000	-0.0000	-0.00
3.945	0.9998	-0.0002	-0.00
3.950	1.0000	+0.0001	-0.00
3.955	0.9998	+0.0001	-0.00
3.960	1.0000	+0.0001	-0.00
3.965	0.9999	-0.0001	-0.00
3.970	0.9998	+0.0002	-0.00
3.975	1.0001	+0.0001	0.00
3.980	1.0000	+0.0001	0.00
3.985	1.0000	+0.0001	0.00
3.990	0.9998	+0.0000	-0.00
3.995	1.0001	+0.0001	0.00
4.000	0.9999	+0.0001	-0.00
4.005	1.0000	+0.0002	0.00
4.010	1.0000	+0.0002	0.00
4.015	0.9999	+0.0001	-0.00
4.020	0.9999	+0.0002	-0.00
4.025	1.0000	-0.0001	-0.00
4.030	1.0002	+0.0003	0.00
4.035	1.0000	+0.0002	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.040	1.0000	+0.0001	0.00
4.045	0.9999	+0.0002	-0.00
4.050	1.0002	-0.0001	0.00
4.055	1.0001	+0.0001	0.00
4.060	1.0001	+0.0002	0.00
4.065	0.9996	+0.0004	-0.00
4.070	0.9999	+0.0003	-0.00
4.075	1.0000	+0.0002	0.00
4.080	0.9999	+0.0001	-0.00
4.085	0.9998	+0.0003	-0.00
4.090	0.9999	+0.0000	-0.00
4.095	1.0000	+0.0003	-0.00
4.100	0.9998	+0.0001	-0.00
4.105	1.0005	+0.0003	0.00
4.110	0.9997	-0.0001	-0.00
4.115	1.0001	+0.0002	0.00
4.120	0.9998	+0.0003	-0.00
4.125	1.0001	-0.0002	0.00
4.130	0.9997	-0.0002	-0.00
4.135	1.0001	+0.0002	0.00
4.140	1.0002	+0.0003	0.00
4.145	1.0004	-0.0006	0.00
4.150	1.0002	+0.0004	0.00
4.155	0.9999	+0.0003	-0.00
4.160	0.9998	-0.0000	-0.00
4.165	1.0001	+0.0001	0.00
4.170	1.0000	+0.0002	-0.00
4.175	0.9998	+0.0002	-0.00
4.180	1.0000	+0.0000	-0.00
4.185	1.0001	+0.0000	0.00
4.190	1.0001	+0.0001	0.00
4.195	1.0001	+0.0001	0.00
4.200	1.0000	+0.0001	0.00
4.205	1.0001	-0.0003	0.00
4.210	1.0002	+0.0002	0.00
4.215	0.9999	+0.0003	-0.00
4.220	1.0000	+0.0001	0.00
4.225	1.0000	+0.0000	-0.00
4.230	0.9999	-0.0000	-0.00
4.235	1.0000	+0.0003	-0.00
4.240	0.9999	+0.0001	-0.00
4.245	1.0001	-0.0003	0.00
4.250	1.0001	+0.0001	0.00
4.255	1.0000	-0.0003	-0.00
4.260	1.0000	+0.0001	-0.00
4.265	1.0002	-0.0001	0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.270	1.0000	+0.0000	-0.00
4.275	1.0000	+0.0001	0.00
4.280	1.0000	+0.0001	-0.00
4.285	0.9999	+0.0001	-0.00
4.290	1.0000	+0.0001	-0.00
4.295	1.0000	+0.0002	0.00
4.300	0.9998	+0.0000	-0.00
4.305	1.0000	+0.0001	0.00
4.310	1.0001	+0.0001	0.00
4.315	1.0000	+0.0001	-0.00
4.320	1.0001	+0.0001	0.00
4.325	1.0000	+0.0002	0.00
4.330	1.0001	+0.0000	0.00
4.335	1.0000	+0.0001	-0.00
4.340	1.0001	+0.0000	0.00
4.345	1.0000	-0.0001	0.00
4.350	1.0000	-0.0001	-0.00
4.355	0.9998	+0.0000	-0.00
4.360	1.0001	+0.0001	0.00
4.365	1.0000	+0.0002	0.00
4.370	0.9999	-0.0000	-0.00
4.375	1.0002	+0.0003	0.00
4.380	1.0001	+0.0000	0.00
4.385	1.0000	+0.0002	0.00
4.390	0.9999	-0.0001	-0.00
4.395	1.0001	+0.0001	0.00
4.400	0.9999	+0.0001	-0.00
4.405	1.0002	-0.0001	0.00
4.410	1.0000	-0.0001	-0.00
4.415	1.0001	-0.0000	0.00
4.420	0.9999	+0.0003	-0.00
4.425	0.9999	+0.0001	-0.00
4.430	1.0000	+0.0001	0.00
4.435	1.0003	+0.0001	0.00
4.440	0.9999	-0.0002	-0.00
4.445	1.0000	+0.0003	0.00
4.450	0.9997	+0.0001	-0.00
4.455	0.9994	+0.0003	-0.01
4.460	0.9998	-0.0000	-0.00
4.465	1.0002	+0.0003	0.00
4.470	1.0000	-0.0002	-0.00
4.475	1.0001	+0.0000	0.00
4.480	0.9999	-0.0000	-0.00
4.485	1.0000	+0.0004	-0.00
4.490	1.0001	+0.0001	0.00
4.495	1.0001	-0.0003	0.00

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.500	1.0000	-0.0001	0.00
4.505	1.0001	+0.0003	0.00
4.510	1.0003	+0.0004	0.00
4.515	0.9998	+0.0002	-0.00
4.520	0.9997	-0.0002	-0.00
4.525	1.0000	-0.0003	0.00
4.530	0.9999	-0.0000	-0.00
4.535	1.0000	-0.0002	0.00
4.540	1.0000	-0.0001	0.00
4.545	0.9999	+0.0000	-0.00
4.550	1.0001	+0.0000	0.00
4.555	1.0000	+0.0001	0.00
4.560	0.9999	+0.0000	-0.00
4.565	0.9999	+0.0001	-0.00
4.570	1.0000	-0.0001	0.00
4.575	1.0000	+0.0001	0.00
4.580	0.9999	+0.0000	-0.00
4.585	0.9999	+0.0001	-0.00
4.590	1.0001	+0.0001	0.00
4.595	1.0000	+0.0000	-0.00
4.600	1.0000	+0.0002	-0.00
4.605	1.0001	+0.0001	0.00
4.610	0.9999	+0.0002	-0.00
4.615	1.0000	+0.0001	-0.00
4.620	1.0000	+0.0001	0.00
4.625	0.9999	-0.0002	-0.00
4.630	1.0000	+0.0001	-0.00
4.635	1.0000	+0.0000	-0.00
4.640	1.0000	+0.0002	-0.00
4.645	0.9998	+0.0003	-0.00
4.650	1.0000	+0.0001	0.00
4.655	1.0001	+0.0003	0.00
4.660	0.9998	-0.0002	-0.00
4.665	0.9999	-0.0002	-0.00
4.670	0.9998	+0.0004	-0.00
4.675	1.0002	+0.0001	0.00
4.680	1.0000	+0.0003	-0.00
4.685	1.0001	+0.0001	0.00
4.690	0.9999	+0.0002	-0.00
4.695	1.0001	+0.0000	0.00
4.700	1.0000	+0.0002	-0.00
4.705	1.0006	-0.0003	0.01
4.710	0.9999	+0.0007	-0.00
4.715	1.0000	+0.0001	0.00
4.720	1.0003	-0.0000	0.00
4.725	0.9996	+0.0003	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.730	1.0000	-0.0000	-0.00
4.735	1.0001	+0.0000	0.00
4.740	1.0002	-0.0001	0.00
4.745	0.9999	+0.0004	-0.00
4.750	1.0004	+0.0001	0.00
4.755	1.0000	-0.0002	-0.00
4.760	1.0003	-0.0001	0.00
4.765	1.0004	-0.0002	0.00
4.770	0.9997	-0.0004	-0.00
4.775	1.0004	+0.0002	0.00
4.780	1.0001	-0.0001	0.00
4.785	0.9990	-0.0002	-0.01
4.790	1.0002	-0.0001	0.00
4.795	1.0001	+0.0001	0.00
4.800	1.0002	-0.0002	0.00
4.805	1.0001	-0.0003	0.00
4.810	1.0001	-0.0002	0.00
4.815	1.0007	-0.0002	0.01
4.820	0.9996	+0.0003	-0.00
4.825	1.0000	+0.0000	-0.00
4.830	0.9997	-0.0000	-0.00
4.835	1.0000	+0.0001	0.00
4.840	1.0001	-0.0001	0.00
4.845	1.0002	-0.0002	0.00
4.850	1.0000	+0.0002	-0.00
4.855	0.9998	+0.0001	-0.00
4.860	1.0000	-0.0001	0.00
4.865	1.0003	+0.0003	0.00
4.870	1.0000	+0.0002	0.00
4.875	1.0000	+0.0001	0.00
4.880	0.9999	-0.0001	-0.00
4.885	1.0000	+0.0001	0.00
4.890	0.9997	-0.0007	-0.00
4.895	0.9999	+0.0003	-0.00
4.900	0.9999	-0.0001	-0.00
4.905	0.9999	-0.0001	-0.00
4.910	1.0000	+0.0002	-0.00
4.915	1.0000	+0.0001	-0.00
4.920	1.0000	+0.0000	-0.00
4.925	1.0000	+0.0000	-0.00
4.930	1.0001	+0.0001	0.00
4.935	1.0000	+0.0001	0.00
4.940	1.0001	-0.0001	0.00
4.945	0.9999	+0.0001	-0.00
4.950	1.0000	-0.0000	0.00
4.955	1.0000	-0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.960	1.0001	-0.0001	0.00
4.965	0.9999	+0.0000	-0.00
4.970	1.0000	+0.0000	-0.00
4.975	1.0000	-0.0001	0.00
4.980	1.0000	+0.0001	0.00
4.985	0.9999	-0.0001	-0.00
4.990	1.0000	-0.0001	-0.00
4.995	1.0000	+0.0002	-0.00
5.000	1.0002	+0.0002	0.00
5.005	1.0000	+0.0001	-0.00
5.010	1.0000	+0.0001	0.00
5.015	0.9998	-0.0000	-0.00
5.020	1.0000	+0.0001	-0.00
5.025	0.9999	+0.0001	-0.00
5.030	1.0000	+0.0003	0.00
5.035	0.9998	+0.0001	-0.00
5.040	0.9999	+0.0001	-0.00
5.045	1.0000	-0.0001	-0.00
5.050	1.0001	+0.0001	0.00
5.055	0.9999	-0.0001	-0.00
5.060	1.0002	+0.0001	0.00
5.065	1.0002	+0.0003	0.00
5.070	0.9998	+0.0001	-0.00
5.075	0.9999	-0.0001	-0.00
5.080	1.0000	+0.0001	0.00
5.085	1.0000	+0.0003	0.00
5.090	1.0001	+0.0001	0.00
5.095	1.0000	-0.0003	-0.00
5.100	0.9999	-0.0002	-0.00
5.105	0.9995	+0.0000	-0.00
5.110	1.0002	-0.0000	0.00
5.115	0.9999	+0.0001	-0.00
5.120	1.0001	+0.0001	0.00
5.125	0.9997	+0.0000	-0.00
5.130	1.0001	-0.0000	0.00
5.135	1.0000	+0.0004	-0.00
5.140	1.0000	+0.0002	-0.00
5.145	1.0002	+0.0001	0.00
5.150	1.0000	+0.0001	0.00
5.155	1.0001	+0.0000	0.00
5.160	1.0001	-0.0000	0.00
5.165	1.0000	+0.0002	-0.00
5.170	1.0000	+0.0000	0.00
5.175	1.0001	+0.0001	0.00
5.180	1.0000	+0.0001	-0.00
5.185	1.0001	+0.0002	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.190	1.0001	+0.0001	0.00
5.195	1.0000	+0.0001	-0.00
5.200	1.0000	+0.0001	-0.00
5.205	1.0000	+0.0000	-0.00
5.210	1.0000	+0.0000	0.00
5.215	1.0000	-0.0000	-0.00
5.220	1.0000	+0.0001	0.00
5.225	1.0000	+0.0001	-0.00
5.230	1.0001	+0.0001	0.00
5.235	1.0000	+0.0000	-0.00
5.240	1.0001	+0.0001	0.00
5.245	1.0000	+0.0001	0.00
5.250	1.0000	+0.0000	0.00
5.255	1.0000	+0.0002	-0.00
5.260	1.0001	-0.0002	0.00
5.265	0.9997	-0.0001	-0.00
5.270	1.0001	+0.0002	0.00
5.275	1.0001	+0.0002	0.00
5.280	1.0000	+0.0001	-0.00
5.285	1.0000	+0.0001	-0.00
5.290	1.0002	+0.0000	0.00
5.295	1.0004	+0.0000	0.00
5.300	1.0000	+0.0001	0.00
5.305	0.9999	-0.0003	-0.00
5.310	1.0001	+0.0003	0.00
5.315	1.0000	-0.0001	-0.00
5.320	1.0000	+0.0000	0.00
5.325	1.0000	+0.0001	-0.00
5.330	0.9999	+0.0000	-0.00
5.335	1.0000	+0.0002	-0.00
5.340	0.9998	+0.0002	-0.00
5.345	1.0001	+0.0002	0.00
5.350	1.0003	-0.0002	0.00
5.355	0.9998	+0.0003	-0.00
5.360	1.0004	+0.0001	0.00
5.365	0.9999	-0.0005	-0.00
5.370	1.0005	+0.0002	0.00
5.375	1.0000	+0.0002	-0.00
5.380	1.0002	-0.0002	0.00
5.385	1.0007	+0.0003	0.01
5.390	1.0000	+0.0004	-0.00
5.395	1.0000	-0.0002	0.00
5.400	1.0001	+0.0003	0.00
5.405	1.0004	-0.0001	0.00
5.410	1.0001	+0.0003	0.00
5.415	0.9999	-0.0005	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.420	1.0000	+0.0002	-0.00
5.425	0.9996	-0.0004	-0.00
5.430	1.0002	-0.0001	0.00
5.435	1.0000	-0.0001	0.00
5.440	0.9999	+0.0003	-0.00
5.445	0.9998	-0.0001	-0.00
5.450	1.0001	+0.0003	0.00
5.455	1.0002	-0.0003	0.00
5.460	1.0000	-0.0000	0.00
5.465	0.9995	-0.0002	-0.00
5.470	1.0001	+0.0002	0.00
5.475	1.0000	-0.0000	0.00
5.480	1.0001	+0.0001	0.00
5.485	1.0003	-0.0001	0.00
5.490	0.9998	+0.0003	-0.00
5.495	1.0002	-0.0003	0.00
5.500	1.0000	+0.0001	-0.00
5.505	1.0000	+0.0000	0.00
5.510	1.0000	-0.0001	-0.00
5.515	1.0000	-0.0001	0.00
5.520	1.0000	+0.0001	-0.00
5.525	1.0000	+0.0000	0.00
5.530	1.0000	+0.0000	0.00
5.535	1.0000	-0.0002	0.00
5.540	0.9999	+0.0001	-0.00
5.545	1.0000	+0.0001	0.00
5.550	0.9999	+0.0000	-0.00
5.555	1.0001	+0.0001	0.00
5.560	1.0000	+0.0000	-0.00
5.565	1.0001	-0.0000	0.00
5.570	1.0001	+0.0001	0.00
5.575	1.0001	+0.0000	0.00
5.580	1.0000	-0.0000	-0.00
5.585	1.0000	+0.0000	0.00
5.590	1.0000	+0.0000	0.00
5.595	1.0000	-0.0000	0.00
5.600	0.9999	+0.0000	-0.00
5.605	1.0001	-0.0001	0.00
5.610	1.0002	+0.0001	0.00
5.615	1.0001	+0.0002	0.00
5.620	1.0000	+0.0000	-0.00
5.625	0.9999	+0.0001	-0.00
5.630	1.0000	+0.0001	-0.00
5.635	1.0001	-0.0001	0.00
5.640	1.0001	-0.0000	0.00
5.645	1.0002	+0.0002	0.00

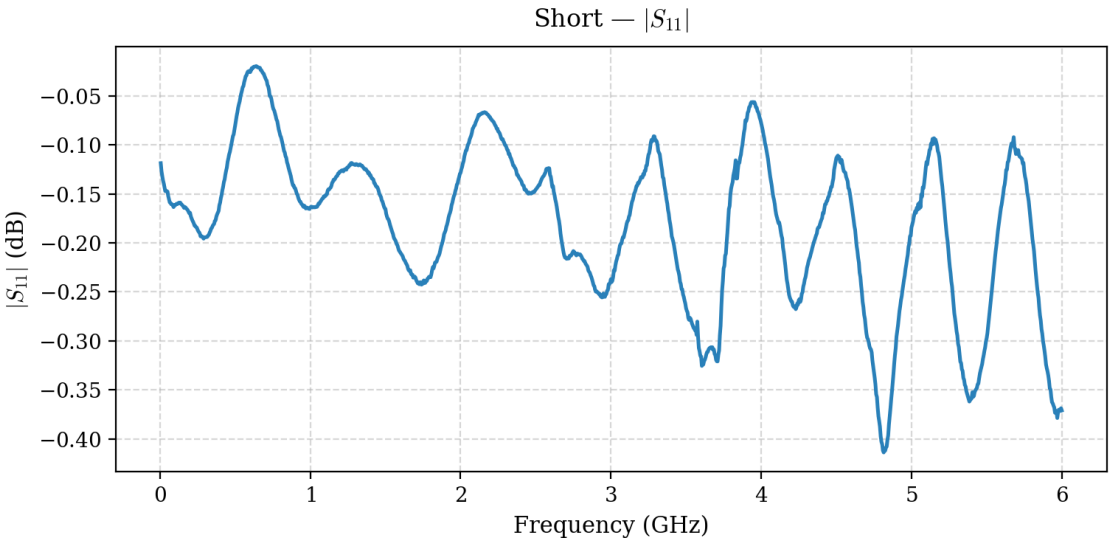
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.650	1.0001	-0.0001	0.00
5.655	0.9998	-0.0000	-0.00
5.660	0.9999	+0.0002	-0.00
5.665	1.0001	-0.0003	0.00
5.670	0.9997	+0.0004	-0.00
5.675	0.9999	+0.0001	-0.00
5.680	0.9999	+0.0003	-0.00
5.685	1.0000	-0.0003	-0.00
5.690	1.0001	-0.0003	0.00
5.695	0.9997	-0.0002	-0.00
5.700	1.0000	-0.0000	0.00
5.705	0.9995	+0.0003	-0.00
5.710	1.0000	-0.0001	0.00
5.715	1.0001	+0.0004	0.00
5.720	0.9999	-0.0001	-0.00
5.725	0.9997	-0.0001	-0.00
5.730	0.9998	-0.0001	-0.00
5.735	0.9999	+0.0001	-0.00
5.740	1.0000	-0.0000	-0.00
5.745	0.9999	-0.0000	-0.00
5.750	1.0002	-0.0000	0.00
5.755	0.9999	-0.0001	-0.00
5.760	1.0001	+0.0003	0.00
5.765	1.0003	-0.0001	0.00
5.770	1.0002	-0.0000	0.00
5.775	1.0001	-0.0000	0.00
5.780	1.0000	-0.0000	-0.00
5.785	1.0002	+0.0001	0.00
5.790	1.0000	+0.0001	0.00
5.795	0.9999	+0.0000	-0.00
5.800	1.0000	-0.0000	0.00
5.805	1.0002	+0.0002	0.00
5.810	1.0000	-0.0001	-0.00
5.815	0.9999	-0.0001	-0.00
5.820	1.0001	+0.0000	0.00
5.825	1.0001	-0.0001	0.00
5.830	1.0000	+0.0001	0.00
5.835	1.0000	+0.0001	-0.00
5.840	1.0000	+0.0001	-0.00
5.845	1.0000	-0.0001	-0.00
5.850	1.0000	+0.0000	0.00
5.855	1.0001	+0.0001	0.00
5.860	1.0000	+0.0001	-0.00
5.865	0.9999	-0.0000	-0.00
5.870	1.0000	+0.0002	-0.00
5.875	1.0000	+0.0000	-0.00

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Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.880	1.0001	+0.0001	0.00
5.885	1.0001	-0.0000	0.00
5.890	0.9999	+0.0002	-0.00
5.895	0.9999	+0.0000	-0.00
5.900	1.0003	+0.0002	0.00
5.905	0.9995	+0.0001	-0.00
5.910	1.0004	+0.0003	0.00
5.915	1.0000	+0.0001	-0.00
5.920	1.0002	-0.0001	0.00
5.925	1.0000	+0.0001	-0.00
5.930	1.0000	+0.0004	0.00
5.935	0.9999	+0.0004	-0.00
5.940	1.0000	+0.0002	-0.00
5.945	1.0005	+0.0001	0.00
5.950	0.9998	-0.0001	-0.00
5.955	0.9998	+0.0003	-0.00
5.960	0.9999	+0.0002	-0.00
5.965	1.0005	+0.0002	0.00
5.970	0.9997	+0.0002	-0.00
5.975	0.9997	-0.0000	-0.00
5.980	1.0002	-0.0005	0.00
5.985	0.9992	+0.0004	-0.01
5.990	0.9999	+0.0003	-0.00
5.995	0.9998	+0.0000	-0.00
6.000	1.0000	-0.0004	0.00

3.2 Short



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	-0.9864	+0.0007	-0.12
0.010	-0.9855	+0.0013	-0.13
0.015	-0.9849	+0.0016	-0.13
0.020	-0.9844	+0.0014	-0.14
0.025	-0.9841	+0.0014	-0.14
0.030	-0.9836	+0.0012	-0.14
0.035	-0.9832	+0.0011	-0.15
0.040	-0.9832	+0.0010	-0.15
0.045	-0.9832	+0.0007	-0.15
0.050	-0.9831	+0.0004	-0.15
0.055	-0.9826	+0.0002	-0.15
0.060	-0.9822	+0.0001	-0.16
0.065	-0.9819	-0.0001	-0.16
0.070	-0.9818	-0.0004	-0.16
0.075	-0.9817	-0.0009	-0.16
0.080	-0.9817	-0.0011	-0.16
0.085	-0.9816	-0.0014	-0.16
0.090	-0.9813	-0.0016	-0.16
0.095	-0.9815	-0.0017	-0.16
0.100	-0.9816	-0.0020	-0.16
0.105	-0.9816	-0.0020	-0.16
0.110	-0.9817	-0.0021	-0.16
0.115	-0.9817	-0.0022	-0.16
0.120	-0.9817	-0.0023	-0.16
0.125	-0.9818	-0.0023	-0.16
0.130	-0.9818	-0.0022	-0.16
0.135	-0.9819	-0.0021	-0.16
0.140	-0.9817	-0.0020	-0.16
0.145	-0.9816	-0.0021	-0.16
0.150	-0.9815	-0.0021	-0.16
0.155	-0.9814	-0.0021	-0.16
0.160	-0.9813	-0.0022	-0.16
0.165	-0.9814	-0.0022	-0.16
0.170	-0.9813	-0.0023	-0.16
0.175	-0.9812	-0.0023	-0.17
0.180	-0.9811	-0.0021	-0.17
0.185	-0.9810	-0.0022	-0.17
0.190	-0.9808	-0.0023	-0.17
0.195	-0.9806	-0.0025	-0.17
0.200	-0.9807	-0.0022	-0.17
0.205	-0.9803	-0.0024	-0.17
0.210	-0.9800	-0.0024	-0.18
0.215	-0.9798	-0.0027	-0.18
0.220	-0.9797	-0.0028	-0.18
0.225	-0.9795	-0.0030	-0.18
0.230	-0.9792	-0.0033	-0.18

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.235	-0.9792	-0.0033	-0.18
0.240	-0.9790	-0.0034	-0.18
0.245	-0.9790	-0.0036	-0.18
0.250	-0.9787	-0.0038	-0.19
0.255	-0.9785	-0.0041	-0.19
0.260	-0.9784	-0.0044	-0.19
0.265	-0.9783	-0.0046	-0.19
0.270	-0.9780	-0.0050	-0.19
0.275	-0.9780	-0.0053	-0.19
0.280	-0.9780	-0.0057	-0.19
0.285	-0.9778	-0.0060	-0.19
0.290	-0.9777	-0.0061	-0.20
0.295	-0.9778	-0.0066	-0.19
0.300	-0.9778	-0.0071	-0.19
0.305	-0.9779	-0.0076	-0.19
0.310	-0.9779	-0.0082	-0.19
0.315	-0.9779	-0.0090	-0.19
0.320	-0.9780	-0.0096	-0.19
0.325	-0.9781	-0.0100	-0.19
0.330	-0.9783	-0.0105	-0.19
0.335	-0.9785	-0.0111	-0.19
0.340	-0.9788	-0.0116	-0.19
0.345	-0.9789	-0.0123	-0.18
0.350	-0.9791	-0.0129	-0.18
0.355	-0.9792	-0.0135	-0.18
0.360	-0.9793	-0.0141	-0.18
0.365	-0.9796	-0.0148	-0.18
0.370	-0.9797	-0.0154	-0.18
0.375	-0.9801	-0.0162	-0.17
0.380	-0.9804	-0.0167	-0.17
0.385	-0.9806	-0.0174	-0.17
0.390	-0.9807	-0.0180	-0.17
0.395	-0.9814	-0.0187	-0.16
0.400	-0.9817	-0.0196	-0.16
0.405	-0.9822	-0.0204	-0.15
0.410	-0.9825	-0.0213	-0.15
0.415	-0.9830	-0.0222	-0.15
0.420	-0.9834	-0.0229	-0.14
0.425	-0.9839	-0.0236	-0.14
0.430	-0.9842	-0.0246	-0.14
0.435	-0.9848	-0.0253	-0.13
0.440	-0.9852	-0.0261	-0.13
0.445	-0.9856	-0.0269	-0.12
0.450	-0.9862	-0.0276	-0.12
0.455	-0.9866	-0.0285	-0.11
0.460	-0.9870	-0.0291	-0.11

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.465	-0.9873	-0.0299	-0.11
0.470	-0.9877	-0.0306	-0.10
0.475	-0.9881	-0.0314	-0.10
0.480	-0.9886	-0.0321	-0.10
0.485	-0.9890	-0.0330	-0.09
0.490	-0.9892	-0.0337	-0.09
0.495	-0.9897	-0.0343	-0.09
0.500	-0.9902	-0.0350	-0.08
0.505	-0.9907	-0.0357	-0.08
0.510	-0.9912	-0.0364	-0.07
0.515	-0.9917	-0.0368	-0.07
0.520	-0.9922	-0.0373	-0.06
0.525	-0.9925	-0.0382	-0.06
0.530	-0.9928	-0.0387	-0.06
0.535	-0.9933	-0.0392	-0.05
0.540	-0.9937	-0.0396	-0.05
0.545	-0.9940	-0.0401	-0.05
0.550	-0.9944	-0.0405	-0.04
0.555	-0.9945	-0.0410	-0.04
0.560	-0.9950	-0.0414	-0.04
0.565	-0.9953	-0.0418	-0.03
0.570	-0.9956	-0.0423	-0.03
0.575	-0.9957	-0.0426	-0.03
0.580	-0.9960	-0.0429	-0.03
0.585	-0.9962	-0.0432	-0.03
0.590	-0.9961	-0.0435	-0.03
0.595	-0.9962	-0.0436	-0.02
0.600	-0.9961	-0.0438	-0.03
0.605	-0.9962	-0.0439	-0.02
0.610	-0.9965	-0.0442	-0.02
0.615	-0.9965	-0.0442	-0.02
0.620	-0.9966	-0.0441	-0.02
0.625	-0.9967	-0.0443	-0.02
0.630	-0.9967	-0.0441	-0.02
0.635	-0.9967	-0.0441	-0.02
0.640	-0.9967	-0.0441	-0.02
0.645	-0.9967	-0.0441	-0.02
0.650	-0.9966	-0.0440	-0.02
0.655	-0.9966	-0.0438	-0.02
0.660	-0.9966	-0.0436	-0.02
0.665	-0.9966	-0.0435	-0.02
0.670	-0.9963	-0.0432	-0.02
0.675	-0.9963	-0.0430	-0.02
0.680	-0.9961	-0.0428	-0.03
0.685	-0.9960	-0.0424	-0.03
0.690	-0.9959	-0.0424	-0.03

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.695	-0.9957	-0.0421	-0.03
0.700	-0.9955	-0.0417	-0.03
0.705	-0.9953	-0.0413	-0.03
0.710	-0.9952	-0.0408	-0.03
0.715	-0.9947	-0.0404	-0.04
0.720	-0.9945	-0.0399	-0.04
0.725	-0.9942	-0.0394	-0.04
0.730	-0.9935	-0.0389	-0.05
0.735	-0.9934	-0.0385	-0.05
0.740	-0.9932	-0.0380	-0.05
0.745	-0.9928	-0.0375	-0.06
0.750	-0.9923	-0.0368	-0.06
0.755	-0.9920	-0.0362	-0.06
0.760	-0.9919	-0.0357	-0.07
0.765	-0.9916	-0.0349	-0.07
0.770	-0.9913	-0.0342	-0.07
0.775	-0.9909	-0.0335	-0.07
0.780	-0.9908	-0.0329	-0.08
0.785	-0.9903	-0.0321	-0.08
0.790	-0.9898	-0.0312	-0.08
0.795	-0.9895	-0.0307	-0.09
0.800	-0.9892	-0.0301	-0.09
0.805	-0.9890	-0.0292	-0.09
0.810	-0.9887	-0.0285	-0.10
0.815	-0.9883	-0.0279	-0.10
0.820	-0.9879	-0.0273	-0.10
0.825	-0.9875	-0.0264	-0.11
0.830	-0.9873	-0.0258	-0.11
0.835	-0.9869	-0.0250	-0.11
0.840	-0.9868	-0.0243	-0.11
0.845	-0.9862	-0.0236	-0.12
0.850	-0.9860	-0.0229	-0.12
0.855	-0.9856	-0.0223	-0.12
0.860	-0.9853	-0.0215	-0.13
0.865	-0.9850	-0.0208	-0.13
0.870	-0.9848	-0.0201	-0.13
0.875	-0.9843	-0.0195	-0.14
0.880	-0.9840	-0.0188	-0.14
0.885	-0.9837	-0.0184	-0.14
0.890	-0.9837	-0.0178	-0.14
0.895	-0.9834	-0.0173	-0.14
0.900	-0.9834	-0.0166	-0.14
0.905	-0.9831	-0.0162	-0.15
0.910	-0.9828	-0.0155	-0.15
0.915	-0.9826	-0.0152	-0.15
0.920	-0.9823	-0.0147	-0.15

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.925	-0.9821	-0.0141	-0.16
0.930	-0.9819	-0.0137	-0.16
0.935	-0.9819	-0.0133	-0.16
0.940	-0.9816	-0.0129	-0.16
0.945	-0.9816	-0.0125	-0.16
0.950	-0.9814	-0.0121	-0.16
0.955	-0.9813	-0.0120	-0.16
0.960	-0.9814	-0.0119	-0.16
0.965	-0.9813	-0.0117	-0.16
0.970	-0.9811	-0.0115	-0.16
0.975	-0.9812	-0.0112	-0.16
0.980	-0.9812	-0.0111	-0.16
0.985	-0.9811	-0.0108	-0.16
0.990	-0.9811	-0.0110	-0.17
0.995	-0.9811	-0.0109	-0.17
1.000	-0.9812	-0.0110	-0.16
1.005	-0.9813	-0.0108	-0.16
1.010	-0.9813	-0.0108	-0.16
1.015	-0.9812	-0.0111	-0.16
1.020	-0.9813	-0.0114	-0.16
1.025	-0.9813	-0.0114	-0.16
1.030	-0.9812	-0.0116	-0.16
1.035	-0.9813	-0.0118	-0.16
1.040	-0.9813	-0.0121	-0.16
1.045	-0.9813	-0.0124	-0.16
1.050	-0.9815	-0.0131	-0.16
1.055	-0.9817	-0.0134	-0.16
1.060	-0.9816	-0.0138	-0.16
1.065	-0.9818	-0.0143	-0.16
1.070	-0.9819	-0.0147	-0.16
1.075	-0.9821	-0.0150	-0.16
1.080	-0.9822	-0.0156	-0.15
1.085	-0.9822	-0.0163	-0.15
1.090	-0.9825	-0.0169	-0.15
1.095	-0.9824	-0.0173	-0.15
1.100	-0.9825	-0.0177	-0.15
1.105	-0.9826	-0.0185	-0.15
1.110	-0.9826	-0.0192	-0.15
1.115	-0.9830	-0.0197	-0.15
1.120	-0.9832	-0.0203	-0.15
1.125	-0.9831	-0.0210	-0.15
1.130	-0.9833	-0.0218	-0.14
1.135	-0.9833	-0.0226	-0.14
1.140	-0.9835	-0.0233	-0.14
1.145	-0.9838	-0.0241	-0.14
1.150	-0.9840	-0.0247	-0.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.155	-0.9842	-0.0254	-0.14
1.160	-0.9841	-0.0262	-0.14
1.165	-0.9843	-0.0267	-0.13
1.170	-0.9844	-0.0275	-0.13
1.175	-0.9845	-0.0282	-0.13
1.180	-0.9845	-0.0290	-0.13
1.185	-0.9848	-0.0297	-0.13
1.190	-0.9848	-0.0305	-0.13
1.195	-0.9849	-0.0310	-0.13
1.200	-0.9849	-0.0315	-0.13
1.205	-0.9850	-0.0324	-0.13
1.210	-0.9850	-0.0330	-0.13
1.215	-0.9849	-0.0337	-0.13
1.220	-0.9849	-0.0342	-0.13
1.225	-0.9850	-0.0348	-0.13
1.230	-0.9850	-0.0354	-0.13
1.235	-0.9851	-0.0358	-0.12
1.240	-0.9850	-0.0362	-0.13
1.245	-0.9852	-0.0366	-0.12
1.250	-0.9852	-0.0371	-0.12
1.255	-0.9853	-0.0376	-0.12
1.260	-0.9855	-0.0379	-0.12
1.265	-0.9854	-0.0382	-0.12
1.270	-0.9857	-0.0386	-0.12
1.275	-0.9857	-0.0386	-0.12
1.280	-0.9856	-0.0388	-0.12
1.285	-0.9855	-0.0389	-0.12
1.290	-0.9855	-0.0390	-0.12
1.295	-0.9855	-0.0388	-0.12
1.300	-0.9855	-0.0386	-0.12
1.305	-0.9856	-0.0385	-0.12
1.310	-0.9855	-0.0387	-0.12
1.315	-0.9855	-0.0385	-0.12
1.320	-0.9854	-0.0382	-0.12
1.325	-0.9855	-0.0383	-0.12
1.330	-0.9855	-0.0380	-0.12
1.335	-0.9855	-0.0376	-0.12
1.340	-0.9854	-0.0372	-0.12
1.345	-0.9854	-0.0368	-0.12
1.350	-0.9853	-0.0364	-0.12
1.355	-0.9852	-0.0358	-0.12
1.360	-0.9851	-0.0351	-0.12
1.365	-0.9851	-0.0344	-0.12
1.370	-0.9851	-0.0338	-0.12
1.375	-0.9850	-0.0331	-0.13
1.380	-0.9849	-0.0324	-0.13

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.385	-0.9847	-0.0316	-0.13
1.390	-0.9849	-0.0307	-0.13
1.395	-0.9846	-0.0297	-0.13
1.400	-0.9846	-0.0286	-0.13
1.405	-0.9844	-0.0277	-0.13
1.410	-0.9843	-0.0267	-0.13
1.415	-0.9843	-0.0258	-0.13
1.420	-0.9842	-0.0246	-0.14
1.425	-0.9838	-0.0235	-0.14
1.430	-0.9838	-0.0221	-0.14
1.435	-0.9836	-0.0209	-0.14
1.440	-0.9835	-0.0198	-0.14
1.445	-0.9835	-0.0185	-0.14
1.450	-0.9833	-0.0173	-0.14
1.455	-0.9831	-0.0160	-0.15
1.460	-0.9827	-0.0146	-0.15
1.465	-0.9826	-0.0135	-0.15
1.470	-0.9824	-0.0121	-0.15
1.475	-0.9821	-0.0106	-0.16
1.480	-0.9820	-0.0092	-0.16
1.485	-0.9817	-0.0078	-0.16
1.490	-0.9814	-0.0068	-0.16
1.495	-0.9812	-0.0051	-0.16
1.500	-0.9811	-0.0037	-0.17
1.505	-0.9806	-0.0023	-0.17
1.510	-0.9803	-0.0010	-0.17
1.515	-0.9800	+0.0003	-0.18
1.520	-0.9799	+0.0015	-0.18
1.525	-0.9797	+0.0030	-0.18
1.530	-0.9793	+0.0043	-0.18
1.535	-0.9790	+0.0058	-0.18
1.540	-0.9787	+0.0071	-0.19
1.545	-0.9785	+0.0083	-0.19
1.550	-0.9782	+0.0095	-0.19
1.555	-0.9778	+0.0107	-0.19
1.560	-0.9776	+0.0118	-0.20
1.565	-0.9775	+0.0129	-0.20
1.570	-0.9773	+0.0139	-0.20
1.575	-0.9770	+0.0149	-0.20
1.580	-0.9768	+0.0157	-0.20
1.585	-0.9765	+0.0167	-0.21
1.590	-0.9762	+0.0175	-0.21
1.595	-0.9761	+0.0185	-0.21
1.600	-0.9759	+0.0191	-0.21
1.605	-0.9758	+0.0199	-0.21
1.610	-0.9756	+0.0207	-0.21

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.615	-0.9755	+0.0215	-0.21
1.620	-0.9752	+0.0223	-0.22
1.625	-0.9750	+0.0230	-0.22
1.630	-0.9749	+0.0237	-0.22
1.635	-0.9747	+0.0242	-0.22
1.640	-0.9745	+0.0247	-0.22
1.645	-0.9743	+0.0252	-0.22
1.650	-0.9740	+0.0254	-0.23
1.655	-0.9737	+0.0257	-0.23
1.660	-0.9737	+0.0261	-0.23
1.665	-0.9737	+0.0263	-0.23
1.670	-0.9733	+0.0263	-0.23
1.675	-0.9732	+0.0263	-0.23
1.680	-0.9732	+0.0265	-0.23
1.685	-0.9729	+0.0265	-0.24
1.690	-0.9730	+0.0262	-0.23
1.695	-0.9730	+0.0259	-0.23
1.700	-0.9729	+0.0257	-0.24
1.705	-0.9726	+0.0254	-0.24
1.710	-0.9725	+0.0252	-0.24
1.715	-0.9723	+0.0249	-0.24
1.720	-0.9724	+0.0243	-0.24
1.725	-0.9722	+0.0237	-0.24
1.730	-0.9724	+0.0231	-0.24
1.735	-0.9725	+0.0226	-0.24
1.740	-0.9722	+0.0218	-0.24
1.745	-0.9725	+0.0209	-0.24
1.750	-0.9724	+0.0198	-0.24
1.755	-0.9728	+0.0190	-0.24
1.760	-0.9725	+0.0180	-0.24
1.765	-0.9725	+0.0169	-0.24
1.770	-0.9728	+0.0162	-0.24
1.775	-0.9727	+0.0151	-0.24
1.780	-0.9730	+0.0139	-0.24
1.785	-0.9731	+0.0131	-0.24
1.790	-0.9733	+0.0121	-0.23
1.795	-0.9733	+0.0108	-0.23
1.800	-0.9733	+0.0094	-0.24
1.805	-0.9736	+0.0082	-0.23
1.810	-0.9740	+0.0072	-0.23
1.815	-0.9744	+0.0061	-0.22
1.820	-0.9743	+0.0048	-0.23
1.825	-0.9746	+0.0038	-0.22
1.830	-0.9750	+0.0023	-0.22
1.835	-0.9750	+0.0010	-0.22
1.840	-0.9751	-0.0004	-0.22

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.845	-0.9754	-0.0015	-0.22
1.850	-0.9756	-0.0029	-0.21
1.855	-0.9758	-0.0041	-0.21
1.860	-0.9760	-0.0054	-0.21
1.865	-0.9763	-0.0066	-0.21
1.870	-0.9767	-0.0079	-0.20
1.875	-0.9770	-0.0091	-0.20
1.880	-0.9773	-0.0104	-0.20
1.885	-0.9777	-0.0115	-0.20
1.890	-0.9781	-0.0129	-0.19
1.895	-0.9783	-0.0140	-0.19
1.900	-0.9785	-0.0151	-0.19
1.905	-0.9789	-0.0162	-0.18
1.910	-0.9792	-0.0172	-0.18
1.915	-0.9795	-0.0182	-0.18
1.920	-0.9800	-0.0193	-0.17
1.925	-0.9803	-0.0203	-0.17
1.930	-0.9805	-0.0213	-0.17
1.935	-0.9808	-0.0221	-0.17
1.940	-0.9813	-0.0227	-0.16
1.945	-0.9816	-0.0238	-0.16
1.950	-0.9819	-0.0244	-0.16
1.955	-0.9823	-0.0251	-0.15
1.960	-0.9825	-0.0257	-0.15
1.965	-0.9829	-0.0263	-0.15
1.970	-0.9831	-0.0270	-0.14
1.975	-0.9835	-0.0273	-0.14
1.980	-0.9837	-0.0278	-0.14
1.985	-0.9841	-0.0283	-0.14
1.990	-0.9844	-0.0286	-0.13
1.995	-0.9848	-0.0289	-0.13
2.000	-0.9849	-0.0289	-0.13
2.005	-0.9853	-0.0289	-0.12
2.010	-0.9856	-0.0289	-0.12
2.015	-0.9860	-0.0290	-0.12
2.020	-0.9862	-0.0288	-0.12
2.025	-0.9868	-0.0285	-0.11
2.030	-0.9871	-0.0282	-0.11
2.035	-0.9875	-0.0281	-0.11
2.040	-0.9875	-0.0277	-0.11
2.045	-0.9880	-0.0273	-0.10
2.050	-0.9882	-0.0267	-0.10
2.055	-0.9884	-0.0263	-0.10
2.060	-0.9888	-0.0258	-0.09
2.065	-0.9891	-0.0250	-0.09
2.070	-0.9893	-0.0244	-0.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.075	-0.9897	-0.0235	-0.09
2.080	-0.9901	-0.0228	-0.08
2.085	-0.9903	-0.0220	-0.08
2.090	-0.9905	-0.0213	-0.08
2.095	-0.9906	-0.0204	-0.08
2.100	-0.9909	-0.0195	-0.08
2.105	-0.9912	-0.0184	-0.08
2.110	-0.9914	-0.0174	-0.07
2.115	-0.9916	-0.0163	-0.07
2.120	-0.9919	-0.0151	-0.07
2.125	-0.9919	-0.0139	-0.07
2.130	-0.9920	-0.0130	-0.07
2.135	-0.9920	-0.0116	-0.07
2.140	-0.9922	-0.0104	-0.07
2.145	-0.9922	-0.0092	-0.07
2.150	-0.9922	-0.0079	-0.07
2.155	-0.9923	-0.0068	-0.07
2.160	-0.9923	-0.0056	-0.07
2.165	-0.9923	-0.0043	-0.07
2.170	-0.9923	-0.0031	-0.07
2.175	-0.9921	-0.0020	-0.07
2.180	-0.9921	-0.0006	-0.07
2.185	-0.9920	+0.0006	-0.07
2.190	-0.9918	+0.0018	-0.07
2.195	-0.9918	+0.0030	-0.07
2.200	-0.9916	+0.0043	-0.07
2.205	-0.9915	+0.0055	-0.07
2.210	-0.9913	+0.0066	-0.08
2.215	-0.9913	+0.0077	-0.08
2.220	-0.9912	+0.0088	-0.08
2.225	-0.9909	+0.0095	-0.08
2.230	-0.9908	+0.0106	-0.08
2.235	-0.9906	+0.0115	-0.08
2.240	-0.9904	+0.0124	-0.08
2.245	-0.9903	+0.0132	-0.08
2.250	-0.9900	+0.0139	-0.09
2.255	-0.9899	+0.0147	-0.09
2.260	-0.9898	+0.0155	-0.09
2.265	-0.9895	+0.0162	-0.09
2.270	-0.9892	+0.0167	-0.09
2.275	-0.9890	+0.0173	-0.09
2.280	-0.9890	+0.0177	-0.09
2.285	-0.9888	+0.0183	-0.10
2.290	-0.9886	+0.0185	-0.10
2.295	-0.9884	+0.0190	-0.10
2.300	-0.9883	+0.0194	-0.10

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.305	-0.9881	+0.0197	-0.10
2.310	-0.9878	+0.0198	-0.11
2.315	-0.9877	+0.0200	-0.11
2.320	-0.9873	+0.0199	-0.11
2.325	-0.9873	+0.0200	-0.11
2.330	-0.9869	+0.0200	-0.11
2.335	-0.9868	+0.0199	-0.11
2.340	-0.9865	+0.0200	-0.12
2.345	-0.9863	+0.0198	-0.12
2.350	-0.9860	+0.0195	-0.12
2.355	-0.9859	+0.0189	-0.12
2.360	-0.9857	+0.0183	-0.12
2.365	-0.9856	+0.0176	-0.12
2.370	-0.9854	+0.0170	-0.13
2.375	-0.9849	+0.0161	-0.13
2.380	-0.9849	+0.0155	-0.13
2.385	-0.9848	+0.0145	-0.13
2.390	-0.9846	+0.0137	-0.13
2.395	-0.9845	+0.0126	-0.14
2.400	-0.9844	+0.0118	-0.14
2.405	-0.9842	+0.0105	-0.14
2.410	-0.9839	+0.0094	-0.14
2.415	-0.9838	+0.0081	-0.14
2.420	-0.9838	+0.0069	-0.14
2.425	-0.9833	+0.0056	-0.15
2.430	-0.9834	+0.0043	-0.15
2.435	-0.9833	+0.0028	-0.15
2.440	-0.9831	+0.0012	-0.15
2.445	-0.9830	-0.0003	-0.15
2.450	-0.9829	-0.0018	-0.15
2.455	-0.9830	-0.0035	-0.15
2.460	-0.9829	-0.0052	-0.15
2.465	-0.9829	-0.0068	-0.15
2.470	-0.9829	-0.0084	-0.15
2.475	-0.9830	-0.0100	-0.15
2.480	-0.9830	-0.0115	-0.15
2.485	-0.9830	-0.0133	-0.15
2.490	-0.9830	-0.0151	-0.15
2.495	-0.9831	-0.0167	-0.15
2.500	-0.9832	-0.0183	-0.15
2.505	-0.9833	-0.0201	-0.14
2.510	-0.9834	-0.0218	-0.14
2.515	-0.9834	-0.0234	-0.14
2.520	-0.9835	-0.0251	-0.14
2.525	-0.9835	-0.0266	-0.14
2.530	-0.9838	-0.0282	-0.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.535	-0.9840	-0.0297	-0.14
2.540	-0.9841	-0.0311	-0.13
2.545	-0.9842	-0.0327	-0.13
2.550	-0.9843	-0.0341	-0.13
2.555	-0.9844	-0.0355	-0.13
2.560	-0.9845	-0.0369	-0.13
2.565	-0.9846	-0.0383	-0.13
2.570	-0.9849	-0.0394	-0.12
2.575	-0.9850	-0.0403	-0.12
2.580	-0.9849	-0.0411	-0.12
2.585	-0.9850	-0.0417	-0.12
2.590	-0.9849	-0.0423	-0.12
2.595	-0.9841	-0.0433	-0.13
2.600	-0.9839	-0.0439	-0.13
2.605	-0.9833	-0.0451	-0.14
2.610	-0.9826	-0.0460	-0.14
2.615	-0.9827	-0.0470	-0.14
2.620	-0.9819	-0.0482	-0.15
2.625	-0.9815	-0.0495	-0.15
2.630	-0.9809	-0.0506	-0.16
2.635	-0.9805	-0.0518	-0.16
2.640	-0.9799	-0.0528	-0.16
2.645	-0.9793	-0.0539	-0.17
2.650	-0.9786	-0.0547	-0.17
2.655	-0.9782	-0.0556	-0.18
2.660	-0.9775	-0.0567	-0.18
2.665	-0.9766	-0.0576	-0.19
2.670	-0.9759	-0.0586	-0.20
2.675	-0.9753	-0.0599	-0.20
2.680	-0.9747	-0.0610	-0.21
2.685	-0.9741	-0.0622	-0.21
2.690	-0.9737	-0.0634	-0.21
2.695	-0.9734	-0.0647	-0.22
2.700	-0.9734	-0.0656	-0.21
2.705	-0.9731	-0.0666	-0.22
2.710	-0.9731	-0.0676	-0.22
2.715	-0.9731	-0.0682	-0.22
2.720	-0.9730	-0.0690	-0.22
2.725	-0.9733	-0.0690	-0.21
2.730	-0.9733	-0.0693	-0.21
2.735	-0.9733	-0.0696	-0.21
2.740	-0.9734	-0.0697	-0.21
2.745	-0.9736	-0.0697	-0.21
2.750	-0.9738	-0.0698	-0.21
2.755	-0.9737	-0.0699	-0.21
2.760	-0.9735	-0.0698	-0.21

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.765	-0.9735	-0.0696	-0.21
2.770	-0.9733	-0.0695	-0.21
2.775	-0.9735	-0.0694	-0.21
2.780	-0.9735	-0.0692	-0.21
2.785	-0.9734	-0.0694	-0.21
2.790	-0.9734	-0.0692	-0.21
2.795	-0.9735	-0.0690	-0.21
2.800	-0.9734	-0.0689	-0.21
2.805	-0.9732	-0.0689	-0.21
2.810	-0.9730	-0.0684	-0.22
2.815	-0.9729	-0.0685	-0.22
2.820	-0.9728	-0.0683	-0.22
2.825	-0.9728	-0.0682	-0.22
2.830	-0.9725	-0.0678	-0.22
2.835	-0.9724	-0.0675	-0.22
2.840	-0.9723	-0.0672	-0.22
2.845	-0.9722	-0.0670	-0.22
2.850	-0.9718	-0.0667	-0.23
2.855	-0.9717	-0.0664	-0.23
2.860	-0.9715	-0.0663	-0.23
2.865	-0.9713	-0.0661	-0.23
2.870	-0.9711	-0.0658	-0.23
2.875	-0.9708	-0.0658	-0.24
2.880	-0.9705	-0.0657	-0.24
2.885	-0.9702	-0.0654	-0.24
2.890	-0.9701	-0.0653	-0.24
2.895	-0.9697	-0.0651	-0.25
2.900	-0.9695	-0.0650	-0.25
2.905	-0.9695	-0.0649	-0.25
2.910	-0.9694	-0.0649	-0.25
2.915	-0.9693	-0.0648	-0.25
2.920	-0.9692	-0.0648	-0.25
2.925	-0.9692	-0.0650	-0.25
2.930	-0.9690	-0.0653	-0.25
2.935	-0.9689	-0.0654	-0.25
2.940	-0.9688	-0.0657	-0.26
2.945	-0.9688	-0.0658	-0.26
2.950	-0.9692	-0.0661	-0.25
2.955	-0.9689	-0.0665	-0.25
2.960	-0.9687	-0.0671	-0.26
2.965	-0.9689	-0.0675	-0.25
2.970	-0.9690	-0.0678	-0.25
2.975	-0.9694	-0.0683	-0.25
2.980	-0.9694	-0.0688	-0.25
2.985	-0.9700	-0.0692	-0.24
2.990	-0.9700	-0.0699	-0.24

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.995	-0.9701	-0.0702	-0.24
3.000	-0.9705	-0.0711	-0.24
3.005	-0.9701	-0.0713	-0.24
3.010	-0.9702	-0.0721	-0.24
3.015	-0.9706	-0.0728	-0.23
3.020	-0.9713	-0.0733	-0.23
3.025	-0.9712	-0.0742	-0.23
3.030	-0.9718	-0.0752	-0.22
3.035	-0.9721	-0.0761	-0.22
3.040	-0.9723	-0.0774	-0.22
3.045	-0.9726	-0.0783	-0.21
3.050	-0.9726	-0.0794	-0.21
3.055	-0.9732	-0.0802	-0.21
3.060	-0.9735	-0.0817	-0.20
3.065	-0.9736	-0.0828	-0.20
3.070	-0.9736	-0.0837	-0.20
3.075	-0.9739	-0.0850	-0.20
3.080	-0.9741	-0.0861	-0.19
3.085	-0.9740	-0.0872	-0.19
3.090	-0.9743	-0.0879	-0.19
3.095	-0.9745	-0.0889	-0.19
3.100	-0.9748	-0.0900	-0.18
3.105	-0.9752	-0.0908	-0.18
3.110	-0.9754	-0.0917	-0.18
3.115	-0.9754	-0.0926	-0.18
3.120	-0.9758	-0.0934	-0.17
3.125	-0.9759	-0.0942	-0.17
3.130	-0.9764	-0.0950	-0.17
3.135	-0.9768	-0.0959	-0.16
3.140	-0.9771	-0.0970	-0.16
3.145	-0.9773	-0.0976	-0.16
3.150	-0.9776	-0.0983	-0.15
3.155	-0.9779	-0.0990	-0.15
3.160	-0.9780	-0.0996	-0.15
3.165	-0.9783	-0.1001	-0.15
3.170	-0.9786	-0.1007	-0.14
3.175	-0.9786	-0.1010	-0.14
3.180	-0.9789	-0.1017	-0.14
3.185	-0.9788	-0.1019	-0.14
3.190	-0.9790	-0.1024	-0.14
3.195	-0.9791	-0.1029	-0.14
3.200	-0.9793	-0.1036	-0.13
3.205	-0.9793	-0.1037	-0.13
3.210	-0.9792	-0.1040	-0.13
3.215	-0.9796	-0.1048	-0.13
3.220	-0.9795	-0.1050	-0.13

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.225	-0.9799	-0.1053	-0.13
3.230	-0.9798	-0.1063	-0.13
3.235	-0.9804	-0.1061	-0.12
3.240	-0.9805	-0.1063	-0.12
3.245	-0.9812	-0.1063	-0.11
3.250	-0.9816	-0.1067	-0.11
3.255	-0.9818	-0.1066	-0.11
3.260	-0.9823	-0.1064	-0.10
3.265	-0.9830	-0.1067	-0.10
3.270	-0.9834	-0.1062	-0.10
3.275	-0.9832	-0.1061	-0.10
3.280	-0.9836	-0.1056	-0.09
3.285	-0.9839	-0.1058	-0.09
3.290	-0.9839	-0.1052	-0.09
3.295	-0.9835	-0.1048	-0.10
3.300	-0.9836	-0.1044	-0.10
3.305	-0.9834	-0.1041	-0.10
3.310	-0.9831	-0.1035	-0.10
3.315	-0.9824	-0.1031	-0.11
3.320	-0.9824	-0.1018	-0.11
3.325	-0.9822	-0.1016	-0.11
3.330	-0.9817	-0.1005	-0.12
3.335	-0.9811	-0.0998	-0.12
3.340	-0.9807	-0.0985	-0.13
3.345	-0.9799	-0.0977	-0.13
3.350	-0.9799	-0.0964	-0.13
3.355	-0.9793	-0.0955	-0.14
3.360	-0.9789	-0.0943	-0.15
3.365	-0.9785	-0.0928	-0.15
3.370	-0.9779	-0.0915	-0.16
3.375	-0.9776	-0.0903	-0.16
3.380	-0.9770	-0.0893	-0.17
3.385	-0.9766	-0.0882	-0.17
3.390	-0.9760	-0.0866	-0.18
3.395	-0.9756	-0.0855	-0.18
3.400	-0.9752	-0.0841	-0.19
3.405	-0.9745	-0.0829	-0.19
3.410	-0.9743	-0.0817	-0.20
3.415	-0.9741	-0.0803	-0.20
3.420	-0.9737	-0.0793	-0.20
3.425	-0.9732	-0.0780	-0.21
3.430	-0.9727	-0.0768	-0.21
3.435	-0.9723	-0.0755	-0.22
3.440	-0.9718	-0.0744	-0.22
3.445	-0.9714	-0.0733	-0.23
3.450	-0.9712	-0.0722	-0.23

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.455	-0.9709	-0.0708	-0.23
3.460	-0.9703	-0.0697	-0.24
3.465	-0.9701	-0.0690	-0.24
3.470	-0.9696	-0.0675	-0.25
3.475	-0.9696	-0.0666	-0.25
3.480	-0.9691	-0.0657	-0.25
3.485	-0.9689	-0.0647	-0.26
3.490	-0.9685	-0.0638	-0.26
3.495	-0.9683	-0.0631	-0.26
3.500	-0.9678	-0.0619	-0.27
3.505	-0.9675	-0.0613	-0.27
3.510	-0.9674	-0.0603	-0.27
3.515	-0.9668	-0.0595	-0.28
3.520	-0.9667	-0.0587	-0.28
3.525	-0.9670	-0.0572	-0.28
3.530	-0.9670	-0.0555	-0.28
3.535	-0.9667	-0.0556	-0.28
3.540	-0.9664	-0.0554	-0.28
3.545	-0.9662	-0.0550	-0.28
3.550	-0.9660	-0.0546	-0.29
3.555	-0.9659	-0.0544	-0.29
3.560	-0.9657	-0.0541	-0.29
3.565	-0.9659	-0.0539	-0.29
3.570	-0.9653	-0.0535	-0.29
3.575	-0.9667	-0.0544	-0.28
3.580	-0.9650	-0.0537	-0.30
3.585	-0.9637	-0.0538	-0.31
3.590	-0.9634	-0.0542	-0.31
3.595	-0.9629	-0.0539	-0.32
3.600	-0.9628	-0.0536	-0.32
3.605	-0.9618	-0.0529	-0.33
3.610	-0.9619	-0.0523	-0.32
3.615	-0.9620	-0.0523	-0.32
3.620	-0.9623	-0.0524	-0.32
3.625	-0.9625	-0.0527	-0.32
3.630	-0.9628	-0.0526	-0.32
3.635	-0.9631	-0.0527	-0.31
3.640	-0.9634	-0.0526	-0.31
3.645	-0.9633	-0.0532	-0.31
3.650	-0.9636	-0.0533	-0.31
3.655	-0.9637	-0.0535	-0.31
3.660	-0.9637	-0.0535	-0.31
3.665	-0.9638	-0.0536	-0.31
3.670	-0.9637	-0.0540	-0.31
3.675	-0.9638	-0.0543	-0.31
3.680	-0.9635	-0.0546	-0.31

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.685	-0.9636	-0.0542	-0.31
3.690	-0.9631	-0.0539	-0.31
3.695	-0.9629	-0.0532	-0.31
3.700	-0.9625	-0.0526	-0.32
3.705	-0.9624	-0.0512	-0.32
3.710	-0.9624	-0.0501	-0.32
3.715	-0.9633	-0.0484	-0.31
3.720	-0.9639	-0.0465	-0.31
3.725	-0.9657	-0.0450	-0.29
3.730	-0.9668	-0.0434	-0.28
3.735	-0.9690	-0.0432	-0.27
3.740	-0.9709	-0.0423	-0.25
3.745	-0.9719	-0.0431	-0.24
3.750	-0.9734	-0.0430	-0.23
3.755	-0.9736	-0.0432	-0.22
3.760	-0.9748	-0.0433	-0.21
3.765	-0.9751	-0.0428	-0.21
3.770	-0.9763	-0.0423	-0.20
3.775	-0.9775	-0.0422	-0.19
3.780	-0.9788	-0.0424	-0.18
3.785	-0.9802	-0.0424	-0.17
3.790	-0.9806	-0.0422	-0.16
3.795	-0.9819	-0.0420	-0.15
3.800	-0.9826	-0.0420	-0.14
3.805	-0.9830	-0.0421	-0.14
3.810	-0.9834	-0.0415	-0.14
3.815	-0.9841	-0.0413	-0.13
3.820	-0.9846	-0.0407	-0.13
3.825	-0.9850	-0.0407	-0.12
3.830	-0.9859	-0.0400	-0.12
3.835	-0.9839	-0.0388	-0.13
3.840	-0.9839	-0.0374	-0.13
3.845	-0.9851	-0.0363	-0.12
3.850	-0.9858	-0.0356	-0.12
3.855	-0.9859	-0.0343	-0.12
3.860	-0.9865	-0.0334	-0.11
3.865	-0.9869	-0.0319	-0.11
3.870	-0.9876	-0.0307	-0.10
3.875	-0.9881	-0.0293	-0.10
3.880	-0.9888	-0.0279	-0.09
3.885	-0.9896	-0.0266	-0.09
3.890	-0.9901	-0.0252	-0.08
3.895	-0.9911	-0.0240	-0.07
3.900	-0.9910	-0.0224	-0.08
3.905	-0.9914	-0.0206	-0.07
3.910	-0.9921	-0.0191	-0.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.915	-0.9923	-0.0175	-0.07
3.920	-0.9928	-0.0158	-0.06
3.925	-0.9929	-0.0140	-0.06
3.930	-0.9933	-0.0131	-0.06
3.935	-0.9934	-0.0115	-0.06
3.940	-0.9934	-0.0099	-0.06
3.945	-0.9935	-0.0083	-0.06
3.950	-0.9934	-0.0067	-0.06
3.955	-0.9935	-0.0050	-0.06
3.960	-0.9933	-0.0035	-0.06
3.965	-0.9929	-0.0020	-0.06
3.970	-0.9929	-0.0007	-0.06
3.975	-0.9925	+0.0011	-0.07
3.980	-0.9923	+0.0027	-0.07
3.985	-0.9921	+0.0042	-0.07
3.990	-0.9917	+0.0055	-0.07
3.995	-0.9914	+0.0071	-0.07
4.000	-0.9910	+0.0087	-0.08
4.005	-0.9906	+0.0099	-0.08
4.010	-0.9902	+0.0113	-0.08
4.015	-0.9896	+0.0125	-0.09
4.020	-0.9892	+0.0137	-0.09
4.025	-0.9886	+0.0146	-0.10
4.030	-0.9883	+0.0159	-0.10
4.035	-0.9878	+0.0169	-0.11
4.040	-0.9872	+0.0181	-0.11
4.045	-0.9867	+0.0191	-0.11
4.050	-0.9860	+0.0198	-0.12
4.055	-0.9853	+0.0206	-0.13
4.060	-0.9846	+0.0213	-0.13
4.065	-0.9841	+0.0216	-0.14
4.070	-0.9836	+0.0221	-0.14
4.075	-0.9829	+0.0227	-0.15
4.080	-0.9823	+0.0229	-0.15
4.085	-0.9819	+0.0230	-0.16
4.090	-0.9814	+0.0231	-0.16
4.095	-0.9807	+0.0232	-0.17
4.100	-0.9802	+0.0233	-0.17
4.105	-0.9799	+0.0234	-0.17
4.110	-0.9795	+0.0232	-0.18
4.115	-0.9792	+0.0232	-0.18
4.120	-0.9793	+0.0232	-0.18
4.125	-0.9789	+0.0232	-0.18
4.130	-0.9783	+0.0229	-0.19
4.135	-0.9778	+0.0232	-0.19
4.140	-0.9775	+0.0232	-0.20

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.145	-0.9764	+0.0230	-0.20
4.150	-0.9759	+0.0226	-0.21
4.155	-0.9752	+0.0223	-0.22
4.160	-0.9745	+0.0217	-0.22
4.165	-0.9737	+0.0203	-0.23
4.170	-0.9728	+0.0193	-0.24
4.175	-0.9724	+0.0176	-0.24
4.180	-0.9718	+0.0165	-0.25
4.185	-0.9712	+0.0151	-0.25
4.190	-0.9709	+0.0134	-0.26
4.195	-0.9707	+0.0117	-0.26
4.200	-0.9705	+0.0100	-0.26
4.205	-0.9705	+0.0082	-0.26
4.210	-0.9701	+0.0062	-0.26
4.215	-0.9699	+0.0041	-0.27
4.220	-0.9699	+0.0021	-0.27
4.225	-0.9700	+0.0000	-0.26
4.230	-0.9697	-0.0021	-0.27
4.235	-0.9699	-0.0045	-0.27
4.240	-0.9703	-0.0068	-0.26
4.245	-0.9701	-0.0093	-0.26
4.250	-0.9706	-0.0112	-0.26
4.255	-0.9707	-0.0137	-0.26
4.260	-0.9704	-0.0158	-0.26
4.265	-0.9706	-0.0181	-0.26
4.270	-0.9712	-0.0203	-0.25
4.275	-0.9713	-0.0228	-0.25
4.280	-0.9719	-0.0252	-0.24
4.285	-0.9722	-0.0276	-0.24
4.290	-0.9726	-0.0300	-0.24
4.295	-0.9728	-0.0322	-0.23
4.300	-0.9732	-0.0346	-0.23
4.305	-0.9737	-0.0370	-0.23
4.310	-0.9739	-0.0394	-0.22
4.315	-0.9742	-0.0416	-0.22
4.320	-0.9744	-0.0439	-0.22
4.325	-0.9745	-0.0463	-0.21
4.330	-0.9748	-0.0487	-0.21
4.335	-0.9750	-0.0509	-0.21
4.340	-0.9752	-0.0530	-0.21
4.345	-0.9754	-0.0552	-0.20
4.350	-0.9756	-0.0575	-0.20
4.355	-0.9757	-0.0594	-0.20
4.360	-0.9758	-0.0619	-0.19
4.365	-0.9760	-0.0640	-0.19
4.370	-0.9761	-0.0660	-0.19

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.375	-0.9765	-0.0681	-0.19
4.380	-0.9765	-0.0700	-0.18
4.385	-0.9764	-0.0720	-0.18
4.390	-0.9765	-0.0740	-0.18
4.395	-0.9767	-0.0757	-0.18
4.400	-0.9768	-0.0772	-0.18
4.405	-0.9769	-0.0789	-0.17
4.410	-0.9773	-0.0810	-0.17
4.415	-0.9771	-0.0826	-0.17
4.420	-0.9775	-0.0841	-0.17
4.425	-0.9779	-0.0854	-0.16
4.430	-0.9778	-0.0874	-0.16
4.435	-0.9780	-0.0887	-0.16
4.440	-0.9780	-0.0899	-0.16
4.445	-0.9782	-0.0918	-0.15
4.450	-0.9782	-0.0927	-0.15
4.455	-0.9785	-0.0945	-0.15
4.460	-0.9785	-0.0960	-0.15
4.465	-0.9788	-0.0978	-0.14
4.470	-0.9786	-0.0994	-0.14
4.475	-0.9790	-0.1015	-0.14
4.480	-0.9792	-0.1033	-0.13
4.485	-0.9800	-0.1049	-0.13
4.490	-0.9804	-0.1059	-0.12
4.495	-0.9807	-0.1074	-0.12
4.500	-0.9811	-0.1082	-0.11
4.505	-0.9811	-0.1090	-0.11
4.510	-0.9812	-0.1093	-0.11
4.515	-0.9809	-0.1096	-0.11
4.520	-0.9810	-0.1101	-0.11
4.525	-0.9804	-0.1102	-0.12
4.530	-0.9803	-0.1100	-0.12
4.535	-0.9807	-0.1099	-0.12
4.540	-0.9802	-0.1096	-0.12
4.545	-0.9798	-0.1089	-0.12
4.550	-0.9794	-0.1092	-0.13
4.555	-0.9794	-0.1089	-0.13
4.560	-0.9793	-0.1085	-0.13
4.565	-0.9789	-0.1081	-0.13
4.570	-0.9787	-0.1076	-0.13
4.575	-0.9785	-0.1072	-0.14
4.580	-0.9777	-0.1066	-0.14
4.585	-0.9777	-0.1060	-0.15
4.590	-0.9772	-0.1056	-0.15
4.595	-0.9767	-0.1051	-0.15
4.600	-0.9762	-0.1045	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.605	-0.9756	-0.1038	-0.17
4.610	-0.9751	-0.1035	-0.17
4.615	-0.9744	-0.1029	-0.18
4.620	-0.9738	-0.1024	-0.18
4.625	-0.9734	-0.1019	-0.19
4.630	-0.9725	-0.1013	-0.20
4.635	-0.9717	-0.1009	-0.20
4.640	-0.9711	-0.1005	-0.21
4.645	-0.9704	-0.0999	-0.22
4.650	-0.9699	-0.0993	-0.22
4.655	-0.9688	-0.0989	-0.23
4.660	-0.9683	-0.0987	-0.23
4.665	-0.9674	-0.0981	-0.24
4.670	-0.9666	-0.0982	-0.25
4.675	-0.9655	-0.0981	-0.26
4.680	-0.9649	-0.0979	-0.27
4.685	-0.9640	-0.0981	-0.27
4.690	-0.9636	-0.0982	-0.28
4.695	-0.9626	-0.0986	-0.29
4.700	-0.9619	-0.0986	-0.29
4.705	-0.9614	-0.0983	-0.30
4.710	-0.9612	-0.0988	-0.30
4.715	-0.9608	-0.0990	-0.30
4.720	-0.9605	-0.0984	-0.31
4.725	-0.9601	-0.0984	-0.31
4.730	-0.9601	-0.0978	-0.31
4.735	-0.9598	-0.0975	-0.31
4.740	-0.9590	-0.0968	-0.32
4.745	-0.9584	-0.0960	-0.33
4.750	-0.9578	-0.0954	-0.33
4.755	-0.9572	-0.0945	-0.34
4.760	-0.9565	-0.0943	-0.34
4.765	-0.9554	-0.0936	-0.35
4.770	-0.9551	-0.0935	-0.36
4.775	-0.9540	-0.0926	-0.37
4.780	-0.9530	-0.0933	-0.38
4.785	-0.9524	-0.0937	-0.38
4.790	-0.9518	-0.0935	-0.39
4.795	-0.9506	-0.0942	-0.40
4.800	-0.9499	-0.0951	-0.40
4.805	-0.9495	-0.0962	-0.41
4.810	-0.9487	-0.0977	-0.41
4.815	-0.9483	-0.0990	-0.41
4.820	-0.9483	-0.1004	-0.41
4.825	-0.9483	-0.1011	-0.41
4.830	-0.9484	-0.1033	-0.41

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.835	-0.9484	-0.1046	-0.41
4.840	-0.9490	-0.1057	-0.40
4.845	-0.9494	-0.1069	-0.40
4.850	-0.9505	-0.1081	-0.39
4.855	-0.9512	-0.1098	-0.38
4.860	-0.9520	-0.1108	-0.37
4.865	-0.9530	-0.1116	-0.36
4.870	-0.9537	-0.1125	-0.35
4.875	-0.9546	-0.1138	-0.34
4.880	-0.9555	-0.1146	-0.33
4.885	-0.9564	-0.1168	-0.32
4.890	-0.9574	-0.1162	-0.31
4.895	-0.9581	-0.1169	-0.31
4.900	-0.9593	-0.1178	-0.30
4.905	-0.9599	-0.1183	-0.29
4.910	-0.9605	-0.1188	-0.28
4.915	-0.9612	-0.1192	-0.28
4.920	-0.9620	-0.1197	-0.27
4.925	-0.9625	-0.1205	-0.26
4.930	-0.9631	-0.1210	-0.26
4.935	-0.9635	-0.1215	-0.25
4.940	-0.9641	-0.1220	-0.25
4.945	-0.9646	-0.1226	-0.24
4.950	-0.9652	-0.1233	-0.24
4.955	-0.9656	-0.1238	-0.23
4.960	-0.9663	-0.1240	-0.23
4.965	-0.9667	-0.1248	-0.22
4.970	-0.9673	-0.1253	-0.22
4.975	-0.9677	-0.1256	-0.21
4.980	-0.9686	-0.1261	-0.20
4.985	-0.9689	-0.1268	-0.20
4.990	-0.9698	-0.1266	-0.19
4.995	-0.9698	-0.1271	-0.19
5.000	-0.9707	-0.1270	-0.18
5.005	-0.9711	-0.1270	-0.18
5.010	-0.9714	-0.1270	-0.18
5.015	-0.9719	-0.1274	-0.17
5.020	-0.9723	-0.1267	-0.17
5.025	-0.9726	-0.1274	-0.17
5.030	-0.9725	-0.1266	-0.17
5.035	-0.9730	-0.1270	-0.16
5.040	-0.9730	-0.1257	-0.17
5.045	-0.9735	-0.1268	-0.16
5.050	-0.9730	-0.1260	-0.17
5.055	-0.9737	-0.1267	-0.16
5.060	-0.9732	-0.1262	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.065	-0.9747	-0.1258	-0.15
5.070	-0.9744	-0.1265	-0.15
5.075	-0.9756	-0.1257	-0.14
5.080	-0.9754	-0.1261	-0.14
5.085	-0.9762	-0.1252	-0.14
5.090	-0.9770	-0.1255	-0.13
5.095	-0.9780	-0.1245	-0.12
5.100	-0.9787	-0.1246	-0.12
5.105	-0.9785	-0.1234	-0.12
5.110	-0.9794	-0.1224	-0.11
5.115	-0.9799	-0.1210	-0.11
5.120	-0.9806	-0.1200	-0.11
5.125	-0.9810	-0.1182	-0.10
5.130	-0.9818	-0.1168	-0.10
5.135	-0.9819	-0.1149	-0.10
5.140	-0.9828	-0.1131	-0.09
5.145	-0.9824	-0.1106	-0.10
5.150	-0.9833	-0.1089	-0.09
5.155	-0.9833	-0.1067	-0.10
5.160	-0.9837	-0.1042	-0.09
5.165	-0.9835	-0.1020	-0.10
5.170	-0.9835	-0.0994	-0.10
5.175	-0.9831	-0.0970	-0.11
5.180	-0.9831	-0.0943	-0.11
5.185	-0.9828	-0.0918	-0.11
5.190	-0.9822	-0.0891	-0.12
5.195	-0.9815	-0.0867	-0.13
5.200	-0.9808	-0.0840	-0.14
5.205	-0.9804	-0.0815	-0.14
5.210	-0.9797	-0.0790	-0.15
5.215	-0.9790	-0.0765	-0.16
5.220	-0.9781	-0.0740	-0.17
5.225	-0.9773	-0.0716	-0.18
5.230	-0.9765	-0.0692	-0.18
5.235	-0.9758	-0.0670	-0.19
5.240	-0.9753	-0.0647	-0.20
5.245	-0.9745	-0.0625	-0.21
5.250	-0.9735	-0.0605	-0.22
5.255	-0.9727	-0.0583	-0.22
5.260	-0.9721	-0.0563	-0.23
5.265	-0.9713	-0.0542	-0.24
5.270	-0.9708	-0.0526	-0.24
5.275	-0.9699	-0.0508	-0.25
5.280	-0.9690	-0.0489	-0.26
5.285	-0.9680	-0.0473	-0.27
5.290	-0.9675	-0.0456	-0.28

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.295	-0.9669	-0.0443	-0.28
5.300	-0.9662	-0.0425	-0.29
5.305	-0.9658	-0.0410	-0.29
5.310	-0.9649	-0.0392	-0.30
5.315	-0.9646	-0.0379	-0.31
5.320	-0.9638	-0.0367	-0.31
5.325	-0.9631	-0.0350	-0.32
5.330	-0.9629	-0.0342	-0.32
5.335	-0.9623	-0.0326	-0.33
5.340	-0.9621	-0.0316	-0.33
5.345	-0.9614	-0.0308	-0.34
5.350	-0.9612	-0.0289	-0.34
5.355	-0.9606	-0.0282	-0.34
5.360	-0.9605	-0.0270	-0.35
5.365	-0.9600	-0.0262	-0.35
5.370	-0.9597	-0.0251	-0.35
5.375	-0.9594	-0.0242	-0.36
5.380	-0.9592	-0.0229	-0.36
5.385	-0.9589	-0.0221	-0.36
5.390	-0.9591	-0.0211	-0.36
5.395	-0.9592	-0.0206	-0.36
5.400	-0.9594	-0.0197	-0.36
5.405	-0.9601	-0.0189	-0.35
5.410	-0.9596	-0.0181	-0.36
5.415	-0.9596	-0.0178	-0.36
5.420	-0.9601	-0.0172	-0.35
5.425	-0.9603	-0.0164	-0.35
5.430	-0.9605	-0.0159	-0.35
5.435	-0.9606	-0.0157	-0.35
5.440	-0.9610	-0.0154	-0.34
5.445	-0.9611	-0.0151	-0.34
5.450	-0.9617	-0.0152	-0.34
5.455	-0.9622	-0.0153	-0.33
5.460	-0.9626	-0.0149	-0.33
5.465	-0.9629	-0.0152	-0.33
5.470	-0.9633	-0.0156	-0.32
5.475	-0.9639	-0.0155	-0.32
5.480	-0.9643	-0.0161	-0.31
5.485	-0.9648	-0.0161	-0.31
5.490	-0.9652	-0.0165	-0.31
5.495	-0.9658	-0.0167	-0.30
5.500	-0.9662	-0.0170	-0.30
5.505	-0.9667	-0.0174	-0.29
5.510	-0.9676	-0.0177	-0.29
5.515	-0.9683	-0.0180	-0.28
5.520	-0.9692	-0.0186	-0.27

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.525	-0.9699	-0.0194	-0.26
5.530	-0.9708	-0.0195	-0.26
5.535	-0.9717	-0.0202	-0.25
5.540	-0.9726	-0.0209	-0.24
5.545	-0.9735	-0.0217	-0.23
5.550	-0.9743	-0.0223	-0.22
5.555	-0.9752	-0.0231	-0.22
5.560	-0.9760	-0.0237	-0.21
5.565	-0.9767	-0.0244	-0.20
5.570	-0.9775	-0.0251	-0.19
5.575	-0.9783	-0.0259	-0.19
5.580	-0.9789	-0.0264	-0.18
5.585	-0.9796	-0.0270	-0.18
5.590	-0.9803	-0.0277	-0.17
5.595	-0.9808	-0.0282	-0.16
5.600	-0.9812	-0.0284	-0.16
5.605	-0.9818	-0.0289	-0.16
5.610	-0.9824	-0.0294	-0.15
5.615	-0.9831	-0.0297	-0.14
5.620	-0.9835	-0.0298	-0.14
5.625	-0.9842	-0.0301	-0.13
5.630	-0.9852	-0.0302	-0.13
5.635	-0.9852	-0.0303	-0.13
5.640	-0.9861	-0.0305	-0.12
5.645	-0.9862	-0.0309	-0.12
5.650	-0.9867	-0.0315	-0.11
5.655	-0.9873	-0.0316	-0.11
5.660	-0.9878	-0.0317	-0.10
5.665	-0.9882	-0.0323	-0.10
5.670	-0.9882	-0.0318	-0.10
5.675	-0.9882	-0.0321	-0.10
5.680	-0.9889	-0.0320	-0.09
5.685	-0.9878	-0.0321	-0.10
5.690	-0.9875	-0.0324	-0.10
5.695	-0.9870	-0.0322	-0.11
5.700	-0.9874	-0.0323	-0.11
5.705	-0.9872	-0.0322	-0.11
5.710	-0.9874	-0.0323	-0.11
5.715	-0.9868	-0.0324	-0.11
5.720	-0.9866	-0.0325	-0.11
5.725	-0.9866	-0.0326	-0.11
5.730	-0.9865	-0.0320	-0.11
5.735	-0.9864	-0.0327	-0.11
5.740	-0.9859	-0.0319	-0.12
5.745	-0.9859	-0.0321	-0.12
5.750	-0.9850	-0.0314	-0.13

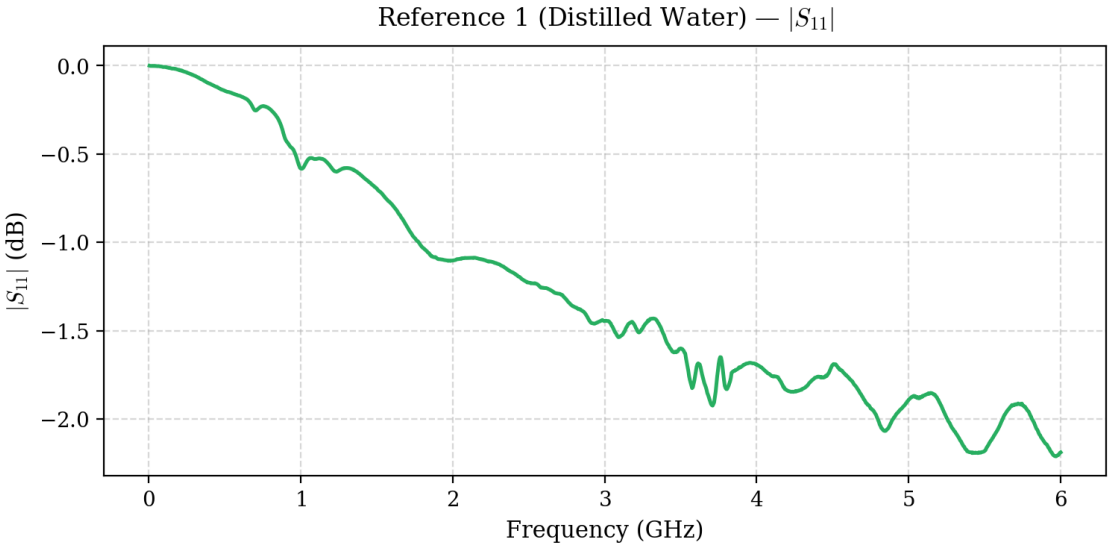
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.755	-0.9844	-0.0313	-0.13
5.760	-0.9840	-0.0311	-0.14
5.765	-0.9834	-0.0307	-0.14
5.770	-0.9825	-0.0305	-0.15
5.775	-0.9821	-0.0299	-0.15
5.780	-0.9816	-0.0295	-0.16
5.785	-0.9807	-0.0286	-0.17
5.790	-0.9799	-0.0287	-0.17
5.795	-0.9791	-0.0280	-0.18
5.800	-0.9782	-0.0275	-0.19
5.805	-0.9773	-0.0271	-0.20
5.810	-0.9767	-0.0265	-0.20
5.815	-0.9760	-0.0257	-0.21
5.820	-0.9749	-0.0255	-0.22
5.825	-0.9738	-0.0249	-0.23
5.830	-0.9729	-0.0242	-0.24
5.835	-0.9720	-0.0237	-0.24
5.840	-0.9713	-0.0233	-0.25
5.845	-0.9707	-0.0227	-0.26
5.850	-0.9696	-0.0221	-0.27
5.855	-0.9687	-0.0217	-0.27
5.860	-0.9681	-0.0212	-0.28
5.865	-0.9672	-0.0207	-0.29
5.870	-0.9667	-0.0202	-0.29
5.875	-0.9658	-0.0198	-0.30
5.880	-0.9653	-0.0195	-0.30
5.885	-0.9647	-0.0191	-0.31
5.890	-0.9641	-0.0188	-0.32
5.895	-0.9632	-0.0188	-0.32
5.900	-0.9628	-0.0186	-0.33
5.905	-0.9624	-0.0186	-0.33
5.910	-0.9620	-0.0185	-0.34
5.915	-0.9614	-0.0185	-0.34
5.920	-0.9605	-0.0185	-0.35
5.925	-0.9604	-0.0183	-0.35
5.930	-0.9595	-0.0188	-0.36
5.935	-0.9589	-0.0187	-0.36
5.940	-0.9588	-0.0188	-0.36
5.945	-0.9585	-0.0192	-0.37
5.950	-0.9583	-0.0195	-0.37
5.955	-0.9577	-0.0196	-0.37
5.960	-0.9579	-0.0197	-0.37
5.965	-0.9578	-0.0201	-0.37
5.970	-0.9571	-0.0209	-0.38
5.975	-0.9577	-0.0213	-0.37
5.980	-0.9580	-0.0225	-0.37

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Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.985	-0.9579	-0.0230	-0.37
5.990	-0.9579	-0.0243	-0.37
5.995	-0.9581	-0.0253	-0.37
6.000	-0.9578	-0.0261	-0.37

3.3 Reference 1 (Distilled Water)



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.005	1.0000	-0.0051	0.00
0.010	0.9998	-0.0101	-0.00
0.015	0.9997	-0.0150	-0.00
0.020	0.9997	-0.0201	-0.00
0.025	0.9995	-0.0252	-0.00
0.030	0.9995	-0.0301	-0.00
0.035	0.9992	-0.0349	-0.00
0.040	0.9991	-0.0401	-0.00
0.045	0.9987	-0.0450	-0.00
0.050	0.9985	-0.0500	-0.00
0.055	0.9981	-0.0550	-0.00
0.060	0.9979	-0.0600	-0.00
0.065	0.9976	-0.0650	-0.00
0.070	0.9972	-0.0699	-0.00
0.075	0.9967	-0.0750	-0.00
0.080	0.9963	-0.0798	-0.00
0.085	0.9958	-0.0848	-0.01
0.090	0.9952	-0.0897	-0.01

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.095	0.9946	-0.0947	-0.01
0.100	0.9942	-0.0996	-0.01
0.105	0.9937	-0.1046	-0.01
0.110	0.9931	-0.1094	-0.01
0.115	0.9925	-0.1144	-0.01
0.120	0.9917	-0.1193	-0.01
0.125	0.9910	-0.1242	-0.01
0.130	0.9902	-0.1292	-0.01
0.135	0.9895	-0.1341	-0.01
0.140	0.9888	-0.1389	-0.01
0.145	0.9879	-0.1440	-0.01
0.150	0.9871	-0.1487	-0.02
0.155	0.9862	-0.1536	-0.02
0.160	0.9855	-0.1583	-0.02
0.165	0.9847	-0.1633	-0.02
0.170	0.9837	-0.1681	-0.02
0.175	0.9829	-0.1730	-0.02
0.180	0.9818	-0.1779	-0.02
0.185	0.9808	-0.1827	-0.02
0.190	0.9797	-0.1876	-0.02
0.195	0.9786	-0.1924	-0.02
0.200	0.9776	-0.1973	-0.02
0.205	0.9763	-0.2020	-0.03
0.210	0.9753	-0.2070	-0.03
0.215	0.9740	-0.2117	-0.03
0.220	0.9730	-0.2166	-0.03
0.225	0.9715	-0.2213	-0.03
0.230	0.9703	-0.2261	-0.03
0.235	0.9689	-0.2308	-0.03
0.240	0.9677	-0.2356	-0.04
0.245	0.9663	-0.2405	-0.04
0.250	0.9650	-0.2453	-0.04
0.255	0.9634	-0.2499	-0.04
0.260	0.9621	-0.2546	-0.04
0.265	0.9606	-0.2594	-0.04
0.270	0.9591	-0.2641	-0.05
0.275	0.9576	-0.2688	-0.05
0.280	0.9561	-0.2735	-0.05
0.285	0.9544	-0.2781	-0.05
0.290	0.9529	-0.2828	-0.05
0.295	0.9513	-0.2874	-0.05
0.300	0.9496	-0.2920	-0.06
0.305	0.9480	-0.2968	-0.06
0.310	0.9463	-0.3013	-0.06
0.315	0.9445	-0.3058	-0.06
0.320	0.9427	-0.3104	-0.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.325	0.9410	-0.3149	-0.07
0.330	0.9393	-0.3194	-0.07
0.335	0.9374	-0.3239	-0.07
0.340	0.9355	-0.3286	-0.07
0.345	0.9337	-0.3328	-0.08
0.350	0.9319	-0.3373	-0.08
0.355	0.9299	-0.3418	-0.08
0.360	0.9280	-0.3461	-0.08
0.365	0.9260	-0.3507	-0.09
0.370	0.9240	-0.3551	-0.09
0.375	0.9221	-0.3595	-0.09
0.380	0.9200	-0.3638	-0.09
0.385	0.9179	-0.3681	-0.10
0.390	0.9160	-0.3725	-0.10
0.395	0.9139	-0.3767	-0.10
0.400	0.9121	-0.3810	-0.10
0.405	0.9100	-0.3853	-0.10
0.410	0.9079	-0.3895	-0.11
0.415	0.9058	-0.3936	-0.11
0.420	0.9037	-0.3978	-0.11
0.425	0.9018	-0.4019	-0.11
0.430	0.8996	-0.4060	-0.11
0.435	0.8974	-0.4102	-0.12
0.440	0.8953	-0.4142	-0.12
0.445	0.8932	-0.4183	-0.12
0.450	0.8912	-0.4225	-0.12
0.455	0.8888	-0.4265	-0.12
0.460	0.8866	-0.4304	-0.13
0.465	0.8843	-0.4344	-0.13
0.470	0.8821	-0.4384	-0.13
0.475	0.8798	-0.4422	-0.13
0.480	0.8776	-0.4463	-0.14
0.485	0.8754	-0.4501	-0.14
0.490	0.8732	-0.4541	-0.14
0.495	0.8710	-0.4579	-0.14
0.500	0.8686	-0.4618	-0.14
0.505	0.8664	-0.4656	-0.14
0.510	0.8641	-0.4694	-0.15
0.515	0.8618	-0.4732	-0.15
0.520	0.8596	-0.4770	-0.15
0.525	0.8574	-0.4809	-0.15
0.530	0.8552	-0.4844	-0.15
0.535	0.8529	-0.4881	-0.15
0.540	0.8506	-0.4919	-0.15
0.545	0.8482	-0.4956	-0.15
0.550	0.8457	-0.4993	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.555	0.8434	-0.5030	-0.16
0.560	0.8409	-0.5067	-0.16
0.565	0.8386	-0.5103	-0.16
0.570	0.8361	-0.5139	-0.16
0.575	0.8337	-0.5176	-0.16
0.580	0.8313	-0.5212	-0.17
0.585	0.8288	-0.5246	-0.17
0.590	0.8264	-0.5282	-0.17
0.595	0.8241	-0.5317	-0.17
0.600	0.8216	-0.5352	-0.17
0.605	0.8190	-0.5387	-0.17
0.610	0.8164	-0.5422	-0.18
0.615	0.8138	-0.5457	-0.18
0.620	0.8113	-0.5492	-0.18
0.625	0.8086	-0.5526	-0.18
0.630	0.8059	-0.5561	-0.18
0.635	0.8033	-0.5595	-0.18
0.640	0.8006	-0.5628	-0.19
0.645	0.7980	-0.5662	-0.19
0.650	0.7952	-0.5694	-0.19
0.655	0.7922	-0.5726	-0.20
0.660	0.7893	-0.5758	-0.20
0.665	0.7864	-0.5789	-0.21
0.670	0.7832	-0.5819	-0.21
0.675	0.7802	-0.5847	-0.22
0.680	0.7773	-0.5872	-0.23
0.685	0.7742	-0.5895	-0.24
0.690	0.7716	-0.5917	-0.24
0.695	0.7690	-0.5937	-0.25
0.700	0.7669	-0.5961	-0.25
0.705	0.7648	-0.5986	-0.25
0.710	0.7630	-0.6012	-0.25
0.715	0.7612	-0.6043	-0.25
0.720	0.7593	-0.6073	-0.24
0.725	0.7571	-0.6107	-0.24
0.730	0.7549	-0.6142	-0.24
0.735	0.7525	-0.6175	-0.23
0.740	0.7500	-0.6210	-0.23
0.745	0.7474	-0.6244	-0.23
0.750	0.7447	-0.6278	-0.23
0.755	0.7420	-0.6310	-0.23
0.760	0.7390	-0.6341	-0.23
0.765	0.7361	-0.6376	-0.23
0.770	0.7331	-0.6409	-0.23
0.775	0.7298	-0.6440	-0.23
0.780	0.7268	-0.6473	-0.24

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.785	0.7235	-0.6506	-0.24
0.790	0.7203	-0.6536	-0.24
0.795	0.7170	-0.6567	-0.24
0.800	0.7138	-0.6596	-0.25
0.805	0.7104	-0.6626	-0.25
0.810	0.7070	-0.6656	-0.26
0.815	0.7036	-0.6685	-0.26
0.820	0.7000	-0.6715	-0.26
0.825	0.6966	-0.6743	-0.27
0.830	0.6929	-0.6772	-0.27
0.835	0.6894	-0.6800	-0.28
0.840	0.6856	-0.6827	-0.29
0.845	0.6821	-0.6854	-0.29
0.850	0.6782	-0.6881	-0.30
0.855	0.6742	-0.6907	-0.31
0.860	0.6702	-0.6932	-0.32
0.865	0.6662	-0.6956	-0.33
0.870	0.6621	-0.6978	-0.34
0.875	0.6579	-0.6999	-0.35
0.880	0.6537	-0.7017	-0.36
0.885	0.6496	-0.7031	-0.38
0.890	0.6459	-0.7045	-0.39
0.895	0.6423	-0.7056	-0.41
0.900	0.6390	-0.7069	-0.42
0.905	0.6362	-0.7084	-0.43
0.910	0.6332	-0.7101	-0.43
0.915	0.6302	-0.7121	-0.44
0.920	0.6270	-0.7140	-0.44
0.925	0.6238	-0.7157	-0.45
0.930	0.6208	-0.7176	-0.46
0.935	0.6177	-0.7195	-0.46
0.940	0.6146	-0.7217	-0.46
0.945	0.6113	-0.7239	-0.47
0.950	0.6077	-0.7263	-0.47
0.955	0.6039	-0.7283	-0.48
0.960	0.6001	-0.7300	-0.49
0.965	0.5964	-0.7315	-0.50
0.970	0.5926	-0.7330	-0.51
0.975	0.5889	-0.7340	-0.53
0.980	0.5855	-0.7348	-0.54
0.985	0.5823	-0.7353	-0.56
0.990	0.5798	-0.7359	-0.57
0.995	0.5771	-0.7364	-0.58
1.000	0.5753	-0.7372	-0.58
1.005	0.5737	-0.7384	-0.58
1.010	0.5719	-0.7398	-0.58

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.015	0.5702	-0.7416	-0.58
1.020	0.5686	-0.7437	-0.57
1.025	0.5668	-0.7461	-0.57
1.030	0.5648	-0.7488	-0.56
1.035	0.5625	-0.7515	-0.55
1.040	0.5602	-0.7544	-0.54
1.045	0.5575	-0.7572	-0.53
1.050	0.5547	-0.7599	-0.53
1.055	0.5518	-0.7626	-0.53
1.060	0.5488	-0.7651	-0.52
1.065	0.5454	-0.7675	-0.52
1.070	0.5422	-0.7698	-0.52
1.075	0.5390	-0.7720	-0.52
1.080	0.5358	-0.7740	-0.53
1.085	0.5327	-0.7759	-0.53
1.090	0.5297	-0.7777	-0.53
1.095	0.5268	-0.7798	-0.53
1.100	0.5237	-0.7817	-0.53
1.105	0.5209	-0.7838	-0.53
1.110	0.5179	-0.7858	-0.53
1.115	0.5149	-0.7879	-0.53
1.120	0.5119	-0.7901	-0.52
1.125	0.5087	-0.7921	-0.53
1.130	0.5054	-0.7941	-0.53
1.135	0.5023	-0.7958	-0.53
1.140	0.4992	-0.7978	-0.53
1.145	0.4959	-0.7997	-0.53
1.150	0.4925	-0.8015	-0.53
1.155	0.4893	-0.8032	-0.53
1.160	0.4861	-0.8047	-0.54
1.165	0.4828	-0.8062	-0.54
1.170	0.4794	-0.8076	-0.55
1.175	0.4764	-0.8089	-0.55
1.180	0.4732	-0.8102	-0.55
1.185	0.4702	-0.8114	-0.56
1.190	0.4673	-0.8123	-0.56
1.195	0.4645	-0.8132	-0.57
1.200	0.4618	-0.8141	-0.57
1.205	0.4590	-0.8149	-0.58
1.210	0.4563	-0.8157	-0.59
1.215	0.4539	-0.8165	-0.59
1.220	0.4515	-0.8172	-0.60
1.225	0.4492	-0.8183	-0.60
1.230	0.4471	-0.8194	-0.60
1.235	0.4448	-0.8204	-0.60
1.240	0.4428	-0.8217	-0.60

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.245	0.4407	-0.8230	-0.60
1.250	0.4385	-0.8246	-0.59
1.255	0.4362	-0.8261	-0.59
1.260	0.4339	-0.8275	-0.59
1.265	0.4316	-0.8290	-0.59
1.270	0.4292	-0.8306	-0.58
1.275	0.4265	-0.8321	-0.58
1.280	0.4241	-0.8337	-0.58
1.285	0.4212	-0.8350	-0.58
1.290	0.4184	-0.8367	-0.58
1.295	0.4156	-0.8381	-0.58
1.300	0.4129	-0.8394	-0.58
1.305	0.4102	-0.8408	-0.58
1.310	0.4076	-0.8420	-0.58
1.315	0.4048	-0.8433	-0.58
1.320	0.4020	-0.8445	-0.58
1.325	0.3993	-0.8458	-0.58
1.330	0.3966	-0.8468	-0.58
1.335	0.3937	-0.8480	-0.58
1.340	0.3909	-0.8491	-0.59
1.345	0.3880	-0.8500	-0.59
1.350	0.3853	-0.8512	-0.59
1.355	0.3823	-0.8521	-0.59
1.360	0.3794	-0.8531	-0.60
1.365	0.3764	-0.8541	-0.60
1.370	0.3736	-0.8551	-0.60
1.375	0.3707	-0.8559	-0.60
1.380	0.3677	-0.8567	-0.61
1.385	0.3648	-0.8577	-0.61
1.390	0.3618	-0.8586	-0.61
1.395	0.3587	-0.8594	-0.62
1.400	0.3557	-0.8603	-0.62
1.405	0.3525	-0.8611	-0.63
1.410	0.3494	-0.8619	-0.63
1.415	0.3463	-0.8627	-0.63
1.420	0.3431	-0.8635	-0.64
1.425	0.3399	-0.8644	-0.64
1.430	0.3368	-0.8652	-0.64
1.435	0.3335	-0.8661	-0.65
1.440	0.3304	-0.8669	-0.65
1.445	0.3273	-0.8677	-0.65
1.450	0.3240	-0.8684	-0.66
1.455	0.3208	-0.8694	-0.66
1.460	0.3175	-0.8701	-0.67
1.465	0.3143	-0.8707	-0.67
1.470	0.3111	-0.8716	-0.67

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.475	0.3077	-0.8723	-0.68
1.480	0.3044	-0.8731	-0.68
1.485	0.3008	-0.8738	-0.69
1.490	0.2975	-0.8747	-0.69
1.495	0.2940	-0.8754	-0.69
1.500	0.2908	-0.8761	-0.70
1.505	0.2872	-0.8766	-0.70
1.510	0.2836	-0.8773	-0.71
1.515	0.2801	-0.8780	-0.71
1.520	0.2767	-0.8787	-0.71
1.525	0.2731	-0.8795	-0.72
1.530	0.2695	-0.8801	-0.72
1.535	0.2658	-0.8808	-0.72
1.540	0.2621	-0.8813	-0.73
1.545	0.2584	-0.8817	-0.74
1.550	0.2549	-0.8823	-0.74
1.555	0.2509	-0.8827	-0.75
1.560	0.2472	-0.8832	-0.75
1.565	0.2434	-0.8838	-0.76
1.570	0.2396	-0.8844	-0.76
1.575	0.2358	-0.8848	-0.77
1.580	0.2323	-0.8853	-0.77
1.585	0.2284	-0.8861	-0.77
1.590	0.2247	-0.8865	-0.78
1.595	0.2209	-0.8870	-0.78
1.600	0.2173	-0.8874	-0.78
1.605	0.2136	-0.8877	-0.79
1.610	0.2099	-0.8879	-0.80
1.615	0.2061	-0.8883	-0.80
1.620	0.2023	-0.8887	-0.81
1.625	0.1985	-0.8890	-0.81
1.630	0.1947	-0.8892	-0.82
1.635	0.1909	-0.8894	-0.82
1.640	0.1869	-0.8893	-0.83
1.645	0.1832	-0.8893	-0.84
1.650	0.1793	-0.8895	-0.84
1.655	0.1756	-0.8898	-0.85
1.660	0.1720	-0.8896	-0.86
1.665	0.1684	-0.8896	-0.86
1.670	0.1648	-0.8898	-0.87
1.675	0.1612	-0.8896	-0.88
1.680	0.1577	-0.8896	-0.88
1.685	0.1541	-0.8893	-0.89
1.690	0.1507	-0.8891	-0.90
1.695	0.1476	-0.8890	-0.90
1.700	0.1441	-0.8887	-0.91

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.705	0.1408	-0.8886	-0.92
1.710	0.1374	-0.8883	-0.93
1.715	0.1340	-0.8880	-0.93
1.720	0.1310	-0.8878	-0.94
1.725	0.1276	-0.8876	-0.95
1.730	0.1247	-0.8875	-0.95
1.735	0.1215	-0.8870	-0.96
1.740	0.1185	-0.8868	-0.97
1.745	0.1157	-0.8867	-0.97
1.750	0.1128	-0.8866	-0.98
1.755	0.1101	-0.8865	-0.98
1.760	0.1074	-0.8861	-0.99
1.765	0.1048	-0.8860	-0.99
1.770	0.1023	-0.8860	-0.99
1.775	0.0999	-0.8858	-1.00
1.780	0.0976	-0.8855	-1.00
1.785	0.0950	-0.8850	-1.01
1.790	0.0923	-0.8846	-1.02
1.795	0.0899	-0.8843	-1.02
1.800	0.0875	-0.8841	-1.03
1.805	0.0852	-0.8839	-1.03
1.810	0.0829	-0.8838	-1.03
1.815	0.0808	-0.8836	-1.04
1.820	0.0785	-0.8834	-1.04
1.825	0.0766	-0.8831	-1.05
1.830	0.0747	-0.8830	-1.05
1.835	0.0725	-0.8826	-1.06
1.840	0.0706	-0.8824	-1.06
1.845	0.0688	-0.8822	-1.06
1.850	0.0667	-0.8815	-1.07
1.855	0.0648	-0.8815	-1.07
1.860	0.0628	-0.8809	-1.08
1.865	0.0608	-0.8811	-1.08
1.870	0.0590	-0.8809	-1.08
1.875	0.0569	-0.8809	-1.08
1.880	0.0551	-0.8809	-1.08
1.885	0.0535	-0.8812	-1.08
1.890	0.0522	-0.8810	-1.09
1.895	0.0504	-0.8807	-1.09
1.900	0.0486	-0.8804	-1.09
1.905	0.0470	-0.8803	-1.09
1.910	0.0452	-0.8804	-1.09
1.915	0.0434	-0.8804	-1.10
1.920	0.0419	-0.8805	-1.10
1.925	0.0403	-0.8804	-1.10
1.930	0.0389	-0.8805	-1.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.935	0.0374	-0.8804	-1.10
1.940	0.0359	-0.8804	-1.10
1.945	0.0345	-0.8804	-1.10
1.950	0.0330	-0.8803	-1.10
1.955	0.0316	-0.8803	-1.10
1.960	0.0302	-0.8802	-1.10
1.965	0.0288	-0.8803	-1.10
1.970	0.0274	-0.8801	-1.10
1.975	0.0262	-0.8803	-1.10
1.980	0.0248	-0.8804	-1.10
1.985	0.0235	-0.8802	-1.10
1.990	0.0220	-0.8804	-1.10
1.995	0.0206	-0.8805	-1.10
2.000	0.0190	-0.8805	-1.10
2.005	0.0175	-0.8807	-1.10
2.010	0.0159	-0.8808	-1.10
2.015	0.0145	-0.8809	-1.10
2.020	0.0130	-0.8810	-1.10
2.025	0.0112	-0.8813	-1.10
2.030	0.0095	-0.8815	-1.09
2.035	0.0079	-0.8815	-1.10
2.040	0.0061	-0.8816	-1.09
2.045	0.0044	-0.8817	-1.09
2.050	0.0026	-0.8817	-1.09
2.055	0.0007	-0.8820	-1.09
2.060	-0.0012	-0.8819	-1.09
2.065	-0.0031	-0.8821	-1.09
2.070	-0.0052	-0.8821	-1.09
2.075	-0.0073	-0.8821	-1.09
2.080	-0.0095	-0.8822	-1.09
2.085	-0.0117	-0.8821	-1.09
2.090	-0.0140	-0.8821	-1.09
2.095	-0.0163	-0.8821	-1.09
2.100	-0.0188	-0.8820	-1.09
2.105	-0.0212	-0.8820	-1.09
2.110	-0.0237	-0.8820	-1.09
2.115	-0.0261	-0.8819	-1.09
2.120	-0.0286	-0.8818	-1.09
2.125	-0.0312	-0.8818	-1.09
2.130	-0.0341	-0.8816	-1.09
2.135	-0.0366	-0.8814	-1.09
2.140	-0.0393	-0.8815	-1.09
2.145	-0.0419	-0.8814	-1.09
2.150	-0.0445	-0.8812	-1.09
2.155	-0.0473	-0.8809	-1.09
2.160	-0.0501	-0.8806	-1.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.165	-0.0529	-0.8804	-1.09
2.170	-0.0558	-0.8800	-1.09
2.175	-0.0585	-0.8798	-1.09
2.180	-0.0613	-0.8795	-1.09
2.185	-0.0641	-0.8791	-1.10
2.190	-0.0671	-0.8790	-1.10
2.195	-0.0700	-0.8785	-1.10
2.200	-0.0729	-0.8783	-1.10
2.205	-0.0759	-0.8778	-1.10
2.210	-0.0788	-0.8775	-1.10
2.215	-0.0817	-0.8770	-1.10
2.220	-0.0846	-0.8766	-1.10
2.225	-0.0877	-0.8761	-1.11
2.230	-0.0906	-0.8756	-1.11
2.235	-0.0936	-0.8752	-1.11
2.240	-0.0967	-0.8750	-1.11
2.245	-0.0995	-0.8745	-1.11
2.250	-0.1025	-0.8742	-1.11
2.255	-0.1056	-0.8737	-1.11
2.260	-0.1083	-0.8733	-1.11
2.265	-0.1114	-0.8728	-1.11
2.270	-0.1141	-0.8723	-1.11
2.275	-0.1170	-0.8716	-1.12
2.280	-0.1199	-0.8711	-1.12
2.285	-0.1228	-0.8707	-1.12
2.290	-0.1255	-0.8700	-1.12
2.295	-0.1283	-0.8695	-1.12
2.300	-0.1312	-0.8689	-1.12
2.305	-0.1339	-0.8684	-1.12
2.310	-0.1369	-0.8677	-1.13
2.315	-0.1394	-0.8669	-1.13
2.320	-0.1423	-0.8662	-1.13
2.325	-0.1449	-0.8656	-1.13
2.330	-0.1474	-0.8649	-1.14
2.335	-0.1503	-0.8643	-1.14
2.340	-0.1526	-0.8635	-1.14
2.345	-0.1553	-0.8627	-1.14
2.350	-0.1577	-0.8620	-1.15
2.355	-0.1601	-0.8612	-1.15
2.360	-0.1625	-0.8603	-1.15
2.365	-0.1646	-0.8598	-1.16
2.370	-0.1667	-0.8589	-1.16
2.375	-0.1690	-0.8584	-1.16
2.380	-0.1711	-0.8576	-1.16
2.385	-0.1732	-0.8570	-1.17
2.390	-0.1754	-0.8564	-1.17

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.395	-0.1773	-0.8557	-1.17
2.400	-0.1792	-0.8551	-1.17
2.405	-0.1813	-0.8544	-1.18
2.410	-0.1831	-0.8536	-1.18
2.415	-0.1849	-0.8531	-1.18
2.420	-0.1866	-0.8523	-1.18
2.425	-0.1882	-0.8517	-1.19
2.430	-0.1899	-0.8508	-1.19
2.435	-0.1914	-0.8503	-1.19
2.440	-0.1929	-0.8498	-1.20
2.445	-0.1945	-0.8490	-1.20
2.450	-0.1958	-0.8483	-1.20
2.455	-0.1969	-0.8476	-1.21
2.460	-0.1982	-0.8471	-1.21
2.465	-0.1995	-0.8464	-1.21
2.470	-0.2007	-0.8459	-1.22
2.475	-0.2018	-0.8453	-1.22
2.480	-0.2033	-0.8448	-1.22
2.485	-0.2044	-0.8443	-1.22
2.490	-0.2055	-0.8436	-1.23
2.495	-0.2066	-0.8434	-1.23
2.500	-0.2080	-0.8429	-1.23
2.505	-0.2090	-0.8426	-1.23
2.510	-0.2101	-0.8422	-1.23
2.515	-0.2111	-0.8417	-1.23
2.520	-0.2123	-0.8415	-1.23
2.525	-0.2133	-0.8411	-1.23
2.530	-0.2144	-0.8408	-1.23
2.535	-0.2155	-0.8405	-1.23
2.540	-0.2166	-0.8404	-1.23
2.545	-0.2174	-0.8400	-1.23
2.550	-0.2183	-0.8396	-1.23
2.555	-0.2192	-0.8392	-1.24
2.560	-0.2200	-0.8386	-1.24
2.565	-0.2204	-0.8382	-1.24
2.570	-0.2209	-0.8375	-1.25
2.575	-0.2213	-0.8371	-1.25
2.580	-0.2216	-0.8365	-1.26
2.585	-0.2221	-0.8364	-1.26
2.590	-0.2225	-0.8361	-1.26
2.595	-0.2229	-0.8361	-1.26
2.600	-0.2234	-0.8358	-1.26
2.605	-0.2241	-0.8356	-1.26
2.610	-0.2246	-0.8354	-1.26
2.615	-0.2253	-0.8353	-1.26
2.620	-0.2258	-0.8350	-1.26

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.625	-0.2263	-0.8346	-1.26
2.630	-0.2268	-0.8342	-1.26
2.635	-0.2273	-0.8339	-1.27
2.640	-0.2280	-0.8335	-1.27
2.645	-0.2285	-0.8331	-1.27
2.650	-0.2291	-0.8328	-1.27
2.655	-0.2297	-0.8323	-1.28
2.660	-0.2303	-0.8317	-1.28
2.665	-0.2309	-0.8314	-1.28
2.670	-0.2317	-0.8307	-1.29
2.675	-0.2321	-0.8305	-1.29
2.680	-0.2327	-0.8302	-1.29
2.685	-0.2330	-0.8300	-1.29
2.690	-0.2336	-0.8298	-1.29
2.695	-0.2341	-0.8296	-1.29
2.700	-0.2344	-0.8294	-1.29
2.705	-0.2349	-0.8292	-1.29
2.710	-0.2350	-0.8290	-1.29
2.715	-0.2357	-0.8284	-1.30
2.720	-0.2362	-0.8282	-1.30
2.725	-0.2369	-0.8276	-1.30
2.730	-0.2374	-0.8268	-1.31
2.735	-0.2383	-0.8261	-1.31
2.740	-0.2391	-0.8253	-1.32
2.745	-0.2402	-0.8247	-1.32
2.750	-0.2409	-0.8239	-1.33
2.755	-0.2419	-0.8230	-1.33
2.760	-0.2432	-0.8221	-1.34
2.765	-0.2444	-0.8213	-1.34
2.770	-0.2456	-0.8205	-1.35
2.775	-0.2467	-0.8196	-1.35
2.780	-0.2481	-0.8189	-1.35
2.785	-0.2493	-0.8181	-1.36
2.790	-0.2508	-0.8173	-1.36
2.795	-0.2522	-0.8168	-1.36
2.800	-0.2533	-0.8161	-1.37
2.805	-0.2550	-0.8155	-1.37
2.810	-0.2563	-0.8148	-1.37
2.815	-0.2576	-0.8143	-1.37
2.820	-0.2590	-0.8136	-1.37
2.825	-0.2603	-0.8127	-1.38
2.830	-0.2617	-0.8123	-1.38
2.835	-0.2630	-0.8116	-1.38
2.840	-0.2644	-0.8112	-1.38
2.845	-0.2657	-0.8103	-1.38
2.850	-0.2670	-0.8096	-1.39

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.855	-0.2682	-0.8089	-1.39
2.860	-0.2694	-0.8082	-1.39
2.865	-0.2705	-0.8074	-1.40
2.870	-0.2715	-0.8066	-1.40
2.875	-0.2724	-0.8057	-1.41
2.880	-0.2733	-0.8046	-1.41
2.885	-0.2745	-0.8035	-1.42
2.890	-0.2758	-0.8022	-1.43
2.895	-0.2773	-0.8009	-1.44
2.900	-0.2786	-0.7996	-1.44
2.905	-0.2803	-0.7985	-1.45
2.910	-0.2817	-0.7975	-1.46
2.915	-0.2836	-0.7966	-1.46
2.920	-0.2853	-0.7958	-1.46
2.925	-0.2870	-0.7952	-1.46
2.930	-0.2887	-0.7944	-1.46
2.935	-0.2900	-0.7940	-1.46
2.940	-0.2914	-0.7937	-1.46
2.945	-0.2929	-0.7934	-1.46
2.950	-0.2940	-0.7931	-1.45
2.955	-0.2953	-0.7929	-1.45
2.960	-0.2960	-0.7928	-1.45
2.965	-0.2971	-0.7926	-1.45
2.970	-0.2978	-0.7925	-1.45
2.975	-0.2985	-0.7925	-1.44
2.980	-0.2992	-0.7925	-1.44
2.985	-0.2998	-0.7925	-1.44
2.990	-0.3000	-0.7921	-1.44
2.995	-0.3003	-0.7916	-1.45
3.000	-0.3008	-0.7915	-1.45
3.005	-0.3044	-0.7903	-1.44
3.010	-0.3045	-0.7902	-1.44
3.015	-0.3047	-0.7900	-1.44
3.020	-0.3049	-0.7899	-1.45
3.025	-0.3046	-0.7900	-1.45
3.030	-0.3048	-0.7895	-1.45
3.035	-0.3048	-0.7890	-1.45
3.040	-0.3047	-0.7885	-1.46
3.045	-0.3047	-0.7874	-1.47
3.050	-0.3048	-0.7868	-1.48
3.055	-0.3049	-0.7859	-1.48
3.060	-0.3048	-0.7849	-1.49
3.065	-0.3054	-0.7837	-1.50
3.070	-0.3057	-0.7827	-1.51
3.075	-0.3065	-0.7819	-1.52
3.080	-0.3070	-0.7808	-1.53

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.085	-0.3079	-0.7797	-1.53
3.090	-0.3092	-0.7787	-1.54
3.095	-0.3105	-0.7783	-1.54
3.100	-0.3117	-0.7779	-1.53
3.105	-0.3131	-0.7779	-1.53
3.110	-0.3143	-0.7777	-1.53
3.115	-0.3156	-0.7774	-1.52
3.120	-0.3168	-0.7774	-1.52
3.125	-0.3180	-0.7778	-1.51
3.130	-0.3188	-0.7780	-1.51
3.135	-0.3196	-0.7788	-1.50
3.140	-0.3202	-0.7797	-1.48
3.145	-0.3210	-0.7800	-1.48
3.150	-0.3213	-0.7808	-1.47
3.155	-0.3217	-0.7814	-1.46
3.160	-0.3218	-0.7817	-1.46
3.165	-0.3219	-0.7820	-1.46
3.170	-0.3220	-0.7822	-1.45
3.175	-0.3220	-0.7825	-1.45
3.180	-0.3214	-0.7829	-1.45
3.185	-0.3216	-0.7824	-1.45
3.190	-0.3213	-0.7819	-1.46
3.195	-0.3210	-0.7813	-1.47
3.200	-0.3215	-0.7803	-1.47
3.205	-0.3214	-0.7792	-1.48
3.210	-0.3218	-0.7784	-1.49
3.215	-0.3221	-0.7772	-1.50
3.220	-0.3232	-0.7760	-1.51
3.225	-0.3243	-0.7753	-1.51
3.230	-0.3258	-0.7749	-1.51
3.235	-0.3269	-0.7750	-1.50
3.240	-0.3281	-0.7750	-1.50
3.245	-0.3291	-0.7754	-1.49
3.250	-0.3305	-0.7752	-1.49
3.255	-0.3315	-0.7758	-1.48
3.260	-0.3325	-0.7760	-1.47
3.265	-0.3338	-0.7761	-1.46
3.270	-0.3347	-0.7762	-1.46
3.275	-0.3355	-0.7765	-1.45
3.280	-0.3364	-0.7764	-1.45
3.285	-0.3374	-0.7770	-1.44
3.290	-0.3383	-0.7764	-1.44
3.295	-0.3394	-0.7765	-1.44
3.300	-0.3401	-0.7767	-1.43
3.305	-0.3404	-0.7766	-1.43
3.310	-0.3414	-0.7763	-1.43

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.315	-0.3421	-0.7760	-1.43
3.320	-0.3425	-0.7759	-1.43
3.325	-0.3434	-0.7754	-1.43
3.330	-0.3437	-0.7752	-1.43
3.335	-0.3442	-0.7747	-1.43
3.340	-0.3448	-0.7740	-1.44
3.345	-0.3457	-0.7729	-1.45
3.350	-0.3462	-0.7719	-1.45
3.355	-0.3467	-0.7710	-1.46
3.360	-0.3474	-0.7698	-1.47
3.365	-0.3478	-0.7688	-1.48
3.370	-0.3490	-0.7674	-1.48
3.375	-0.3494	-0.7660	-1.49
3.380	-0.3506	-0.7644	-1.50
3.385	-0.3515	-0.7628	-1.52
3.390	-0.3526	-0.7614	-1.52
3.395	-0.3537	-0.7595	-1.54
3.400	-0.3549	-0.7577	-1.55
3.405	-0.3560	-0.7560	-1.56
3.410	-0.3576	-0.7542	-1.57
3.415	-0.3591	-0.7527	-1.58
3.420	-0.3607	-0.7515	-1.58
3.425	-0.3621	-0.7497	-1.59
3.430	-0.3637	-0.7481	-1.60
3.435	-0.3654	-0.7468	-1.60
3.440	-0.3671	-0.7452	-1.61
3.445	-0.3693	-0.7434	-1.62
3.450	-0.3713	-0.7422	-1.62
3.455	-0.3729	-0.7410	-1.62
3.460	-0.3748	-0.7402	-1.62
3.465	-0.3768	-0.7396	-1.62
3.470	-0.3784	-0.7383	-1.62
3.475	-0.3801	-0.7377	-1.62
3.480	-0.3815	-0.7372	-1.62
3.485	-0.3836	-0.7370	-1.61
3.490	-0.3849	-0.7370	-1.60
3.495	-0.3859	-0.7365	-1.60
3.500	-0.3866	-0.7365	-1.60
3.505	-0.3870	-0.7360	-1.60
3.510	-0.3877	-0.7355	-1.60
3.515	-0.3879	-0.7348	-1.61
3.520	-0.3877	-0.7343	-1.61
3.525	-0.3879	-0.7332	-1.62
3.530	-0.3875	-0.7328	-1.63
3.535	-0.3867	-0.7303	-1.66
3.540	-0.3866	-0.7279	-1.68

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.545	-0.3866	-0.7253	-1.70
3.550	-0.3873	-0.7224	-1.73
3.555	-0.3877	-0.7190	-1.76
3.560	-0.3894	-0.7159	-1.78
3.565	-0.3909	-0.7134	-1.79
3.570	-0.3936	-0.7102	-1.81
3.575	-0.3960	-0.7071	-1.83
3.580	-0.4000	-0.7054	-1.82
3.585	-0.4046	-0.7048	-1.80
3.590	-0.4077	-0.7051	-1.78
3.595	-0.4106	-0.7063	-1.76
3.600	-0.4126	-0.7081	-1.73
3.605	-0.4139	-0.7100	-1.70
3.610	-0.4137	-0.7118	-1.69
3.615	-0.4130	-0.7127	-1.68
3.620	-0.4117	-0.7129	-1.69
3.625	-0.4102	-0.7136	-1.69
3.630	-0.4091	-0.7128	-1.70
3.635	-0.4074	-0.7115	-1.73
3.640	-0.4067	-0.7098	-1.74
3.645	-0.4064	-0.7082	-1.76
3.650	-0.4063	-0.7063	-1.78
3.655	-0.4066	-0.7043	-1.80
3.660	-0.4070	-0.7029	-1.81
3.665	-0.4075	-0.7013	-1.82
3.670	-0.4080	-0.6994	-1.83
3.675	-0.4083	-0.6971	-1.85
3.680	-0.4090	-0.6956	-1.86
3.685	-0.4095	-0.6939	-1.88
3.690	-0.4106	-0.6917	-1.89
3.695	-0.4123	-0.6893	-1.90
3.700	-0.4140	-0.6870	-1.92
3.705	-0.4166	-0.6848	-1.92
3.710	-0.4195	-0.6827	-1.92
3.715	-0.4231	-0.6814	-1.92
3.720	-0.4272	-0.6800	-1.91
3.725	-0.4314	-0.6805	-1.88
3.730	-0.4369	-0.6803	-1.85
3.735	-0.4404	-0.6828	-1.80
3.740	-0.4438	-0.6855	-1.76
3.745	-0.4451	-0.6885	-1.73
3.750	-0.4457	-0.6924	-1.69
3.755	-0.4450	-0.6956	-1.66
3.760	-0.4432	-0.6982	-1.65
3.765	-0.4403	-0.6999	-1.65
3.770	-0.4369	-0.6998	-1.67

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.775	-0.4339	-0.6980	-1.70
3.780	-0.4311	-0.6954	-1.74
3.785	-0.4307	-0.6915	-1.78
3.790	-0.4317	-0.6874	-1.81
3.795	-0.4343	-0.6844	-1.82
3.800	-0.4370	-0.6820	-1.83
3.805	-0.4394	-0.6804	-1.83
3.810	-0.4425	-0.6794	-1.82
3.815	-0.4452	-0.6787	-1.81
3.820	-0.4479	-0.6784	-1.80
3.825	-0.4503	-0.6782	-1.79
3.830	-0.4526	-0.6788	-1.77
3.835	-0.4534	-0.6816	-1.74
3.840	-0.4549	-0.6811	-1.73
3.845	-0.4559	-0.6808	-1.73
3.850	-0.4572	-0.6803	-1.73
3.855	-0.4583	-0.6795	-1.73
3.860	-0.4599	-0.6791	-1.72
3.865	-0.4613	-0.6782	-1.72
3.870	-0.4627	-0.6775	-1.72
3.875	-0.4642	-0.6768	-1.72
3.880	-0.4657	-0.6761	-1.71
3.885	-0.4677	-0.6755	-1.71
3.890	-0.4691	-0.6742	-1.71
3.895	-0.4708	-0.6734	-1.71
3.900	-0.4726	-0.6725	-1.70
3.905	-0.4743	-0.6716	-1.70
3.910	-0.4761	-0.6708	-1.70
3.915	-0.4782	-0.6698	-1.69
3.920	-0.4798	-0.6687	-1.69
3.925	-0.4818	-0.6675	-1.69
3.930	-0.4834	-0.6667	-1.69
3.935	-0.4855	-0.6652	-1.69
3.940	-0.4869	-0.6644	-1.68
3.945	-0.4889	-0.6630	-1.68
3.950	-0.4906	-0.6620	-1.68
3.955	-0.4925	-0.6606	-1.68
3.960	-0.4942	-0.6593	-1.68
3.965	-0.4959	-0.6579	-1.68
3.970	-0.4975	-0.6566	-1.68
3.975	-0.4991	-0.6552	-1.69
3.980	-0.5012	-0.6536	-1.68
3.985	-0.5028	-0.6523	-1.69
3.990	-0.5046	-0.6508	-1.69
3.995	-0.5061	-0.6491	-1.69
4.000	-0.5079	-0.6477	-1.69

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.005	-0.5095	-0.6459	-1.70
4.010	-0.5114	-0.6442	-1.70
4.015	-0.5127	-0.6430	-1.70
4.020	-0.5145	-0.6412	-1.70
4.025	-0.5158	-0.6397	-1.71
4.030	-0.5174	-0.6383	-1.71
4.035	-0.5191	-0.6364	-1.71
4.040	-0.5205	-0.6347	-1.71
4.045	-0.5219	-0.6331	-1.72
4.050	-0.5233	-0.6314	-1.72
4.055	-0.5247	-0.6298	-1.73
4.060	-0.5259	-0.6282	-1.73
4.065	-0.5273	-0.6268	-1.73
4.070	-0.5285	-0.6251	-1.74
4.075	-0.5296	-0.6238	-1.74
4.080	-0.5309	-0.6225	-1.74
4.085	-0.5322	-0.6210	-1.75
4.090	-0.5331	-0.6198	-1.75
4.095	-0.5340	-0.6185	-1.75
4.100	-0.5353	-0.6170	-1.76
4.105	-0.5364	-0.6159	-1.76
4.110	-0.5375	-0.6148	-1.76
4.115	-0.5389	-0.6138	-1.76
4.120	-0.5399	-0.6128	-1.76
4.125	-0.5416	-0.6113	-1.76
4.130	-0.5426	-0.6103	-1.76
4.135	-0.5438	-0.6089	-1.76
4.140	-0.5450	-0.6077	-1.76
4.145	-0.5458	-0.6060	-1.77
4.150	-0.5468	-0.6043	-1.78
4.155	-0.5475	-0.6029	-1.78
4.160	-0.5480	-0.6016	-1.79
4.165	-0.5480	-0.6004	-1.80
4.170	-0.5481	-0.5994	-1.81
4.175	-0.5476	-0.5984	-1.82
4.180	-0.5479	-0.5979	-1.82
4.185	-0.5474	-0.5971	-1.83
4.190	-0.5476	-0.5965	-1.83
4.195	-0.5473	-0.5966	-1.83
4.200	-0.5471	-0.5963	-1.84
4.205	-0.5469	-0.5962	-1.84
4.210	-0.5468	-0.5961	-1.84
4.215	-0.5468	-0.5960	-1.84
4.220	-0.5464	-0.5962	-1.84
4.225	-0.5463	-0.5960	-1.85
4.230	-0.5460	-0.5964	-1.85

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.235	-0.5459	-0.5966	-1.84
4.240	-0.5456	-0.5967	-1.85
4.245	-0.5452	-0.5971	-1.85
4.250	-0.5450	-0.5976	-1.84
4.255	-0.5447	-0.5980	-1.84
4.260	-0.5442	-0.5982	-1.84
4.265	-0.5438	-0.5988	-1.84
4.270	-0.5437	-0.5992	-1.84
4.275	-0.5431	-0.5996	-1.84
4.280	-0.5431	-0.6000	-1.84
4.285	-0.5425	-0.6007	-1.84
4.290	-0.5423	-0.6010	-1.84
4.295	-0.5418	-0.6015	-1.83
4.300	-0.5418	-0.6022	-1.83
4.305	-0.5416	-0.6026	-1.83
4.310	-0.5413	-0.6032	-1.83
4.315	-0.5410	-0.6037	-1.82
4.320	-0.5406	-0.6043	-1.82
4.325	-0.5405	-0.6051	-1.82
4.330	-0.5401	-0.6056	-1.81
4.335	-0.5400	-0.6064	-1.81
4.340	-0.5399	-0.6072	-1.80
4.345	-0.5396	-0.6082	-1.80
4.350	-0.5395	-0.6089	-1.79
4.355	-0.5390	-0.6098	-1.79
4.360	-0.5386	-0.6107	-1.78
4.365	-0.5381	-0.6117	-1.78
4.370	-0.5378	-0.6124	-1.78
4.375	-0.5370	-0.6135	-1.77
4.380	-0.5367	-0.6143	-1.77
4.385	-0.5362	-0.6152	-1.77
4.390	-0.5353	-0.6161	-1.76
4.395	-0.5347	-0.6168	-1.76
4.400	-0.5343	-0.6175	-1.76
4.405	-0.5337	-0.6180	-1.76
4.410	-0.5330	-0.6186	-1.76
4.415	-0.5322	-0.6190	-1.76
4.420	-0.5321	-0.6192	-1.76
4.425	-0.5315	-0.6193	-1.77
4.430	-0.5311	-0.6200	-1.76
4.435	-0.5308	-0.6199	-1.77
4.440	-0.5307	-0.6200	-1.77
4.445	-0.5304	-0.6206	-1.76
4.450	-0.5304	-0.6205	-1.76
4.455	-0.5306	-0.6214	-1.75
4.460	-0.5297	-0.6218	-1.76

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.465	-0.5295	-0.6227	-1.75
4.470	-0.5292	-0.6239	-1.74
4.475	-0.5289	-0.6248	-1.74
4.480	-0.5285	-0.6263	-1.73
4.485	-0.5285	-0.6275	-1.72
4.490	-0.5279	-0.6293	-1.71
4.495	-0.5279	-0.6306	-1.70
4.500	-0.5274	-0.6315	-1.69
4.505	-0.5270	-0.6326	-1.69
4.510	-0.5266	-0.6327	-1.69
4.515	-0.5259	-0.6331	-1.69
4.520	-0.5255	-0.6337	-1.69
4.525	-0.5246	-0.6335	-1.70
4.530	-0.5248	-0.6328	-1.70
4.535	-0.5244	-0.6322	-1.71
4.540	-0.5246	-0.6318	-1.71
4.545	-0.5245	-0.6307	-1.72
4.550	-0.5249	-0.6304	-1.72
4.555	-0.5251	-0.6297	-1.72
4.560	-0.5255	-0.6293	-1.73
4.565	-0.5255	-0.6286	-1.73
4.570	-0.5258	-0.6277	-1.74
4.575	-0.5263	-0.6268	-1.74
4.580	-0.5264	-0.6261	-1.75
4.585	-0.5269	-0.6250	-1.75
4.590	-0.5273	-0.6245	-1.75
4.595	-0.5275	-0.6237	-1.76
4.600	-0.5282	-0.6227	-1.76
4.605	-0.5283	-0.6218	-1.77
4.610	-0.5289	-0.6209	-1.77
4.615	-0.5295	-0.6196	-1.78
4.620	-0.5299	-0.6189	-1.78
4.625	-0.5305	-0.6178	-1.78
4.630	-0.5307	-0.6172	-1.79
4.635	-0.5309	-0.6161	-1.80
4.640	-0.5309	-0.6151	-1.80
4.645	-0.5312	-0.6143	-1.81
4.650	-0.5314	-0.6136	-1.81
4.655	-0.5318	-0.6126	-1.82
4.660	-0.5317	-0.6118	-1.82
4.665	-0.5317	-0.6109	-1.83
4.670	-0.5317	-0.6102	-1.84
4.675	-0.5315	-0.6095	-1.84
4.680	-0.5313	-0.6086	-1.85
4.685	-0.5309	-0.6083	-1.86
4.690	-0.5307	-0.6076	-1.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.695	-0.5305	-0.6068	-1.87
4.700	-0.5301	-0.6063	-1.88
4.705	-0.5299	-0.6054	-1.89
4.710	-0.5300	-0.6048	-1.89
4.715	-0.5299	-0.6039	-1.90
4.720	-0.5299	-0.6028	-1.91
4.725	-0.5303	-0.6021	-1.91
4.730	-0.5310	-0.6007	-1.92
4.735	-0.5316	-0.6000	-1.92
4.740	-0.5325	-0.5987	-1.92
4.745	-0.5329	-0.5975	-1.93
4.750	-0.5337	-0.5957	-1.94
4.755	-0.5348	-0.5945	-1.94
4.760	-0.5354	-0.5933	-1.95
4.765	-0.5361	-0.5924	-1.95
4.770	-0.5363	-0.5916	-1.95
4.775	-0.5361	-0.5908	-1.96
4.780	-0.5363	-0.5898	-1.97
4.785	-0.5361	-0.5891	-1.98
4.790	-0.5351	-0.5887	-1.99
4.795	-0.5345	-0.5880	-2.00
4.800	-0.5337	-0.5874	-2.01
4.805	-0.5329	-0.5868	-2.02
4.810	-0.5320	-0.5858	-2.03
4.815	-0.5312	-0.5858	-2.04
4.820	-0.5301	-0.5853	-2.05
4.825	-0.5298	-0.5852	-2.05
4.830	-0.5291	-0.5850	-2.06
4.835	-0.5287	-0.5849	-2.07
4.840	-0.5284	-0.5849	-2.07
4.845	-0.5285	-0.5848	-2.07
4.850	-0.5283	-0.5851	-2.07
4.855	-0.5284	-0.5856	-2.06
4.860	-0.5286	-0.5860	-2.06
4.865	-0.5292	-0.5859	-2.05
4.870	-0.5293	-0.5867	-2.05
4.875	-0.5298	-0.5874	-2.04
4.880	-0.5302	-0.5880	-2.03
4.885	-0.5308	-0.5885	-2.02
4.890	-0.5315	-0.5892	-2.01
4.895	-0.5319	-0.5897	-2.00
4.900	-0.5325	-0.5899	-2.00
4.905	-0.5329	-0.5904	-1.99
4.910	-0.5335	-0.5903	-1.99
4.915	-0.5341	-0.5908	-1.98
4.920	-0.5348	-0.5910	-1.97

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.925	-0.5352	-0.5911	-1.97
4.930	-0.5356	-0.5913	-1.96
4.935	-0.5361	-0.5916	-1.96
4.940	-0.5366	-0.5917	-1.95
4.945	-0.5370	-0.5920	-1.95
4.950	-0.5371	-0.5922	-1.94
4.955	-0.5377	-0.5923	-1.94
4.960	-0.5381	-0.5925	-1.93
4.965	-0.5384	-0.5928	-1.93
4.970	-0.5387	-0.5931	-1.92
4.975	-0.5392	-0.5933	-1.92
4.980	-0.5397	-0.5936	-1.91
4.985	-0.5399	-0.5939	-1.91
4.990	-0.5405	-0.5944	-1.90
4.995	-0.5408	-0.5946	-1.90
5.000	-0.5414	-0.5949	-1.89
5.005	-0.5418	-0.5951	-1.89
5.010	-0.5425	-0.5951	-1.88
5.015	-0.5426	-0.5953	-1.88
5.020	-0.5431	-0.5954	-1.87
5.025	-0.5432	-0.5956	-1.87
5.030	-0.5437	-0.5950	-1.87
5.035	-0.5438	-0.5955	-1.87
5.040	-0.5439	-0.5948	-1.87
5.045	-0.5438	-0.5952	-1.87
5.050	-0.5436	-0.5945	-1.88
5.055	-0.5436	-0.5949	-1.88
5.060	-0.5433	-0.5943	-1.88
5.065	-0.5438	-0.5945	-1.88
5.070	-0.5434	-0.5941	-1.88
5.075	-0.5444	-0.5940	-1.88
5.080	-0.5440	-0.5941	-1.88
5.085	-0.5452	-0.5936	-1.87
5.090	-0.5452	-0.5935	-1.87
5.095	-0.5464	-0.5933	-1.87
5.100	-0.5468	-0.5929	-1.87
5.105	-0.5477	-0.5922	-1.87
5.110	-0.5484	-0.5921	-1.86
5.115	-0.5493	-0.5913	-1.86
5.120	-0.5500	-0.5909	-1.86
5.125	-0.5512	-0.5900	-1.86
5.130	-0.5520	-0.5898	-1.85
5.135	-0.5527	-0.5888	-1.86
5.140	-0.5538	-0.5879	-1.86
5.145	-0.5552	-0.5870	-1.85
5.150	-0.5559	-0.5862	-1.85

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.155	-0.5573	-0.5846	-1.85
5.160	-0.5585	-0.5832	-1.86
5.165	-0.5595	-0.5817	-1.86
5.170	-0.5609	-0.5804	-1.86
5.175	-0.5623	-0.5785	-1.87
5.180	-0.5632	-0.5766	-1.87
5.185	-0.5643	-0.5752	-1.88
5.190	-0.5655	-0.5727	-1.89
5.195	-0.5665	-0.5710	-1.89
5.200	-0.5676	-0.5685	-1.90
5.205	-0.5687	-0.5665	-1.91
5.210	-0.5695	-0.5645	-1.92
5.215	-0.5706	-0.5623	-1.93
5.220	-0.5715	-0.5600	-1.94
5.225	-0.5728	-0.5574	-1.95
5.230	-0.5736	-0.5553	-1.96
5.235	-0.5745	-0.5531	-1.97
5.240	-0.5755	-0.5509	-1.97
5.245	-0.5764	-0.5489	-1.98
5.250	-0.5772	-0.5466	-1.99
5.255	-0.5784	-0.5443	-2.00
5.260	-0.5792	-0.5421	-2.01
5.265	-0.5803	-0.5401	-2.02
5.270	-0.5814	-0.5381	-2.02
5.275	-0.5822	-0.5360	-2.03
5.280	-0.5829	-0.5341	-2.04
5.285	-0.5837	-0.5320	-2.05
5.290	-0.5848	-0.5303	-2.05
5.295	-0.5856	-0.5283	-2.06
5.300	-0.5863	-0.5267	-2.07
5.305	-0.5874	-0.5250	-2.07
5.310	-0.5878	-0.5233	-2.08
5.315	-0.5886	-0.5217	-2.09
5.320	-0.5894	-0.5198	-2.09
5.325	-0.5900	-0.5182	-2.10
5.330	-0.5905	-0.5167	-2.11
5.335	-0.5910	-0.5150	-2.11
5.340	-0.5915	-0.5133	-2.12
5.345	-0.5925	-0.5116	-2.13
5.350	-0.5926	-0.5101	-2.14
5.355	-0.5933	-0.5083	-2.14
5.360	-0.5937	-0.5067	-2.15
5.365	-0.5946	-0.5051	-2.16
5.370	-0.5950	-0.5034	-2.16
5.375	-0.5959	-0.5014	-2.17
5.380	-0.5967	-0.5001	-2.17

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.385	-0.5972	-0.4983	-2.18
5.390	-0.5981	-0.4971	-2.18
5.395	-0.5988	-0.4959	-2.19
5.400	-0.5999	-0.4944	-2.19
5.405	-0.6007	-0.4932	-2.19
5.410	-0.6017	-0.4920	-2.19
5.415	-0.6026	-0.4909	-2.19
5.420	-0.6033	-0.4895	-2.19
5.425	-0.6046	-0.4884	-2.19
5.430	-0.6051	-0.4874	-2.19
5.435	-0.6057	-0.4866	-2.19
5.440	-0.6066	-0.4858	-2.19
5.445	-0.6072	-0.4848	-2.19
5.450	-0.6079	-0.4839	-2.19
5.455	-0.6084	-0.4831	-2.19
5.460	-0.6093	-0.4825	-2.19
5.465	-0.6098	-0.4816	-2.19
5.470	-0.6104	-0.4811	-2.19
5.475	-0.6109	-0.4803	-2.19
5.480	-0.6117	-0.4797	-2.19
5.485	-0.6125	-0.4788	-2.19
5.490	-0.6131	-0.4782	-2.19
5.495	-0.6139	-0.4775	-2.18
5.500	-0.6147	-0.4769	-2.18
5.505	-0.6154	-0.4764	-2.18
5.510	-0.6168	-0.4762	-2.17
5.515	-0.6177	-0.4761	-2.16
5.520	-0.6186	-0.4760	-2.15
5.525	-0.6193	-0.4760	-2.15
5.530	-0.6203	-0.4760	-2.14
5.535	-0.6210	-0.4761	-2.13
5.540	-0.6217	-0.4762	-2.12
5.545	-0.6221	-0.4763	-2.12
5.550	-0.6229	-0.4765	-2.11
5.555	-0.6236	-0.4766	-2.10
5.560	-0.6243	-0.4768	-2.10
5.565	-0.6250	-0.4768	-2.09
5.570	-0.6259	-0.4768	-2.08
5.575	-0.6266	-0.4770	-2.08
5.580	-0.6273	-0.4771	-2.07
5.585	-0.6281	-0.4770	-2.06
5.590	-0.6290	-0.4771	-2.05
5.595	-0.6299	-0.4771	-2.04
5.600	-0.6309	-0.4772	-2.04
5.605	-0.6319	-0.4774	-2.03
5.610	-0.6330	-0.4775	-2.02

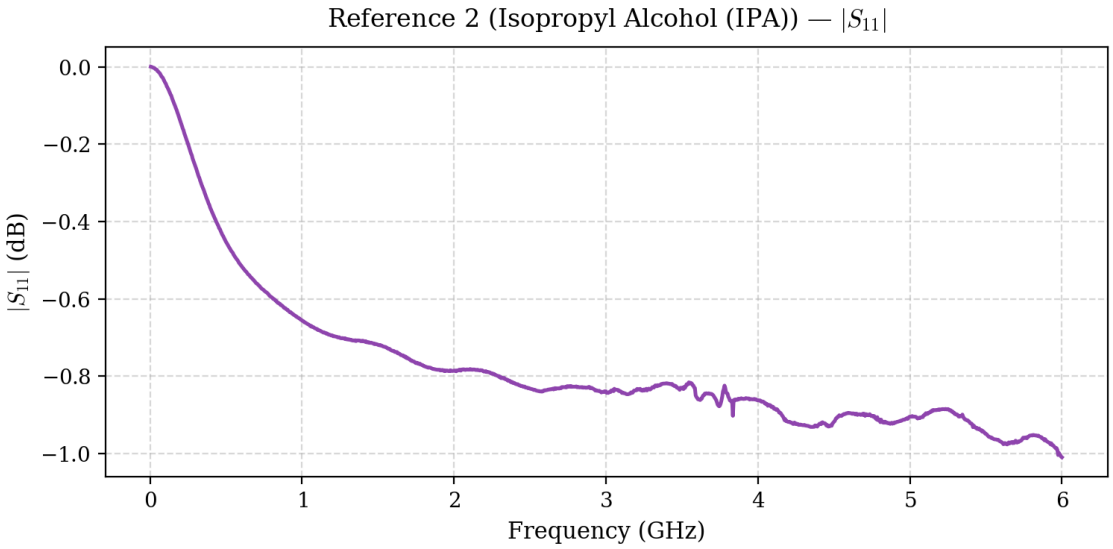
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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.615	-0.6337	-0.4779	-2.01
5.620	-0.6348	-0.4780	-2.00
5.625	-0.6356	-0.4785	-1.99
5.630	-0.6364	-0.4787	-1.98
5.635	-0.6370	-0.4789	-1.97
5.640	-0.6376	-0.4791	-1.96
5.645	-0.6382	-0.4794	-1.96
5.650	-0.6389	-0.4794	-1.95
5.655	-0.6393	-0.4795	-1.95
5.660	-0.6399	-0.4795	-1.94
5.665	-0.6404	-0.4797	-1.94
5.670	-0.6410	-0.4798	-1.93
5.675	-0.6415	-0.4797	-1.93
5.680	-0.6424	-0.4796	-1.92
5.685	-0.6421	-0.4794	-1.92
5.690	-0.6425	-0.4791	-1.92
5.695	-0.6430	-0.4791	-1.92
5.700	-0.6430	-0.4789	-1.92
5.705	-0.6436	-0.4786	-1.92
5.710	-0.6435	-0.4784	-1.92
5.715	-0.6439	-0.4784	-1.92
5.720	-0.6442	-0.4784	-1.91
5.725	-0.6441	-0.4781	-1.92
5.730	-0.6445	-0.4775	-1.92
5.735	-0.6446	-0.4774	-1.91
5.740	-0.6447	-0.4767	-1.92
5.745	-0.6453	-0.4768	-1.91
5.750	-0.6450	-0.4758	-1.92
5.755	-0.6450	-0.4754	-1.92
5.760	-0.6448	-0.4748	-1.93
5.765	-0.6454	-0.4741	-1.93
5.770	-0.6450	-0.4735	-1.94
5.775	-0.6453	-0.4721	-1.94
5.780	-0.6448	-0.4717	-1.95
5.785	-0.6452	-0.4705	-1.95
5.790	-0.6452	-0.4698	-1.96
5.795	-0.6450	-0.4685	-1.97
5.800	-0.6448	-0.4671	-1.98
5.805	-0.6448	-0.4661	-1.99
5.810	-0.6443	-0.4649	-2.00
5.815	-0.6444	-0.4637	-2.00
5.820	-0.6445	-0.4623	-2.01
5.825	-0.6443	-0.4613	-2.02
5.830	-0.6442	-0.4598	-2.03
5.835	-0.6441	-0.4584	-2.04
5.840	-0.6438	-0.4569	-2.05

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.845	-0.6441	-0.4555	-2.06
5.850	-0.6441	-0.4544	-2.07
5.855	-0.6438	-0.4530	-2.08
5.860	-0.6440	-0.4514	-2.09
5.865	-0.6441	-0.4504	-2.09
5.870	-0.6442	-0.4487	-2.10
5.875	-0.6443	-0.4477	-2.11
5.880	-0.6446	-0.4463	-2.11
5.885	-0.6447	-0.4450	-2.12
5.890	-0.6447	-0.4439	-2.13
5.895	-0.6447	-0.4429	-2.13
5.900	-0.6449	-0.4419	-2.14
5.905	-0.6451	-0.4410	-2.14
5.910	-0.6452	-0.4404	-2.15
5.915	-0.6449	-0.4393	-2.15
5.920	-0.6449	-0.4381	-2.16
5.925	-0.6450	-0.4373	-2.17
5.930	-0.6447	-0.4361	-2.18
5.935	-0.6446	-0.4351	-2.18
5.940	-0.6448	-0.4341	-2.19
5.945	-0.6444	-0.4334	-2.20
5.950	-0.6445	-0.4322	-2.20
5.955	-0.6448	-0.4312	-2.21
5.960	-0.6451	-0.4303	-2.21
5.965	-0.6454	-0.4296	-2.21
5.970	-0.6460	-0.4286	-2.21
5.975	-0.6468	-0.4279	-2.21
5.980	-0.6469	-0.4282	-2.20
5.985	-0.6475	-0.4276	-2.20
5.990	-0.6480	-0.4277	-2.20
5.995	-0.6486	-0.4279	-2.19
6.000	-0.6488	-0.4281	-2.19

3.4 Reference 2 (Isopropyl Alcohol (IPA))



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0001	-0.0016	0.00
0.010	1.0000	-0.0028	0.00
0.015	0.9999	-0.0040	-0.00
0.020	0.9998	-0.0053	-0.00
0.025	0.9997	-0.0064	-0.00
0.030	0.9996	-0.0077	-0.00
0.035	0.9993	-0.0089	-0.01
0.040	0.9992	-0.0102	-0.01
0.045	0.9991	-0.0112	-0.01
0.050	0.9986	-0.0125	-0.01
0.055	0.9985	-0.0137	-0.01
0.060	0.9982	-0.0148	-0.01
0.065	0.9980	-0.0161	-0.02
0.070	0.9975	-0.0172	-0.02
0.075	0.9970	-0.0183	-0.02
0.080	0.9968	-0.0194	-0.03
0.085	0.9965	-0.0206	-0.03
0.090	0.9959	-0.0216	-0.03
0.095	0.9955	-0.0225	-0.04
0.100	0.9950	-0.0237	-0.04
0.105	0.9946	-0.0248	-0.04
0.110	0.9939	-0.0257	-0.05
0.115	0.9935	-0.0268	-0.05
0.120	0.9931	-0.0278	-0.06
0.125	0.9925	-0.0287	-0.06
0.130	0.9919	-0.0297	-0.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.135	0.9914	-0.0305	-0.07
0.140	0.9910	-0.0315	-0.07
0.145	0.9902	-0.0325	-0.08
0.150	0.9896	-0.0334	-0.09
0.155	0.9890	-0.0341	-0.09
0.160	0.9883	-0.0351	-0.10
0.165	0.9878	-0.0358	-0.10
0.170	0.9871	-0.0366	-0.11
0.175	0.9866	-0.0374	-0.11
0.180	0.9858	-0.0382	-0.12
0.185	0.9852	-0.0388	-0.12
0.190	0.9844	-0.0397	-0.13
0.195	0.9837	-0.0405	-0.14
0.200	0.9831	-0.0412	-0.14
0.205	0.9824	-0.0418	-0.15
0.210	0.9817	-0.0424	-0.15
0.215	0.9809	-0.0433	-0.16
0.220	0.9804	-0.0438	-0.16
0.225	0.9795	-0.0444	-0.17
0.230	0.9788	-0.0449	-0.18
0.235	0.9782	-0.0455	-0.18
0.240	0.9775	-0.0461	-0.19
0.245	0.9768	-0.0468	-0.19
0.250	0.9762	-0.0473	-0.20
0.255	0.9754	-0.0477	-0.21
0.260	0.9746	-0.0482	-0.21
0.265	0.9740	-0.0487	-0.22
0.270	0.9733	-0.0493	-0.22
0.275	0.9728	-0.0496	-0.23
0.280	0.9718	-0.0502	-0.24
0.285	0.9712	-0.0505	-0.24
0.290	0.9705	-0.0509	-0.25
0.295	0.9699	-0.0514	-0.25
0.300	0.9692	-0.0517	-0.26
0.305	0.9686	-0.0521	-0.26
0.310	0.9679	-0.0525	-0.27
0.315	0.9672	-0.0527	-0.28
0.320	0.9665	-0.0530	-0.28
0.325	0.9660	-0.0535	-0.29
0.330	0.9652	-0.0538	-0.29
0.335	0.9645	-0.0541	-0.30
0.340	0.9639	-0.0543	-0.31
0.345	0.9634	-0.0545	-0.31
0.350	0.9627	-0.0548	-0.32
0.355	0.9622	-0.0552	-0.32
0.360	0.9615	-0.0554	-0.33

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.365	0.9609	-0.0556	-0.33
0.370	0.9603	-0.0558	-0.34
0.375	0.9597	-0.0560	-0.34
0.380	0.9592	-0.0563	-0.35
0.385	0.9585	-0.0564	-0.35
0.390	0.9581	-0.0566	-0.36
0.395	0.9573	-0.0569	-0.36
0.400	0.9568	-0.0570	-0.37
0.405	0.9564	-0.0572	-0.37
0.410	0.9557	-0.0572	-0.38
0.415	0.9553	-0.0574	-0.38
0.420	0.9548	-0.0576	-0.39
0.425	0.9543	-0.0577	-0.39
0.430	0.9538	-0.0579	-0.39
0.435	0.9532	-0.0580	-0.40
0.440	0.9528	-0.0581	-0.40
0.445	0.9524	-0.0583	-0.41
0.450	0.9519	-0.0584	-0.41
0.455	0.9513	-0.0584	-0.42
0.460	0.9510	-0.0585	-0.42
0.465	0.9504	-0.0586	-0.42
0.470	0.9500	-0.0587	-0.43
0.475	0.9495	-0.0589	-0.43
0.480	0.9491	-0.0589	-0.44
0.485	0.9487	-0.0590	-0.44
0.490	0.9482	-0.0590	-0.44
0.495	0.9478	-0.0592	-0.45
0.500	0.9474	-0.0592	-0.45
0.505	0.9470	-0.0592	-0.46
0.510	0.9466	-0.0592	-0.46
0.515	0.9462	-0.0594	-0.46
0.520	0.9458	-0.0594	-0.47
0.525	0.9455	-0.0594	-0.47
0.530	0.9451	-0.0595	-0.47
0.535	0.9448	-0.0596	-0.48
0.540	0.9445	-0.0596	-0.48
0.545	0.9442	-0.0597	-0.48
0.550	0.9437	-0.0597	-0.49
0.555	0.9434	-0.0598	-0.49
0.560	0.9430	-0.0599	-0.49
0.565	0.9427	-0.0599	-0.49
0.570	0.9424	-0.0600	-0.50
0.575	0.9422	-0.0599	-0.50
0.580	0.9419	-0.0600	-0.50
0.585	0.9414	-0.0599	-0.51
0.590	0.9412	-0.0600	-0.51

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.595	0.9409	-0.0601	-0.51
0.600	0.9406	-0.0601	-0.51
0.605	0.9403	-0.0602	-0.52
0.610	0.9400	-0.0602	-0.52
0.615	0.9398	-0.0602	-0.52
0.620	0.9395	-0.0602	-0.52
0.625	0.9392	-0.0603	-0.53
0.630	0.9390	-0.0602	-0.53
0.635	0.9388	-0.0603	-0.53
0.640	0.9385	-0.0604	-0.53
0.645	0.9382	-0.0604	-0.54
0.650	0.9380	-0.0605	-0.54
0.655	0.9378	-0.0604	-0.54
0.660	0.9375	-0.0604	-0.54
0.665	0.9373	-0.0604	-0.54
0.670	0.9370	-0.0605	-0.55
0.675	0.9368	-0.0605	-0.55
0.680	0.9366	-0.0607	-0.55
0.685	0.9363	-0.0606	-0.55
0.690	0.9361	-0.0606	-0.56
0.695	0.9358	-0.0606	-0.56
0.700	0.9356	-0.0606	-0.56
0.705	0.9355	-0.0607	-0.56
0.710	0.9352	-0.0608	-0.56
0.715	0.9350	-0.0608	-0.57
0.720	0.9349	-0.0608	-0.57
0.725	0.9345	-0.0608	-0.57
0.730	0.9343	-0.0607	-0.57
0.735	0.9341	-0.0609	-0.57
0.740	0.9340	-0.0608	-0.58
0.745	0.9338	-0.0609	-0.58
0.750	0.9336	-0.0610	-0.58
0.755	0.9334	-0.0610	-0.58
0.760	0.9332	-0.0610	-0.58
0.765	0.9331	-0.0610	-0.58
0.770	0.9329	-0.0611	-0.58
0.775	0.9327	-0.0610	-0.59
0.780	0.9325	-0.0612	-0.59
0.785	0.9322	-0.0612	-0.59
0.790	0.9320	-0.0611	-0.59
0.795	0.9318	-0.0613	-0.59
0.800	0.9317	-0.0613	-0.60
0.805	0.9315	-0.0614	-0.60
0.810	0.9312	-0.0614	-0.60
0.815	0.9312	-0.0614	-0.60
0.820	0.9310	-0.0615	-0.60

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.825	0.9307	-0.0615	-0.60
0.830	0.9307	-0.0615	-0.60
0.835	0.9303	-0.0617	-0.61
0.840	0.9301	-0.0617	-0.61
0.845	0.9301	-0.0616	-0.61
0.850	0.9299	-0.0617	-0.61
0.855	0.9298	-0.0619	-0.61
0.860	0.9295	-0.0618	-0.62
0.865	0.9294	-0.0621	-0.62
0.870	0.9293	-0.0621	-0.62
0.875	0.9291	-0.0620	-0.62
0.880	0.9287	-0.0620	-0.62
0.885	0.9286	-0.0621	-0.62
0.890	0.9286	-0.0622	-0.62
0.895	0.9283	-0.0623	-0.63
0.900	0.9282	-0.0621	-0.63
0.905	0.9281	-0.0623	-0.63
0.910	0.9278	-0.0624	-0.63
0.915	0.9277	-0.0625	-0.63
0.920	0.9274	-0.0624	-0.63
0.925	0.9275	-0.0625	-0.63
0.930	0.9273	-0.0626	-0.64
0.935	0.9270	-0.0627	-0.64
0.940	0.9269	-0.0627	-0.64
0.945	0.9267	-0.0628	-0.64
0.950	0.9266	-0.0627	-0.64
0.955	0.9265	-0.0628	-0.64
0.960	0.9262	-0.0629	-0.65
0.965	0.9262	-0.0630	-0.65
0.970	0.9260	-0.0629	-0.65
0.975	0.9258	-0.0630	-0.65
0.980	0.9258	-0.0630	-0.65
0.985	0.9256	-0.0631	-0.65
0.990	0.9254	-0.0632	-0.65
0.995	0.9253	-0.0633	-0.65
1.000	0.9251	-0.0633	-0.66
1.005	0.9250	-0.0634	-0.66
1.010	0.9249	-0.0636	-0.66
1.015	0.9246	-0.0636	-0.66
1.020	0.9245	-0.0636	-0.66
1.025	0.9244	-0.0636	-0.66
1.030	0.9243	-0.0637	-0.66
1.035	0.9241	-0.0638	-0.66
1.040	0.9240	-0.0638	-0.67
1.045	0.9239	-0.0639	-0.67
1.050	0.9237	-0.0639	-0.67

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.055	0.9235	-0.0640	-0.67
1.060	0.9235	-0.0640	-0.67
1.065	0.9233	-0.0640	-0.67
1.070	0.9233	-0.0642	-0.67
1.075	0.9232	-0.0643	-0.67
1.080	0.9230	-0.0643	-0.67
1.085	0.9229	-0.0643	-0.68
1.090	0.9228	-0.0644	-0.68
1.095	0.9226	-0.0645	-0.68
1.100	0.9226	-0.0646	-0.68
1.105	0.9223	-0.0647	-0.68
1.110	0.9223	-0.0647	-0.68
1.115	0.9222	-0.0647	-0.68
1.120	0.9222	-0.0648	-0.68
1.125	0.9220	-0.0648	-0.68
1.130	0.9219	-0.0649	-0.68
1.135	0.9218	-0.0651	-0.69
1.140	0.9216	-0.0650	-0.69
1.145	0.9217	-0.0651	-0.69
1.150	0.9216	-0.0650	-0.69
1.155	0.9215	-0.0652	-0.69
1.160	0.9213	-0.0653	-0.69
1.165	0.9214	-0.0654	-0.69
1.170	0.9212	-0.0654	-0.69
1.175	0.9210	-0.0655	-0.69
1.180	0.9210	-0.0657	-0.69
1.185	0.9210	-0.0655	-0.69
1.190	0.9208	-0.0656	-0.69
1.195	0.9208	-0.0657	-0.69
1.200	0.9208	-0.0657	-0.69
1.205	0.9206	-0.0658	-0.70
1.210	0.9206	-0.0658	-0.70
1.215	0.9205	-0.0659	-0.70
1.220	0.9205	-0.0658	-0.70
1.225	0.9204	-0.0659	-0.70
1.230	0.9204	-0.0660	-0.70
1.235	0.9202	-0.0661	-0.70
1.240	0.9202	-0.0661	-0.70
1.245	0.9203	-0.0661	-0.70
1.250	0.9201	-0.0663	-0.70
1.255	0.9200	-0.0663	-0.70
1.260	0.9200	-0.0663	-0.70
1.265	0.9200	-0.0664	-0.70
1.270	0.9199	-0.0664	-0.70
1.275	0.9199	-0.0665	-0.70
1.280	0.9198	-0.0665	-0.70

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.285	0.9199	-0.0666	-0.70
1.290	0.9197	-0.0666	-0.70
1.295	0.9197	-0.0667	-0.70
1.300	0.9196	-0.0669	-0.71
1.305	0.9197	-0.0669	-0.70
1.310	0.9196	-0.0669	-0.71
1.315	0.9196	-0.0669	-0.71
1.320	0.9194	-0.0670	-0.71
1.325	0.9195	-0.0671	-0.71
1.330	0.9193	-0.0671	-0.71
1.335	0.9193	-0.0672	-0.71
1.340	0.9193	-0.0671	-0.71
1.345	0.9192	-0.0672	-0.71
1.350	0.9193	-0.0674	-0.71
1.355	0.9191	-0.0673	-0.71
1.360	0.9193	-0.0674	-0.71
1.365	0.9193	-0.0675	-0.71
1.370	0.9193	-0.0675	-0.71
1.375	0.9193	-0.0675	-0.71
1.380	0.9192	-0.0676	-0.71
1.385	0.9191	-0.0677	-0.71
1.390	0.9190	-0.0679	-0.71
1.395	0.9192	-0.0680	-0.71
1.400	0.9192	-0.0680	-0.71
1.405	0.9190	-0.0680	-0.71
1.410	0.9190	-0.0681	-0.71
1.415	0.9189	-0.0681	-0.71
1.420	0.9190	-0.0683	-0.71
1.425	0.9187	-0.0682	-0.71
1.430	0.9187	-0.0684	-0.71
1.435	0.9187	-0.0683	-0.71
1.440	0.9187	-0.0685	-0.71
1.445	0.9186	-0.0687	-0.71
1.450	0.9185	-0.0687	-0.71
1.455	0.9186	-0.0689	-0.71
1.460	0.9185	-0.0691	-0.71
1.465	0.9184	-0.0691	-0.71
1.470	0.9183	-0.0691	-0.72
1.475	0.9182	-0.0692	-0.72
1.480	0.9182	-0.0692	-0.72
1.485	0.9181	-0.0695	-0.72
1.490	0.9181	-0.0695	-0.72
1.495	0.9180	-0.0696	-0.72
1.500	0.9179	-0.0698	-0.72
1.505	0.9179	-0.0699	-0.72
1.510	0.9178	-0.0699	-0.72

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.515	0.9178	-0.0701	-0.72
1.520	0.9177	-0.0703	-0.72
1.525	0.9176	-0.0704	-0.72
1.530	0.9175	-0.0704	-0.72
1.535	0.9174	-0.0705	-0.72
1.540	0.9173	-0.0707	-0.72
1.545	0.9172	-0.0709	-0.72
1.550	0.9172	-0.0709	-0.72
1.555	0.9171	-0.0710	-0.73
1.560	0.9169	-0.0712	-0.73
1.565	0.9168	-0.0714	-0.73
1.570	0.9166	-0.0714	-0.73
1.575	0.9167	-0.0716	-0.73
1.580	0.9165	-0.0715	-0.73
1.585	0.9163	-0.0718	-0.73
1.590	0.9162	-0.0719	-0.73
1.595	0.9161	-0.0721	-0.73
1.600	0.9161	-0.0721	-0.73
1.605	0.9160	-0.0723	-0.74
1.610	0.9158	-0.0723	-0.74
1.615	0.9157	-0.0725	-0.74
1.620	0.9157	-0.0727	-0.74
1.625	0.9157	-0.0729	-0.74
1.630	0.9153	-0.0730	-0.74
1.635	0.9153	-0.0731	-0.74
1.640	0.9152	-0.0731	-0.74
1.645	0.9150	-0.0734	-0.74
1.650	0.9148	-0.0736	-0.75
1.655	0.9148	-0.0736	-0.75
1.660	0.9147	-0.0738	-0.75
1.665	0.9147	-0.0740	-0.75
1.670	0.9143	-0.0741	-0.75
1.675	0.9143	-0.0743	-0.75
1.680	0.9143	-0.0743	-0.75
1.685	0.9139	-0.0744	-0.75
1.690	0.9141	-0.0746	-0.75
1.695	0.9138	-0.0746	-0.75
1.700	0.9137	-0.0748	-0.75
1.705	0.9137	-0.0749	-0.76
1.710	0.9134	-0.0750	-0.76
1.715	0.9132	-0.0750	-0.76
1.720	0.9132	-0.0754	-0.76
1.725	0.9130	-0.0753	-0.76
1.730	0.9130	-0.0754	-0.76
1.735	0.9129	-0.0756	-0.76
1.740	0.9128	-0.0757	-0.76

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.745	0.9125	-0.0760	-0.77
1.750	0.9124	-0.0759	-0.77
1.755	0.9125	-0.0761	-0.77
1.760	0.9123	-0.0762	-0.77
1.765	0.9123	-0.0763	-0.77
1.770	0.9121	-0.0764	-0.77
1.775	0.9122	-0.0766	-0.77
1.780	0.9120	-0.0766	-0.77
1.785	0.9119	-0.0768	-0.77
1.790	0.9118	-0.0768	-0.77
1.795	0.9117	-0.0769	-0.77
1.800	0.9115	-0.0770	-0.77
1.805	0.9115	-0.0772	-0.77
1.810	0.9113	-0.0772	-0.78
1.815	0.9113	-0.0774	-0.78
1.820	0.9112	-0.0774	-0.78
1.825	0.9111	-0.0776	-0.78
1.830	0.9110	-0.0777	-0.78
1.835	0.9110	-0.0779	-0.78
1.840	0.9110	-0.0780	-0.78
1.845	0.9109	-0.0781	-0.78
1.850	0.9108	-0.0780	-0.78
1.855	0.9107	-0.0782	-0.78
1.860	0.9106	-0.0780	-0.78
1.865	0.9106	-0.0781	-0.78
1.870	0.9106	-0.0782	-0.78
1.875	0.9105	-0.0784	-0.78
1.880	0.9104	-0.0786	-0.78
1.885	0.9104	-0.0787	-0.78
1.890	0.9104	-0.0788	-0.78
1.895	0.9103	-0.0789	-0.78
1.900	0.9103	-0.0790	-0.78
1.905	0.9103	-0.0790	-0.78
1.910	0.9101	-0.0791	-0.79
1.915	0.9102	-0.0793	-0.78
1.920	0.9101	-0.0793	-0.79
1.925	0.9102	-0.0795	-0.78
1.930	0.9100	-0.0794	-0.79
1.935	0.9100	-0.0796	-0.79
1.940	0.9099	-0.0797	-0.79
1.945	0.9100	-0.0799	-0.79
1.950	0.9100	-0.0799	-0.79
1.955	0.9100	-0.0800	-0.79
1.960	0.9099	-0.0800	-0.79
1.965	0.9098	-0.0802	-0.79
1.970	0.9099	-0.0803	-0.79

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.975	0.9100	-0.0804	-0.79
1.980	0.9099	-0.0804	-0.79
1.985	0.9099	-0.0804	-0.79
1.990	0.9098	-0.0806	-0.79
1.995	0.9099	-0.0806	-0.79
2.000	0.9099	-0.0808	-0.79
2.005	0.9099	-0.0807	-0.79
2.010	0.9099	-0.0809	-0.79
2.015	0.9097	-0.0810	-0.79
2.020	0.9099	-0.0811	-0.79
2.025	0.9100	-0.0812	-0.79
2.030	0.9099	-0.0812	-0.79
2.035	0.9101	-0.0814	-0.78
2.040	0.9101	-0.0814	-0.78
2.045	0.9099	-0.0816	-0.79
2.050	0.9101	-0.0816	-0.78
2.055	0.9102	-0.0816	-0.78
2.060	0.9101	-0.0819	-0.78
2.065	0.9102	-0.0818	-0.78
2.070	0.9100	-0.0821	-0.78
2.075	0.9100	-0.0821	-0.78
2.080	0.9101	-0.0822	-0.78
2.085	0.9101	-0.0824	-0.78
2.090	0.9101	-0.0824	-0.78
2.095	0.9100	-0.0825	-0.78
2.100	0.9100	-0.0828	-0.78
2.105	0.9102	-0.0829	-0.78
2.110	0.9100	-0.0828	-0.78
2.115	0.9101	-0.0830	-0.78
2.120	0.9099	-0.0831	-0.78
2.125	0.9100	-0.0833	-0.78
2.130	0.9100	-0.0835	-0.78
2.135	0.9100	-0.0835	-0.78
2.140	0.9099	-0.0837	-0.78
2.145	0.9098	-0.0839	-0.78
2.150	0.9098	-0.0840	-0.78
2.155	0.9099	-0.0842	-0.78
2.160	0.9097	-0.0843	-0.78
2.165	0.9097	-0.0844	-0.78
2.170	0.9096	-0.0846	-0.79
2.175	0.9097	-0.0847	-0.78
2.180	0.9096	-0.0849	-0.79
2.185	0.9096	-0.0850	-0.79
2.190	0.9094	-0.0852	-0.79
2.195	0.9093	-0.0855	-0.79
2.200	0.9092	-0.0856	-0.79

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.205	0.9092	-0.0857	-0.79
2.210	0.9093	-0.0859	-0.79
2.215	0.9091	-0.0860	-0.79
2.220	0.9090	-0.0862	-0.79
2.225	0.9090	-0.0864	-0.79
2.230	0.9089	-0.0866	-0.79
2.235	0.9089	-0.0869	-0.79
2.240	0.9087	-0.0869	-0.79
2.245	0.9086	-0.0872	-0.79
2.250	0.9084	-0.0874	-0.79
2.255	0.9085	-0.0874	-0.79
2.260	0.9084	-0.0877	-0.79
2.265	0.9083	-0.0879	-0.79
2.270	0.9081	-0.0880	-0.80
2.275	0.9080	-0.0882	-0.80
2.280	0.9079	-0.0884	-0.80
2.285	0.9078	-0.0886	-0.80
2.290	0.9077	-0.0888	-0.80
2.295	0.9077	-0.0888	-0.80
2.300	0.9073	-0.0889	-0.80
2.305	0.9075	-0.0894	-0.80
2.310	0.9071	-0.0893	-0.81
2.315	0.9072	-0.0895	-0.80
2.320	0.9070	-0.0898	-0.81
2.325	0.9069	-0.0901	-0.81
2.330	0.9070	-0.0903	-0.80
2.335	0.9068	-0.0904	-0.81
2.340	0.9066	-0.0904	-0.81
2.345	0.9066	-0.0908	-0.81
2.350	0.9065	-0.0910	-0.81
2.355	0.9062	-0.0912	-0.81
2.360	0.9062	-0.0913	-0.81
2.365	0.9062	-0.0914	-0.81
2.370	0.9059	-0.0916	-0.81
2.375	0.9057	-0.0916	-0.82
2.380	0.9058	-0.0920	-0.81
2.385	0.9055	-0.0921	-0.82
2.390	0.9056	-0.0923	-0.82
2.395	0.9054	-0.0926	-0.82
2.400	0.9053	-0.0929	-0.82
2.405	0.9051	-0.0929	-0.82
2.410	0.9050	-0.0931	-0.82
2.415	0.9050	-0.0933	-0.82
2.420	0.9050	-0.0935	-0.82
2.425	0.9047	-0.0936	-0.82
2.430	0.9047	-0.0936	-0.82

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.435	0.9046	-0.0938	-0.82
2.440	0.9044	-0.0940	-0.83
2.445	0.9044	-0.0942	-0.83
2.450	0.9044	-0.0943	-0.83
2.455	0.9042	-0.0945	-0.83
2.460	0.9042	-0.0946	-0.83
2.465	0.9042	-0.0947	-0.83
2.470	0.9040	-0.0948	-0.83
2.475	0.9040	-0.0949	-0.83
2.480	0.9039	-0.0951	-0.83
2.485	0.9039	-0.0952	-0.83
2.490	0.9037	-0.0953	-0.83
2.495	0.9037	-0.0955	-0.83
2.500	0.9036	-0.0954	-0.83
2.505	0.9035	-0.0957	-0.83
2.510	0.9034	-0.0958	-0.83
2.515	0.9034	-0.0959	-0.83
2.520	0.9033	-0.0961	-0.83
2.525	0.9031	-0.0962	-0.84
2.530	0.9032	-0.0964	-0.84
2.535	0.9031	-0.0964	-0.84
2.540	0.9030	-0.0965	-0.84
2.545	0.9029	-0.0967	-0.84
2.550	0.9028	-0.0968	-0.84
2.555	0.9028	-0.0968	-0.84
2.560	0.9028	-0.0970	-0.84
2.565	0.9027	-0.0970	-0.84
2.570	0.9027	-0.0970	-0.84
2.575	0.9027	-0.0971	-0.84
2.580	0.9026	-0.0971	-0.84
2.585	0.9028	-0.0972	-0.84
2.590	0.9028	-0.0972	-0.84
2.595	0.9029	-0.0975	-0.84
2.600	0.9029	-0.0976	-0.84
2.605	0.9029	-0.0976	-0.84
2.610	0.9029	-0.0977	-0.84
2.615	0.9030	-0.0979	-0.84
2.620	0.9031	-0.0981	-0.83
2.625	0.9030	-0.0982	-0.83
2.630	0.9031	-0.0984	-0.83
2.635	0.9032	-0.0981	-0.83
2.640	0.9031	-0.0985	-0.83
2.645	0.9031	-0.0985	-0.83
2.650	0.9032	-0.0987	-0.83
2.655	0.9030	-0.0987	-0.83
2.660	0.9032	-0.0990	-0.83

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.665	0.9033	-0.0989	-0.83
2.670	0.9031	-0.0988	-0.83
2.675	0.9032	-0.0993	-0.83
2.680	0.9032	-0.0993	-0.83
2.685	0.9032	-0.0995	-0.83
2.690	0.9032	-0.0995	-0.83
2.695	0.9032	-0.0997	-0.83
2.700	0.9033	-0.0999	-0.83
2.705	0.9032	-0.0999	-0.83
2.710	0.9035	-0.1001	-0.83
2.715	0.9035	-0.1002	-0.83
2.720	0.9035	-0.1003	-0.83
2.725	0.9034	-0.1003	-0.83
2.730	0.9034	-0.1004	-0.83
2.735	0.9035	-0.1005	-0.83
2.740	0.9034	-0.1005	-0.83
2.745	0.9037	-0.1008	-0.83
2.750	0.9036	-0.1007	-0.83
2.755	0.9036	-0.1008	-0.83
2.760	0.9037	-0.1008	-0.83
2.765	0.9035	-0.1010	-0.83
2.770	0.9037	-0.1010	-0.83
2.775	0.9034	-0.1011	-0.83
2.780	0.9035	-0.1011	-0.83
2.785	0.9035	-0.1013	-0.83
2.790	0.9035	-0.1012	-0.83
2.795	0.9035	-0.1015	-0.83
2.800	0.9035	-0.1015	-0.83
2.805	0.9035	-0.1017	-0.83
2.810	0.9033	-0.1018	-0.83
2.815	0.9034	-0.1020	-0.83
2.820	0.9033	-0.1020	-0.83
2.825	0.9032	-0.1020	-0.83
2.830	0.9032	-0.1021	-0.83
2.835	0.9033	-0.1022	-0.83
2.840	0.9032	-0.1024	-0.83
2.845	0.9032	-0.1027	-0.83
2.850	0.9031	-0.1026	-0.83
2.855	0.9031	-0.1027	-0.83
2.860	0.9031	-0.1030	-0.83
2.865	0.9030	-0.1031	-0.83
2.870	0.9030	-0.1033	-0.83
2.875	0.9030	-0.1033	-0.83
2.880	0.9030	-0.1034	-0.83
2.885	0.9031	-0.1036	-0.83
2.890	0.9032	-0.1036	-0.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.895	0.9028	-0.1036	-0.83
2.900	0.9028	-0.1037	-0.83
2.905	0.9029	-0.1038	-0.83
2.910	0.9027	-0.1039	-0.83
2.915	0.9026	-0.1041	-0.83
2.920	0.9025	-0.1041	-0.83
2.925	0.9023	-0.1042	-0.84
2.930	0.9021	-0.1042	-0.84
2.935	0.9022	-0.1045	-0.84
2.940	0.9020	-0.1045	-0.84
2.945	0.9019	-0.1049	-0.84
2.950	0.9019	-0.1051	-0.84
2.955	0.9020	-0.1049	-0.84
2.960	0.9018	-0.1057	-0.84
2.965	0.9016	-0.1053	-0.84
2.970	0.9014	-0.1056	-0.84
2.975	0.9016	-0.1058	-0.84
2.980	0.9015	-0.1060	-0.84
2.985	0.9015	-0.1060	-0.84
2.990	0.9017	-0.1063	-0.84
2.995	0.9016	-0.1063	-0.84
3.000	0.9016	-0.1066	-0.84
3.005	0.9012	-0.1064	-0.84
3.010	0.9012	-0.1069	-0.84
3.015	0.9011	-0.1068	-0.84
3.020	0.9013	-0.1071	-0.84
3.025	0.9014	-0.1073	-0.84
3.030	0.9014	-0.1071	-0.84
3.035	0.9018	-0.1073	-0.84
3.040	0.9017	-0.1073	-0.84
3.045	0.9019	-0.1075	-0.84
3.050	0.9015	-0.1076	-0.84
3.055	0.9022	-0.1075	-0.83
3.060	0.9019	-0.1074	-0.84
3.065	0.9020	-0.1072	-0.84
3.070	0.9020	-0.1072	-0.84
3.075	0.9019	-0.1071	-0.84
3.080	0.9019	-0.1071	-0.84
3.085	0.9016	-0.1072	-0.84
3.090	0.9017	-0.1070	-0.84
3.095	0.9016	-0.1070	-0.84
3.100	0.9014	-0.1070	-0.84
3.105	0.9011	-0.1072	-0.84
3.110	0.9012	-0.1073	-0.84
3.115	0.9010	-0.1073	-0.84
3.120	0.9010	-0.1074	-0.84

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.125	0.9009	-0.1076	-0.84
3.130	0.9008	-0.1077	-0.85
3.135	0.9006	-0.1081	-0.85
3.140	0.9007	-0.1081	-0.85
3.145	0.9006	-0.1084	-0.85
3.150	0.9006	-0.1085	-0.85
3.155	0.9007	-0.1088	-0.85
3.160	0.9007	-0.1088	-0.85
3.165	0.9010	-0.1090	-0.84
3.170	0.9009	-0.1090	-0.84
3.175	0.9009	-0.1092	-0.84
3.180	0.9012	-0.1095	-0.84
3.185	0.9014	-0.1098	-0.84
3.190	0.9015	-0.1093	-0.84
3.195	0.9015	-0.1096	-0.84
3.200	0.9015	-0.1097	-0.84
3.205	0.9019	-0.1097	-0.83
3.210	0.9018	-0.1095	-0.83
3.215	0.9018	-0.1094	-0.83
3.220	0.9019	-0.1090	-0.83
3.225	0.9019	-0.1095	-0.83
3.230	0.9015	-0.1094	-0.84
3.235	0.9016	-0.1096	-0.84
3.240	0.9014	-0.1098	-0.84
3.245	0.9015	-0.1097	-0.84
3.250	0.9017	-0.1103	-0.83
3.255	0.9014	-0.1103	-0.84
3.260	0.9016	-0.1104	-0.83
3.265	0.9018	-0.1106	-0.83
3.270	0.9014	-0.1106	-0.84
3.275	0.9016	-0.1108	-0.83
3.280	0.9018	-0.1109	-0.83
3.285	0.9017	-0.1116	-0.83
3.290	0.9015	-0.1111	-0.83
3.295	0.9016	-0.1115	-0.83
3.300	0.9019	-0.1115	-0.83
3.305	0.9021	-0.1116	-0.83
3.310	0.9021	-0.1117	-0.83
3.315	0.9020	-0.1120	-0.83
3.320	0.9023	-0.1120	-0.83
3.325	0.9026	-0.1121	-0.82
3.330	0.9023	-0.1125	-0.83
3.335	0.9023	-0.1126	-0.83
3.340	0.9025	-0.1127	-0.82
3.345	0.9024	-0.1128	-0.82
3.350	0.9024	-0.1130	-0.82

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.355	0.9026	-0.1129	-0.82
3.360	0.9026	-0.1131	-0.82
3.365	0.9028	-0.1131	-0.82
3.370	0.9028	-0.1135	-0.82
3.375	0.9029	-0.1131	-0.82
3.380	0.9029	-0.1132	-0.82
3.385	0.9029	-0.1135	-0.82
3.390	0.9029	-0.1135	-0.82
3.395	0.9029	-0.1136	-0.82
3.400	0.9029	-0.1138	-0.82
3.405	0.9028	-0.1137	-0.82
3.410	0.9029	-0.1137	-0.82
3.415	0.9028	-0.1139	-0.82
3.420	0.9027	-0.1138	-0.82
3.425	0.9027	-0.1138	-0.82
3.430	0.9027	-0.1139	-0.82
3.435	0.9027	-0.1140	-0.82
3.440	0.9024	-0.1139	-0.82
3.445	0.9023	-0.1143	-0.82
3.450	0.9020	-0.1147	-0.83
3.455	0.9021	-0.1145	-0.83
3.460	0.9020	-0.1148	-0.83
3.465	0.9016	-0.1149	-0.83
3.470	0.9016	-0.1151	-0.83
3.475	0.9014	-0.1152	-0.83
3.480	0.9012	-0.1153	-0.83
3.485	0.9009	-0.1154	-0.84
3.490	0.9014	-0.1162	-0.83
3.495	0.9013	-0.1166	-0.83
3.500	0.9011	-0.1166	-0.83
3.505	0.9014	-0.1165	-0.83
3.510	0.9010	-0.1167	-0.83
3.515	0.9015	-0.1171	-0.83
3.520	0.9012	-0.1172	-0.83
3.525	0.9016	-0.1181	-0.83
3.530	0.9021	-0.1180	-0.82
3.535	0.9020	-0.1180	-0.82
3.540	0.9022	-0.1179	-0.82
3.545	0.9027	-0.1177	-0.82
3.550	0.9023	-0.1175	-0.82
3.555	0.9026	-0.1175	-0.82
3.560	0.9026	-0.1169	-0.82
3.565	0.9025	-0.1167	-0.82
3.570	0.9020	-0.1167	-0.82
3.575	0.9020	-0.1163	-0.82
3.580	0.9016	-0.1155	-0.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.585	0.9014	-0.1153	-0.83
3.590	0.8994	-0.1153	-0.85
3.595	0.8989	-0.1157	-0.85
3.600	0.8986	-0.1162	-0.86
3.605	0.8982	-0.1166	-0.86
3.610	0.8981	-0.1171	-0.86
3.615	0.8979	-0.1183	-0.86
3.620	0.8977	-0.1189	-0.86
3.625	0.8980	-0.1191	-0.86
3.630	0.8979	-0.1198	-0.86
3.635	0.8985	-0.1198	-0.85
3.640	0.8989	-0.1196	-0.85
3.645	0.8989	-0.1197	-0.85
3.650	0.8993	-0.1197	-0.85
3.655	0.8992	-0.1199	-0.85
3.660	0.8994	-0.1197	-0.84
3.665	0.8994	-0.1197	-0.84
3.670	0.8993	-0.1196	-0.85
3.675	0.8995	-0.1195	-0.84
3.680	0.8993	-0.1194	-0.85
3.685	0.8993	-0.1193	-0.85
3.690	0.8993	-0.1194	-0.85
3.695	0.8993	-0.1194	-0.85
3.700	0.8992	-0.1190	-0.85
3.705	0.8990	-0.1189	-0.85
3.710	0.8987	-0.1190	-0.85
3.715	0.8984	-0.1187	-0.86
3.720	0.8980	-0.1185	-0.86
3.725	0.8977	-0.1188	-0.86
3.730	0.8970	-0.1189	-0.87
3.735	0.8964	-0.1194	-0.87
3.740	0.8960	-0.1198	-0.88
3.745	0.8958	-0.1207	-0.88
3.750	0.8958	-0.1217	-0.88
3.755	0.8963	-0.1227	-0.87
3.760	0.8970	-0.1235	-0.86
3.765	0.8980	-0.1239	-0.85
3.770	0.8995	-0.1235	-0.84
3.775	0.9001	-0.1229	-0.83
3.780	0.9011	-0.1229	-0.82
3.785	0.9007	-0.1210	-0.83
3.790	0.8996	-0.1202	-0.84
3.795	0.8994	-0.1202	-0.84
3.800	0.8993	-0.1202	-0.85
3.805	0.8984	-0.1200	-0.85
3.810	0.8980	-0.1203	-0.86

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.815	0.8973	-0.1207	-0.86
3.820	0.8974	-0.1206	-0.86
3.825	0.8970	-0.1210	-0.87
3.830	0.8970	-0.1214	-0.87
3.835	0.8927	-0.1242	-0.90
3.840	0.8971	-0.1227	-0.86
3.845	0.8973	-0.1228	-0.86
3.850	0.8970	-0.1230	-0.86
3.855	0.8973	-0.1231	-0.86
3.860	0.8972	-0.1233	-0.86
3.865	0.8971	-0.1236	-0.86
3.870	0.8974	-0.1236	-0.86
3.875	0.8972	-0.1239	-0.86
3.880	0.8973	-0.1238	-0.86
3.885	0.8973	-0.1242	-0.86
3.890	0.8974	-0.1244	-0.86
3.895	0.8973	-0.1244	-0.86
3.900	0.8973	-0.1245	-0.86
3.905	0.8972	-0.1249	-0.86
3.910	0.8972	-0.1247	-0.86
3.915	0.8972	-0.1252	-0.86
3.920	0.8973	-0.1254	-0.86
3.925	0.8972	-0.1256	-0.86
3.930	0.8972	-0.1258	-0.86
3.935	0.8971	-0.1258	-0.86
3.940	0.8971	-0.1263	-0.86
3.945	0.8968	-0.1265	-0.86
3.950	0.8968	-0.1267	-0.86
3.955	0.8969	-0.1269	-0.86
3.960	0.8968	-0.1271	-0.86
3.965	0.8967	-0.1273	-0.86
3.970	0.8967	-0.1275	-0.86
3.975	0.8968	-0.1279	-0.86
3.980	0.8966	-0.1280	-0.86
3.985	0.8966	-0.1282	-0.86
3.990	0.8964	-0.1284	-0.86
3.995	0.8963	-0.1288	-0.86
4.000	0.8962	-0.1290	-0.86
4.005	0.8962	-0.1293	-0.86
4.010	0.8961	-0.1297	-0.86
4.015	0.8961	-0.1298	-0.86
4.020	0.8958	-0.1300	-0.87
4.025	0.8956	-0.1301	-0.87
4.030	0.8957	-0.1306	-0.87
4.035	0.8956	-0.1309	-0.87
4.040	0.8954	-0.1310	-0.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.045	0.8950	-0.1311	-0.87
4.050	0.8950	-0.1317	-0.87
4.055	0.8947	-0.1319	-0.87
4.060	0.8947	-0.1323	-0.87
4.065	0.8942	-0.1324	-0.88
4.070	0.8944	-0.1326	-0.87
4.075	0.8942	-0.1330	-0.88
4.080	0.8943	-0.1334	-0.88
4.085	0.8939	-0.1334	-0.88
4.090	0.8937	-0.1338	-0.88
4.095	0.8938	-0.1342	-0.88
4.100	0.8936	-0.1343	-0.88
4.105	0.8931	-0.1346	-0.88
4.110	0.8930	-0.1350	-0.88
4.115	0.8930	-0.1353	-0.88
4.120	0.8926	-0.1355	-0.89
4.125	0.8928	-0.1358	-0.89
4.130	0.8925	-0.1362	-0.89
4.135	0.8921	-0.1366	-0.89
4.140	0.8917	-0.1365	-0.89
4.145	0.8917	-0.1369	-0.89
4.150	0.8914	-0.1375	-0.90
4.155	0.8910	-0.1378	-0.90
4.160	0.8907	-0.1378	-0.90
4.165	0.8904	-0.1383	-0.91
4.170	0.8900	-0.1384	-0.91
4.175	0.8894	-0.1382	-0.91
4.180	0.8897	-0.1385	-0.91
4.185	0.8891	-0.1387	-0.92
4.190	0.8890	-0.1390	-0.92
4.195	0.8889	-0.1392	-0.92
4.200	0.8888	-0.1390	-0.92
4.205	0.8888	-0.1395	-0.92
4.210	0.8887	-0.1399	-0.92
4.215	0.8886	-0.1394	-0.92
4.220	0.8885	-0.1399	-0.92
4.225	0.8884	-0.1395	-0.92
4.230	0.8881	-0.1401	-0.92
4.235	0.8881	-0.1402	-0.92
4.240	0.8884	-0.1405	-0.92
4.245	0.8882	-0.1406	-0.92
4.250	0.8880	-0.1407	-0.92
4.255	0.8881	-0.1409	-0.92
4.260	0.8878	-0.1411	-0.93
4.265	0.8879	-0.1409	-0.92
4.270	0.8877	-0.1413	-0.93

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.275	0.8877	-0.1414	-0.93
4.280	0.8877	-0.1415	-0.93
4.285	0.8876	-0.1417	-0.93
4.290	0.8876	-0.1418	-0.93
4.295	0.8875	-0.1418	-0.93
4.300	0.8875	-0.1420	-0.93
4.305	0.8875	-0.1421	-0.93
4.310	0.8874	-0.1422	-0.93
4.315	0.8873	-0.1423	-0.93
4.320	0.8873	-0.1423	-0.93
4.325	0.8872	-0.1424	-0.93
4.330	0.8872	-0.1425	-0.93
4.335	0.8871	-0.1426	-0.93
4.340	0.8870	-0.1427	-0.93
4.345	0.8869	-0.1430	-0.93
4.350	0.8868	-0.1431	-0.93
4.355	0.8868	-0.1432	-0.93
4.360	0.8868	-0.1434	-0.93
4.365	0.8868	-0.1437	-0.93
4.370	0.8868	-0.1436	-0.93
4.375	0.8871	-0.1440	-0.93
4.380	0.8871	-0.1441	-0.93
4.385	0.8869	-0.1445	-0.93
4.390	0.8867	-0.1444	-0.93
4.395	0.8870	-0.1447	-0.93
4.400	0.8872	-0.1448	-0.93
4.405	0.8874	-0.1453	-0.92
4.410	0.8874	-0.1451	-0.92
4.415	0.8873	-0.1450	-0.92
4.420	0.8877	-0.1450	-0.92
4.425	0.8878	-0.1449	-0.92
4.430	0.8875	-0.1451	-0.92
4.435	0.8874	-0.1454	-0.92
4.440	0.8874	-0.1454	-0.92
4.445	0.8873	-0.1455	-0.92
4.450	0.8868	-0.1454	-0.93
4.455	0.8870	-0.1457	-0.93
4.460	0.8866	-0.1456	-0.93
4.465	0.8866	-0.1453	-0.93
4.470	0.8868	-0.1454	-0.93
4.475	0.8868	-0.1450	-0.93
4.480	0.8870	-0.1447	-0.93
4.485	0.8868	-0.1451	-0.93
4.490	0.8874	-0.1452	-0.92
4.495	0.8876	-0.1454	-0.92
4.500	0.8878	-0.1456	-0.92

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.505	0.8880	-0.1456	-0.92
4.510	0.8884	-0.1461	-0.91
4.515	0.8887	-0.1461	-0.91
4.520	0.8888	-0.1463	-0.91
4.525	0.8891	-0.1461	-0.90
4.530	0.8891	-0.1464	-0.91
4.535	0.8895	-0.1465	-0.90
4.540	0.8893	-0.1462	-0.90
4.545	0.8893	-0.1464	-0.90
4.550	0.8893	-0.1463	-0.90
4.555	0.8895	-0.1465	-0.90
4.560	0.8896	-0.1465	-0.90
4.565	0.8898	-0.1465	-0.90
4.570	0.8897	-0.1467	-0.90
4.575	0.8898	-0.1466	-0.90
4.580	0.8899	-0.1466	-0.90
4.585	0.8899	-0.1468	-0.90
4.590	0.8899	-0.1468	-0.90
4.595	0.8900	-0.1470	-0.90
4.600	0.8900	-0.1469	-0.90
4.605	0.8899	-0.1472	-0.90
4.610	0.8899	-0.1471	-0.90
4.615	0.8898	-0.1474	-0.90
4.620	0.8897	-0.1473	-0.90
4.625	0.8898	-0.1477	-0.90
4.630	0.8897	-0.1476	-0.90
4.635	0.8895	-0.1478	-0.90
4.640	0.8896	-0.1480	-0.90
4.645	0.8894	-0.1478	-0.90
4.650	0.8896	-0.1481	-0.90
4.655	0.8893	-0.1481	-0.90
4.660	0.8894	-0.1482	-0.90
4.665	0.8894	-0.1484	-0.90
4.670	0.8894	-0.1487	-0.90
4.675	0.8892	-0.1488	-0.90
4.680	0.8892	-0.1486	-0.90
4.685	0.8892	-0.1487	-0.90
4.690	0.8895	-0.1487	-0.90
4.695	0.8891	-0.1488	-0.90
4.700	0.8894	-0.1490	-0.90
4.705	0.8888	-0.1496	-0.90
4.710	0.8892	-0.1495	-0.90
4.715	0.8892	-0.1494	-0.90
4.720	0.8890	-0.1496	-0.90
4.725	0.8890	-0.1493	-0.90
4.730	0.8890	-0.1493	-0.90

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.735	0.8893	-0.1492	-0.90
4.740	0.8891	-0.1493	-0.90
4.745	0.8887	-0.1495	-0.90
4.750	0.8888	-0.1494	-0.90
4.755	0.8888	-0.1497	-0.90
4.760	0.8884	-0.1500	-0.91
4.765	0.8887	-0.1497	-0.90
4.770	0.8882	-0.1501	-0.91
4.775	0.8878	-0.1507	-0.91
4.780	0.8876	-0.1506	-0.91
4.785	0.8882	-0.1509	-0.91
4.790	0.8874	-0.1508	-0.91
4.795	0.8874	-0.1511	-0.91
4.800	0.8874	-0.1512	-0.91
4.805	0.8868	-0.1510	-0.92
4.810	0.8875	-0.1513	-0.91
4.815	0.8870	-0.1509	-0.92
4.820	0.8867	-0.1511	-0.92
4.825	0.8870	-0.1508	-0.92
4.830	0.8866	-0.1512	-0.92
4.835	0.8870	-0.1510	-0.92
4.840	0.8869	-0.1508	-0.92
4.845	0.8867	-0.1507	-0.92
4.850	0.8867	-0.1508	-0.92
4.855	0.8867	-0.1503	-0.92
4.860	0.8866	-0.1505	-0.92
4.865	0.8868	-0.1499	-0.92
4.870	0.8865	-0.1503	-0.92
4.875	0.8867	-0.1503	-0.92
4.880	0.8868	-0.1504	-0.92
4.885	0.8865	-0.1504	-0.92
4.890	0.8867	-0.1505	-0.92
4.895	0.8866	-0.1504	-0.92
4.900	0.8868	-0.1505	-0.92
4.905	0.8868	-0.1505	-0.92
4.910	0.8868	-0.1507	-0.92
4.915	0.8869	-0.1508	-0.92
4.920	0.8870	-0.1509	-0.92
4.925	0.8870	-0.1510	-0.92
4.930	0.8870	-0.1512	-0.92
4.935	0.8870	-0.1513	-0.92
4.940	0.8872	-0.1514	-0.92
4.945	0.8871	-0.1516	-0.92
4.950	0.8872	-0.1517	-0.91
4.955	0.8874	-0.1518	-0.91
4.960	0.8873	-0.1520	-0.91

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.965	0.8873	-0.1520	-0.91
4.970	0.8875	-0.1522	-0.91
4.975	0.8876	-0.1522	-0.91
4.980	0.8876	-0.1526	-0.91
4.985	0.8874	-0.1528	-0.91
4.990	0.8877	-0.1529	-0.91
4.995	0.8878	-0.1533	-0.91
5.000	0.8879	-0.1534	-0.90
5.005	0.8879	-0.1537	-0.90
5.010	0.8878	-0.1540	-0.91
5.015	0.8879	-0.1542	-0.90
5.020	0.8877	-0.1543	-0.91
5.025	0.8877	-0.1549	-0.90
5.030	0.8876	-0.1551	-0.91
5.035	0.8876	-0.1549	-0.90
5.040	0.8875	-0.1552	-0.91
5.045	0.8873	-0.1551	-0.91
5.050	0.8871	-0.1551	-0.91
5.055	0.8872	-0.1547	-0.91
5.060	0.8874	-0.1548	-0.91
5.065	0.8873	-0.1544	-0.91
5.070	0.8875	-0.1545	-0.91
5.075	0.8876	-0.1545	-0.91
5.080	0.8876	-0.1545	-0.91
5.085	0.8876	-0.1543	-0.91
5.090	0.8881	-0.1543	-0.90
5.095	0.8882	-0.1543	-0.90
5.100	0.8883	-0.1542	-0.90
5.105	0.8887	-0.1542	-0.90
5.110	0.8886	-0.1544	-0.90
5.115	0.8884	-0.1545	-0.90
5.120	0.8888	-0.1545	-0.89
5.125	0.8887	-0.1546	-0.90
5.130	0.8889	-0.1546	-0.89
5.135	0.8889	-0.1546	-0.89
5.140	0.8892	-0.1547	-0.89
5.145	0.8891	-0.1550	-0.89
5.150	0.8894	-0.1550	-0.89
5.155	0.8893	-0.1552	-0.89
5.160	0.8893	-0.1553	-0.89
5.165	0.8894	-0.1554	-0.89
5.170	0.8893	-0.1555	-0.89
5.175	0.8893	-0.1557	-0.89
5.180	0.8893	-0.1556	-0.89
5.185	0.8894	-0.1558	-0.89
5.190	0.8895	-0.1558	-0.89

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.195	0.8893	-0.1559	-0.89
5.200	0.8894	-0.1559	-0.89
5.205	0.8894	-0.1560	-0.89
5.210	0.8894	-0.1562	-0.89
5.215	0.8895	-0.1563	-0.89
5.220	0.8894	-0.1564	-0.89
5.225	0.8894	-0.1566	-0.89
5.230	0.8893	-0.1566	-0.89
5.235	0.8894	-0.1569	-0.88
5.240	0.8892	-0.1571	-0.89
5.245	0.8893	-0.1570	-0.89
5.250	0.8893	-0.1573	-0.89
5.255	0.8892	-0.1575	-0.89
5.260	0.8889	-0.1574	-0.89
5.265	0.8890	-0.1574	-0.89
5.270	0.8888	-0.1577	-0.89
5.275	0.8885	-0.1579	-0.89
5.280	0.8885	-0.1580	-0.89
5.285	0.8882	-0.1580	-0.89
5.290	0.8881	-0.1585	-0.89
5.295	0.8878	-0.1583	-0.90
5.300	0.8877	-0.1587	-0.90
5.305	0.8875	-0.1588	-0.90
5.310	0.8874	-0.1593	-0.90
5.315	0.8873	-0.1593	-0.90
5.320	0.8868	-0.1596	-0.90
5.325	0.8870	-0.1595	-0.90
5.330	0.8868	-0.1598	-0.90
5.335	0.8868	-0.1602	-0.90
5.340	0.8868	-0.1603	-0.90
5.345	0.8872	-0.1606	-0.90
5.350	0.8866	-0.1604	-0.91
5.355	0.8863	-0.1606	-0.91
5.360	0.8859	-0.1610	-0.91
5.365	0.8858	-0.1607	-0.91
5.370	0.8857	-0.1612	-0.91
5.375	0.8855	-0.1609	-0.91
5.380	0.8855	-0.1612	-0.91
5.385	0.8853	-0.1615	-0.92
5.390	0.8851	-0.1614	-0.92
5.395	0.8847	-0.1616	-0.92
5.400	0.8846	-0.1617	-0.92
5.405	0.8844	-0.1623	-0.92
5.410	0.8842	-0.1622	-0.93
5.415	0.8834	-0.1628	-0.93
5.420	0.8838	-0.1628	-0.93

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.425	0.8834	-0.1633	-0.93
5.430	0.8833	-0.1630	-0.93
5.435	0.8830	-0.1633	-0.93
5.440	0.8831	-0.1636	-0.93
5.445	0.8827	-0.1635	-0.94
5.450	0.8827	-0.1640	-0.94
5.455	0.8828	-0.1641	-0.94
5.460	0.8824	-0.1642	-0.94
5.465	0.8823	-0.1641	-0.94
5.470	0.8822	-0.1647	-0.94
5.475	0.8818	-0.1646	-0.94
5.480	0.8819	-0.1648	-0.94
5.485	0.8816	-0.1649	-0.94
5.490	0.8815	-0.1652	-0.95
5.495	0.8812	-0.1652	-0.95
5.500	0.8811	-0.1653	-0.95
5.505	0.8809	-0.1654	-0.95
5.510	0.8806	-0.1656	-0.95
5.515	0.8804	-0.1657	-0.96
5.520	0.8801	-0.1660	-0.96
5.525	0.8801	-0.1659	-0.96
5.530	0.8800	-0.1662	-0.96
5.535	0.8798	-0.1663	-0.96
5.540	0.8798	-0.1665	-0.96
5.545	0.8795	-0.1668	-0.96
5.550	0.8794	-0.1668	-0.96
5.555	0.8793	-0.1669	-0.96
5.560	0.8792	-0.1671	-0.96
5.565	0.8792	-0.1671	-0.96
5.570	0.8792	-0.1672	-0.96
5.575	0.8791	-0.1673	-0.97
5.580	0.8791	-0.1674	-0.96
5.585	0.8789	-0.1676	-0.97
5.590	0.8787	-0.1676	-0.97
5.595	0.8785	-0.1679	-0.97
5.600	0.8784	-0.1679	-0.97
5.605	0.8782	-0.1681	-0.97
5.610	0.8780	-0.1685	-0.97
5.615	0.8776	-0.1687	-0.98
5.620	0.8776	-0.1691	-0.98
5.625	0.8778	-0.1693	-0.97
5.630	0.8775	-0.1697	-0.98
5.635	0.8775	-0.1700	-0.98
5.640	0.8773	-0.1702	-0.98
5.645	0.8774	-0.1706	-0.98
5.650	0.8777	-0.1709	-0.97

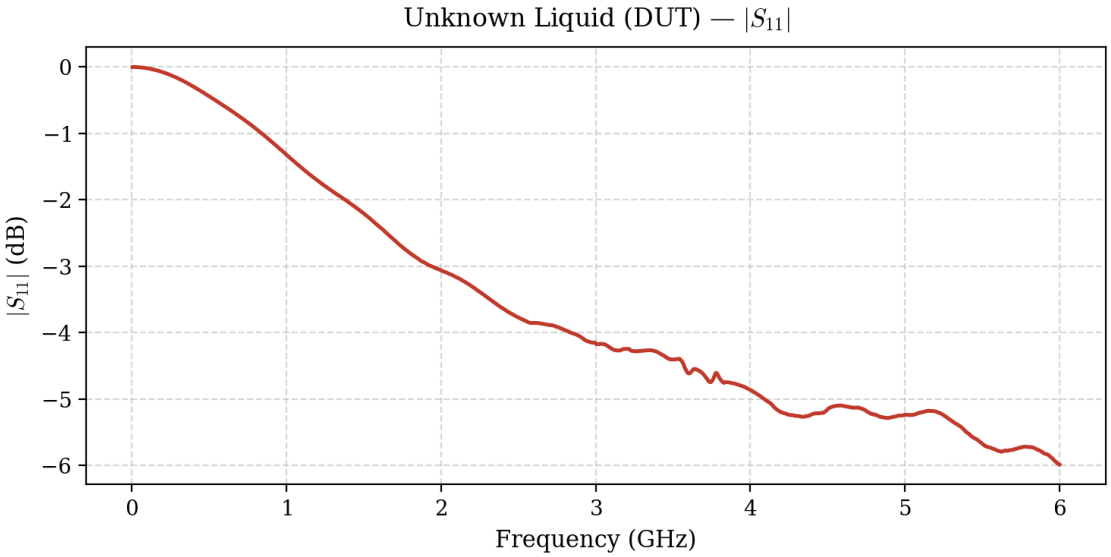
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.655	0.8773	-0.1712	-0.97
5.660	0.8776	-0.1713	-0.97
5.665	0.8776	-0.1712	-0.97
5.670	0.8775	-0.1721	-0.97
5.675	0.8773	-0.1720	-0.97
5.680	0.8775	-0.1724	-0.97
5.685	0.8772	-0.1726	-0.97
5.690	0.8773	-0.1732	-0.97
5.695	0.8777	-0.1734	-0.97
5.700	0.8772	-0.1737	-0.97
5.705	0.8770	-0.1741	-0.97
5.710	0.8773	-0.1740	-0.97
5.715	0.8771	-0.1742	-0.97
5.720	0.8773	-0.1747	-0.97
5.725	0.8772	-0.1752	-0.97
5.730	0.8772	-0.1754	-0.97
5.735	0.8774	-0.1754	-0.97
5.740	0.8777	-0.1759	-0.96
5.745	0.8780	-0.1759	-0.96
5.750	0.8779	-0.1765	-0.96
5.755	0.8778	-0.1764	-0.96
5.760	0.8777	-0.1768	-0.96
5.765	0.8779	-0.1768	-0.96
5.770	0.8776	-0.1770	-0.96
5.775	0.8779	-0.1771	-0.96
5.780	0.8779	-0.1776	-0.96
5.785	0.8781	-0.1779	-0.95
5.790	0.8781	-0.1781	-0.95
5.795	0.8781	-0.1781	-0.95
5.800	0.8781	-0.1786	-0.95
5.805	0.8780	-0.1785	-0.95
5.810	0.8781	-0.1788	-0.95
5.815	0.8781	-0.1789	-0.95
5.820	0.8780	-0.1790	-0.95
5.825	0.8779	-0.1793	-0.95
5.830	0.8779	-0.1795	-0.95
5.835	0.8778	-0.1795	-0.95
5.840	0.8777	-0.1797	-0.95
5.845	0.8777	-0.1801	-0.95
5.850	0.8775	-0.1800	-0.96
5.855	0.8774	-0.1802	-0.96
5.860	0.8772	-0.1805	-0.96
5.865	0.8772	-0.1805	-0.96
5.870	0.8768	-0.1807	-0.96
5.875	0.8767	-0.1809	-0.96
5.880	0.8766	-0.1811	-0.96

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Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.885	0.8762	-0.1812	-0.97
5.890	0.8764	-0.1817	-0.96
5.895	0.8760	-0.1818	-0.97
5.900	0.8758	-0.1818	-0.97
5.905	0.8757	-0.1820	-0.97
5.910	0.8754	-0.1823	-0.97
5.915	0.8753	-0.1825	-0.97
5.920	0.8751	-0.1828	-0.97
5.925	0.8747	-0.1829	-0.98
5.930	0.8749	-0.1833	-0.97
5.935	0.8744	-0.1833	-0.98
5.940	0.8744	-0.1835	-0.98
5.945	0.8743	-0.1831	-0.98
5.950	0.8739	-0.1838	-0.98
5.955	0.8737	-0.1835	-0.99
5.960	0.8734	-0.1838	-0.99
5.965	0.8735	-0.1840	-0.99
5.970	0.8729	-0.1842	-0.99
5.975	0.8716	-0.1840	-1.00
5.980	0.8722	-0.1839	-1.00
5.985	0.8715	-0.1848	-1.00
5.990	0.8713	-0.1851	-1.01
5.995	0.8709	-0.1854	-1.01
6.000	0.8707	-0.1854	-1.01

3.5 Unknown Liquid (DUT)



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	0.9999	-0.0023	-0.00
0.010	1.0000	-0.0044	0.00
0.015	1.0000	-0.0066	-0.00
0.020	0.9999	-0.0089	-0.00
0.025	0.9997	-0.0111	-0.00
0.030	0.9997	-0.0134	-0.00
0.035	0.9997	-0.0158	-0.00
0.040	0.9995	-0.0179	-0.00
0.045	0.9993	-0.0201	-0.00
0.050	0.9992	-0.0223	-0.00
0.055	0.9989	-0.0246	-0.01
0.060	0.9988	-0.0268	-0.01
0.065	0.9986	-0.0291	-0.01
0.070	0.9984	-0.0313	-0.01
0.075	0.9983	-0.0333	-0.01
0.080	0.9979	-0.0356	-0.01
0.085	0.9977	-0.0380	-0.01
0.090	0.9973	-0.0402	-0.02
0.095	0.9970	-0.0423	-0.02
0.100	0.9967	-0.0445	-0.02
0.105	0.9963	-0.0468	-0.02
0.110	0.9961	-0.0490	-0.02
0.115	0.9957	-0.0512	-0.03
0.120	0.9955	-0.0533	-0.03
0.125	0.9949	-0.0556	-0.03
0.130	0.9944	-0.0576	-0.03
0.135	0.9941	-0.0599	-0.04
0.140	0.9937	-0.0622	-0.04
0.145	0.9932	-0.0643	-0.04
0.150	0.9927	-0.0664	-0.04
0.155	0.9925	-0.0685	-0.05
0.160	0.9918	-0.0707	-0.05
0.165	0.9912	-0.0727	-0.05
0.170	0.9908	-0.0749	-0.06
0.175	0.9902	-0.0771	-0.06
0.180	0.9895	-0.0792	-0.06
0.185	0.9891	-0.0813	-0.07
0.190	0.9884	-0.0835	-0.07
0.195	0.9878	-0.0855	-0.07
0.200	0.9872	-0.0876	-0.08
0.205	0.9866	-0.0898	-0.08
0.210	0.9860	-0.0918	-0.09
0.215	0.9853	-0.0939	-0.09
0.220	0.9845	-0.0960	-0.09
0.225	0.9840	-0.0981	-0.10
0.230	0.9833	-0.1003	-0.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.235	0.9826	-0.1024	-0.11
0.240	0.9818	-0.1043	-0.11
0.245	0.9810	-0.1063	-0.12
0.250	0.9803	-0.1083	-0.12
0.255	0.9795	-0.1104	-0.13
0.260	0.9787	-0.1125	-0.13
0.265	0.9778	-0.1145	-0.14
0.270	0.9771	-0.1165	-0.14
0.275	0.9763	-0.1186	-0.14
0.280	0.9754	-0.1205	-0.15
0.285	0.9745	-0.1225	-0.16
0.290	0.9739	-0.1244	-0.16
0.295	0.9728	-0.1263	-0.17
0.300	0.9720	-0.1282	-0.17
0.305	0.9710	-0.1303	-0.18
0.310	0.9702	-0.1322	-0.18
0.315	0.9693	-0.1342	-0.19
0.320	0.9682	-0.1361	-0.20
0.325	0.9674	-0.1380	-0.20
0.330	0.9664	-0.1399	-0.21
0.335	0.9655	-0.1417	-0.21
0.340	0.9644	-0.1435	-0.22
0.345	0.9635	-0.1454	-0.23
0.350	0.9625	-0.1472	-0.23
0.355	0.9616	-0.1491	-0.24
0.360	0.9605	-0.1509	-0.24
0.365	0.9595	-0.1526	-0.25
0.370	0.9584	-0.1545	-0.26
0.375	0.9573	-0.1563	-0.26
0.380	0.9564	-0.1581	-0.27
0.385	0.9553	-0.1598	-0.28
0.390	0.9543	-0.1615	-0.28
0.395	0.9532	-0.1632	-0.29
0.400	0.9522	-0.1650	-0.30
0.405	0.9512	-0.1668	-0.30
0.410	0.9501	-0.1684	-0.31
0.415	0.9490	-0.1702	-0.32
0.420	0.9478	-0.1718	-0.32
0.425	0.9468	-0.1734	-0.33
0.430	0.9457	-0.1748	-0.34
0.435	0.9447	-0.1764	-0.35
0.440	0.9435	-0.1781	-0.35
0.445	0.9425	-0.1796	-0.36
0.450	0.9413	-0.1814	-0.37
0.455	0.9402	-0.1830	-0.37
0.460	0.9391	-0.1845	-0.38

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.465	0.9382	-0.1861	-0.39
0.470	0.9369	-0.1875	-0.40
0.475	0.9357	-0.1891	-0.40
0.480	0.9345	-0.1904	-0.41
0.485	0.9335	-0.1920	-0.42
0.490	0.9323	-0.1935	-0.43
0.495	0.9312	-0.1951	-0.43
0.500	0.9300	-0.1965	-0.44
0.505	0.9289	-0.1979	-0.45
0.510	0.9277	-0.1993	-0.46
0.515	0.9267	-0.2007	-0.46
0.520	0.9255	-0.2021	-0.47
0.525	0.9244	-0.2034	-0.48
0.530	0.9234	-0.2047	-0.48
0.535	0.9222	-0.2062	-0.49
0.540	0.9211	-0.2075	-0.50
0.545	0.9199	-0.2090	-0.51
0.550	0.9187	-0.2103	-0.51
0.555	0.9176	-0.2118	-0.52
0.560	0.9164	-0.2130	-0.53
0.565	0.9152	-0.2143	-0.54
0.570	0.9142	-0.2155	-0.54
0.575	0.9130	-0.2168	-0.55
0.580	0.9119	-0.2180	-0.56
0.585	0.9108	-0.2193	-0.57
0.590	0.9096	-0.2207	-0.58
0.595	0.9083	-0.2218	-0.58
0.600	0.9072	-0.2231	-0.59
0.605	0.9060	-0.2243	-0.60
0.610	0.9050	-0.2255	-0.61
0.615	0.9037	-0.2266	-0.61
0.620	0.9027	-0.2278	-0.62
0.625	0.9015	-0.2289	-0.63
0.630	0.9003	-0.2301	-0.64
0.635	0.8991	-0.2313	-0.65
0.640	0.8979	-0.2325	-0.65
0.645	0.8968	-0.2337	-0.66
0.650	0.8956	-0.2348	-0.67
0.655	0.8943	-0.2358	-0.68
0.660	0.8934	-0.2369	-0.68
0.665	0.8922	-0.2379	-0.69
0.670	0.8909	-0.2390	-0.70
0.675	0.8898	-0.2401	-0.71
0.680	0.8886	-0.2410	-0.72
0.685	0.8874	-0.2422	-0.73
0.690	0.8862	-0.2433	-0.73

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.695	0.8851	-0.2442	-0.74
0.700	0.8839	-0.2451	-0.75
0.705	0.8826	-0.2462	-0.76
0.710	0.8816	-0.2472	-0.77
0.715	0.8803	-0.2483	-0.78
0.720	0.8791	-0.2492	-0.78
0.725	0.8778	-0.2502	-0.79
0.730	0.8766	-0.2513	-0.80
0.735	0.8754	-0.2522	-0.81
0.740	0.8741	-0.2531	-0.82
0.745	0.8730	-0.2540	-0.83
0.750	0.8719	-0.2549	-0.83
0.755	0.8707	-0.2557	-0.84
0.760	0.8694	-0.2566	-0.85
0.765	0.8682	-0.2576	-0.86
0.770	0.8668	-0.2586	-0.87
0.775	0.8657	-0.2595	-0.88
0.780	0.8645	-0.2605	-0.89
0.785	0.8631	-0.2614	-0.90
0.790	0.8620	-0.2624	-0.91
0.795	0.8607	-0.2631	-0.91
0.800	0.8595	-0.2639	-0.92
0.805	0.8581	-0.2648	-0.93
0.810	0.8568	-0.2656	-0.94
0.815	0.8556	-0.2664	-0.95
0.820	0.8544	-0.2674	-0.96
0.825	0.8530	-0.2683	-0.97
0.830	0.8518	-0.2691	-0.98
0.835	0.8505	-0.2699	-0.99
0.840	0.8492	-0.2707	-1.00
0.845	0.8480	-0.2716	-1.01
0.850	0.8467	-0.2723	-1.02
0.855	0.8455	-0.2730	-1.03
0.860	0.8441	-0.2739	-1.04
0.865	0.8428	-0.2747	-1.05
0.870	0.8415	-0.2756	-1.06
0.875	0.8401	-0.2764	-1.07
0.880	0.8389	-0.2771	-1.08
0.885	0.8376	-0.2778	-1.09
0.890	0.8361	-0.2786	-1.10
0.895	0.8349	-0.2794	-1.11
0.900	0.8335	-0.2800	-1.12
0.905	0.8323	-0.2809	-1.13
0.910	0.8310	-0.2817	-1.14
0.915	0.8297	-0.2824	-1.15
0.920	0.8284	-0.2831	-1.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.925	0.8270	-0.2838	-1.17
0.930	0.8257	-0.2843	-1.18
0.935	0.8245	-0.2851	-1.19
0.940	0.8230	-0.2859	-1.20
0.945	0.8218	-0.2865	-1.21
0.950	0.8204	-0.2872	-1.22
0.955	0.8191	-0.2880	-1.23
0.960	0.8178	-0.2886	-1.24
0.965	0.8165	-0.2892	-1.25
0.970	0.8153	-0.2899	-1.26
0.975	0.8139	-0.2905	-1.27
0.980	0.8125	-0.2911	-1.28
0.985	0.8112	-0.2918	-1.29
0.990	0.8098	-0.2925	-1.30
0.995	0.8086	-0.2932	-1.31
1.000	0.8072	-0.2937	-1.32
1.005	0.8058	-0.2944	-1.33
1.010	0.8047	-0.2949	-1.34
1.015	0.8034	-0.2954	-1.35
1.020	0.8022	-0.2960	-1.36
1.025	0.8007	-0.2965	-1.37
1.030	0.7995	-0.2972	-1.38
1.035	0.7982	-0.2977	-1.39
1.040	0.7969	-0.2983	-1.40
1.045	0.7956	-0.2989	-1.41
1.050	0.7943	-0.2994	-1.42
1.055	0.7930	-0.2999	-1.43
1.060	0.7918	-0.3004	-1.44
1.065	0.7906	-0.3008	-1.45
1.070	0.7895	-0.3014	-1.46
1.075	0.7884	-0.3019	-1.47
1.080	0.7871	-0.3023	-1.48
1.085	0.7859	-0.3028	-1.49
1.090	0.7845	-0.3033	-1.50
1.095	0.7833	-0.3038	-1.51
1.100	0.7821	-0.3043	-1.52
1.105	0.7810	-0.3047	-1.53
1.110	0.7798	-0.3051	-1.54
1.115	0.7787	-0.3054	-1.55
1.120	0.7776	-0.3059	-1.56
1.125	0.7763	-0.3064	-1.57
1.130	0.7752	-0.3068	-1.58
1.135	0.7740	-0.3072	-1.59
1.140	0.7728	-0.3076	-1.60
1.145	0.7717	-0.3080	-1.61
1.150	0.7706	-0.3083	-1.62

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.155	0.7696	-0.3086	-1.63
1.160	0.7687	-0.3089	-1.64
1.165	0.7676	-0.3092	-1.64
1.170	0.7666	-0.3096	-1.65
1.175	0.7655	-0.3099	-1.66
1.180	0.7642	-0.3104	-1.67
1.185	0.7632	-0.3107	-1.68
1.190	0.7622	-0.3109	-1.69
1.195	0.7612	-0.3111	-1.70
1.200	0.7603	-0.3114	-1.71
1.205	0.7594	-0.3116	-1.71
1.210	0.7584	-0.3118	-1.72
1.215	0.7572	-0.3120	-1.73
1.220	0.7562	-0.3123	-1.74
1.225	0.7553	-0.3127	-1.75
1.230	0.7542	-0.3129	-1.76
1.235	0.7532	-0.3132	-1.77
1.240	0.7524	-0.3133	-1.78
1.245	0.7515	-0.3135	-1.78
1.250	0.7506	-0.3137	-1.79
1.255	0.7497	-0.3139	-1.80
1.260	0.7488	-0.3140	-1.81
1.265	0.7479	-0.3141	-1.82
1.270	0.7470	-0.3144	-1.83
1.275	0.7459	-0.3145	-1.84
1.280	0.7451	-0.3148	-1.84
1.285	0.7442	-0.3150	-1.85
1.290	0.7433	-0.3152	-1.86
1.295	0.7423	-0.3153	-1.87
1.300	0.7415	-0.3154	-1.88
1.305	0.7406	-0.3156	-1.88
1.310	0.7398	-0.3158	-1.89
1.315	0.7389	-0.3160	-1.90
1.320	0.7379	-0.3162	-1.91
1.325	0.7370	-0.3162	-1.92
1.330	0.7362	-0.3164	-1.92
1.335	0.7356	-0.3164	-1.93
1.340	0.7348	-0.3165	-1.94
1.345	0.7340	-0.3167	-1.94
1.350	0.7332	-0.3167	-1.95
1.355	0.7324	-0.3168	-1.96
1.360	0.7314	-0.3170	-1.97
1.365	0.7304	-0.3171	-1.98
1.370	0.7296	-0.3172	-1.99
1.375	0.7289	-0.3174	-1.99
1.380	0.7281	-0.3174	-2.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.385	0.7273	-0.3174	-2.01
1.390	0.7266	-0.3176	-2.01
1.395	0.7257	-0.3175	-2.02
1.400	0.7249	-0.3177	-2.03
1.405	0.7238	-0.3180	-2.04
1.410	0.7228	-0.3181	-2.05
1.415	0.7219	-0.3182	-2.06
1.420	0.7209	-0.3183	-2.07
1.425	0.7203	-0.3184	-2.07
1.430	0.7194	-0.3185	-2.08
1.435	0.7185	-0.3186	-2.09
1.440	0.7178	-0.3187	-2.10
1.445	0.7168	-0.3190	-2.11
1.450	0.7158	-0.3191	-2.12
1.455	0.7149	-0.3194	-2.12
1.460	0.7139	-0.3194	-2.13
1.465	0.7129	-0.3196	-2.14
1.470	0.7121	-0.3198	-2.15
1.475	0.7112	-0.3199	-2.16
1.480	0.7103	-0.3200	-2.17
1.485	0.7094	-0.3201	-2.18
1.490	0.7084	-0.3204	-2.19
1.495	0.7074	-0.3205	-2.20
1.500	0.7064	-0.3207	-2.21
1.505	0.7055	-0.3208	-2.21
1.510	0.7043	-0.3210	-2.23
1.515	0.7033	-0.3212	-2.23
1.520	0.7024	-0.3214	-2.24
1.525	0.7014	-0.3215	-2.25
1.530	0.7005	-0.3216	-2.26
1.535	0.6995	-0.3218	-2.27
1.540	0.6984	-0.3220	-2.28
1.545	0.6973	-0.3221	-2.29
1.550	0.6963	-0.3223	-2.30
1.555	0.6951	-0.3225	-2.31
1.560	0.6940	-0.3227	-2.32
1.565	0.6929	-0.3230	-2.33
1.570	0.6919	-0.3230	-2.34
1.575	0.6909	-0.3233	-2.35
1.580	0.6898	-0.3234	-2.36
1.585	0.6886	-0.3236	-2.37
1.590	0.6875	-0.3239	-2.38
1.595	0.6865	-0.3240	-2.39
1.600	0.6855	-0.3243	-2.40
1.605	0.6845	-0.3244	-2.41
1.610	0.6835	-0.3245	-2.42

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.615	0.6825	-0.3247	-2.43
1.620	0.6814	-0.3249	-2.44
1.625	0.6804	-0.3252	-2.45
1.630	0.6793	-0.3252	-2.46
1.635	0.6781	-0.3253	-2.47
1.640	0.6769	-0.3256	-2.49
1.645	0.6758	-0.3257	-2.50
1.650	0.6747	-0.3260	-2.51
1.655	0.6737	-0.3261	-2.52
1.660	0.6726	-0.3263	-2.53
1.665	0.6714	-0.3264	-2.54
1.670	0.6704	-0.3266	-2.55
1.675	0.6694	-0.3268	-2.56
1.680	0.6682	-0.3270	-2.57
1.685	0.6672	-0.3273	-2.58
1.690	0.6660	-0.3273	-2.59
1.695	0.6652	-0.3273	-2.60
1.700	0.6640	-0.3275	-2.61
1.705	0.6631	-0.3275	-2.62
1.710	0.6621	-0.3277	-2.63
1.715	0.6609	-0.3278	-2.64
1.720	0.6598	-0.3279	-2.65
1.725	0.6586	-0.3282	-2.66
1.730	0.6577	-0.3282	-2.67
1.735	0.6566	-0.3283	-2.69
1.740	0.6556	-0.3285	-2.69
1.745	0.6545	-0.3287	-2.70
1.750	0.6537	-0.3287	-2.71
1.755	0.6528	-0.3288	-2.72
1.760	0.6519	-0.3288	-2.73
1.765	0.6509	-0.3290	-2.74
1.770	0.6501	-0.3291	-2.75
1.775	0.6492	-0.3291	-2.76
1.780	0.6483	-0.3292	-2.77
1.785	0.6475	-0.3290	-2.78
1.790	0.6466	-0.3292	-2.79
1.795	0.6457	-0.3290	-2.80
1.800	0.6449	-0.3291	-2.81
1.805	0.6442	-0.3293	-2.81
1.810	0.6432	-0.3293	-2.82
1.815	0.6425	-0.3295	-2.83
1.820	0.6416	-0.3296	-2.84
1.825	0.6408	-0.3296	-2.85
1.830	0.6401	-0.3296	-2.85
1.835	0.6394	-0.3295	-2.86
1.840	0.6387	-0.3296	-2.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.845	0.6381	-0.3295	-2.88
1.850	0.6374	-0.3293	-2.88
1.855	0.6367	-0.3292	-2.89
1.860	0.6358	-0.3292	-2.90
1.865	0.6350	-0.3292	-2.91
1.870	0.6342	-0.3293	-2.92
1.875	0.6333	-0.3293	-2.93
1.880	0.6329	-0.3294	-2.93
1.885	0.6324	-0.3294	-2.94
1.890	0.6321	-0.3293	-2.94
1.895	0.6315	-0.3292	-2.95
1.900	0.6309	-0.3291	-2.96
1.905	0.6302	-0.3291	-2.96
1.910	0.6295	-0.3292	-2.97
1.915	0.6288	-0.3293	-2.98
1.920	0.6282	-0.3293	-2.98
1.925	0.6277	-0.3292	-2.99
1.930	0.6274	-0.3292	-2.99
1.935	0.6270	-0.3291	-3.00
1.940	0.6266	-0.3290	-3.00
1.945	0.6262	-0.3290	-3.01
1.950	0.6258	-0.3290	-3.01
1.955	0.6251	-0.3290	-3.02
1.960	0.6247	-0.3289	-3.02
1.965	0.6242	-0.3289	-3.03
1.970	0.6237	-0.3288	-3.04
1.975	0.6234	-0.3288	-3.04
1.980	0.6231	-0.3286	-3.04
1.985	0.6228	-0.3284	-3.05
1.990	0.6225	-0.3283	-3.05
1.995	0.6220	-0.3283	-3.06
2.000	0.6214	-0.3282	-3.06
2.005	0.6210	-0.3282	-3.07
2.010	0.6205	-0.3281	-3.07
2.015	0.6202	-0.3280	-3.08
2.020	0.6198	-0.3279	-3.08
2.025	0.6194	-0.3277	-3.09
2.030	0.6191	-0.3278	-3.09
2.035	0.6188	-0.3278	-3.10
2.040	0.6182	-0.3276	-3.10
2.045	0.6177	-0.3276	-3.11
2.050	0.6173	-0.3276	-3.11
2.055	0.6169	-0.3276	-3.12
2.060	0.6164	-0.3274	-3.12
2.065	0.6161	-0.3275	-3.13
2.070	0.6157	-0.3272	-3.13

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.075	0.6154	-0.3271	-3.14
2.080	0.6148	-0.3271	-3.14
2.085	0.6143	-0.3271	-3.15
2.090	0.6136	-0.3271	-3.16
2.095	0.6130	-0.3271	-3.16
2.100	0.6125	-0.3271	-3.17
2.105	0.6121	-0.3271	-3.17
2.110	0.6116	-0.3270	-3.18
2.115	0.6110	-0.3269	-3.19
2.120	0.6105	-0.3270	-3.19
2.125	0.6101	-0.3269	-3.20
2.130	0.6093	-0.3269	-3.20
2.135	0.6087	-0.3270	-3.21
2.140	0.6080	-0.3270	-3.22
2.145	0.6074	-0.3270	-3.22
2.150	0.6068	-0.3271	-3.23
2.155	0.6062	-0.3271	-3.24
2.160	0.6054	-0.3272	-3.25
2.165	0.6049	-0.3272	-3.25
2.170	0.6043	-0.3272	-3.26
2.175	0.6036	-0.3273	-3.27
2.180	0.6030	-0.3272	-3.27
2.185	0.6022	-0.3274	-3.28
2.190	0.6013	-0.3274	-3.29
2.195	0.6006	-0.3277	-3.30
2.200	0.5997	-0.3277	-3.31
2.205	0.5991	-0.3278	-3.31
2.210	0.5983	-0.3278	-3.32
2.215	0.5976	-0.3278	-3.33
2.220	0.5970	-0.3279	-3.34
2.225	0.5961	-0.3279	-3.35
2.230	0.5953	-0.3280	-3.35
2.235	0.5943	-0.3282	-3.36
2.240	0.5936	-0.3284	-3.37
2.245	0.5928	-0.3282	-3.38
2.250	0.5920	-0.3285	-3.39
2.255	0.5914	-0.3285	-3.39
2.260	0.5906	-0.3285	-3.40
2.265	0.5898	-0.3287	-3.41
2.270	0.5890	-0.3288	-3.42
2.275	0.5881	-0.3289	-3.43
2.280	0.5872	-0.3291	-3.44
2.285	0.5864	-0.3293	-3.45
2.290	0.5855	-0.3292	-3.46
2.295	0.5848	-0.3295	-3.46
2.300	0.5841	-0.3294	-3.47

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.305	0.5832	-0.3295	-3.48
2.310	0.5824	-0.3295	-3.49
2.315	0.5816	-0.3297	-3.50
2.320	0.5807	-0.3299	-3.51
2.325	0.5799	-0.3301	-3.51
2.330	0.5790	-0.3301	-3.52
2.335	0.5781	-0.3303	-3.53
2.340	0.5774	-0.3305	-3.54
2.345	0.5766	-0.3307	-3.55
2.350	0.5759	-0.3308	-3.56
2.355	0.5750	-0.3308	-3.56
2.360	0.5743	-0.3308	-3.57
2.365	0.5733	-0.3309	-3.58
2.370	0.5726	-0.3310	-3.59
2.375	0.5719	-0.3312	-3.60
2.380	0.5710	-0.3311	-3.61
2.385	0.5703	-0.3313	-3.62
2.390	0.5696	-0.3314	-3.62
2.395	0.5689	-0.3316	-3.63
2.400	0.5681	-0.3317	-3.64
2.405	0.5675	-0.3319	-3.64
2.410	0.5668	-0.3318	-3.65
2.415	0.5661	-0.3319	-3.66
2.420	0.5652	-0.3320	-3.67
2.425	0.5646	-0.3321	-3.67
2.430	0.5639	-0.3322	-3.68
2.435	0.5632	-0.3322	-3.69
2.440	0.5627	-0.3322	-3.70
2.445	0.5622	-0.3323	-3.70
2.450	0.5617	-0.3322	-3.71
2.455	0.5611	-0.3321	-3.72
2.460	0.5604	-0.3322	-3.72
2.465	0.5598	-0.3322	-3.73
2.470	0.5591	-0.3322	-3.74
2.475	0.5586	-0.3322	-3.74
2.480	0.5581	-0.3322	-3.75
2.485	0.5576	-0.3321	-3.75
2.490	0.5571	-0.3320	-3.76
2.495	0.5566	-0.3320	-3.77
2.500	0.5561	-0.3320	-3.77
2.505	0.5557	-0.3321	-3.78
2.510	0.5551	-0.3320	-3.78
2.515	0.5546	-0.3320	-3.79
2.520	0.5542	-0.3320	-3.79
2.525	0.5538	-0.3320	-3.80
2.530	0.5532	-0.3319	-3.81

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.535	0.5528	-0.3319	-3.81
2.540	0.5523	-0.3319	-3.82
2.545	0.5518	-0.3320	-3.82
2.550	0.5514	-0.3319	-3.83
2.555	0.5510	-0.3318	-3.83
2.560	0.5505	-0.3317	-3.84
2.565	0.5502	-0.3314	-3.85
2.570	0.5499	-0.3311	-3.85
2.575	0.5499	-0.3307	-3.85
2.580	0.5499	-0.3305	-3.86
2.585	0.5500	-0.3304	-3.85
2.590	0.5500	-0.3301	-3.86
2.595	0.5499	-0.3302	-3.86
2.600	0.5500	-0.3304	-3.85
2.605	0.5499	-0.3305	-3.86
2.610	0.5499	-0.3306	-3.86
2.615	0.5498	-0.3305	-3.86
2.620	0.5499	-0.3304	-3.86
2.625	0.5499	-0.3305	-3.86
2.630	0.5497	-0.3304	-3.86
2.635	0.5496	-0.3303	-3.86
2.640	0.5493	-0.3303	-3.86
2.645	0.5491	-0.3303	-3.87
2.650	0.5491	-0.3303	-3.87
2.655	0.5490	-0.3303	-3.87
2.660	0.5487	-0.3301	-3.87
2.665	0.5487	-0.3301	-3.87
2.670	0.5485	-0.3301	-3.87
2.675	0.5484	-0.3299	-3.88
2.680	0.5483	-0.3300	-3.88
2.685	0.5480	-0.3299	-3.88
2.690	0.5478	-0.3301	-3.88
2.695	0.5475	-0.3300	-3.89
2.700	0.5475	-0.3300	-3.89
2.705	0.5474	-0.3299	-3.89
2.710	0.5474	-0.3299	-3.89
2.715	0.5475	-0.3297	-3.89
2.720	0.5473	-0.3296	-3.89
2.725	0.5472	-0.3293	-3.89
2.730	0.5470	-0.3293	-3.90
2.735	0.5468	-0.3290	-3.90
2.740	0.5465	-0.3288	-3.91
2.745	0.5463	-0.3286	-3.91
2.750	0.5462	-0.3283	-3.91
2.755	0.5460	-0.3281	-3.92
2.760	0.5457	-0.3279	-3.92

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.765	0.5454	-0.3277	-3.93
2.770	0.5452	-0.3275	-3.93
2.775	0.5448	-0.3274	-3.94
2.780	0.5445	-0.3271	-3.94
2.785	0.5440	-0.3270	-3.95
2.790	0.5438	-0.3269	-3.95
2.795	0.5435	-0.3266	-3.96
2.800	0.5432	-0.3264	-3.96
2.805	0.5429	-0.3263	-3.97
2.810	0.5425	-0.3261	-3.97
2.815	0.5422	-0.3261	-3.98
2.820	0.5418	-0.3261	-3.98
2.825	0.5413	-0.3260	-3.99
2.830	0.5409	-0.3260	-3.99
2.835	0.5404	-0.3259	-4.00
2.840	0.5401	-0.3259	-4.00
2.845	0.5397	-0.3258	-4.01
2.850	0.5394	-0.3258	-4.01
2.855	0.5392	-0.3257	-4.01
2.860	0.5389	-0.3256	-4.02
2.865	0.5387	-0.3255	-4.02
2.870	0.5384	-0.3254	-4.03
2.875	0.5380	-0.3253	-4.03
2.880	0.5375	-0.3251	-4.04
2.885	0.5374	-0.3248	-4.04
2.890	0.5369	-0.3247	-4.05
2.895	0.5367	-0.3242	-4.05
2.900	0.5361	-0.3241	-4.06
2.905	0.5356	-0.3237	-4.07
2.910	0.5353	-0.3234	-4.08
2.915	0.5346	-0.3235	-4.08
2.920	0.5338	-0.3234	-4.09
2.925	0.5331	-0.3235	-4.10
2.930	0.5324	-0.3231	-4.11
2.935	0.5321	-0.3232	-4.12
2.940	0.5315	-0.3233	-4.12
2.945	0.5310	-0.3235	-4.13
2.950	0.5306	-0.3234	-4.13
2.955	0.5300	-0.3237	-4.14
2.960	0.5296	-0.3237	-4.14
2.965	0.5291	-0.3240	-4.15
2.970	0.5290	-0.3240	-4.15
2.975	0.5286	-0.3242	-4.15
2.980	0.5284	-0.3243	-4.15
2.985	0.5285	-0.3244	-4.15
2.990	0.5285	-0.3243	-4.15

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.995	0.5283	-0.3244	-4.15
3.000	0.5281	-0.3243	-4.16
3.005	0.5266	-0.3240	-4.18
3.010	0.5264	-0.3243	-4.18
3.015	0.5264	-0.3243	-4.18
3.020	0.5264	-0.3244	-4.18
3.025	0.5265	-0.3245	-4.17
3.030	0.5267	-0.3245	-4.17
3.035	0.5269	-0.3245	-4.17
3.040	0.5270	-0.3242	-4.17
3.045	0.5270	-0.3240	-4.17
3.050	0.5271	-0.3234	-4.17
3.055	0.5272	-0.3232	-4.17
3.060	0.5270	-0.3228	-4.18
3.065	0.5268	-0.3222	-4.19
3.070	0.5267	-0.3218	-4.19
3.075	0.5264	-0.3212	-4.20
3.080	0.5262	-0.3209	-4.20
3.085	0.5258	-0.3203	-4.21
3.090	0.5253	-0.3199	-4.22
3.095	0.5247	-0.3196	-4.23
3.100	0.5242	-0.3196	-4.24
3.105	0.5237	-0.3193	-4.25
3.110	0.5233	-0.3193	-4.25
3.115	0.5227	-0.3192	-4.26
3.120	0.5223	-0.3192	-4.26
3.125	0.5219	-0.3195	-4.27
3.130	0.5217	-0.3196	-4.27
3.135	0.5214	-0.3199	-4.27
3.140	0.5212	-0.3201	-4.27
3.145	0.5209	-0.3204	-4.27
3.150	0.5207	-0.3207	-4.27
3.155	0.5206	-0.3210	-4.27
3.160	0.5208	-0.3213	-4.27
3.165	0.5209	-0.3215	-4.26
3.170	0.5213	-0.3214	-4.26
3.175	0.5217	-0.3217	-4.25
3.180	0.5219	-0.3219	-4.25
3.185	0.5220	-0.3216	-4.25
3.190	0.5222	-0.3216	-4.25
3.195	0.5224	-0.3214	-4.25
3.200	0.5224	-0.3211	-4.25
3.205	0.5228	-0.3207	-4.25
3.210	0.5230	-0.3201	-4.25
3.215	0.5233	-0.3197	-4.25
3.220	0.5230	-0.3191	-4.26

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.225	0.5224	-0.3187	-4.27
3.230	0.5220	-0.3187	-4.27
3.235	0.5214	-0.3187	-4.28
3.240	0.5213	-0.3189	-4.28
3.245	0.5212	-0.3187	-4.28
3.250	0.5209	-0.3187	-4.28
3.255	0.5208	-0.3188	-4.28
3.260	0.5209	-0.3194	-4.28
3.265	0.5208	-0.3192	-4.28
3.270	0.5207	-0.3192	-4.28
3.275	0.5208	-0.3198	-4.28
3.280	0.5206	-0.3197	-4.28
3.285	0.5209	-0.3198	-4.28
3.290	0.5206	-0.3201	-4.28
3.295	0.5209	-0.3202	-4.27
3.300	0.5207	-0.3201	-4.28
3.305	0.5207	-0.3202	-4.27
3.310	0.5206	-0.3204	-4.27
3.315	0.5208	-0.3204	-4.27
3.320	0.5209	-0.3205	-4.27
3.325	0.5209	-0.3208	-4.27
3.330	0.5210	-0.3206	-4.27
3.335	0.5210	-0.3206	-4.27
3.340	0.5211	-0.3206	-4.27
3.345	0.5212	-0.3204	-4.27
3.350	0.5213	-0.3203	-4.27
3.355	0.5214	-0.3201	-4.27
3.360	0.5215	-0.3200	-4.27
3.365	0.5215	-0.3198	-4.27
3.370	0.5213	-0.3197	-4.27
3.375	0.5212	-0.3194	-4.28
3.380	0.5210	-0.3192	-4.28
3.385	0.5211	-0.3189	-4.28
3.390	0.5208	-0.3187	-4.29
3.395	0.5208	-0.3183	-4.29
3.400	0.5205	-0.3179	-4.29
3.405	0.5203	-0.3176	-4.30
3.410	0.5199	-0.3173	-4.31
3.415	0.5195	-0.3169	-4.31
3.420	0.5190	-0.3168	-4.32
3.425	0.5188	-0.3164	-4.33
3.430	0.5183	-0.3163	-4.33
3.435	0.5178	-0.3159	-4.34
3.440	0.5174	-0.3160	-4.35
3.445	0.5170	-0.3157	-4.35
3.450	0.5164	-0.3153	-4.36

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.455	0.5159	-0.3154	-4.37
3.460	0.5152	-0.3154	-4.38
3.465	0.5145	-0.3153	-4.39
3.470	0.5138	-0.3158	-4.39
3.475	0.5135	-0.3156	-4.40
3.480	0.5128	-0.3160	-4.40
3.485	0.5129	-0.3162	-4.40
3.490	0.5123	-0.3165	-4.41
3.495	0.5121	-0.3165	-4.41
3.500	0.5119	-0.3170	-4.41
3.505	0.5119	-0.3174	-4.40
3.510	0.5117	-0.3173	-4.41
3.515	0.5118	-0.3174	-4.40
3.520	0.5118	-0.3175	-4.40
3.525	0.5120	-0.3175	-4.40
3.530	0.5120	-0.3178	-4.40
3.535	0.5131	-0.3166	-4.39
3.540	0.5132	-0.3160	-4.40
3.545	0.5136	-0.3153	-4.40
3.550	0.5132	-0.3140	-4.41
3.555	0.5131	-0.3128	-4.42
3.560	0.5123	-0.3116	-4.44
3.565	0.5116	-0.3105	-4.46
3.570	0.5102	-0.3092	-4.49
3.575	0.5083	-0.3072	-4.53
3.580	0.5071	-0.3069	-4.54
3.585	0.5051	-0.3067	-4.57
3.590	0.5033	-0.3071	-4.59
3.595	0.5015	-0.3079	-4.61
3.600	0.5001	-0.3090	-4.61
3.605	0.4990	-0.3105	-4.62
3.610	0.4984	-0.3118	-4.61
3.615	0.4988	-0.3129	-4.60
3.620	0.4993	-0.3138	-4.59
3.625	0.5000	-0.3144	-4.57
3.630	0.5008	-0.3146	-4.56
3.635	0.5020	-0.3144	-4.55
3.640	0.5022	-0.3139	-4.55
3.645	0.5024	-0.3135	-4.55
3.650	0.5027	-0.3129	-4.55
3.655	0.5026	-0.3120	-4.56
3.660	0.5026	-0.3114	-4.57
3.665	0.5024	-0.3109	-4.57
3.670	0.5022	-0.3101	-4.58
3.675	0.5021	-0.3097	-4.58
3.680	0.5020	-0.3088	-4.59

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.685	0.5017	-0.3083	-4.60
3.690	0.5011	-0.3077	-4.61
3.695	0.5006	-0.3070	-4.62
3.700	0.4999	-0.3064	-4.64
3.705	0.4992	-0.3059	-4.65
3.710	0.4981	-0.3055	-4.67
3.715	0.4972	-0.3050	-4.68
3.720	0.4961	-0.3049	-4.70
3.725	0.4948	-0.3051	-4.71
3.730	0.4932	-0.3052	-4.73
3.735	0.4919	-0.3059	-4.74
3.740	0.4908	-0.3071	-4.75
3.745	0.4900	-0.3084	-4.75
3.750	0.4897	-0.3099	-4.74
3.755	0.4898	-0.3115	-4.72
3.760	0.4907	-0.3131	-4.70
3.765	0.4921	-0.3144	-4.67
3.770	0.4938	-0.3146	-4.65
3.775	0.4960	-0.3153	-4.62
3.780	0.4982	-0.3130	-4.61
3.785	0.4984	-0.3111	-4.62
3.790	0.4985	-0.3091	-4.63
3.795	0.4976	-0.3075	-4.66
3.800	0.4963	-0.3063	-4.68
3.805	0.4955	-0.3057	-4.70
3.810	0.4945	-0.3054	-4.71
3.815	0.4932	-0.3053	-4.73
3.820	0.4923	-0.3054	-4.74
3.825	0.4912	-0.3058	-4.75
3.830	0.4904	-0.3062	-4.76
3.835	0.4899	-0.3084	-4.75
3.840	0.4897	-0.3086	-4.75
3.845	0.4898	-0.3086	-4.75
3.850	0.4896	-0.3088	-4.75
3.855	0.4894	-0.3090	-4.75
3.860	0.4894	-0.3089	-4.75
3.865	0.4891	-0.3089	-4.75
3.870	0.4888	-0.3092	-4.76
3.875	0.4886	-0.3092	-4.76
3.880	0.4883	-0.3093	-4.76
3.885	0.4880	-0.3094	-4.76
3.890	0.4878	-0.3096	-4.77
3.895	0.4876	-0.3095	-4.77
3.900	0.4874	-0.3097	-4.77
3.905	0.4871	-0.3096	-4.77
3.910	0.4867	-0.3098	-4.78

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.915	0.4862	-0.3100	-4.78
3.920	0.4858	-0.3103	-4.79
3.925	0.4854	-0.3104	-4.79
3.930	0.4851	-0.3105	-4.79
3.935	0.4848	-0.3106	-4.79
3.940	0.4843	-0.3107	-4.80
3.945	0.4839	-0.3107	-4.81
3.950	0.4835	-0.3110	-4.81
3.955	0.4831	-0.3110	-4.81
3.960	0.4827	-0.3112	-4.82
3.965	0.4822	-0.3113	-4.82
3.970	0.4817	-0.3114	-4.83
3.975	0.4812	-0.3115	-4.83
3.980	0.4806	-0.3116	-4.84
3.985	0.4799	-0.3118	-4.85
3.990	0.4794	-0.3119	-4.85
3.995	0.4788	-0.3121	-4.86
4.000	0.4783	-0.3123	-4.86
4.005	0.4776	-0.3126	-4.87
4.010	0.4772	-0.3125	-4.88
4.015	0.4765	-0.3125	-4.88
4.020	0.4758	-0.3127	-4.89
4.025	0.4751	-0.3128	-4.90
4.030	0.4745	-0.3130	-4.91
4.035	0.4739	-0.3131	-4.91
4.040	0.4732	-0.3131	-4.92
4.045	0.4726	-0.3132	-4.93
4.050	0.4721	-0.3133	-4.93
4.055	0.4713	-0.3136	-4.94
4.060	0.4706	-0.3136	-4.95
4.065	0.4700	-0.3135	-4.96
4.070	0.4691	-0.3138	-4.97
4.075	0.4685	-0.3138	-4.98
4.080	0.4679	-0.3139	-4.98
4.085	0.4670	-0.3139	-4.99
4.090	0.4664	-0.3139	-5.00
4.095	0.4657	-0.3142	-5.01
4.100	0.4650	-0.3142	-5.02
4.105	0.4640	-0.3147	-5.03
4.110	0.4634	-0.3148	-5.03
4.115	0.4626	-0.3148	-5.04
4.120	0.4619	-0.3151	-5.05
4.125	0.4613	-0.3151	-5.06
4.130	0.4603	-0.3157	-5.07
4.135	0.4596	-0.3158	-5.07
4.140	0.4586	-0.3158	-5.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.145	0.4572	-0.3161	-5.10
4.150	0.4562	-0.3161	-5.11
4.155	0.4554	-0.3159	-5.13
4.160	0.4545	-0.3158	-5.14
4.165	0.4539	-0.3156	-5.15
4.170	0.4533	-0.3156	-5.16
4.175	0.4527	-0.3156	-5.16
4.180	0.4522	-0.3153	-5.17
4.185	0.4516	-0.3152	-5.18
4.190	0.4512	-0.3150	-5.19
4.195	0.4507	-0.3151	-5.19
4.200	0.4501	-0.3151	-5.20
4.205	0.4498	-0.3153	-5.20
4.210	0.4494	-0.3152	-5.21
4.215	0.4491	-0.3155	-5.21
4.220	0.4488	-0.3153	-5.22
4.225	0.4484	-0.3158	-5.22
4.230	0.4482	-0.3156	-5.22
4.235	0.4478	-0.3156	-5.23
4.240	0.4472	-0.3158	-5.23
4.245	0.4468	-0.3160	-5.24
4.250	0.4466	-0.3160	-5.24
4.255	0.4465	-0.3161	-5.24
4.260	0.4461	-0.3161	-5.24
4.265	0.4460	-0.3163	-5.24
4.270	0.4460	-0.3164	-5.24
4.275	0.4458	-0.3165	-5.24
4.280	0.4455	-0.3166	-5.25
4.285	0.4452	-0.3168	-5.25
4.290	0.4450	-0.3167	-5.25
4.295	0.4448	-0.3168	-5.26
4.300	0.4446	-0.3168	-5.26
4.305	0.4446	-0.3170	-5.26
4.310	0.4445	-0.3168	-5.26
4.315	0.4445	-0.3168	-5.26
4.320	0.4443	-0.3168	-5.26
4.325	0.4440	-0.3169	-5.26
4.330	0.4436	-0.3171	-5.27
4.335	0.4433	-0.3173	-5.27
4.340	0.4431	-0.3175	-5.27
4.345	0.4431	-0.3176	-5.27
4.350	0.4432	-0.3178	-5.27
4.355	0.4433	-0.3179	-5.26
4.360	0.4433	-0.3182	-5.26
4.365	0.4433	-0.3185	-5.26
4.370	0.4432	-0.3186	-5.26

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.375	0.4432	-0.3191	-5.25
4.380	0.4431	-0.3194	-5.25
4.385	0.4434	-0.3197	-5.25
4.390	0.4436	-0.3197	-5.24
4.395	0.4440	-0.3199	-5.24
4.400	0.4445	-0.3200	-5.23
4.405	0.4448	-0.3199	-5.23
4.410	0.4450	-0.3201	-5.22
4.415	0.4451	-0.3200	-5.22
4.420	0.4454	-0.3201	-5.22
4.425	0.4454	-0.3200	-5.22
4.430	0.4454	-0.3201	-5.22
4.435	0.4454	-0.3199	-5.22
4.440	0.4455	-0.3199	-5.22
4.445	0.4457	-0.3197	-5.22
4.450	0.4461	-0.3196	-5.21
4.455	0.4463	-0.3193	-5.21
4.460	0.4463	-0.3193	-5.21
4.465	0.4465	-0.3193	-5.21
4.470	0.4465	-0.3191	-5.21
4.475	0.4468	-0.3194	-5.21
4.480	0.4472	-0.3194	-5.20
4.485	0.4477	-0.3198	-5.19
4.490	0.4482	-0.3200	-5.18
4.495	0.4488	-0.3204	-5.17
4.500	0.4496	-0.3207	-5.16
4.505	0.4501	-0.3211	-5.15
4.510	0.4506	-0.3213	-5.14
4.515	0.4510	-0.3215	-5.13
4.520	0.4514	-0.3216	-5.13
4.525	0.4516	-0.3216	-5.12
4.530	0.4521	-0.3216	-5.12
4.535	0.4524	-0.3213	-5.12
4.540	0.4527	-0.3212	-5.11
4.545	0.4531	-0.3211	-5.11
4.550	0.4534	-0.3211	-5.11
4.555	0.4535	-0.3211	-5.10
4.560	0.4536	-0.3209	-5.10
4.565	0.4537	-0.3209	-5.10
4.570	0.4539	-0.3209	-5.10
4.575	0.4540	-0.3206	-5.10
4.580	0.4541	-0.3205	-5.10
4.585	0.4543	-0.3205	-5.10
4.590	0.4544	-0.3203	-5.10
4.595	0.4542	-0.3202	-5.10
4.600	0.4542	-0.3201	-5.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.605	0.4539	-0.3201	-5.11
4.610	0.4539	-0.3199	-5.11
4.615	0.4538	-0.3199	-5.11
4.620	0.4537	-0.3196	-5.12
4.625	0.4536	-0.3196	-5.12
4.630	0.4537	-0.3196	-5.12
4.635	0.4535	-0.3194	-5.12
4.640	0.4533	-0.3193	-5.12
4.645	0.4532	-0.3192	-5.12
4.650	0.4529	-0.3191	-5.13
4.655	0.4528	-0.3194	-5.13
4.660	0.4528	-0.3190	-5.13
4.665	0.4527	-0.3189	-5.13
4.670	0.4530	-0.3189	-5.13
4.675	0.4530	-0.3186	-5.13
4.680	0.4530	-0.3187	-5.13
4.685	0.4531	-0.3185	-5.13
4.690	0.4530	-0.3187	-5.13
4.695	0.4534	-0.3185	-5.13
4.700	0.4534	-0.3183	-5.13
4.705	0.4527	-0.3184	-5.14
4.710	0.4531	-0.3178	-5.14
4.715	0.4529	-0.3177	-5.14
4.720	0.4526	-0.3174	-5.15
4.725	0.4526	-0.3173	-5.15
4.730	0.4527	-0.3167	-5.15
4.735	0.4523	-0.3166	-5.16
4.740	0.4521	-0.3162	-5.17
4.745	0.4518	-0.3162	-5.17
4.750	0.4512	-0.3159	-5.18
4.755	0.4506	-0.3157	-5.19
4.760	0.4500	-0.3157	-5.20
4.765	0.4496	-0.3155	-5.20
4.770	0.4490	-0.3157	-5.21
4.775	0.4488	-0.3155	-5.21
4.780	0.4479	-0.3156	-5.23
4.785	0.4479	-0.3159	-5.22
4.790	0.4477	-0.3157	-5.23
4.795	0.4473	-0.3155	-5.23
4.800	0.4472	-0.3154	-5.24
4.805	0.4477	-0.3151	-5.23
4.810	0.4475	-0.3144	-5.24
4.815	0.4476	-0.3142	-5.24
4.820	0.4474	-0.3137	-5.25
4.825	0.4473	-0.3135	-5.25
4.830	0.4470	-0.3130	-5.26

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.835	0.4469	-0.3125	-5.27
4.840	0.4471	-0.3121	-5.27
4.845	0.4471	-0.3117	-5.27
4.850	0.4471	-0.3115	-5.27
4.855	0.4473	-0.3111	-5.27
4.860	0.4471	-0.3108	-5.28
4.865	0.4469	-0.3109	-5.28
4.870	0.4469	-0.3107	-5.28
4.875	0.4469	-0.3106	-5.28
4.880	0.4467	-0.3105	-5.29
4.885	0.4467	-0.3105	-5.29
4.890	0.4466	-0.3107	-5.29
4.895	0.4467	-0.3107	-5.29
4.900	0.4470	-0.3107	-5.28
4.905	0.4469	-0.3108	-5.28
4.910	0.4472	-0.3109	-5.28
4.915	0.4473	-0.3111	-5.28
4.920	0.4474	-0.3112	-5.27
4.925	0.4474	-0.3115	-5.27
4.930	0.4472	-0.3117	-5.27
4.935	0.4472	-0.3121	-5.27
4.940	0.4473	-0.3123	-5.26
4.945	0.4473	-0.3126	-5.26
4.950	0.4473	-0.3129	-5.26
4.955	0.4474	-0.3131	-5.25
4.960	0.4475	-0.3134	-5.25
4.965	0.4475	-0.3136	-5.25
4.970	0.4473	-0.3139	-5.25
4.975	0.4470	-0.3139	-5.25
4.980	0.4469	-0.3143	-5.25
4.985	0.4470	-0.3147	-5.25
4.990	0.4469	-0.3149	-5.24
4.995	0.4471	-0.3150	-5.24
5.000	0.4471	-0.3152	-5.24
5.005	0.4470	-0.3156	-5.24
5.010	0.4467	-0.3158	-5.24
5.015	0.4464	-0.3159	-5.24
5.020	0.4464	-0.3161	-5.24
5.025	0.4463	-0.3162	-5.24
5.030	0.4463	-0.3162	-5.24
5.035	0.4463	-0.3158	-5.24
5.040	0.4466	-0.3155	-5.24
5.045	0.4469	-0.3152	-5.24
5.050	0.4472	-0.3150	-5.24
5.055	0.4473	-0.3148	-5.24
5.060	0.4477	-0.3146	-5.24

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.065	0.4480	-0.3147	-5.23
5.070	0.4486	-0.3142	-5.23
5.075	0.4490	-0.3142	-5.22
5.080	0.4496	-0.3138	-5.22
5.085	0.4501	-0.3138	-5.21
5.090	0.4506	-0.3135	-5.21
5.095	0.4508	-0.3134	-5.21
5.100	0.4512	-0.3135	-5.20
5.105	0.4512	-0.3133	-5.20
5.110	0.4513	-0.3135	-5.20
5.115	0.4513	-0.3135	-5.20
5.120	0.4517	-0.3136	-5.20
5.125	0.4518	-0.3136	-5.19
5.130	0.4521	-0.3137	-5.19
5.135	0.4523	-0.3135	-5.19
5.140	0.4526	-0.3136	-5.18
5.145	0.4527	-0.3137	-5.18
5.150	0.4528	-0.3137	-5.18
5.155	0.4528	-0.3137	-5.18
5.160	0.4527	-0.3139	-5.18
5.165	0.4525	-0.3138	-5.18
5.170	0.4525	-0.3139	-5.18
5.175	0.4525	-0.3138	-5.18
5.180	0.4525	-0.3137	-5.18
5.185	0.4526	-0.3137	-5.18
5.190	0.4525	-0.3136	-5.18
5.195	0.4523	-0.3135	-5.19
5.200	0.4520	-0.3135	-5.19
5.205	0.4519	-0.3135	-5.19
5.210	0.4515	-0.3134	-5.20
5.215	0.4510	-0.3134	-5.21
5.220	0.4510	-0.3132	-5.21
5.225	0.4508	-0.3131	-5.21
5.230	0.4503	-0.3130	-5.22
5.235	0.4502	-0.3127	-5.22
5.240	0.4497	-0.3126	-5.23
5.245	0.4493	-0.3123	-5.24
5.250	0.4489	-0.3122	-5.24
5.255	0.4485	-0.3119	-5.25
5.260	0.4481	-0.3117	-5.26
5.265	0.4477	-0.3115	-5.27
5.270	0.4471	-0.3113	-5.28
5.275	0.4466	-0.3110	-5.28
5.280	0.4459	-0.3107	-5.30
5.285	0.4455	-0.3106	-5.30
5.290	0.4449	-0.3106	-5.31

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.295	0.4444	-0.3106	-5.32
5.300	0.4439	-0.3102	-5.33
5.305	0.4433	-0.3104	-5.33
5.310	0.4427	-0.3101	-5.34
5.315	0.4423	-0.3101	-5.35
5.320	0.4417	-0.3098	-5.36
5.325	0.4414	-0.3097	-5.36
5.330	0.4408	-0.3096	-5.37
5.335	0.4402	-0.3097	-5.38
5.340	0.4398	-0.3096	-5.39
5.345	0.4395	-0.3094	-5.39
5.350	0.4390	-0.3091	-5.40
5.355	0.4387	-0.3088	-5.41
5.360	0.4384	-0.3086	-5.41
5.365	0.4378	-0.3082	-5.43
5.370	0.4373	-0.3078	-5.44
5.375	0.4367	-0.3074	-5.45
5.380	0.4360	-0.3072	-5.46
5.385	0.4351	-0.3071	-5.47
5.390	0.4344	-0.3068	-5.48
5.395	0.4337	-0.3068	-5.49
5.400	0.4332	-0.3064	-5.50
5.405	0.4324	-0.3065	-5.51
5.410	0.4319	-0.3065	-5.52
5.415	0.4316	-0.3065	-5.53
5.420	0.4305	-0.3063	-5.54
5.425	0.4297	-0.3065	-5.55
5.430	0.4289	-0.3063	-5.56
5.435	0.4284	-0.3064	-5.57
5.440	0.4277	-0.3063	-5.58
5.445	0.4272	-0.3064	-5.59
5.450	0.4269	-0.3063	-5.59
5.455	0.4264	-0.3061	-5.60
5.460	0.4262	-0.3062	-5.60
5.465	0.4254	-0.3061	-5.61
5.470	0.4251	-0.3060	-5.62
5.475	0.4245	-0.3057	-5.63
5.480	0.4239	-0.3057	-5.64
5.485	0.4231	-0.3056	-5.65
5.490	0.4227	-0.3054	-5.66
5.495	0.4222	-0.3052	-5.66
5.500	0.4216	-0.3049	-5.67
5.505	0.4209	-0.3049	-5.69
5.510	0.4203	-0.3050	-5.69
5.515	0.4195	-0.3050	-5.70
5.520	0.4187	-0.3052	-5.71

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.525	0.4184	-0.3052	-5.72
5.530	0.4179	-0.3055	-5.72
5.535	0.4175	-0.3055	-5.72
5.540	0.4172	-0.3056	-5.73
5.545	0.4169	-0.3058	-5.73
5.550	0.4166	-0.3058	-5.73
5.555	0.4162	-0.3060	-5.74
5.560	0.4158	-0.3062	-5.74
5.565	0.4154	-0.3062	-5.75
5.570	0.4151	-0.3063	-5.75
5.575	0.4147	-0.3064	-5.75
5.580	0.4143	-0.3065	-5.76
5.585	0.4137	-0.3066	-5.76
5.590	0.4132	-0.3067	-5.77
5.595	0.4127	-0.3070	-5.78
5.600	0.4123	-0.3072	-5.78
5.605	0.4117	-0.3076	-5.78
5.610	0.4111	-0.3078	-5.79
5.615	0.4105	-0.3083	-5.79
5.620	0.4099	-0.3088	-5.79
5.625	0.4094	-0.3094	-5.80
5.630	0.4094	-0.3098	-5.79
5.635	0.4093	-0.3102	-5.79
5.640	0.4092	-0.3108	-5.78
5.645	0.4090	-0.3111	-5.78
5.650	0.4090	-0.3117	-5.78
5.655	0.4086	-0.3119	-5.78
5.660	0.4082	-0.3124	-5.78
5.665	0.4079	-0.3127	-5.78
5.670	0.4078	-0.3133	-5.78
5.675	0.4076	-0.3137	-5.78
5.680	0.4075	-0.3143	-5.77
5.685	0.4073	-0.3144	-5.77
5.690	0.4073	-0.3145	-5.77
5.695	0.4070	-0.3151	-5.77
5.700	0.4069	-0.3156	-5.76
5.705	0.4070	-0.3160	-5.76
5.710	0.4067	-0.3164	-5.76
5.715	0.4068	-0.3168	-5.75
5.720	0.4067	-0.3169	-5.75
5.725	0.4069	-0.3173	-5.75
5.730	0.4068	-0.3176	-5.74
5.735	0.4068	-0.3181	-5.74
5.740	0.4067	-0.3185	-5.74
5.745	0.4068	-0.3189	-5.73
5.750	0.4065	-0.3191	-5.73

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.755	0.4066	-0.3193	-5.73
5.760	0.4067	-0.3195	-5.73
5.765	0.4068	-0.3198	-5.72
5.770	0.4068	-0.3199	-5.72
5.775	0.4069	-0.3200	-5.72
5.780	0.4067	-0.3202	-5.72
5.785	0.4065	-0.3204	-5.72
5.790	0.4063	-0.3204	-5.72
5.795	0.4061	-0.3205	-5.72
5.800	0.4060	-0.3205	-5.73
5.805	0.4060	-0.3206	-5.72
5.810	0.4059	-0.3206	-5.73
5.815	0.4059	-0.3205	-5.73
5.820	0.4059	-0.3205	-5.73
5.825	0.4058	-0.3203	-5.73
5.830	0.4056	-0.3203	-5.73
5.835	0.4053	-0.3203	-5.74
5.840	0.4048	-0.3202	-5.74
5.845	0.4045	-0.3201	-5.75
5.850	0.4041	-0.3200	-5.76
5.855	0.4038	-0.3200	-5.76
5.860	0.4036	-0.3195	-5.77
5.865	0.4034	-0.3196	-5.77
5.870	0.4031	-0.3193	-5.78
5.875	0.4028	-0.3192	-5.78
5.880	0.4024	-0.3191	-5.79
5.885	0.4019	-0.3191	-5.80
5.890	0.4013	-0.3189	-5.81
5.895	0.4009	-0.3187	-5.81
5.900	0.4002	-0.3187	-5.82
5.905	0.3998	-0.3188	-5.83
5.910	0.3997	-0.3188	-5.83
5.915	0.3995	-0.3186	-5.83
5.920	0.3992	-0.3186	-5.84
5.925	0.3988	-0.3186	-5.84
5.930	0.3985	-0.3183	-5.85
5.935	0.3982	-0.3180	-5.86
5.940	0.3974	-0.3176	-5.87
5.945	0.3970	-0.3172	-5.88
5.950	0.3966	-0.3171	-5.89
5.955	0.3959	-0.3166	-5.90
5.960	0.3956	-0.3162	-5.91
5.965	0.3948	-0.3157	-5.93
5.970	0.3940	-0.3159	-5.93
5.975	0.3933	-0.3154	-5.95
5.980	0.3927	-0.3157	-5.95

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.985	0.3922	-0.3152	-5.97
5.990	0.3913	-0.3153	-5.98
5.995	0.3908	-0.3157	-5.98
6.000	0.3902	-0.3156	-5.99